



Efficiency and productivity in container terminal operation:
A case study for the Hamburg – Le Havre range

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Sisangile Nduna, February 2022

Abstract

The container sector's significance dates back to the 1950s. Times of economic turmoil, such as the 2008 financial crisis, the COVID-19 Pandemic, the 2021 Suez Canal blockade, are recent phenomena where the importance of the maritime and port sector became visible to the general public.

As an exchange interface, terminals handle hundreds of millions of containers every year¹ to facilitate trade and globalisation. Efficiency and productivity at container terminals are highly driven by frequent port calls, resilient infrastructure and adequate accessibility through the hinterlands and waterside. Investments in the infrastructure and the superstructure by terminals can save customers a large amount of costs.

This thesis addresses the role of infrastructure and superstructure in evaluating efficiency in container terminals. The role of port competition is addressed to indicate implications of closeby terminals and ports to performance. The implications due to performance can also be observed through costs, an element also investigated in this thesis. The methodological approach used for performance includes non-parametric and parametric methods to conclude. The analysis for productivity and efficiency is through the use of the Malmquist productivity Index (MPI) and the data envelopment analysis (DEA) (Charnes, Cooper and Rhodes (CCR) and Banker, Charnes and Cooper (BCC)). To evaluate the association of the external factors with the relative technical efficiency of the terminals, the Tobit regression and the Kruskal – Wallis models are used. The methods are applied in the Hamburg Le-Havre (HLH) context considering two periods, 2013 and 2018. For the costs analysis, only the terminals at the port of Antwerp are evaluated.

The emphasis in efficient handling of resources is identified as key in the functioning of the business. The external factors are recognised as additional major contributing factors to the container handling business. Through efficient handling of containers, terminal operators can save costs, not only for themselves but also for the rest of the stakeholders. The outcomes indicate that relative efficiency is vital to evaluate, especially in a business environment where competition is high. The HLH is found to maintain a positive trend of productivity and efficiency over the time measured. Scale

¹ In 2020 815.6 million TEUs were handled globally, based on UNCTAD.

inefficiencies contribute negatively to the overall efficiency in the region. The high efficiency and productivity in the region indicate proper management of resources by the terminal managers, which suggests that the capacity expansion investments are paying off. Although it is proven that environmental factors are not highly significant in terminals in close proximity, it is also emphasised that the external factors are generally substantial in shaping cargo flows, which is vital in determining efficiency.

In addressing the measurement and importance of terminal efficiency, this research provides insight to container terminal managers on effective ways to manage resources at their terminals. The research contributes by providing a holistic view of terminal handling to policymakers for large investment decision-making purposes. Research focusing on the terminal level, especially in a region with business operation characteristics such as the HLH, is limited thus this work helps to reduce that gap.

Nederlandstalig Abstract

Het belang van de container sector gaat terug tot de jaren vijftig. Tijden van economische onrust, zoals de financiële crisis van 2008, de COVID-19-pandemie, de blokkade van het Suezkanaal in 2021, zijn recente fenomenen waarbij het belang van de maritieme en havensector zichtbaar werd voor het grote publiek.

Als facilitator voor wereldhandel en globalisatie, behandelen containerterminals jaarlijks honderden miljoenen containers². De efficiëntie en productiviteit van deze containerterminals worden sterk gedreven door de frequentie van havenaanlopen, veerkrach van de infrastructuur en de bereikbaarheid via het achterland. Investerings in de infra- en suprastructuur door terminals kunnen klanten veel kosten besparen.

Dit proefschrift behandelt de rol van infra- en suprastructuur bij het evalueren van efficiëntie in containerterminals. Ook de rol van havenconcurrentie wordt bekeken, om de effecten van de nabijheid van andere terminals en havens voor de prestaties aan te geven. De gebruikte methodologische benadering omvat zowel niet-parametrische als parametrische methoden. De analyse voor productiviteit en efficiëntie gebeurt door het gebruik van de Malmquist-productiviteitsindex (MPI) en de data-envelopmentanalyse (DEA) (Charnes, Cooper en Rhodes (CCR) en Banker, Charnes en Cooper (BCC)). Om de associatie van de externe factoren met de relatieve technische efficiëntie van de terminals te evalueren, worden de Tobit-regressie en de Kruskal – Wallis-modellen gebruikt. De methoden worden toegepast in de Hamburg Le-Havre (HLH)-context over twee periodes, 2013 en 2018. Voor de kostenanalyse worden enkel de terminals in de Antwerpse haven geëvalueerd.

De nadruk op een efficiënte omgang met middelen wordt gezien als de sleutel tot het functioneren van het bedrijf. De externe factoren worden erkend als bijkomende belangrijke factoren die bijdragen aan de containeroverslagactiviteiten. Door een efficiënte afhandeling van containers kunnen terminaloperators kosten besparen, niet alleen voor zichzelf, maar ook voor de rest van de stakeholders. De resultaten

² In 2020 werden wereldwijd 815,6 miljoen TEU's behandeld, op basis van UNCTAD.

geven aan dat het van vitaal belang is om de relatieve efficiëntie te evalueren in een zakelijke omgeving waar de concurrentie hoog is. De HLH blijkt een positieve trend van productiviteit en efficiëntie te handhaven gedurende de gemeten tijd. Schaallinefficiënties dragen negatief bij aan de algehele efficiëntie in de regio. De hoge efficiëntie en productiviteit in de regio wijzen op een goed beheer van middelen door de terminalmanagers, wat erop wijst dat de investeringen in capaciteitsuitbreiding vruchten afwerpen. Hoewel is bewezen dat omgevingsfactoren niet significant zijn in terminals in de buurt, wordt ook benadrukt dat de externe factoren over het algemeen substantieel zijn bij het vormgeven van ladingstromen, wat van vitaal belang is bij het bepalen van de efficiëntie.

Door de meting en het belang van havenefficiëntie aan te pakken, biedt dit onderzoek containerhaven- en terminalmanagers inzicht in effectieve manieren om middelen op hun terminals te beheren. Het onderzoek draagt bij door beleidsmakers een holistisch beeld te geven van de afhandeling van terminals bij het nemen van beslissingen over grote investeringen. Onderzoek gericht op het terminalniveau, vooral in een regio met kenmerken van de bedrijfsvoering zoals de HLH, is beperkt, dus dit werk helpt om die kloof te verkleinen.

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Abbreviations

ABC	activity-based costing
ACT	automated container terminal
ADD	additive model
AE	allocative efficiency
AGV	Automated guided vehicle
AI	artificial intelligence
ALV	automated lifting vehicles
ANOVA	Analysis of the variance
ARMGC	automated rail-mounted gantry cranes
ARTG	Tyred gantries
AutoStrad	Automated straddles
BCC	Banker, Charnes and Cooper
BL	Berth length
CAGR	global Compound annual growth rate
CCR	Charnes, Cooper and Rhodes
CD	Cobb-Douglas
CDEA	centralized development envelopment analysis
CE	cost-efficiency
CES	Constant Elasticity of Substitution
CHE	Container yard handling equipment
COLS	corrected ordinary least squares
COVID-19	Coronavirus disease 2019

CRS constant returns to scale

CRSTE constant returns to scale technical efficiency

CT container terminals

CTO container terminal operators

DEA data envelopment analysis

DMU Decision making unit

DRS decreasing returns to scale

DRSTE decreasing returns to scale technical efficiency

EDI Electronic Data Interchange

GC gantry crane

GCTs global container terminals

GTO global terminal operators

GDP Gross domestic product

GPS global positioning systems

HLH Hamburg Le-Havre

ICS International Chamber of Shipping

IMO International Maritime organization

IRS increasing returns to scale

ITO international terminal operator

MAUT multi-attribute utility theory

MCDM multi-criteria decision making

MCMC Markov chain Monte Carlo

MFP multifactor productivity

MI Malmquist index

MPI Malmquist productivity index

MSW Maritime Single Windows

MTS Multi Trailer System trains

NIRS TE non-increasing returns to scale technical efficiency

OCR optical character recognition

OHBC Overhead bridge crane

OLS ordinary least squares

ONE Ocean Network Express

PCS port community system

PoA port of Antwerp

PTE pure technical efficiency

QC quay crane

RMGCs Rail mounted gantry cranes

RTGCs Rubber-tyred gantry cranes

RTLS real-time locating systems

SBM slack-based models

SC Straddle carriers

SEC Scale efficiency changes

SFA stochastic frontier analysis

ShC shuttle carrier

SL shipping lines

STS ship to shore

TCA total cost analysis

TE technical efficiency

TEU twenty-foot equivalent unit

TFP Total factor productivity

TIL Terminal Investment Limited

TO terminal operators

TOS terminal operating system

TTU tractor-trailer unit

UNCTAD United Nations Conference on Trade and Development

VAL value-added

VRS Variable returns to scale

VRSTE Variable returns to scale technical efficiency

XT external trucks

YC yard crane

*A dedication to Lunda Village, OoBhadela, OoMchumane, and my late grandfather,
Mhlangabezi Captain Nduna...*

Chapter 1: Introduction

There is no doubt regarding the significant evolution in the container sector for ports, terminals, shipping lines, freight forwarders, and all other stakeholders along the container supply chain. Although evolution and change culminated in a considerable amount of development in the sector, many challenges resulted. Today, container terminals continuously seek solutions to their challenges while diligently attempting to maintain improved performance for survival.

This chapter opens by clarifying key concepts to help understand container terminal operation. Following that, a synopsis highlighting the significance of ports and trade to the economy (and vice versa) is presented. The challenges faced by terminal operators, specifically Western-European, are discussed. It becomes clear that although many of the challenges facing container terminals are similar, they also differ to some extent, affecting terminal operation and, consequently, the terminal's performance. The efficiency of container terminals thus becomes vital for investigation given these challenges. Finally, the thesis outline is structured, taking into consideration the objectives, research questions and expected output.

1.1 Definition and terminology of key concepts

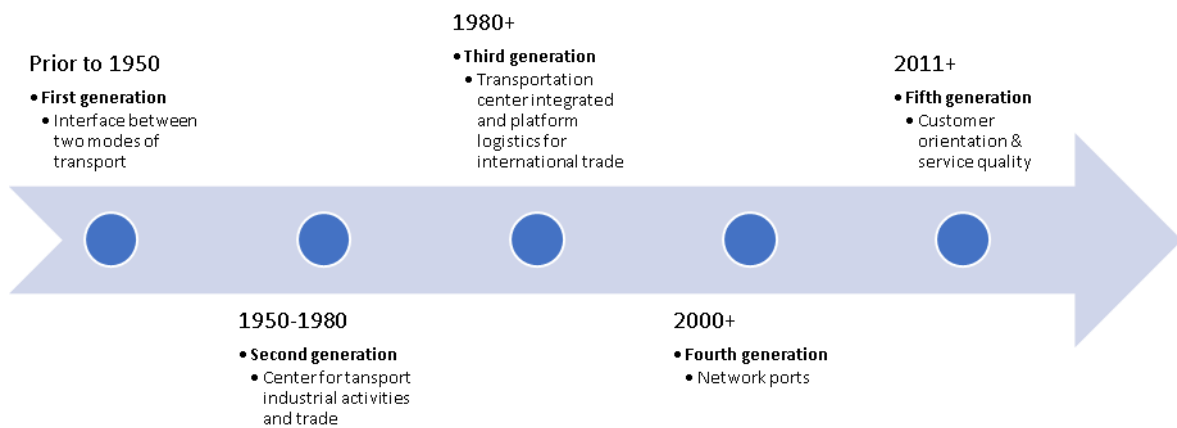
In maritime research, there are some concepts closely related, used interchangeably, or occasionally used in a manner that causes confusion. To avoid this confusion, it is best to define some of these terms before delving into the discussion.

1.1.1 A Seaport

Hlali & Hammami (2017) discuss the evolution of the 'seaport' concept over time. In general, a seaport serves as a point of contact where exchange occurs between sea and land; it is a geographical area where vessels and cargo can be received. Hlali & Hammami (2017) recognize that the definition of a port can be related to categories such as economic, geographical, legal, and institutional dimensions. It is evident that the seaport definition has greatly evolved from how it was defined from the 1950s till today, demonstrated in *Figure 1-1*. The 1950s (first generation) defined a seaport simply by considering loading/unloading, storage, and delivery/receipt of cargo. The

definition transitioned to what is currently observable (fifth generation), defined by Lee, Lam, Lin, Hu, & Cheong (2018) as bodies primarily concerned about customer requirements, throughput, supply chain logistics, and networks. Innovation is also among the important factors in the definition of ports in the new era.

Figure 1-1: Evolution of the seaport concept



Source: Hlali & Hammami (2017)

1.1.2 A Terminal

It happens that the 'port' and 'terminal' terms are in many cases not treated differently. A seaport comprises many players of which one of them is a container terminal operator³. A container terminal as described by Steenken, Voß, & Stahlbock (2004) is a system of material flow with two external interfaces, the quayside (for loading and unloading water transport) and the landside (loading and unloading land transport) using shared and reusable resources and equipment (elaborated in chapter 4). These actions are often referred to as waterside and landside transshipment processes. The main elements of a terminal comprise the docking area; container crane; un/loading area; gate; on-dock rail terminal; administration; repair/maintenance and chassis storage. According to Bearzotti, Gonzalez, & Miranda (2013), the operations carried out at a container terminal constitute the most complex environments in the transport industry. To perform these operations, generally, a CTO relies on intermodal equipment; specifics of which depend on the level of automation.

³ Sea ports, being multifunctional entities have main types of terminals including: break-bulk, dry bulk, liquid bulk, containers, ro-ro, passengers.

1.1.3 Performance

In ports and terminals, one of the most well-used measures in classifying firms is referred to as '*performance*'. A concept that has been frequently defined but whose translation remains ambiguous. Generally, firm performance is the ability of a firm to carry out its functions. Taouab & Issor (2019) show the definition of firm performance from the 1950s where performance was defined as equivalent to firm efficiency. In many studies, performance is associated with shareholder value which historically falls under the classical and financial school of thought. During the 1950s, performance was measured using productivity, flexibility, and inter-organizational tensions. In the 21st century, performance has been defined such that it considers items such as piloting, evaluation, efficiency, effectiveness, and quality. Strictly, Siminică, Berceanu, & Cîrciumaru (2008) argue that for performance to take place, at least, a firm has to be productive, efficient, and effective. This resonates with the following quote in business practice:

"Get it done" (productivity) by "Doing the right things!" (effectiveness) and "Doing things right!" (efficiency).

Peter F. Drucker

(1967)

The above description thus breaks down performance into productivity and efficiency, described as follows:

1.1.3.1 Productivity

Productivity is defined as the ratio between the volume produced and the resources employed to produce the output. Roghanian, Rasli, & Gheysari (2012) confirm that productivity is a measure of effectiveness and efficiency. A thorough definition and measurement of productivity is given in Linna, Pekkola, Ukko, & Melkas (2010). It is stressed that productivity should not be confused with performance but should be viewed from different angles. In the study of Linna et al. (2010), three categories of definitions for productivity are given. First, the techno-economic approach where productivity is the proportion of output given the input. Secondly, productivity as a combination of efficiency and effectiveness; and lastly, as everything that helps to improve an organization. The main difference between how productivity is measured lies in how the output and the inputs are aggregated into the scalars that demonstrate ratios as a measure of productivity. When productivity only improves in

a quantitative sense, it may lead to a greater share of low-quality performance (Drucker, 1963).

1.1.3.2 Efficiency

Productivity measures performance and efficiency is a component of productivity (Guerrero & Rivera, 2009). Productivity evaluates the ratio, whereas efficiency evaluates the observed output in comparison with its optimal values. Efficiency is rooted in microeconomics, where the firm or industry is assessed to whether/not or to what extent the limited resources of production are being 'wasted'. Efficiency refers to how a firm achieves its goals given its limited resources and using the resources in a non-excessive and economical manner (Taouab & Issor, 2019). A majority of economic issues arise due to the scarcity of resources, which firms seek to distribute in the best possible manner. Efficiency can further be evaluated from differing but related aspects.

A firm is said to be *technically efficient* if a maximum amount of output is achieved with the given resources. *Productive efficiency* is reached when a firm achieves the maximum output for a given cost or the minimal cost for a given output. *Allocative efficiency* is a broader term of efficiency, rooted in welfare economics and implicated in opportunity costs. Allocative efficiency takes into account productive efficiency as well as the efficiency over which the outcomes are distributed to the economy. When a firm is said to be *scale efficient* then the size of its operations are optimal such that if any changes are made on the size, inefficiency occurs, Palmer & Torgerson (1999).

1.2 The evolution: Developments & Trends

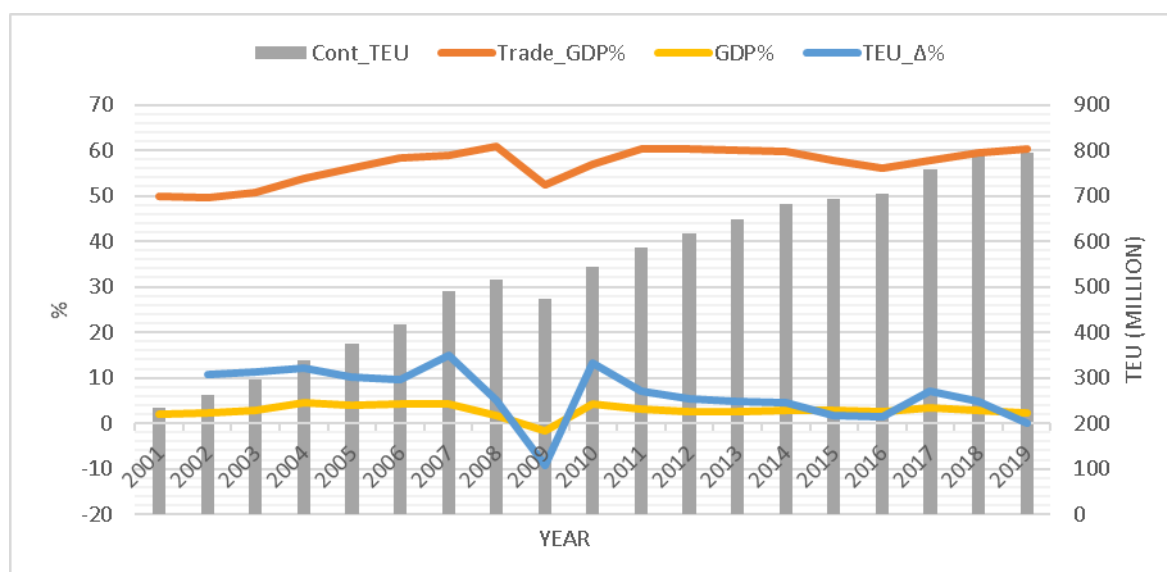
In the book titled - "The box", Levinson (2016) describes how the container, occasionally referred to as "the metal box", or "the humble hero" UNCTAD (2019), became popular. A wooden container was made prevalent already in the 18th century as a means of coal transportation. The efficiency a container engendered presented to the globe, among many other benefits, the lowered costs along the entire supply chain. The container allowed for the possibility of loading and unloading cargo from various transport modes, from one corner of the world to the other in the void of tempering with the cargo. This convenience later allowed the entire supply chain to reap the fruits borne from the metal box.

Port performance, economic growth, and trade are not mutually exclusive, Walsh & Mander (2016). In 2020, more than 80% (volume) and 73% (value) vs in the year 2000, 70% (volume) and 50% (value) of the cargo was traded via shipping according to the International Chamber of Shipping (ICS), International Maritime Organization (IMO), UNCTAD. Container traded cargo has been growing steadily through history, with an annual growth rate of about 8% between 1980 to 2018, according to the review of the maritime report by the UNCTAD (2019).

International trade has since its recognition evolved and affected through ages by many events such as wars, economic depressions, recessions, politics, to name a few events. The financial crisis of 2008 and the recent COVID-19 pandemic are the two most recent global level catastrophes where international trade was affected. Given the COVID-19 pandemic, the uncertainty it brings, the disruptions to supply chains, and demand contractions, UNCTAD (2020) predicts the global trade to drop by 4.1% in 2020. The growth in international maritime trade reached its lowest level since 2008, further affected by trade tensions between China and the United States of America (USA) (UNCTAD 2020). Since 2008, the container sector has faced many changes in how business is conducted generally.

In 2019, the global container traffic reached 796 million TEUs, which is about 0.007% of an increase from the previous year. Although the rate is still positive (compared to the -9.71% of 2009), it is much lower than the average growth of 5.57% between 2010 and 2018. The rate highlights the impact of economic disruptions on container trade on the one hand. On the other hand, the contribution of total trade to GDP has for the past 20 years been above 50%, and although the contribution to GDP has remained high, there is an indication of lesser and even a negative year-on-year change between 2013 and 2016. These trade indicators are demonstrated in Figure 1-2.

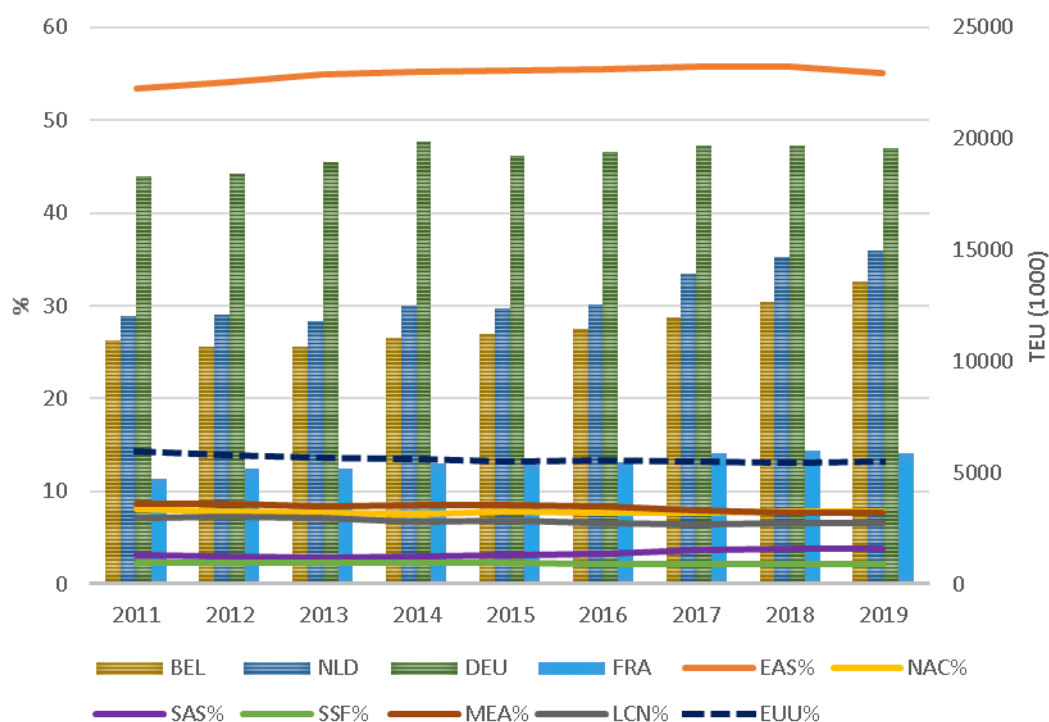
Figure 1-2 *Trade indicators*



Source: Own compilation based on UNCTADstat (2021) and World Bank (2021)

Figure 1-3 illustrates the regional contribution to world volumes of containers (TEU). It is visible over the entire period that Europe comes second following the Asian region in terms of container trade at the world level, which marks the significance of the region to container trade. Within the HLH countries, it can be seen that Germany handles most (36%) of the container cargo and representing the 7th highest country handling containers in the year 2019 on a global scale. The Netherlands, Belgium, and France are rated number 13, 16, and 32 out of all countries. These rates indicate the activity level in the ports and terminals within these countries.

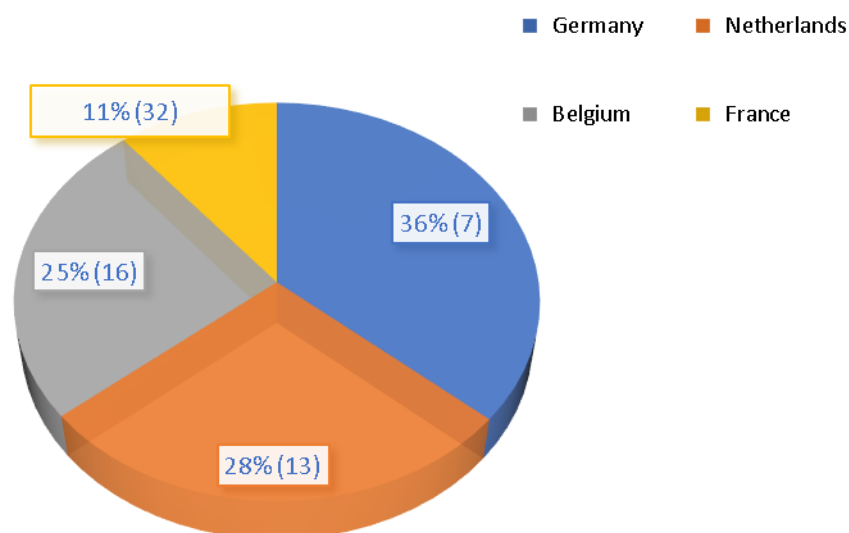
Figure 1-3: (a) Percentage (%) contribution by region & the TEU (1000) of HLH countries
(b) Share (%) between the HLH countries and global rating of countries by TEU (2019).



(a)

***(Graph (a)) - The bars represent the throughput (TEUs) and the lines represent the percentages (%).**

East Asia & Pacific (EAS); European Union (EUU); North America (NAC); South Asia (SAS); Sub-Saharan Africa (SSF); the Middle East & North Africa (MEA); Latin America & Caribbean (LCN); Belgium (BEL); Netherlands (NLD); Germany (DEU); France (FRA)



(b)

Source: Based on UNCTAD (2021) data

According to UNCTAD (2019), Europe (the 28) in 2018 indicated the most intra-regional trade where 69% of all trade was exported within the region. In Plasschaert, Derudder, Dullaert, & Witlox (2011) the level of competition that could occur between ports in close proximity is demonstrated through the HLH ports with immediate effect on practices such as pricing (and costs), efficiency, regulation, etc.

The UNCTAD (2019) report specifies the world container throughput contribution from Europe to be 16%, the second-largest container traffic contributor, following Asia, handling 55% of world cargo in 2018. Figure 1-4 demonstrates the container traffic in TEUs which the ports in the HLH region handled over the past 20 years. At port level (vs country level), the chronology is to some extent different in that Rotterdam in the Netherlands takes the 1st position and the Le Havre port in France maintains the position of France in handling the lowest number of containers for the year 2019 among the 5 ports taken into account.

On average, the HLH ports (Zeebrugge and Hamburg ⁴excluded), recorded positive annual throughput (TEUs) over the years observed in Figure 1-4. The seaports and terminals which were unable to cope with the changes and changing business environment continued to lose volumes. In 2001, Zeebrugge saw a major decrease in container cargo. The limited inland navigation services in comparison to competitors (especially Antwerp) could be the reason behind the poor cargo generation. There have also been strategic changes such as changes of terminal ownership in the Zeebrugge container terminal. In 2017 for example in Zeebrugge Cosco Shipping Ports (CSP) took over the APM terminal (and its other partner Shanghai International Port Group (SIPG)); then CMA CGM acquired 10% of CSP in 2018 also in Zeebrugge. The downfall of Hanjin had a major effect on Hamburg and could have affected the growth rate of traffic. These occurrences have an impact on the vessel alliance(s) calling at the port and thus steering the container traffic coming in and out via the port.

⁴ More the discussions related to strategic changes are discussed in chapter 2 of this thesis.

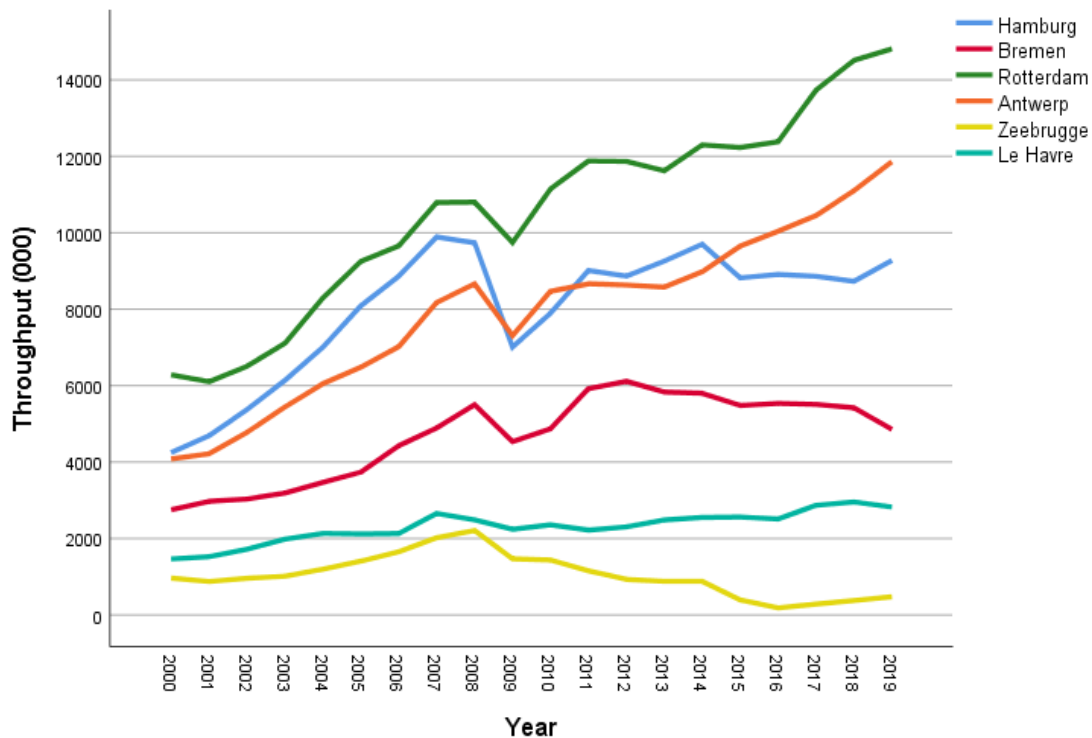


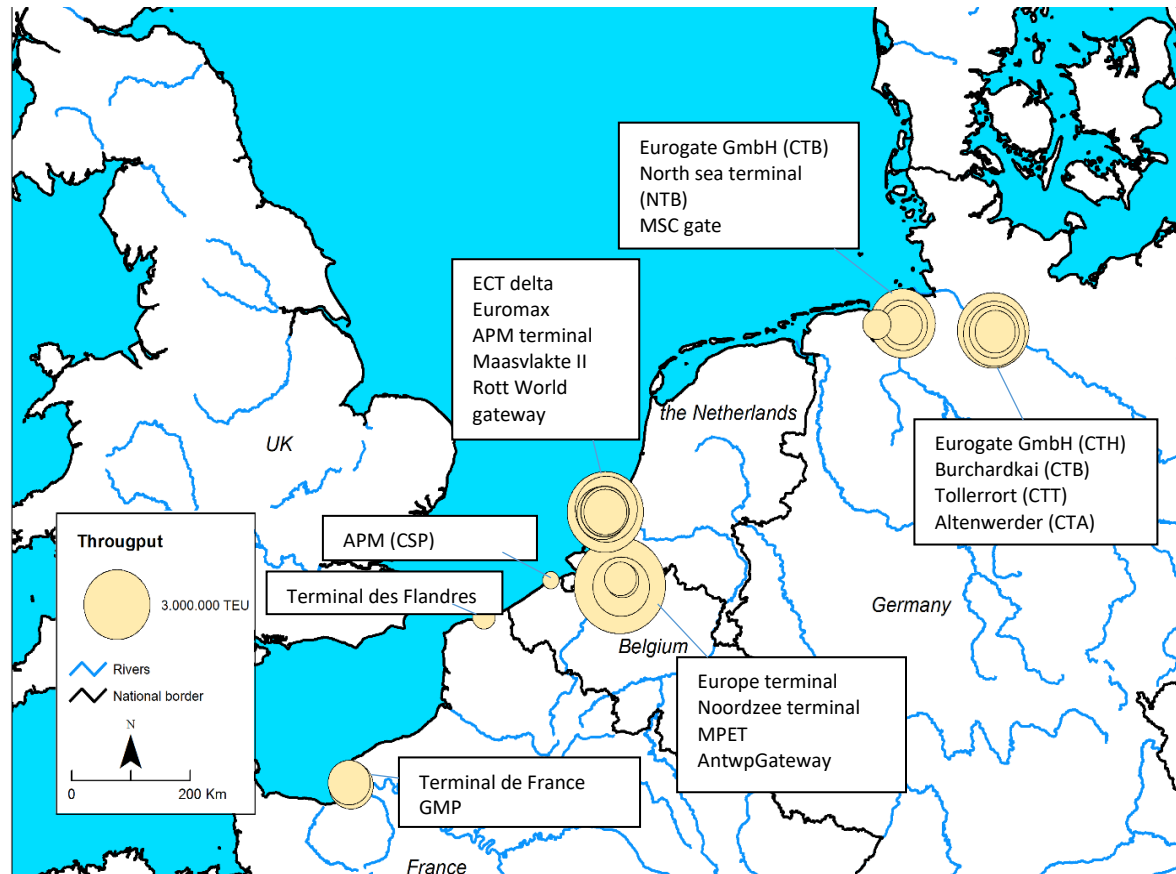
Figure 1-4: Container port throughput (1000 TEUs) of selected ports in the HLH region

Source: Based on the port authorities and Lloyd's list reports

Although there may be similarities to some extent, e.g in the hinterlands, modes of transport, port structure, infrastructure standards, market competition, regulations, currency, business practices, shared historical impacts, Meersman, Strandenes, & van de Voorde (2014) point out that no ports are physically and economically similar. The HLH range of container terminals, located in the north of Europe, functions as an integrated economic-geographic entity. The region includes busy deep sea ports such as the port of Hamburg, Bremerhaven, Amsterdam, Rotterdam, Antwerp, Zeebrugge, Dunkirk, Le Havre, and other smaller (by TEUs) ports. The sample of ports and the container terminals are demonstrated in Figure 1-5. These ports are among the busiest and most concentrated ports in the world, handling approximately 50% of the European region container volumes. These ports operate as landlord ports, which means that the port authority owns the land and infrastructure and the private sector is given an opportunity through lease and concession agreements to operate the terminals. The private company is expected to provide and maintain the equipment and employ the labour for cargo handling, (Liu, 2010). The case of Antwerp is a unique

case where the requirements of the landlord port are met, according to the definition above, but also differs in the element of the port labour. In the port of Antwerp, the port authority manages and maintains the infrastructure but it is not involved in the dock labour matters or port labour organization and negotiations (Bottalico, 2017).

Figure 1-5 The Hamburg - Le Havre container terminals throughput (TEUs) 2018



Source: Owner's compilation

Faced with the continuous growth of container trade and the significant impact trade has on the economy (and vice versa), it is no wonder why it becomes a priority to maintain continuous improvement in return at the maritime supply chain, including in ports and terminals. In as much as this comes along with many benefits and developments, it also comes with challenges that impact the performance and costs of container terminals.

1.3 Challenges facing the container terminal sector

Ports, as well as players within the port, were previously known for their monopolistic component. internationalization and globalization (also facilitated by

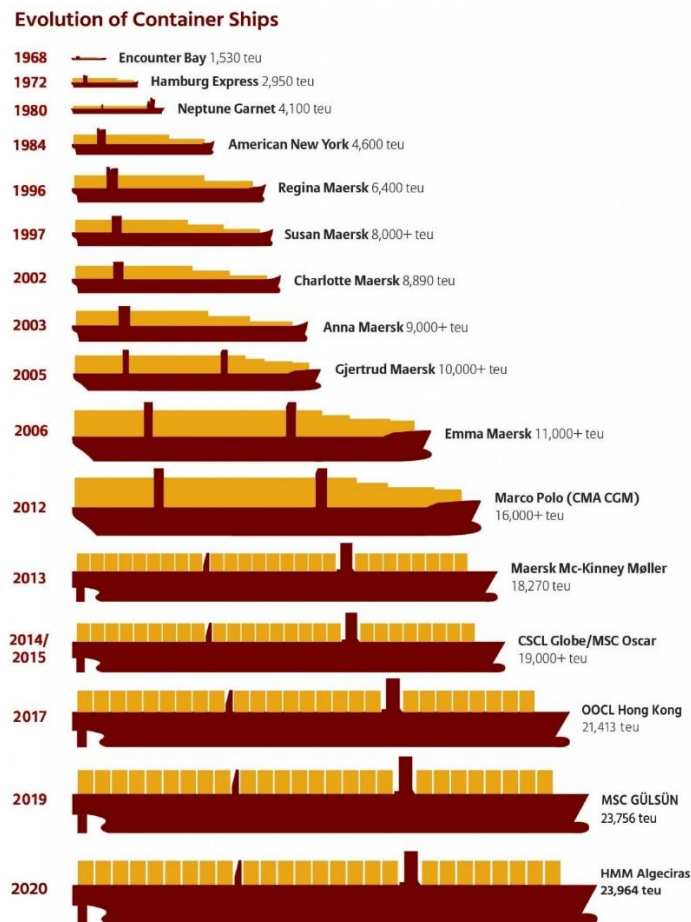
containerization) have influenced patterns in the way in which trade commences (Cullinane, Wang, Song, & Ji, 2006). This has led to various kinds of business relations and strategies along the maritime supply chain (elaborated in chapter 2). According to (J. W. Böse, 2020), there are various complex facets of changes over which ports and terminals go through, affecting how they perform(ed). Specifically, in the container terminal sector, many changes lead to challenges that affected the manner of 'doing business' from many perspectives such as customer needs (i.e. vessel sizes), infrastructure, labour, technology, safety, container traffic, environment, competition, to name a few. It had become apparent that to remain competitive in the industry, one has to improve on performance and in so doing, maintain innovation and remain agile enough to deal with problems as they come. These challenges often faced by the terminal operators are discussed below.

1.3.1 Challenges from shipping lines

Shipping lines and terminal operators are dependent on each other to survive. This, to some extent, ⁵makes the terminal operator interlinked with the shipping line's economic health. The spread of globalization and competition has resulted in shipping lines surging on their vessel size capacities. The massive size of the new ships directly impacts the efficiency of container terminals and ports (Liu, 2010). One of the driving forces to the changes in the market was introduced by the shipping lines who introduced larger shipping vessels which required a changed manner in the service delivery from terminals and ports. As demonstrated in Figure 1-6, vessel capacity has grown by approximately 1466% since the year 1968.

⁵ More about vertical and horizontal integration in Chapter 2

Figure 1-6: The evolution of container vessel capacity (TEUs)



Source: Manaadiar (2021)

Ships started to grow in size, from 500-800 TEU vessels in the 1950s to the nowadays mega vessels with a carrying capacity of 23 964 TEUs (built-in 2020 owned by Hyundai Merchant Marine (HMM))), others surpassing 20,000 TEUs include MOL triumph, Madrid Maersk, OOCL Hong Kong. It follows then that a wave of consolidation hit the container shipping industry. As it stands, in 2021, three shipping line alliances that dominate constitute 2M, the Ocean Alliance, and THE Alliance, bringing together the top ten maritime container carriers and a total of twelve carriers. Some of the top carriers did not survive, e.g. Hanjin, a South Korean carrier declared bankruptcy in 2017. The alliance strategy is to assist the problem of slot under-utilization and is a way to gain more significant economies of scale from the shipping lines' point of view (Wilmsmeier & Monios, 2020).

Table 1-1 demonstrates a general performance of shipping lines for 2019 based on the countries where the ports of interest for this research are located. The liner shipping

connectivity index captures how well the countries are connected to global shipping networks taking into account the number of ships, vessel carrying capacity, maximum vessel size, number of services, and the number of companies who deploy container vessels in the ports of the country (World bank 2021). At a country level, the ports are well connected with the shipping lines, with Belgium achieving the highest index of 88.4 and France with the lowest index of 72.6. This is also an indication of how the terminals within these countries can cater to the shipping lines arriving at their ports. Although Belgium has achieved the highest index, the country's ports keep the shipping lines longer in the port compared to the other countries (longer turnaround times). Since terminal operators occupy between 75% and 85% of the port time, these measures indicate port and terminal performance direction (especially given the size and number of port calls). The performance direction can result from many other issues, including the number of resources available, management of the terminal given the amount of container to be handled (efficiency).

Table 1-1: Container ships performance (2019)

ECONOMY	Median time in port (days)	Average age of vessels	Average size (GT) of vessels	Average carrying capacity (TEU) per vessel	Number of port calls	Liner shipping connectivity index (maximum value in 2004 = 100)
Belgium	1.0	14	52 967	4 795	5 190	88.4
France	0.7	13	56 344	5 060	5 447	72.6
Germany	0.7	13	42 018	3 855	9 543	82.8
Netherlands	0.8	13	32 385	2 990	12 155	88.0

Source: UNCTADstat (2021) and World bank (2021)

Belgium, the Netherlands, and the United Kingdom are identified as the sub-regional leaders in liner shipping connectivity in the UNCTADstats (2019) report. In 2020, Spain is added to the category, UNCTAD (2021). Within the region, the pair between the port (Antwerp – Rotterdam (24 companies)) and (Hamburg -Rotterdam (23 companies)) being the most connected, and makes it to the top 50 most connected ports.

The advancement in vessel sizes has changed the traditional norms for many operators along the supply chain, including how terminals are operated and the structure of the company's costs. Sys et al. (2008) mention various impacts to

terminal operations from the increases in the sizes of vessels. Larger vessels often require additional cargo handling services (such as labor costs related to loading/unloading, stevedoring), infrastructure, and superstructure, which also need increased funding. At times, other ports (and therefore terminals) may be called at less due to the inability to cater to the large ships.

Shipping lines are the main customers of CTOs, therefore meeting the demands from these SLs is essential for the competitiveness and survival of the terminal. As dominant players in the maritime chain, it has for a long time been the case that shipping lines have had a significant influence on container terminal operators. Therefore, this entails that it is not new that terminal operators have had to attend to the demands of shipping lines. On the landside, the same types of challenges are expected in comparison to the waterside, the hinterland transportation requires a well-organized slot management system for operational efficiency.

1.3.2 Challenges from environmental concerns

The issue centered around the implication of scale increase in container vessel sizes is multi-faceted and complex to conclude on. According to the study by Wilmsmeier & Monios (2020), suggestions lead to concluding that the container shipping industry has come to its maturity stages. Although there is no certainty regarding the direction of the sector in the near future, there is a prediction of what challenges to expect concerning the entire supply chain and the external sector, e.g the impact of environmental regulations on the globalized production and consumption system.

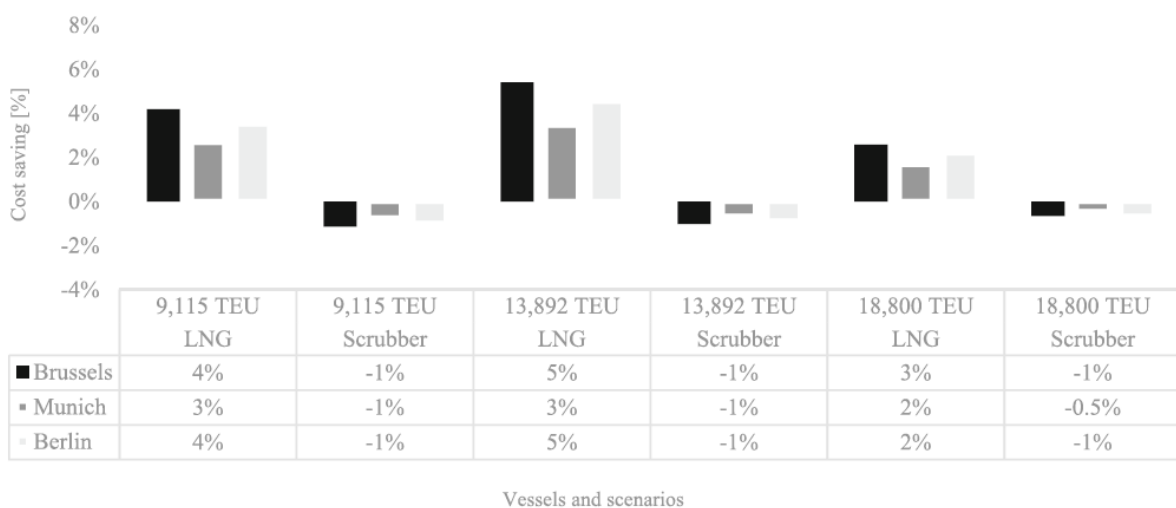
About 70% of maritime emissions occur within 400 km of land, causing problems to the quality of air in coastal areas and ports (Svindland, 2018). Port authorities and companies at the port have had to demonstrate high levels of environmental concern to protect the environments in which they operate. This entails paying attention to environmental-related activities such as dredging, air emissions, water pollution, congestion, loss of green areas, light and noise externalities, conflicts with fishing, etc. The issue of air emissions especially is mainly a challenge to shipping lines but also affects many other players along the maritime supply chain, including terminal operators.

Not only did shipping lines introduce mega large vessels, but they also brought to light an increased awareness of the environmental impact of the shipping industry, which brought about many conventions (e.g the Prevention of Pollution from Ships

(MARPOL)) and laws which the shipping lines would need to be mindful of in their business operations.

Mohseni, van Hassel, Sys, & Vanelslander (2019) demonstrates the potential cost impact pending the environmental regulations to port selection. Where the supply chain total costs are compared from China to the EU (in Figure 1-7). In this route, choosing Brussels as a destination point shows the most cost-effective solution versus Munich and Berlin. This indicates the potential of changing destination preferences when costs increase given the new regulations, affecting the volumes handled in the particular ports.

Figure 1-7: Relative changes for the supply chain costs from Shanghai to the EU for different ship types and LNG and Scrubber scenarios



Source: Mohseni, van Hassel, Sys, & Vanelslander (2019)

Air emissions are another concern to CTOs. A majority of CTOs make use of diesel or diesel-hydraulic engines which harm the environment (hazardous and greenhouse gases). It has been demonstrated in Chang et al.,(2018) that the regulations introduced in the ports have a negative impact on productivity. The regulations can amount to an average reduction of about 22-27% of the emission control area (ECA) port's efficiencies in 58 European container ports.

Shipping lines operating in the Baltic Sea raised concerns regarding the competitive impact of short sea shipping as a shift to cleaner and more expensive solutions would be necessary (Svindland, 2018). According to Svindland (2018), in 2020, Sulphur

emissions in maritime are expected to be equal to those of the other land modes of transport. Aside from aspiring to beat their competitors in providing a better service, selected terminal operators in the HLH region have confirmed to invest in newer requirements that are in line with the required standards to protect the environment.

1.3.3 Challenges from technologies

The change in the manner of service provision, the environmental regulations, and conventions meant that the container terminals would need to select their terminal handling equipment carefully taking into account the latest technological inventions and not ignoring the standards expected into protecting the environment. New opportunities coming with automation contributed to driving the need for innovation in the container terminal sector. Although the new technologies at the terminals promise to improve competitiveness, one of the challenges faced by many of the operators in this regard is integrating the new processes with the already existing processes without affecting the day-to-day operations, (Böse, 2020). Innovation is also very much associated with deeper knowledge and experience, which is another challenge this element introduces to the labour force. According to Böse (2020), challenges associated with technological changes have less of an impact on the sector in comparison to the global crisis of 2008, for example. The extent of the impact, however does vary. In Gharehgozli, Roy, & De Koster (2016), it has been noted that many terminals have been facing challenges especially related to the conversion of the yard from using manual equipment such as rubber-tyred gantries (RTGs) to automated RTGs. The ‘clash’ between these types of equipment in the container yard has been a challenge for many terminal operators. Although many of the terminals currently use advanced TOSs, there are still some challenges related to planning and integrating terminal plans. For example, the Bremerhaven terminal operator North Sea Terminal complains about the time it takes to build scenarios and to run simulations when planning daily shifts, (Holger Schütt, 2020).

The disruptions caused by new technologies to the traditional organization of work and the displacement of employees are emphasized in Gekara & Thanh Nguyen (2018). There has been a change to the skills and education required from a terminal worker, which is different from traditional. Whereas in the past, stevedoring workers did not necessarily need to have the technological know-how, today, it is often necessary to have the skills at least to some level. This, therefore, means that the

terminals would need to offer training and learning to their employees to move along with the technologies, requiring finances.

1.3.4 Labour challenges

The gradual shift from the traditional break-bulk terminals to container terminals is among other developments introduced into the container system. A majority of break bulk cargo saw changes from the manual work completed by dock workers to the current state where the cargo is containerized and handled by various equipment at the terminals. How labour performed their work reformed and modernized, instigating changes to the labour structure (including unions, working conditions, work equipment, etc.) 2018, UNCTAD (2019).

Influenced by the strategies of the maritime logistics chain, terminals have been affected in terms of dock labour systems. According to Bottalico (2018), port labour systems and schemes have to deal with national regulations, neoliberal policies, and regulations at the European level which at times have contradicting objectives. With this regard, Bottalico (2018) identified two main dynamics of challenges for European ports, namely the Institutional regulatory challenges which are concerned about the incompatibility between supranational and national regulations. The other category of the challenges is concerned with organizational issues that occur due to endogenous and exogenous factors introduced by labour pools, labour settings, and continuous changes.

In as much as there is much to benefit from technology, the introduction of information and communication technology (ICT) is one of the challenges facing many sectors, not excluding the port sector. Many of the terminals along the HLH range offer training for their labour to reduce the challenges of the transition with the new systems. According to Esser, Sys, Vanelslander, & Verhetsel (2020) in the port of Antwerp, the identified problem concerning technological advancements is the polarization of the job market where an observed reduction of middle-paid paperwork jobs begin to disappear. Another finding of the study is that basic jobs on the floor will be taken over by robotics and artificial intelligence, whereas higher-level management jobs will require those who have multi-skills. Women and non-natives are identified areas that need special attention.

1.3.5 Challenges associated with infrastructure

To achieve efficient cargo handling operations, infrastructural development is necessary (Acciaro & McKinnon, 2015). To avoid congestion, to foster trade

development, to secure deep-sea container connectivity, it is necessary to invest in infrastructure, which is to be complemented with efficient hinterland connections.

It is not uncommon for container terminals to handle about 6000 to 8000 containers per vessel (Holger Schütt, 2020). It means that not only is it enough to improve the handling equipment but it also means that the infrastructure of the port and the terminal should be able to accommodate such a change. A continuous improvement (if even possible) to the container yards could easily result in high costs and a lower utilization rate of the capacity. As highlighted by Moschouli, Soecipto, & Vanelslander (2019), in 2015 alone, infrastructural investments in maritime ports amounted to about 1.9 billion euros, 1.9 billion euros in inland waterways, 33 billion euro in rail, and about 46 billion in road projects in Europe. The capacity available in the terminals often contributes to resolving or worsening the challenges faced by the port indicating an increase in financial pressure.

It is therefore important for terminals (and ports) to secure funding to invest (as appropriate) in their infrastructures (and superstructure) and that the hinterland connections are also taken into account. The decisions taken in this regard will influence the performance of the ports and terminals.

1.3.6 Challenges related to competition

The inability (ability) to cope with the changes in the sector has an obvious impact on the competitiveness of the terminal. Container terminals nowadays have to compete for cargo in comparison to ports (Liu, 2010). Due to high pressure from inter-port competitors, terminals may increase their efficiency by applying strategies that work for them. On the other hand, competition can also lead to over-investment which threatens efficiency. De Oliveira & Cariou (2015) finds that at the regional level with a range of 400-800 km, competition intensity reduces efficiency. At a local level measured within a range of 300 km as well as at a global level with a range of more than 800 km, the effect of competition is insignificant.

Many factors are to be considered when assessing the competitiveness of a terminal. According to Heaver, Meersman, & Van De Voorde (2001) and Meersman, Pauwels, Van De Voorde, & Vanelslander (2008) the available intermodal services affect competition between ports and terminals. Increased possibilities of connectivity create an extended hinterland area causing an overlap in the different port's hinterland areas and therefore increased competitive pressures. Wang & Cullinane (2015) mention that, given the relative proximity of competitor ports and advanced

transportation systems, European container terminals face fiercer competition in comparison to the rest of the world.

Co-operation also introduces competitive pressures which were not in existence before. For terminal operators, given location differences and regulations to competition, cooperation is difficult to obtain. Often, the possibilities occur vertically⁶ with other parties along the chain. Where possible, ports and terminals in close proximity do co-operate to enhance their competition (among other reasons), e.g. merger between the port of Antwerp and the port of Zeebrugge in 2021; Copenhagen and Malmo in 2001. According to Haezendonck & Langenus (2019), retaining competitive advantage allows ports and terminals to retain their competitive strength. Ports therefore should continuously consider options to assist them to form strategic partnerships, collaborations to assist in the creation of unique resources to reap the benefits of competitive advantage.

1.3.7 Challenges of empty containers

The transportation of cargo between China and Europe is mainly through containers (Kuzmicz, 2020). It is estimated that about 20% of the containers are transported empty in the maritime lag and 40-50% in the inland lag (Kuzmicz & Pesch, 2019). The structure of global supply chains has resulted in trade imbalances between China and the rest of the world. The proportional differences between the incoming imports versus outgoing exports in Europe have resulted in a problem of empty container handling in many ports. This means that many CT handlers have to find solutions for operational costs and environmental issues arising due to circulating empty containers.

Although this is mainly a problem affecting shipping lines and hinterland transportation, some of the manners in which this problem affects terminal operators is that it contributes to terminal congestion, re-handling, higher dwell time, which lowers efficiency. Basarici & Satir (2019) provide several solutions to this problem such as establishing a common container pool by ship operators, container leasing, making use of foldable containers, optimizing voyage planning, making use of twenty containers, optimizing container movement techniques, to name a few. One of the big initiatives taken to resolve this is the One Belt One Road (OBOR) also known as the New Silk Road (NSR) to develop rail connections between China and Europe.

⁶ In Chapter 2

There is anticipation that the initiative will contribute to trade patterns between the regions.

1.3.8 The re-handling problem

How a terminal is managed is significantly important in determining its efficiency and the efficiency of the port. According to Caserta, Schwarze, & Voß (2020), there are generally three types of problems identified post stacking. One is identified as a pre-marshalling problem; another is the re-marshalling and the location problem. Optimizing how operations are carried out at a terminal has always been a key factor since the 1960s. Given the vessel size increases, it has meant that CTOs have to handle much more of a number of containers in a few hours than it was unimagined in the past (while the available surface for managing remains unchanged), generating a challenge for terminal operators. Although many researchers have attempted to bring solutions to resolving this challenge, there is still more room for further research, especially on optimization techniques.

1.3.9 Challenges related to congestion

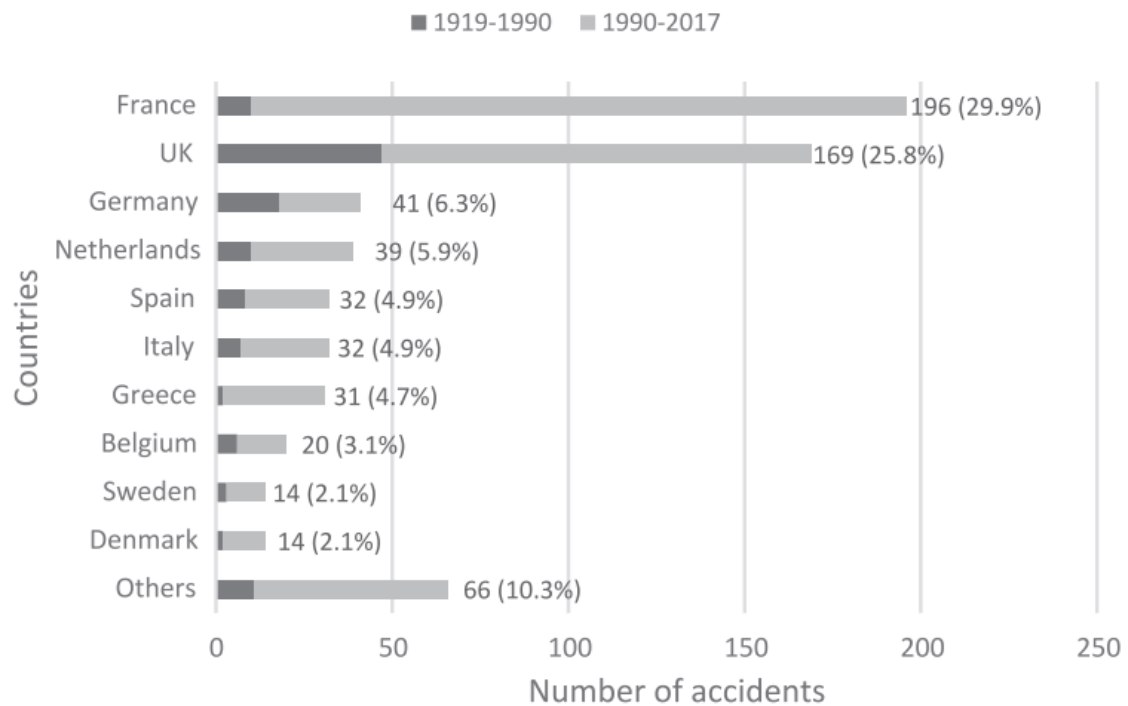
Congestion in ports and terminals can (although not limited to), be a result of many issues such as low capacity, surges from large vessels, labour unrest, infrastructural needs, equipment shortages, marine management, among some. These disruptions lead to higher costs, spoiled perishable products, disruption to production lines, and delay to retail goods. Congestion also causes direct costs such as demurrage costs and costs increase to the rest of the supply chain (Lange et al., 2017). Struyf, Sys, Van de Voorde, & Vanelslander (2020) demonstrate a model to calculate the operational and social costs of congestion in transport. The authors emphasize that congestion can, on the one hand, be a reflection of a lack of a match between demand and supply of transport capacity, on the other hand, congestion can be a reflection of strong economic growth. European seaports have generally been facing an increase in container throughput. The increase in cargo throughput can in some cases also lead to a reduction to efficiency, especially when the infrastructures and capacity is not able to cater for the growth (Simões & Marques, 2010). The container terminals in the HLH range generally do not have a capacity utilization rate exceeding 100%, with the exception of some terminals in Rotterdam for the year 2018. The port of Antwerp according to Rashed, Meersman, Sys, Van de Voorde, & Vanelslander (2018) faced congestion problems in 2003 -2004; similarly, the port of Hamburg faced congestion from 2005 through 2008. Congestion in and around ports in the ports around the

Hamburg-Le Havre and rapid growth in freight is one of what leads to an increase in advanced commercial services offering value-added logistics in the ports of the region, to bring answers to many questions (B. Wiegmans & Dekker, 2014). Among other solutions for congestion in ports, Zhang, Zeng, & Yang (2019) suggests that congestion in container terminals can be efficiently reduced through managing the truck appointment system. Yard congestion caused by external trucks tends to also affect other operations when the yard cranes are utilized for other priority operations.

1.3.10 Challenges related to safety and security

The safety of ships when arriving at terminal is one of the important elements. Risk assessment methodologies and waterborne infrastructure had to greatly change as ship sizes grew. One of the challenges facing terminal operators is ensuring safe ship navigation and manoeuvring in the fast-changing sector, (Burmeister, 2020). A declining trend of navigation-related accidents has been observed, with an estimate of about 100 accidents per annum of which 20% is associated with grounding accidents that occur closer to shore or port approaches. A majority of the accidents are linked to human error given that fairways and turning basins are less fault-tolerant than operation at open seas. Real-life data is what promises to assist in improving this challenge for terminals. Figure 1-8 indicates that France and the UK face the highest number of port accidents at present in the European region (Lecue & Darbra, 2019). The Internet of things (IoT), especially through Automatic Identification Systems (AIS) has assisted in reducing accidents and improving safety in recent years (Belfkih, Duvallet, & Sadeg, 2017; Rajabi, Khodadad Saryazdi, Belfkih, & Duvallet, 2019).

Figure 1-8: Location of accidents



Source: (Lecue & Darbra, 2019)

The impact of security in container terminals has been a centre of discussion. Terrorism, piracy, smuggling of drugs and stowaways, cargo theft and fraud, cyber security, bribery, lack of well-trained security personnel, extortion are recent examples of security threats faced by container terminals worldwide (Aamlid, 2019). Challenges linked to these security threats inconvenience the terminals and their customers and increase costs due to the inconveniences and implementation of effective measures. Many ports and terminals globally adhere to the application of the International Ship and Port Security (ISPS) Code. The ISPS Code aims to protect seaports from potential threats due to terrorism and criminal activity. The code has introduced challenges and vulnerabilities to the terminals and the employees (Senarak, 2021).

1.3.11 Challenges related to digitization

Digitization is one of the developing trends in the current times. It is not an exception in the container terminal context where business analytics are often referred to make a daily business decision, requiring that the data is in real-time, to take into account all activities, including those of external parties such as shipping lines,

hinterland transportation, environmental emissions, etc. This is one of the challenges to the sector. According to Heilig, Stahlbock, & Voß (2020), the lack of data-driven approaches to help resolve container terminal problems (planning and management) is identified as one of the sector's challenges. Some initiatives taken are for example by the Maritime Port Authority of Singapore who collaborated with IBM to make use of big data to improve maritime and port operations; Kalmar has developed a cloud-based platform for tracking real-life data; Navis launched a terminal business intelligence for historical and real-time performance-related data; DP World Antwerp became the first terminal operator to sign an agreement to share its data through NxtPort (port community data sharing platform) in 2018. Although research concerning data mining in maritime already exists, it is still in its infancy stage and more still is to be done.

1.4 Research scope, objective & question

A supply chain is a fragile network, where weak links on one point can introduce problems on another point (chapter 2). Holding the interface position along the supply chain, the performance of terminals is crucial as problems at the terminals can spread to a larger amount of the port stakeholders, where costs implications become crucial. The other way around is also true; problems faced at the waterside and at the port's hinterlands can become problematic for the terminal operators, affecting efficiency, productivity and costs (Chapter 5, 6 and 7). Given the economic role played by ports, as public assets in many countries, the research addresses crucial problems faced today.

Addressing the productivity and efficiency of terminals is important for cost implications (chapter 7). Still, it is also important to address the challenges that terminals meet in their business operation (as indicated in the earlier sections of this chapter and chapter 4). Specifically, this research observes container terminals operating in a single region. The regional focus aims to address efficiency and productivity without much external environmental influence as in operations from different locations. This helps to understand the main drivers of efficiency and productivity and the external factors that should be considered for supplementary effects.

Achieving high productivity and efficiency, terminals worldwide tend to turn towards capacity building, which requires heavy financing. When measuring efficiency and productivity, firms aim to achieve their goals with the given limited resources in a non-excessive and economical manner. Resources in economics refer to the factors of production which take into account land, capital and labour. All these are considered given their importance in the generation of the output. Although all the internal and external factors of production are realised in this research, given data limitations and methodology implications, some of the elements are not considered for evaluation (labour and energy use, for example). This research aims to address the levels to which capacity building is adequate given a specific terminal size while also highlighting the impact of resource management and scale on efficiency and productivity. It is important to note that although recognised, the container terminal's economic efficiency and economies of scale are not estimated in the analyses of this thesis.

The scope of focus in the analysis of this research is on container terminals whose primary mandate is to handle containers and deep-sea going vessels solely. The geographical scope for the analysis is limited to the Hamburg Le-Havre (HLH) region, which is home to a range of relatively large, medium to small terminals and thus covers a wide range of terminals that relate to many other terminals globally.

Although the focus is in one region, many of these terminal operators run terminals in many other world regions. The research primarily bases itself on non-parametric approaches, supplemented by parametric approaches for performance evaluation, for the years 2013 and 2018. As will be clarified later, this research evaluates container terminals at a 'terminal' level (not the company level).

As highlighted in this chapter, the maritime supply chain is progressing at a fast pace. Through the lens of the challenges faced by the container terminals and changes due to port reform, container terminal management impact is realised and discussed. The thesis draws attention to attributes controllable by terminal managers and those external in nature, which can significantly impact on technical efficiency, productivity, and costs.

The ultimate objective of this thesis therefore, is:

- To investigate container terminals' productivity and technical efficiency and suggest potential manners of improvement.
- To highlight important factors in determining performance, some of which are under the control of terminal operators, and others which are external.
- To develop some recommendations which policymakers can consider.

This thesis seeks to answer the following questions:

What are the measured technical efficiencies and productivities in container terminals? What can be observed in the container terminal cost structures?

The main questions will be answered through the following steps:

- What is the level of play between competition or cooperation in the Hamburg-Le Havre container terminals? (chapter 2)
- What efficiency and productivity methods have been applied previously? (Chapter 3)
- What are the various container terminal characteristics (performance affecting) that can be identified along the region? (Chapter 4)
- How has productivity changed in the HLH over the previous years? (Chapter 5)
- What is the level of technical and scale efficiency in the HLH container terminals? (Chapter 6)
- What is the observed cost structure of the container terminals? (chapter 7)

1.5 Expected output

The thesis will fill the gap in the existing literature by explicitly focusing on container terminal operation rather than the entire port operation. The research will contribute by exploring various methods available to measure performance, taking into account various characteristics of the various CTs that exist today.

The main goal is to measure container terminal performance through productivity (through the DEA Malmquist) technical and scale efficiency (through DEA CCR and BCC). The output of this research should highlight exactly the important factors to help improve the performance of a CTO. Identifying such factors will assist CTOs to reduce unnecessary costs and wastage of resources. Given the state ownership, governance and interface role of many ports worldwide, which involve the private sector (including many container terminals), this research impacts on numerous

economic players. For instance, ports and policymakers make ongoing capacity building and concessionaire decisions. For these decision-makers, the primary goal is to achieve efficiency, which also reduces costs for the terminals and stakeholders. The lessons from this research can bring awareness of issues around performance to other players along the chain which can assist in competition and/or cooperation.

The industry specificity of this research can also assist the CTOs to stay aware of the significant aspects of their terminals and thus focus on what is essential.

1.6 Research outline

To address the research questions, various activities are carried out and outlined below.

Given port reform, many ports and terminals changed their strategies. Chapter 2 elaborates on the specific changes in maritime-related consolidations and competition, specifically of firms forming relations with container handling companies, e.g. shipping lines and hinterland transportation. Chapter 2 seeks to outline the changes in the business relations in the container sector and the impact this has had in forming the container terminal firms as they stand currently.

Following that, past literature in the context of terminal efficiency is explored to indicate the development of literature at a standpoint. The literature shows the underlying theories related to performance and reveals methods that have been applied so far for efficiency, focusing (but not exclusively) on container ports and terminals, and focusing on specific differing components of the previous literature. The aim is to deduct and learn from past literature, information that can be useful to the current thesis. The literature is collected mainly from scientific journals, books, and reports. This exercise contributes to chapter 3 of the thesis (literature review).

Next, chapter 4 identifies the characteristics of terminals to demonstrate the differences existing between terminals. The differences identified must be affecting the efficiency of the container terminal, whether directly or indirectly. This contributes to chapters 5 and 6 of the thesis and hence forms a basis for the empirical chapters of the thesis.

Furthermore, productivity (and efficiency) over a longer period is estimated in chapter 5. This is to give an overview of performance in the HLH range over two years. This will help explain the efficiency scores which are analyzed in chapter 6.

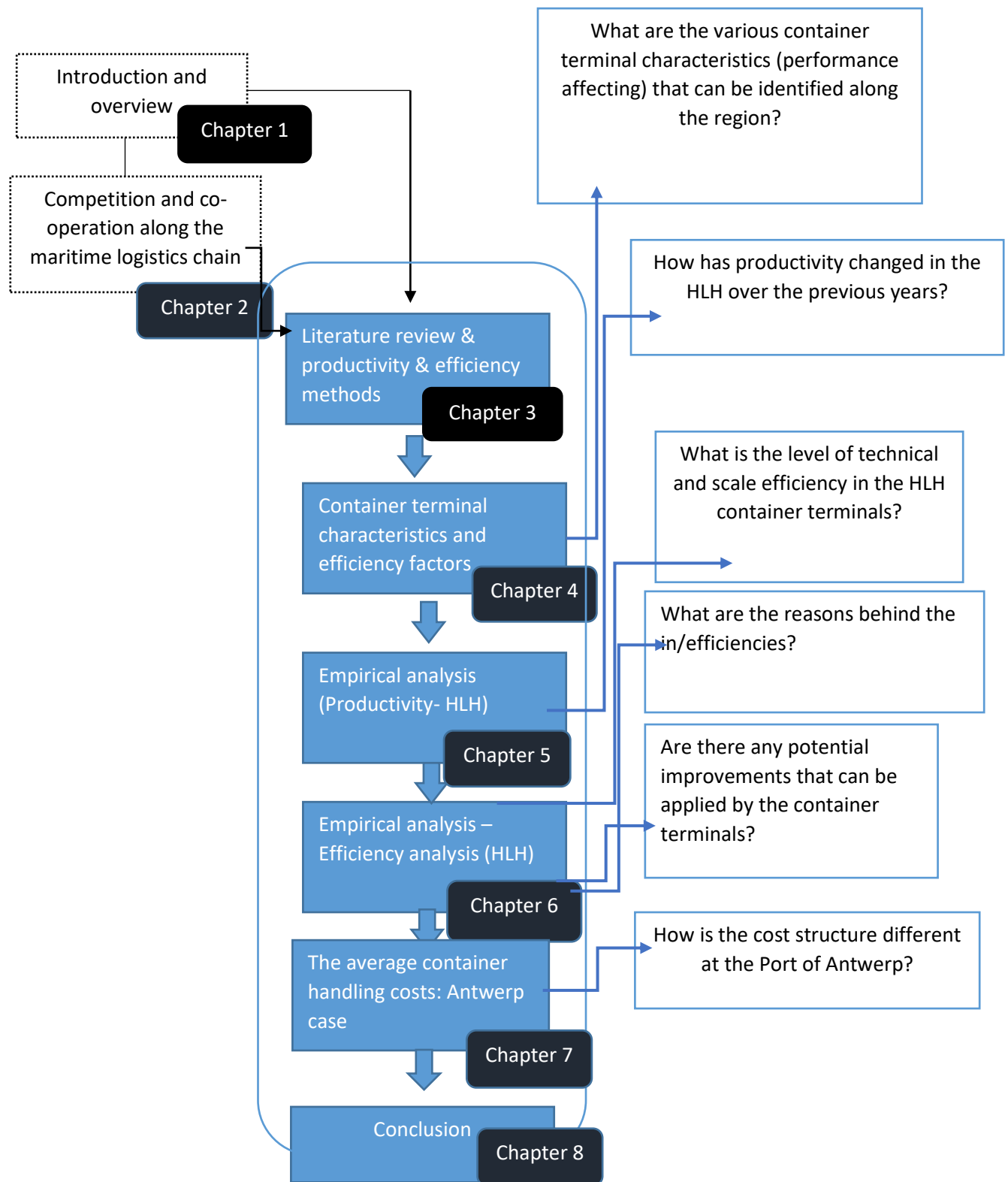
Sixthly, an efficiency method is applied in chapter 6, taking into account the characteristics identified in chapter 4. The technical and scale efficiencies are measured and quantified after that suggestions to improve efficiency in container terminals are given. This chapter addresses the gap in literature where there are few studies measuring the efficiency of especially terminals (as opposed to ports) and in the Hamburg Le Havre. Focusing on a certain region gives the ability to eliminate potential regional differences that would exist when terminals in different geographical locations would be compared.

Given that there are various factors of production involved in handling containers and for achieving efficiency, it becomes interesting to identify the costs associated with handling containers and relate this to efficiency. This analysis is compiled in chapter 7. This estimation is based on the analysis of container terminals in the Port of Antwerp.

Chapter 8 is a reflection of the entire thesis, giving conclusions, recommendations, limitations, and suggestions for future research.

A schematic presentation of the methodology and the outline of the thesis is demonstrated below.

Figure 1-9: Research method and structure



There is a total of *eight* chapters. Chapters 3,4,5,6 and 7 directly contribute to answering the research questions. While 1 and 2 provide a critical formulation of the background and chapter 8 reflects the results of the thesis

Chapter 2: Competition and co-operation along the maritime logistics chain

The container terminal challenges highlighted in Chapter 1 depict the effect of changes in business activities in the sector. These changes may affect the performance of a terminal operator, affecting the level playing field of competition and/or co-operation of terminal operators. The first section illustrates the container terminal market in the Hamburg- Le Havre (HLH) range as it currently stands. As mentioned in chapter 1, the supply chain consists of various players that are interlinked. The interrelations indicate how one even in one link can affect the other. Therefore, the second section demonstrates the connection between the maritime logistics chain players. Port reform is one important element responsible for especially the involvement of the private sector in the port sector. The third section explains why and how ports nowadays have turned to the private sector in the hope of performance improvement. The fourth section discusses the drivers of consolidation. The fifth section addresses how the port reform in the industry shaped up through various strategies, horizontally, vertically and organically. As expected, the impact of such strategies has implications for terminal operators, including capital investments, for capacity building; this aspect is treated in section 6. Finally, the last section closes the chapter with a conclusion.

2.1 The level of play between competition or cooperation in container terminals

While competition (i.e., inter-port) may improve efficiency due to pressures from nearby terminals, it could also lead to reduced efficiencies due to overinvesting. Either way, there is an indication that terminals need to remain strategically prepared. The port authorities in the HLH range act as landlords to private firms within their ports, instigating limitations on areas of direct influence by port authorities. Together as a region, the HLH range of ports forms a solid muscle to represent Europe in the container sector at a global level. Along the 'coastline' of the HLH region, four of the world's 25 largest ports are located. Over the years, market changes that led to increased privatization and diverted focus to advanced commercial activities increased the level of competition between deep-sea ports in

the HLH range (Wiegman & Dekker, 2014). However, maintaining cargo volumes would require container terminals within these ports to improve their delivery services continuously.

Table 2-1 gives an overview of the market share of the five principal ports at the HLH range in terms of container trade. It can be seen that when the market share reduces in some ports (Le Havre, Bremerhaven, and Hamburg), it increases in other ports (Antwerp and Rotterdam). The increase in the market share of Antwerp in comparison to the other ports is the largest over the years. This indicates to some level the play of competition between the ports. For instance, market share total losses from 2018 to 2019 amount to 2.10%, while the total gains in the market share amount to 2%. Similarly, between 2019 and 2020, the total losses amount to 1.80%, while the total gains amount to 1.70%. Although shares and losses don't need to be the same, occurrence indicates a shift of cargo between the HLH ports and thus between terminals.

Table 2-1: Market share of main ports of the Hamburg-Le Havre

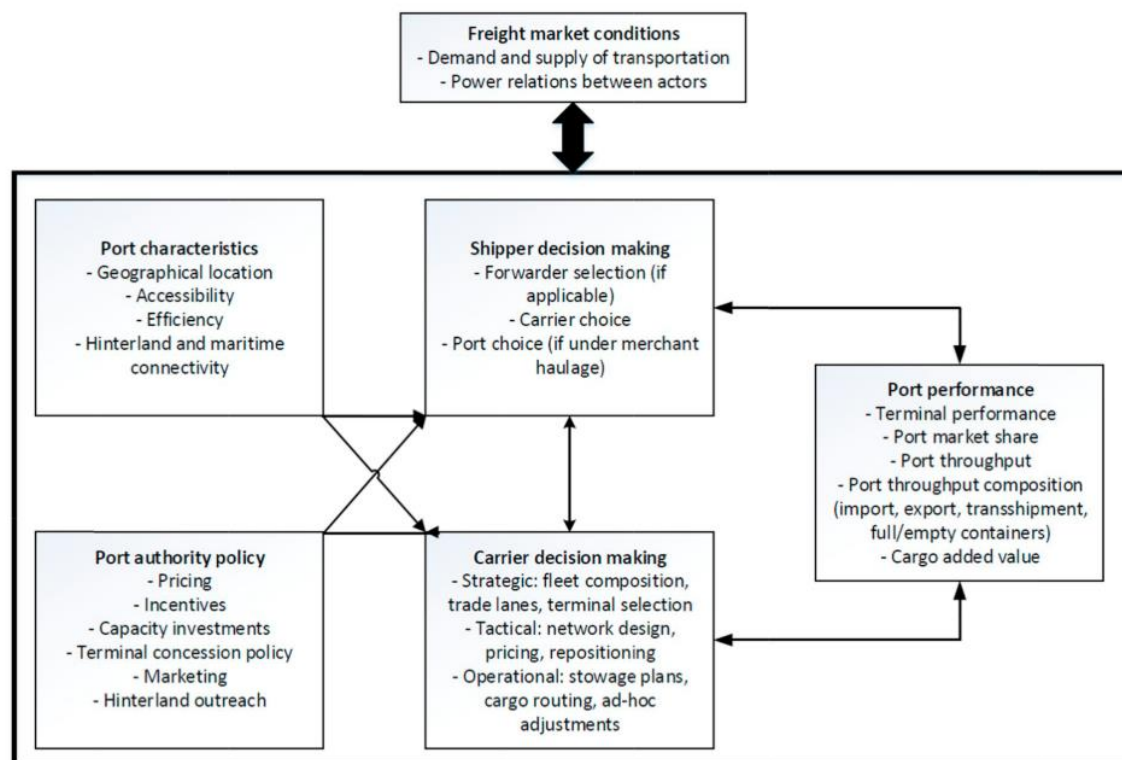
Market share				
	Growth 2020/2019	2020	2019	2018
Le Havre	-12%	5.60%	6.40%	6.80%
Antwerp	5%	28.60%	27.20%	26.00%
Rotterdam	0%	34.10%	34.00%	34.00%
Bremerhaven	2%	11.30%	11.10%	12.80%
Hamburg	-5%	20.30%	21.30%	20.50%
Total TEUs		42 041 300	43 568 900	42 721 300
Changes		2019/2020	2018/2019	
Le Havre		-0.80%	-0.40%	
Antwerp		1.40%	1.20%	
Rotterdam		0.10%	0.00%	
Bremerhaven		0.20%	-1.70%	
Hamburg		-1.00%	0.80%	
Total losses		-1.80%	-2.10%	
Total gains		1.70%	2.00%	

Source: Own composition based on DynaLiners weekly 09/2021

The decision-making choice by port stakeholders is often an ambiguous one. Castelein, Geerlings, & van Duin (2019) demonstrate in Figure 2-1 the various criteria for port selection, port user behaviour, and implications to ports. One of the principal decision-makers of terminal choice is a shipping line based on accessibility (among

many other factors). One of the challenges of the port of Antwerp and the port of Hamburg in this regard is the inability to accommodate vessels of 20,000 TEU and, more significantly, due to water depth and draft restrictions. To mitigate such issues, the port of Rotterdam created the Maasvlakte II, which also assists with capacity availability, factors that affect competition, and efficiency. Although highly accessible due to its coastline location, the port of Bremerhaven has restrictions to extend its 5km quay wall as the city-state limit has been reached (Castelein et al., 2019).

Figure 2-1: Port choice criteria and implications



Source : Castelein, Geerlings, & van Duin (2019)

By comparing the terminal handling charges from the perspective of three shipping lines (Maersk, MSC, and Hamburg Sud), Antwerp shows to be the cheapest in container handling, the German ports tend to be most expensive, and the port of Rotterdam positioned in between (Castelein et al., 2019). The rates charged to carriers by CTOs can vary depending on the negotiation. However, these are passed on to the shipper, who also decides on the carrier and thus affects the choice of a terminal. According to Castelein et al. (2019), when it comes to port dues, quay, buoy, dolphin dues, mooring services, tugs, and pilotage, Rotterdam is the most generous, followed by Bremerhaven, Antwerp, and then Hamburg. These factors all influence the choice

of port and terminal. As can be seen in Figure 2-1, there are many contributing factors to port and terminal choice and thus scoring the highest points among all aspects of consideration in comparison to competing terminals is not an easy task and even then would not guarantee the terminal the most cargo as there are also other external factors not considered in these factors mentioned here.

This chapter analyses how the players in the maritime chain interact with each other and thus how competition and/or co-operation are likely to unfold. Eventually, the match between these players will affect the cost structure and pricing decisions along the supply chain.

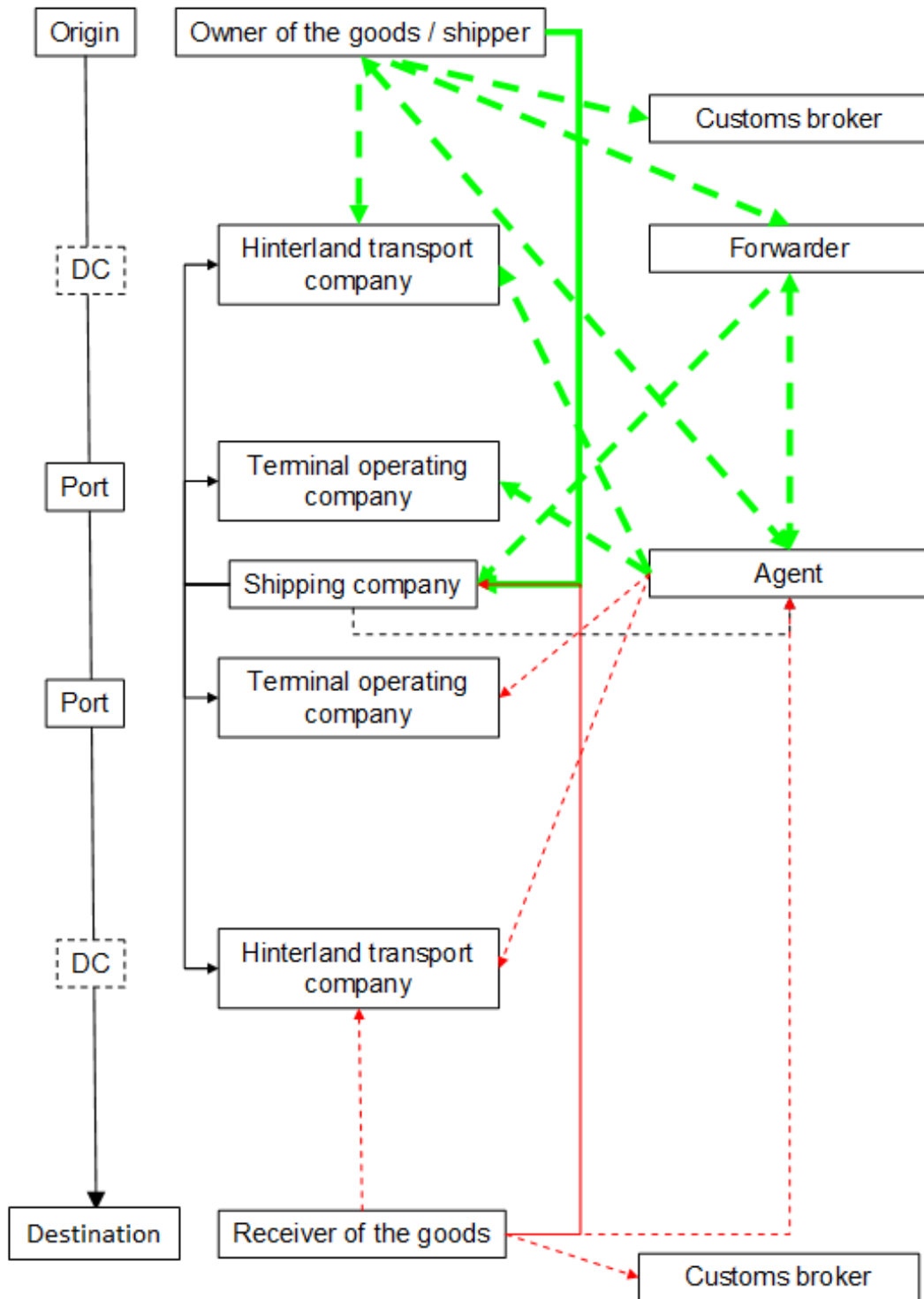
Section 2.1 gives an overview of the maritime logistics chain. Section 2.2 discusses the maritime logistics chain and the players involved. Section 2.3 discusses the drivers of consolidation, and in section 2.4, the dynamics in which mergers in the sector occur, including horizontal and vertical integration. Section 2.5 finally discusses the effects of consolidations on the market players.

2.2 The maritime logistics chain

The maritime industry has changed from an environment where the same type and level of entities would compete together to an environment where the entire chain connecting the alpha and omega (beginning and the end) is now competing. It is common to evaluate the performance of a port and how that affects the other players in that particular port. The opposite is true that the port is also a function of its players and which thus brings interest to want to understand who the players in the port area are and how the maritime chain players are integrated.

There are three large sections to the maritime chain, as recognized by Van De Voorde & Vanelander, (2010) consisting of – pure maritime activities, goods handling, and the hinterland. A sequence showing the maritime logistics chain is demonstrated in *Figure 2-2*:

Figure 2-2: Maritime logistics chain



Note: DC- Distribution center

Source: Meersman, Van de Voorde, & Vanelander (2010)

The players along the chain have differing levels of influence on the facilitation of the chain. Meersman et al. (2010) identify that the container's owner, shipper, forwarder, and shipping lines (SL) significantly influence what should happen to the cargo. *Figure 2-2* demonstrates the process followed by the goods from the sender to the receiver. Dashed lines in the figure indicate alternatives to the direct path (solid lines). The green lines show export flows, and the red lines show import flows.

Container terminals naturally stand at a position of connecting all the modes of transportation and information systems. They are in an excellent position to enhance a change in the logistics chain and stand in a better position to add value (VAL). Improving logistics is now widely accepted to improve competitiveness (Carlan et al., 2018). When developing a successful logistics chain, the costs and operation behavior of the entire chain must be known (Van Hassel et al., 2016). It is also vital that ways to build strong relations between the players are developed, that flexible transportation means (such as containers) stand in place, to ensure information sharing, which is still one of the challenges the logistics chain system faces.

2.3 Container port and terminal management

It cannot be denied that ports nowadays are much different from ports many years ago. While in the 19th-century, ports tended to be instruments of the state or colonial powers, ports today are divided into various categories with varying service offers (The World Bank, 2007a). Today's changes have made efficiency more critical than it ever was in the past. However, this change means that ports and port service providers have to face rivalry from existing competitors, the threat of new competitors, global substitutes, the bargaining power of port users and service providers.

Mainly due to transshipment, ports have become competitors not only of nearby ports but even of ports further away distances. According to Liu (2010), innovation in the maritime and port sector is considered conservative due to its slow transition in cooperative interactions between players involved (Acciaro et al., 2018). Innovation is an essential driver to gaining competitiveness in any firm nowadays. However, innovation in processes is not as visible in the port sector as in other sectors. This is primarily due to the players' poor collaborative relations, making technology-related innovation challenging to achieve. By making efficient use of a comprehensive tool (port community system (PCS) and maritime single windows (MSW)) to facilitate

communication, players to the chain can increase their competitiveness. Carlan, Sys, & Vanelslander (2016) and Carlan et al. (2018) emphasize the synchronization between the players to the maritime supply chain to achieve collaborative innovation. Financial and operational benefits flowing from the private sector to the ports are one of the forces that have invigorated the appetite of the private sector presence in ports which is visible nowadays.

Table 2-2 demonstrates how port roles can be distributed in the different port categories. These services also expand the role of the service providers along the chain, giving the particular port an advantage to differentiate themselves from competitors by other means other than price and turnaround times (The World Bank, 2007b).

Worldwide, port authorities have received much pressure to shift some or all of their service provision to the private sector for competition purposes. It has been proven that privatizing port operations has led to changes in bureaucratic procedures and controls and has led to improvements in the modernization of facilities and equipment.

To improve external trade competitiveness, ports attempt to reduce the costs of port services and enhance the efficiency of the sea and land interface. Governments may also prefer to transfer some port activities to the private sector to reduce stress on the national financial budget and raise revenues (The World Bank, 2007b).

There are many examples where privatization of terminals has been carried out successfully. A classic example is that in Buenos Aires, where the involvement of the private sector successfully brought down handling charging fees through labor productivity. Another case is the Kipevu terminal operation in Mombasa, which was transferred to a private operator where the terminal's productivity massively increased thereafter (The World Bank, 2007b). The ports in the United Kingdom are also another example of a pure private port.

Table 2-2: Public-Private roles in port management

Public-private roles in port management	Port/activity	Public service port		Tool port		Landlord port		Private service port	
	Port administration								
	Nautical management								
	Port infrastructure								
	Superstructure (equipment)								
	Superstructure (buildings)								
	Cargo handling activities								
	Pilotage								
	Towage								
	Mooring services								
	Dredging								
	Other functions								



Private responsibility



Public responsibility

Source: The World Bank (2007a)

It is estimated that 80-85% of ports worldwide apply the landlord port authority structure, accounting for about 60-65% of global container throughput. State-owned and state-run terminals (e.g., Transnet South Africa) handle approximately 35-40% of the global container throughput. This indicator is not clear-cut in some cases: for instance, DP World is almost entirely owned by the state and operates Jebel Ali port in Dubai. At the same time, the operator is considered private outside Dubai, similar to

PSA in Singapore. In the US, it is typical that the state invests in both the infrastructure and equipment, allowing stevedores or terminal operators (TOs) to use common berths and, in some cases, a common yard. These dynamics blur the lines of division regarding the exact profiling into strict categories (Drewry, 2019). *Table 2-3* shows the current and forecast future capacity by ownership type. This gives an idea of the split of the ownership of container capacity between the different port players and possible future changes.

Table 2-3: Forecast of cont. port and terminal capacity by ownership (by million TEU and percentage share)

	2018	2019	2020	2021	2022	2023	Av growth
Global/International Operators	694.6	710.8	728.9	741.7	747.3	749.6	1.5%
	61.9%	61.7%	61.3%	61.1%	60.9%	60.9%	
Other Private Sector	196.6	200.0	205.5	209.2	213.0	213.9	1.7%
	17.5%	17.4%	17.3%	17.2%	17.3%	17.4%	
State Sector	210.2	217.5	227.7	234.4	238.3	238.3	2.5%
	18.7%	18.9%	19.2%	19.3%	19.4%	19.3%	
Other *	21.2	22.9	26.6	28.8	29.4	30.0	7.2%
	1.9%	2.0%	2.2%	2.4%	2.4%	2.4%	
Total	1122.5	1151.2	1188.8	1214.0	1228.0	1231.8	1.9%

Source: Drewry (2019)

There are many terminals whose ownership structure is divided or not clear, these are recorded in the 'other' category. Where there is no clarity, the judgment here is based on the party with the most significant stake in each terminal. As it is now, it seems that the majority of container capacity will remain with the international terminal operators (ITOs) (62%), followed by the state sector (19%). The Global Compound Annual Growth Rate (CAGR) is 1.9% for capacity expansion, implying that the ITOs are growing slower than average. This can be due to the lower risk appetite in the ITOs than in the state sector.

2.4 Drivers of consolidation

The world economies have become more interrelated and closer than ever, thanks to globalization and growth in trade. Consolidation has become increasingly prevalent. Firms generally cooperate to take advantage of scale and scope economies, taking into account antitrust legislation. Innovation, in especially the shipping sector, is

changing rapidly, creating the need to find more sources of finance. This is one of the factors bringing rise to consolidations. In Crotti, Ferrari, & Tei (2019), it is speculated on one hand that merging between carriers proves to be more beneficial than not joining. Non-merging carriers are observed to be more likely to enjoy fewer profits. On the other hand, an alliance-based vertical integration between carriers and terminal handlers is seen to be unavoidable, at least in the near future, as it promises cost savings.

According to Meersman et al. (2010), Nedlloyd and P&O in 1996 merged solely for scale benefits and cost reductions, albeit they were not achieved in the end, and another company, later on, took over the firm. In addition to the strategy that could differ regarding consolidation, the player consolidating may also do so for a different reason than the other. The CTOs are confronted by larger, fewer, and more demanding SLs requiring that they change their ordinary ways of doing things, i.e., pure stevedoring. The capacity of TOs is very abundant (subjective), throughput per annum continues to grow, and ships are also getting larger. The continuous increase in global annual throughputs, the development of hub and spoke networks, SL alliances; all impact the changes of terminals and have caused a rise in the interest of terminal operation (Crotti et al., 2019). This has led to both horizontal and vertical integration of the CTOs in all parts of the world (Franc & Van der Horst, 2010).

Obtaining gains in efficiencies, increased customer responsiveness, gaining a competitive edge, and achieving lower costs have been some of the apparent driving forces leading to consolidation in the sector. Consolidation also allows for access to other international firms, which introduces the opportunity to transmit skills from those with better know-how. Innovative systems and new technologies have changed port infrastructure and superstructure such that there are many financial investments at stake, driving the realignments and consolidations in the sector (The World Bank, 2007a).

According to Liu (2010), some of the reasons for the increase in the number of terminal operators include – the increase in stevedoring costs, port privatization, and an increase in transshipment traffic. Crotti, Ferrari, & Tei (2019) also observe the same and confirm that mainly the liner sector has aroused cooperation in the container sector, including ship sizes, falling freight rates, the decreasing importance of conference agreements, and privatization of terminals. Ocean carriers have resorted to strategic alliances since the 1990s to exploit global demand opportunities,

achieve joint efficiencies, and introduce changes to the TO and port sector. Economies of scale are usually realized from the perspective of ocean carriers. For economies of scale for the ocean carriers to be discovered, terminals at the ports should also benefit from the economies of scale. For this reason, vertical integrations between liners and terminals operators were induced. In extreme cases, dedicated terminals resulted (Parola et al., 2015).

Heaver, Meersman, & Van De Voorde (2001) confirm that the decision to consolidate in the port sector is driven by many forces, especially those affecting the organizations' power and the efficiency and profits of the companies. The inability to specify the exact factors lies in the fact that the financial strength of a particular firm can be affected by many factors, which indicates that firms can take on various paths to reduce their costs and attain their objectives. Similarly, when dealing with competition, the different players face different environments; for instance, an SL may be faced with a larger variety of ports to call at while ports may not be faced with such a wide choice of SLs, resulting in power imbalances.

What can be drawn from literature and according to Vanelslander (2005) is that although there are many driving forces, consolidation is very central to economic, financial, efficiency, strategic, and market-related decisions. Because the decisions taken are likely to affect the public interests, anti-trust laws are also fundamental to the decisions. Consolidation can be considered anti-competitive and thus is monitored, e.g., the Competition directorate of the European Commission limits approved mergers subject to certain conditions. Musso et al. (2000) mention some conditions to be met to indicate that a particular co-operation is not anti-competitive. The co-operation should intend to expand the firm's borders; it should seek to achieve scales necessary to compete in the global market and allow the firm entry into a market while maximizing the output.

2.5 The dynamics of consolidation

Buckley (1992) defines a co-operation scheme as an "inter-firm collaboration over a given economic space and time for attaining mutually defined goals."

Gaining control over the maritime logistics chain is vital for many marine and port players, introducing vertical and horizontal co-operations, mergers, and acquisitions.

Players are interested in expanding their operations at the same service level and/or up or down their service level.

Shipping companies have taken both horizontal (alliances) and vertical (acquisitions and mergers with TOs, or even further to hinterland transportation) collaborations to optimize the chain as a whole. Van De Voorde & Vanelslander (2010) give an overview of the maritime chain players and how the co-operation strategies have been redirected. Following the horizontal integration strategy, shipping lines have taken: vessel sharing, joint venture, consortia, alliances, mergers/acquisition, and conferences as strategies.

Vertically with SL, stevedores take on joint ventures, dedicated companies, and shares. Horizontally, the firms have transformed through mergers/acquisitions and joint ventures.

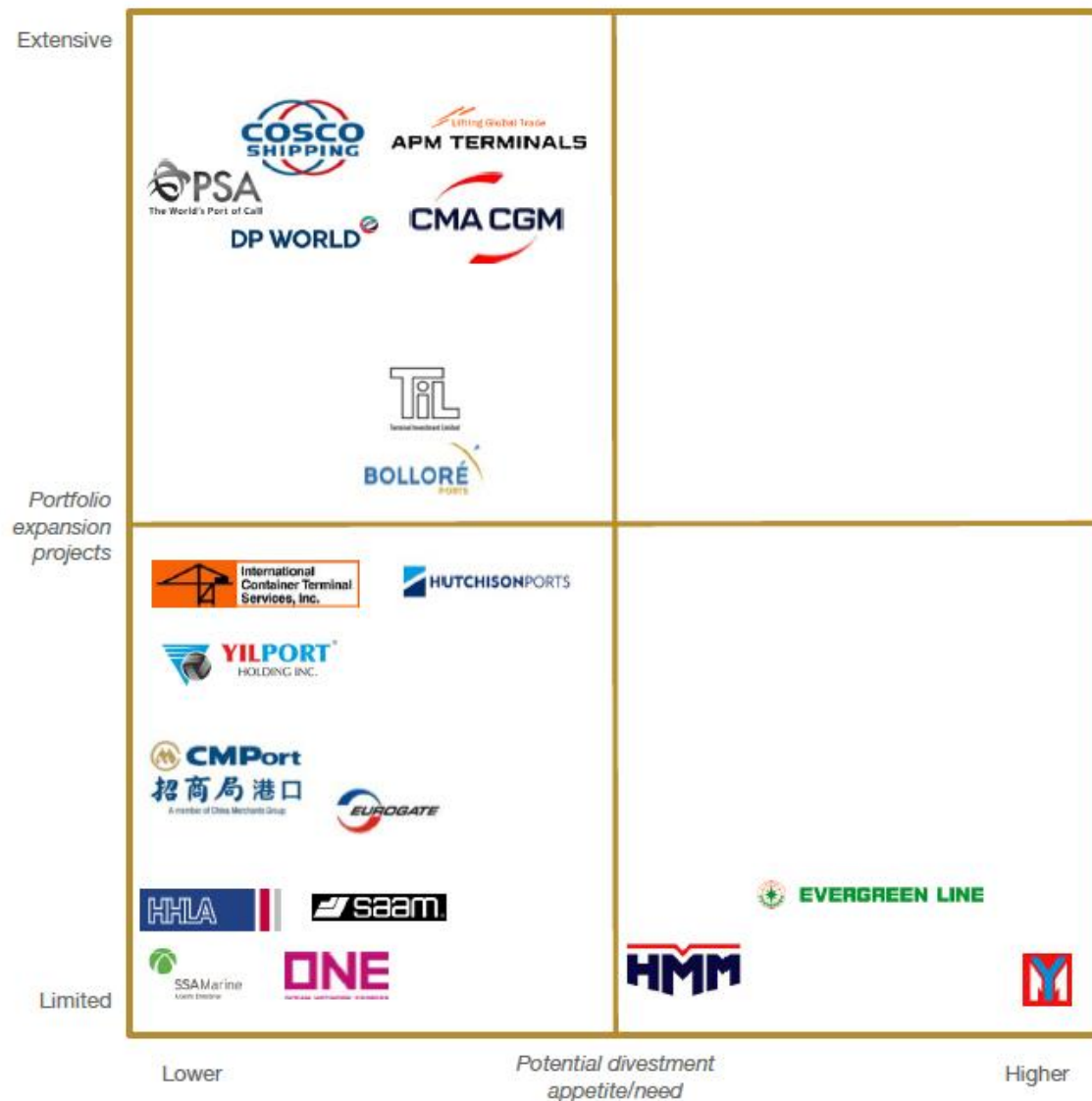
Port authorities have formed partnerships with shipping lines and given various concessions for dedicated terminals. With TOs apart from concession agreements, there are also cases where joint ventures have been created. Port authorities among themselves have also been observed to form alliances and mergers.

Hinterland operations have formed agreements such as block trains, capacity sharing, and acquisitions with SL. With CTOs, joint ventures were created, and alliances were formed with other hinterland transportation players.

Figure 2-3 shows expansion plans⁷ (from limited to extensive) of TOs as well as the appetite or the need to divest (from low to high) of the terminal operator, Drewry (2019). Most CTOs are willing to expand and have a low appetite to divest assets, thus occupying the left-hand quadrant. Drewry (2019) states that the present picture is very different from five years ago. SL were in the past feeling increased financial pressure compared to the present moment, and stevedores are moving towards a 'portfolio management style' vs. the previous style, which was more towards adding to the portfolio. AMPT and COSCO have the highest number (six) of expansions projects followed by CMA CMG (5) and PSA and DP World, each with four projects lined up. Only three TOs (which are SLs) have a higher need to divest assets – Evergreen, Yang Ming, and Hyundai.

⁷ The projects are based on currently confirmed plans.

Figure 2-3: Strategic focus by expansion plans and divestment

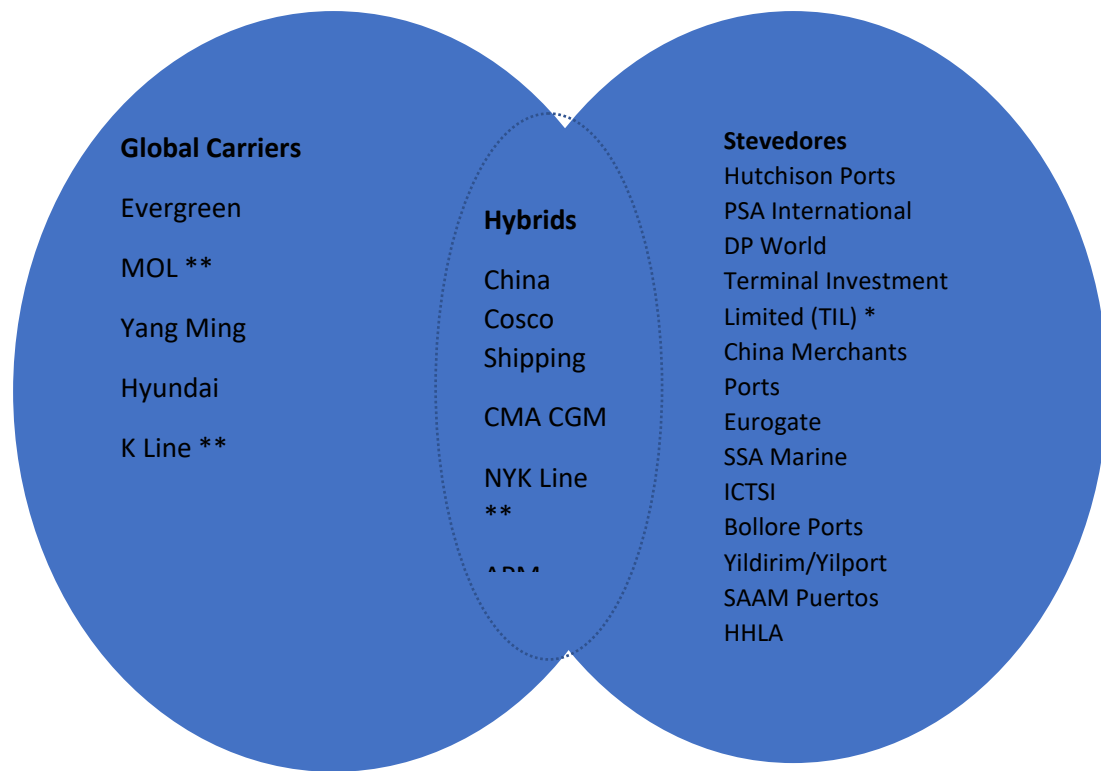


Source: Based on Drewry (2019)

2.5.1 Nature of GTOs and ITOs

Twenty-one companies in Drewry (2019) qualify as global/international terminal operators (GTOs/ITOs). Purely regional operators do not qualify in the Drewry reports. These GTOs can be categorized into stevedoring; global carriers and hybrids as demonstrated in Figure 2-4.

Figure 2-4: Global/international stevedores, carriers and hybrids



* Terminal Investment Limited has a strategic relationship with MSC

** The international terminal portfolios of the three Japanese lines are merged under ONE as from July 7, 2017 and operations from April 1, 2018

*** New corporate strategy by AP Moller Group has moved APMT and Maersk Line closer together

Source: Adaptation from Drewry (2019)

The stevedore companies focus on terminal operation, the global carriers focus mainly on shipping, and the hybrids primarily on liner shipping and have the terminal operation as a separate business unit. Strategically, stevedores run terminals as profit centres while the carriers are likely to run terminals as cost centres. While stevedores put systems to improve productivity and efficiency throughout the terminal, carriers intend to integrate the terminal with the shipping line. The hybrid terminal strategies vary between the two extreme cases.

A distinction is made between a 'stevedore' and 'stevedoring' (Ferrari, Puliafito, & Tei, 2018; Vanelslander, 2005). Stevedore indicates the provision of labour and crane operation and all aspects of the service. The majority, as observed, belong to the stevedore category. APM terminals are in the stevedore category but have formed

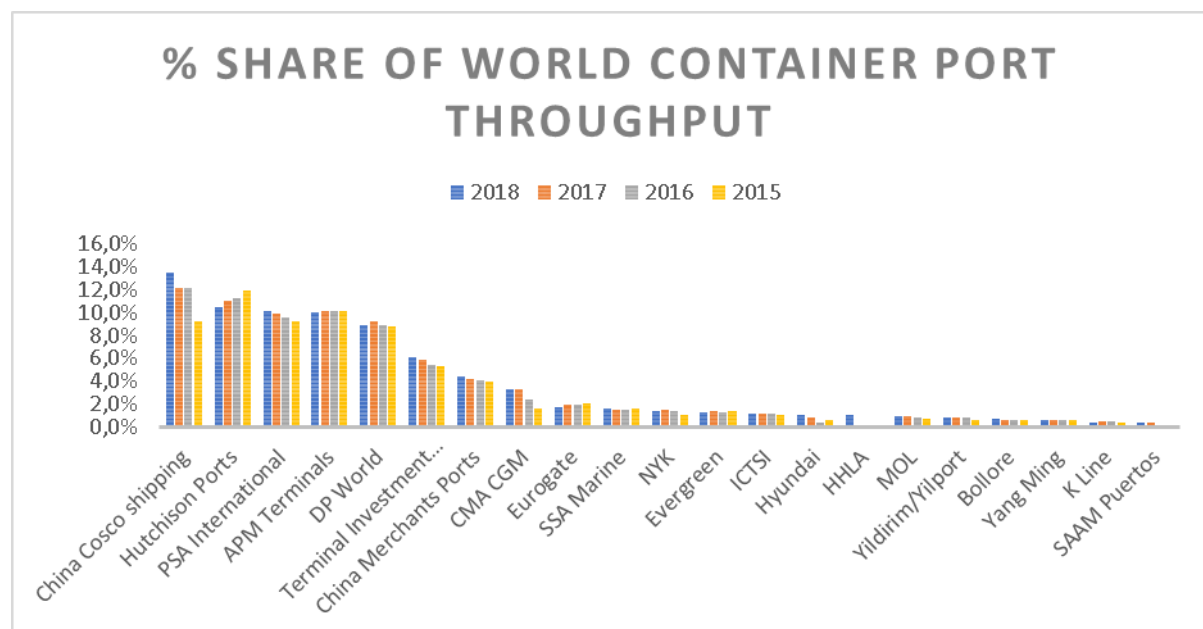
strong relations with the Maersk line, falling under the hybrid group. TIL can also be viewed similarly as they have closer ties with MSC. This shows that the division is not a clear black and white one.

GTOs handled a share of 65% regarding throughput share, which stood at only 18% in 1996. The state-managed 35%, which has now reduced to 20%, and the private sector fell from 46% to 14%.

Considering the sample of 21 GTOs, between 2016 and 2018, the stevedore throughput share fell from 70% to 60%; the hybrid group jumped from 20% share to 35%, and the global carriers shrunk from 10% to 5%. These changes are not driven solely by natural demand/supply forces but also by strategic changes such as the CEPA relations to Maersk line and some events such as the downfall of Hanjin in 2016.

To avoid the issue of double counting, Drewry expresses throughput by the equity stake of the operator. Figure 2-5 shows how throughput is distributed between the GTOs in the container sector. From 2015-2018.

Figure 2-5: GTO's percentage share of throughput



Source: Based on Drewry (2019,2018,2017,2016)

Sociedad Matriz SAAM Ports and HHLA are newly added to quality as GTOs. SAAM Ports is included in the sample because its throughput by equity from its terminals

has accounted for around 10% of the overall equity, one of Drewry's requirements for inclusion into this group. The main driver of SAAM Ports' figures is the acquisition of a majority stake in Puerto Caldera, which occurred in 2016. HHLA has been added to the sample to its new addition of terminal operation in Estonia and Odessa and home terminals in Germany (Hamburg). One other requirement for the addition of terminal operators in the GTOs by Drewry is that they should at least have 5% of their throughput outside the home country.

Cosco handled the highest throughput share of 13,5% (16% growth from 2017) in 2018 and, most of the time, evaluated, followed by Hutchison and PSA (both with 8% growth from 2017) with a 10,5% 10,2% share, respectively. Each of these top 4 players has handled about 10% (or more) of the world's throughput since 2015. In 2018 DP World achieved a 9% share, roughly the same over the years. Collectively the 21 GTOs handled 627 million TEUs in 2018, representing 80% of global throughput. This increased from 69.6% in 2015 and 75.9% in 2016.

Figure 2-6 shows the evolution of how the top 10 ITOs have held their positions from 2008 to 2018 by equity TEU shares.

Figure 2-6: Evolution of the top 10 ITO by equity TEUs 2008-2018

Rank	2008	2011	2014	2017	2018
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					

Source: Drewry (2019)

2.5.2 Terminal operators and vertical integration

When the expansion occurs upstream or downstream, it is referred to as vertical integration. This type of co-operation takes various forms in maritime, from controlled market transactions to complete vertical integration. Vertical integration can be observed in the terminals operated for a specific ocean carrier/s, known as dedicated carriers. The ocean carriers may also serve other carriers on their terminals (non-exclusive terminals); inland transportation may also be encompassed in this strategy to gain greater control within the chain. The central question to vertical integration is finding a way to organize the chain efficiently. The effect of vertical integration on competition has received much attention in maritime, especially regarding the shipping line and terminal relations. For regulating authorities, difficulties in recognizing the balance between anticompetition and efficiency gains still give challenges. As vertical integration assists the buyer to transition into internalizing the services, increasing efficiency, it also raises costs for rivals (Loertscher & Reisinger, 2014).

Although dedicated terminals are much more common in the US and Asia and much less in Europe, co-operation in the shipping market existed even years before containerization, as observed in 1875 in the Calcutta Conference and some stabilization agreements, whose aim was to reduce competitive downwards pressures on prices (Musso et al., 2000). One of the most influential sectors in the supply chain and a sector which container terminal operators strongly depend on is the shipping line industry, where vessels have been expanded more and more with time, under the assumption that larger scale leads to greater profits (largest container vessel to date is 23 964 TEUs large, built-in 2020 and owned by Hyundai Merchant Marine (HMM)). However (Lim, 1998) provided an eye-opening idea in the container business. The study of Lim (1998) warns that enlarging the ship sizes will not always be beneficial; it may drive costs down but may also reduce rates. Drewry (2019) shows how the current SL sector is now characterized by low freight rates, low profits, bankruptcies, and volatilities. The effects of larger vessels on terminal operators is also studied by Sys, Blauwens, Omeij, Van de Voorde, & Witlox (2008). The additional services and infrastructure often required by larger vessels require additional funding and investments. Other ports may be called at less at times due to the inability to cater to the large ships. Unloading and loading these ships also require an additional amount of time, costs, etc., and to some extent leads to diseconomies of scale. Sys et al. (2008) found the optimal ship size to be 12 500 TEU at the study time. Still, the possibility of even larger ships was simulated, and the findings were that even larger vessels would be more optimal in the future (considering the growth of developing nations). However, it is Important to take into consideration is that the development of the ships does not solely affect the terminal operators but also the intermodal and commercial sectors.

Long before, Lim (1998) had predicted that because of the rise in competition as well as cooperative strategies, there would be a decline in the number of existing businesses in container shipping lines. The three dominant shipping alliances (2M, Ocean, and THE) operate three main east-west routes, dominating the container shipping industry and the terminal sector. While some have various portfolios of terminals and would thus prefer making use of their terminals, there are several carriers per alliance and at times with different interests at the same port, in other ports, the liners have no terminals and thus make use of independent terminals, or terminals by liners outside their alliance. This indicates the dominance and the complexity that exists in the relations. In 2018, the 2M alliance had more presence in

terminals than the other two alliances. 2M had a total of 104 terminal interests (with average equity TEU of 0.71 per terminal); the Ocean alliance with a total of 97 terminal interests (with average equity TEU of 0.70) per terminal and THE alliance with the lowest interest adding up to 38 terminals (with average equity TEU of 0.32 per terminal) (Drewry, 2019a). These are the terminals affiliated with the shipping lines. Such terminals can either be hybrids or carrier liners, as discussed earlier.

Crotti et al. (2019) show how the vertical integration between carriers and terminal operators can affect the merger waves. It is concluded that integration between terminal operators and container liners is unavoidable at the current moment due to the cost savings offered. When the co-operation is alliance-based, merger waves are more sustainable. In the port and shipping line industry, larger scales are associated with increases in profits and reduction in costs, while others may disagree ((Wu & Lin, 2015). To elaborate on these differences, a study by Seo & Park (2016) states that companies are likely to face increasing economies of scale early on the production (throughput in the case of container terminal operators). Later economies associated with the production begin to stabilize, which continues if production is increased and becomes negative returns to scale. Therefore, larger ships are hypothesized to have more potential for cost-saving than smaller ships due to these reasons. Seo & Park (2016) estimate minimum efficient scales determine whether there are increasing, decreasing, or constant returns to scale in the Korean container ports. According to Drewry (2016), a 12000 TEU vessel would save up to 11 percent per cent per cent of the total cost compared to an 8000 TEU ship and up to 23 percent per cent per cent compared to a 4000 TEU vessel. The consequences of the excess capacity, expansion, cost advantages, as well as challenges in container liner shipping, are also looked at by Tran & Haasis (2015), who built multiple regression models to compare revenues and costs and the changes in carrying capacity, ship sizes, slot utilization, etc. The study attests to several effects, but there is no evidence pointing to the impact of increasing vessel sizes on the financial indicators. It is concluded that total costs increase at a decreasing rate compared to capacity, leading to even smaller per-unit costs. There is no doubt about the economies of scale in larger ships, but it becomes less noticeable considering additional costs and significant investments.

A confirmation of the existence of scale effects is demonstrated in Van Hassel, Meersman, Van de Voorde, Vanelslander, & Van Voorde (2016). They evaluate the impact of the increase in ship sizes on the rest of the chain. Conclusions reveal that

the total generalized chain costs decrease with an increase in ship sizes in the container sector. The benefits are realized more in the Far East - Europe route when the costs are determined by the maritime part of the whole chain. The study reveals that the importance of maritime costs is reduced when vessels are larger, whereas the importance of the port and the hinterland increases. However, due to the limitations on the hinterland costs, the port costs become the most important. According to this study, the chain with the lowest generalized costs will be the most successful.

It is not by chance that very likely, ports receiving and handling goods smoothly are often considered to be efficient (more in Chapter 6) (Benacchio & Ferrari, 2002). For a successful vertical integration chain, how the players are integrated is essential. The hinterland transportation availability is one factor that influences the compatibility of the port and the level of co-operation between players (Heaver et al., 2001). Franc & Van der Horst (2010) find that SLs and TOs enlarge their scopes to reduce uncertainty. SLs expand to the hinterland services specifically due to unreliable services, while the TO's motive is mainly due to space concerns and to ease coordination with hinterland transport providers. Zhu et al. (2019) develop a model that shows that when authorities and regulators regularly review the competition status, vertical integration can lead to higher capacity, market output, consumer surplus, social welfare, and reduced costs related to delays. According to the Port of Antwerp (2020), vertical integration has led to more successful firms in the logistics sector. Such collaborations have been achieved much through data integrating, making decision making faster and more efficient.

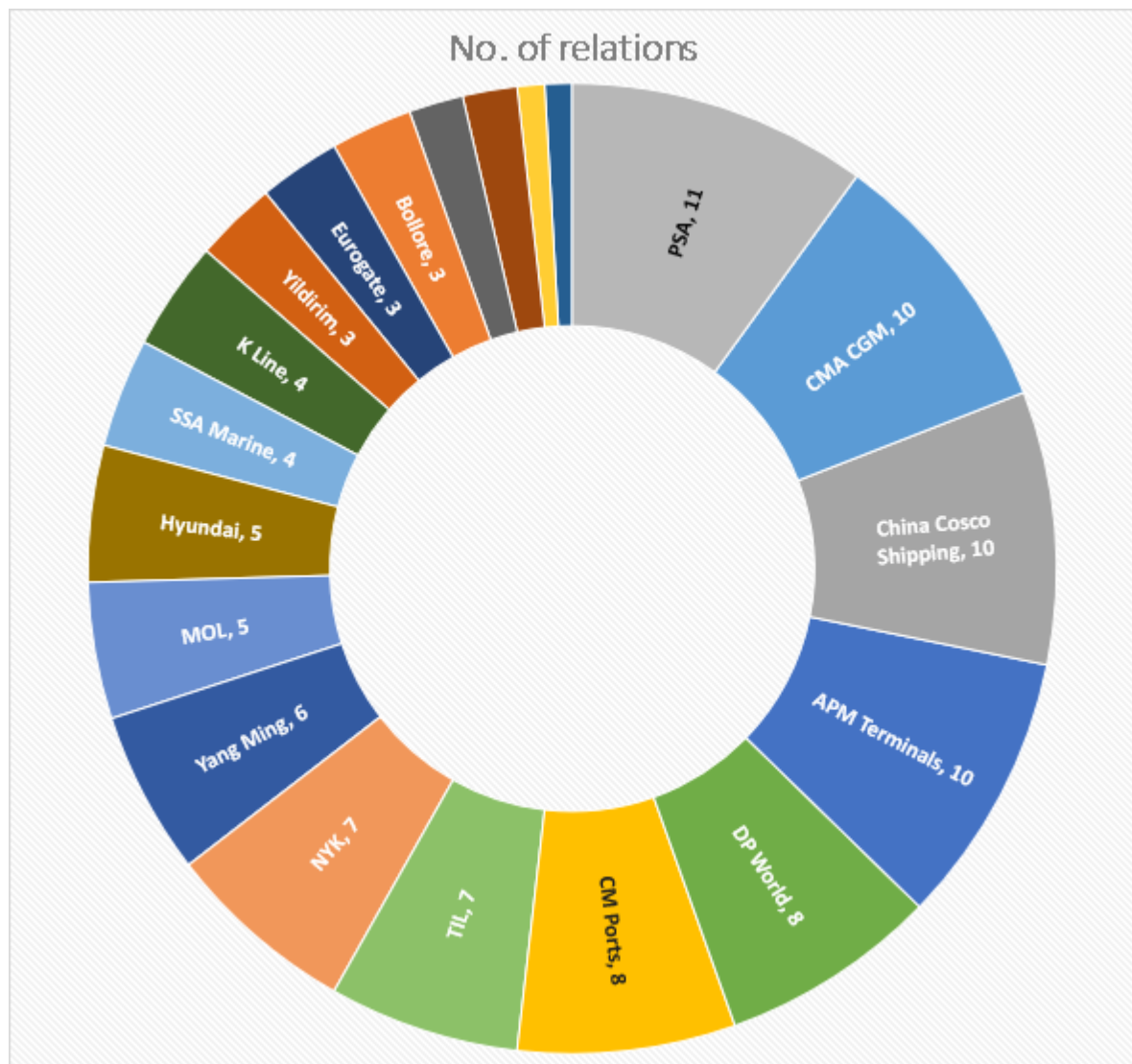
2.5.3 Terminal operators and horizontal integration

TOs, SLs, port authorities, and hinterland transportation can form horizontal integration by acquiring additional business activities at the same level to gain economies of scope, maintain competition, and gain bargaining power. According to Van De Voorde & Vanellander (2010), the scale and scope effects are necessary for shaping the company's merging or diversification strategies. Mergers and acquisitions between terminals are not very prevalent in the sector, similarly between shipping lines (alliances are more visible). Port authority's co-operation seems to be even more visible. Van de Voorde & Vanellander (2010) mention that mergers and acquisitions between TOs lost much of the momentum in the 1990s. In Antwerp, one of the most prominent mergers between terminals was between

Hessenatie and Noordnatie. The aim was for scale economies purpose and to gain negotiating power with SLs and the port authority. However, some relations still exist between TOs.

Terminal ownership is quite complex. The TOs have some form of relation among each other as indicated in *Figure 2-7*. The player with the most relations is PSA (11), most of which are joint ventures. Other terminals with numerous entanglements include APMT (10), Cosco (10), and CMA CGM (10). CM Ports (8) and DP World (8) are not far behind. HHLA is the only GTO in the sample of 21 with no ownership connections with any of the terminals. Most of these terminals have at least a joint venture with another terminal. Only CMA CGM & CM Ports, CMA CGM & Yildirim, and Hutchison ports & PSA have shareholder relations. The most recent merger is between three operators (MOL, K-Line, and NYK) to form Ocean Network Express (ONE). The company policy of TIL is much based on developing relations with other terminals, it would be expected to have the highest of such ties, but this is not the case, although they are not very far behind.

Figure 2-7: The number of relations between GTOs (2017 and 2018)



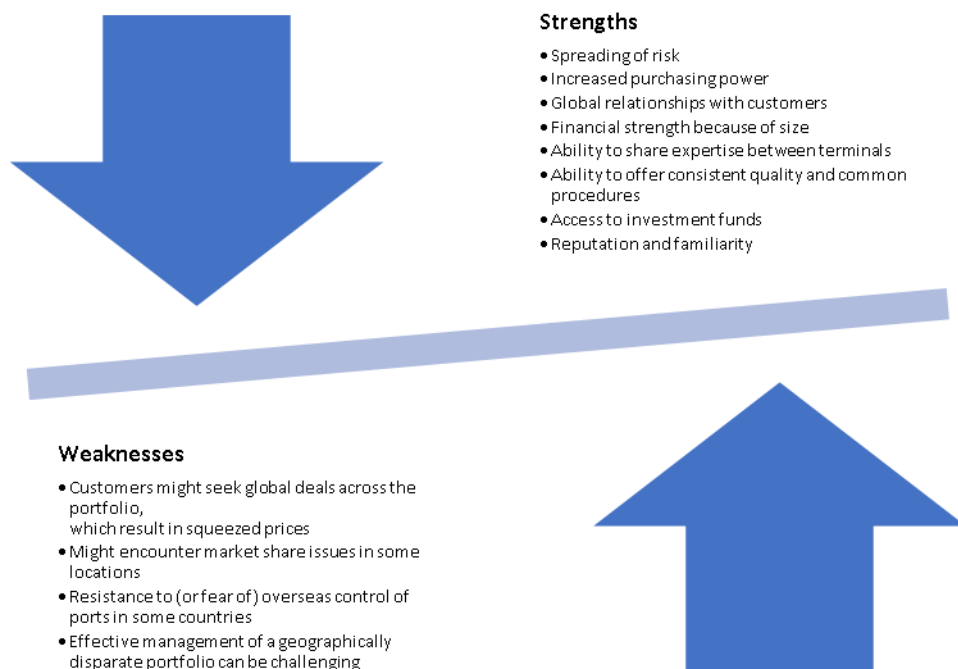
Source: Based on Drewry (2019,2018)

2.6 Effects of consolidations on the maritime supply chain players

In Vanelslander (2005), the effects of various forms of co-operation are reported. The effects are analyzed differently between horizontal and vertical integration. While vertical integration affects more efficiency and strategic moves, horizontal integration is more concerned with economic, financial, and/or market-related issues. Capital requirements and sunk costs show the scale diverted to investments in the CT sector. Economies of scale have allowed companies some power in the market, provided increased experience, and saved costs (Musso et al., 2000).

Co-operation between terminal operators and shipping lines is beneficial to the terminals operator on one hand because it increases the certainty in volumes; it provides the TO with more information regarding the plans and objectives (including long-term goals) of the shipping line; the shipping line can influence how the terminal is developed, and in the case of entire dedication, there is a guarantee of capacity in the terminal for the shipping line. On the other hand, instead of being profit centres, terminals dedicated to shipping lines are usually run as cost centres, resulting in the underutilization of the terminal resources. *Figure 2-8* shows attractions and weaknesses in being a GTO.

Figure 2-8: Strengths and weaknesses of GTOs/ITOs



Source: Based on Drewry (2019)

While port strategies shape (govern) the overall port operations, they also affect the terminal operator's productivity, efficiency (chapter 5 and 6), costs and pricing (chapter 7). In such a port setting, the flow of costs, benefits, and revenue is crucial. Container terminals strive towards providing efficiency to their customers. This indicates that capital utilisation should be optimal while value is created (operational flexibility) for the port users and while terminal costs are covered.

Different strategic moves of different terminals can affect capacity utilisation, pricing, concessions and cost distribution, which can affect performance (UNCTAD,

1975). The desirable utilisation levels determined by a terminal will impact on the efficiency and productivity of each of the assets at the terminal. For instance, a terminal could decide between a single or a double work shift per day. The decision will affect the utilisation levels of the terminal's assets (quay wall, quay cranes, yard cranes, storage facilities, etc.). In some cases, the terminal can improve the utilisation level by setting its prices to consider utilisation elements. While the motive for a terminal operator is supply-related, for a port user, it is demand-related.

While efficiency lies in the optimal use of assets, it doesn't mean that assets must be fully utilised. Asset underutilisation is expected at initial use; at some point, the utilisation levels are expected to increase up to a certain level. Some level of utilisation rate must be determined, balancing supply and demand. Terminals may provide excess capacity to provide a cushion for times of changes in demand and supply, weather disruption, faults, or other unpredictable events. This would certainly affect short term efficiency and productivity and can also be considered when pricing the terminal's services.

2.7 Conclusions

The container terminal operators in Hamburg – Le Havre face a unique situation in that the hinterlands are close to each other, and thus container cargo can easily enter or exit through almost either one of the ports with ease. Therefore, container terminals need to continuously review their management strategies to remain competitive. And at the same time, they should be aware of any decision-making-factors that their customers may depend on in terms of port selection.

This chapter examined the maritime logistics chain to determine the changes that have resulted in the types of terminals we see today. With all these market changes, the efficiency of terminals as a connection between various modes of transport became of paramount importance. The chapter highlights that the strategic position of a terminal can also influence costs and prices of the terminal, affecting efficiency.

The dynamics in the maritime supply chain through time are highlighted. The chapter gives a clear idea of the value-adding position and role of a terminal operator among the other players. The services offered through a terminal operator are shown to depend on the port management strategy of the operator. It is shown that various factors can drive consolidation and competition. Through mergers, takeovers, and

alliances, co-operation and competition have been established to influence the sector significantly. Competition has led to the high prevalence of horizontal integration, especially between container carriers and between terminal operators, to gain market power and scale economies, among other reasons. Vertical integration has also been observed throughout the chain, especially between carriers and terminals in the container sector, to gain reliability and efficiency. These types of relationships are expected to continue at least in the near future.

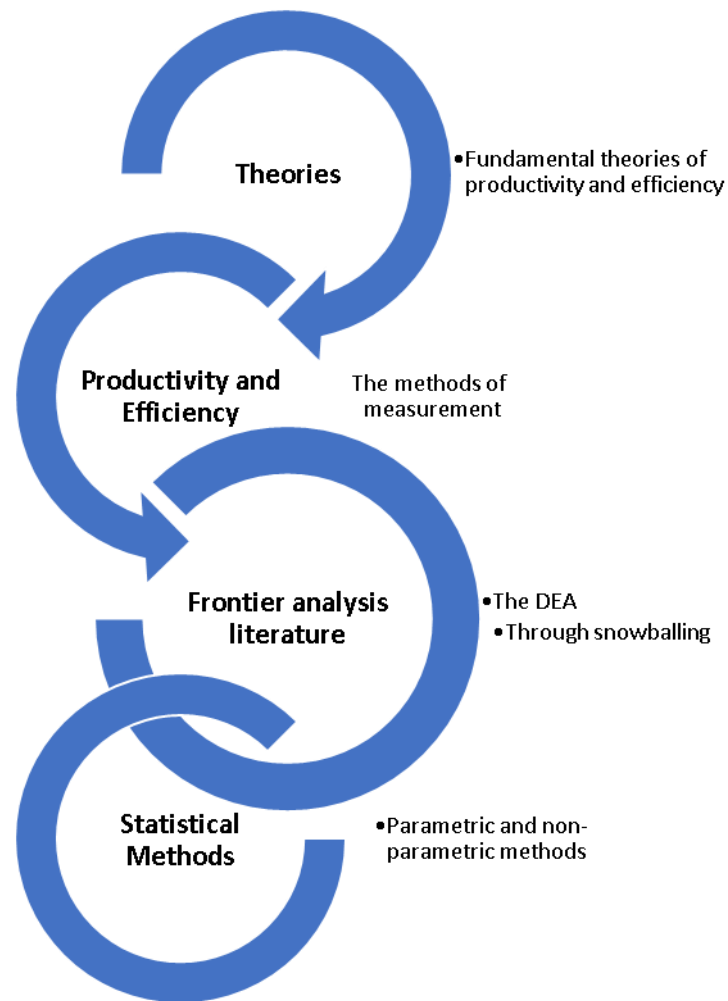
The global economy continues to be the engine of the maritime sector, indicating that phenomena such as the 2008 global crisis and the 2019/20 Coronavirus (COVID-19) pandemic will significantly shape the trends and changes in the economy. The outbreak of the Coronavirus has tested the fragility of the supply chain. Although the impact of the virus is already hard felt by the economy, the extent of the effect at this moment cannot be determined.

Chapter 3: A critical review of efficiency and productivity literature

The previous two chapters have demonstrated the importance of measuring performance in the container terminal sector. In this chapter, the methods of measurement are elaborated. The main objective of this chapter is to deduct and learn from past data envelopment analysis (DEA) literature in the container terminal sector. The lessons are then used as the building blocks of the analyses in this thesis. The gaps existing in productivity and efficiency analysis in the container terminal sector are addressed to contribute to the body of knowledge in this domain.

The literature review process followed in this chapter is illustrated in Figure 3-1 . First, the early works of productivity and efficiency are reviewed. This leads to the discussion of the various methods available based on past literature to measure efficiency and productivity, assessing their advantages and disadvantages in the second section. The third section discusses the DEA literature reviewed in the container port and terminal sector and expands on the attributes of the body of literature. Statistical analyses often complement the mathematical DEA. The fourth section is dedicated to explaining the parametric and non-parametric statistical approaches which can be supplemented with the DEA.

Figure 3-1: The process of the literature review in this chapter



3.1 The early research work in productivity and efficiency

A natural starting point in discussing the efficient frontier analysis is with the work of Farrell (1957) who seemed to have been forgotten until the publication of the Charnes, Cooper & Rhodes (CCR) (1978) paper. Farrell's (1957) seminal paper was presented with the concepts of efficiency and influenced much of the DEA work of today, especially DEA models of radial efficiency (explained in chapter 5). As discussed in the theories section earlier in this chapter, the work of Farrell was not the only one that led to the CCR publication and thereafter, certain other works stemming from the CCR have also made a significant impact on the topic of research. The work of Fare and Lovell (1978) later introduced non-radial efficiency. Forsund & Hjalmarsson (1974) lay a good foundation of efficiency terminologies in an attempt to resolve ambiguities and unanswered questions based on the work of Farrell (1957). The

efficient frontier topic was work adapted from multi-disciplines which inspired works of authors such as Aigner and Chu (1968), Seitz (1970 and 1971); Timmer (1971), Carlsson (1972); Lovell (1993); Fare et al. (1994); Greene (1997) among others.

Other research work with contributions to the frontier topic include Aigner, Lovell, and Schmidt (1977); Meeusen and Van den Broeck (1977), who specifically laid the foundations for the SFA. De Borger, Kerstens, & Costas (2002) give a comprehensive review of production and cost frontiers in the public transport sector, referencing the early works dedicated to the transport sector. The review mentions influential research such as Viton (1997), who studied technical efficiency in the US bus industry; Good et al. (1993), where the US and European airlines are compared; De Borger (1993), whose focus was on the Belgian railroads; Oum et al. (1999) in railroads.

The construction and analysis of Pareto-efficient frontier production functions proposed by the DEA are based on new theoretical characterizations (Charnes et al., 1985). For a detailed discussion on production economics, readers can refer to Coelli et al. (2005). The concept of Pareto-efficiency is heavily based on economic theory, making use of the work of Shephard (Shephard, 1953; Shephard, 1970) where production possibility sets and costs are evaluated, developing what Charnes et al. (1985) refer to as the 'transform' theory.

Although based on a single output scenario, the efforts of (Afriat, 1972; Aigner et al., 1977; Forsund & Hjalmarsson, 1979) have a major influence on the statistical estimations of parameters. (Farrell, 1957) whose work is based on Debreu (1951) and Koopmans (1951), then later developed these and referred to it as the 'activity analysis' making use of a single output scenario, and later also the multiple output case. Farrell also recognized the measure of economic efficiency which was derived from technical as well as allocative efficiency. Applying the method of Farrell in the service sector grew the need for the multi-variable approach. Building on this idea, the concept of a Decision-Making Unit (DMU) was coined in the birth of the DEA by (Charnes et al., 1978a).

Dyckhoff & Souren (2020) recognize that most evaluation methods such as those of cost accounting, life cycle assessment, and the DEA, as tools to assess the performance of human behaviour and as tools to assist with the improvement of performance itself. With this philosophy, Dyckhoff & Souren (2020) associate these methods with natural science research in a way that justifies and develops theories by explaining or predicting a human or an organization's behaviour. On the other

hand, the methods are also associated with blends of engineering and arts, seeking to create useful innovative artefacts. The authors relate this to Kernel theories of non-financial performance evaluation. With the understanding that a DMU controls the use of resources to achieve the desired ends, Dyckhoff & Souren (2020) base their approach on the decision theory and propose a Kernel theory named 'decision-based production theory'. In contrast to the traditional theories based on Pareto-Koopman's principles, where more output and less input are preferred, the decision-based production theory is derived from assessing the consequences based on the DMU's preferences.

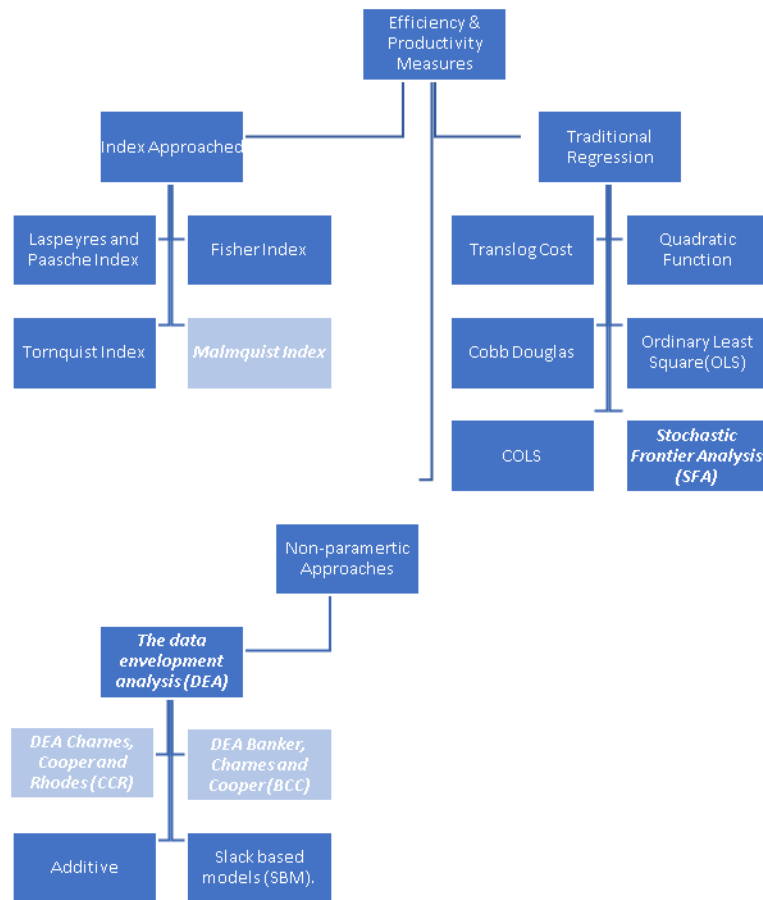
As a method based on the production theory, Dyckhoff & Souren (2020) develop a multi-criteria production theory (MCPT). The technique is a decision-theoretical generalization of the traditional production theories (such as the activity analysis of production and allocation of Koopmans (1951), the theory of cost and production functions of Shephard (1970) or the business-oriented production theory of Gutenberg (1951)). The theory addresses modern-day management science and economics concerns, applying theorems from multi-criteria decision making (MCDM) and multi-attribute utility theory (MAUT). The MCPT distinguishes the inputs and outputs for the DEA as basic technological entities and costs & benefits as performance attributes.

3.2 Efficiency and productivity methods of measurement

Based on the early research work, efficiency and productivity can be measured in several ways, summarized in Figure 3-2.

According to past literature, in measuring productivity and efficiency, alternatives exist between index approaches (Laspeyres and Paasche Index; Fisher Index; Tornquist Index; Malmquist Index); traditional regression approaches (Translog Cost; Quadratic Function; Cobb Douglas; Ordinary Least Square(OLS); COLS; Stochastic Frontier Analysis (SFA)) and non-parametric approaches (The data envelopment analysis (DEA); DEA Charnes, Cooper and Rhodes (CCR); DEA Banker, Charnes and Cooper (BCC); Additive; Slack based models (SBM).

Figure 3-2: Efficiency and productivity measurement approaches



Source: Own compilation

3.2.1 The index approaches

In economics, the use of index numbers was already made famous in the 19th century by contributors such as Laspeyres and Paasche, whose formulas are still made use of today. The Fisher (1922) and Tornquist (1936) index play a significant role in productivity measurement. The historical background of indexing is fully discussed in Diewert and Nakamura (1993). An index number is designed to measure comparison over time, space, or both (Coelli et al., 2005). In the economic-theoretic approach, index numbers rely on well-defined properties, tests, and axioms, whereas, in productivity measurement, the microeconomic theory of a firm is more relevant. The index approach involves constructing aggregates using a certain index formula. According to Liu (2010), the indices created are based on economic assumptions of production.

Total factor productivity (TFP) or multifactor productivity (MFP) ⁸ approaches can be based on Hicks-Moorsteen, Malmquist, profitability ratios, and component-based approaches. The purpose of measuring efficiency determines which approach to use. While the Hicks-Moorsteen and Malmquist approach are appropriate in a situation where productivity change is to be determined without the need to specify sources of productivity, the profitability ratios are more appropriate for a more business-oriented approach (Coelli et al., 2005), this is thus the reason for less preference of these methods compared to the Malmquist.

All these methods have their strengths and weaknesses. To measure efficiency, Estache, de la Fé, & Trujillo, 2004 use an index approach specifically due to the non-requirement of the method for economic assumption and input prices. The major strength of the index approaches, such as the TFP approach, is the ease in which calculations can be carried, while one of its weaknesses is the inability to differentiate between technical changes and the scale economies (Liu, 2010). The integration of the TFP with the DEA resolves this problem, which is later discussed in this chapter.

3.2.2 Traditional regression analysis

Various studies in the literature have evaluated cost efficiencies employing various econometric and mathematical regression models. The most common functions applied for such estimation are the translog cost function as well as the quadratic function (Caves, Christiansen, & Trethway, 1984)(Pulley & Braunstein, 1992); Faust & Baranzini, 2014)(Caves et al., 1984; Onghena, 2013). In regression analysis, distance functions can also be utilized. Coelli & Perelman (1996) list five alternatives of distance functions in estimating the efficient frontier. The parametric frontier estimation through the corrected ordinary least squares (COLS), favoured by authors such as Lovell et al. (1994) and Grosskopf et al. (1996).

Cubbin & Tzanidakis (1998) give a comparison between regression analysis (RA) and the DEA as substitutes in measuring comparative efficiency. In the study of Cubbin & Tzanidakis, the highlighted advantage of the RA is its built-in checks in ensuring objectivity and comparability between the firms under observation. RA is strongly reliant on functional forms and tests of significance, something which the DEA lacks.

⁸ There is a subtle difference between TFP and MFP, grounded in the question of whether or not it can ever be possible to take into account all the total factors, or perhaps a multiple of factors, often, these terms are used interchangeably.

This then assists in ensuring that factors are included and excluded appropriately – a condition that usually depends on the extent of the normality assumption of errors. In the regression approach, when an irrelevant variable is included in the analysis, it will have a low or even a zero weight, making its impact negligible, again something different from the DEA.

According to Cubbin & Tzanidakis (1998), to get to the frontier, at least one of the firms is assumed to be fully efficient. According to this author, the cost function of a firm is not only affected by inefficiency but also by measurement error. This element is usually expressed in the efficiency field of research by making use of the SFA approach.

Banker & Natarajan (2008) mention that the OLS regression method, which fits an average line through data, has the advantage that it is based on economic theory and therefore offers a reliable explanation of the cost or production structures and it is also able to distinguish between the variable's role to output. However, the method's rootedness to the average line assumption indicates that the firms are rational and efficient at all times, which does not always reflect reality and thus introduces inaccuracy. The COLS approach has a similar approach to the OLS; it estimates a line through the data and creates a corrected line that includes all the data, which then becomes the efficiency line; taking it a step further. One of the advantages of the method is that it accounts for relative efficiency and how the input variables affect the output. The drawback of the method is that it requires a priori assumptions; it cannot measure the error term and other statistical noises and it is also sensitive to outliers (Banker & Natarajan, 2008).

3.2.2.1 Stochastic frontier analysis (SFA)

The SFA evolves from regression as well as frontier analysis. Production and cost functions are very significant aspects of the SFA. Although the similarity is very strong between the traditional regression analysis and the SFA, there is also a major distinction to be made. In the traditional regression analysis, while the assumption is that firms can obtain the maximum output, in the stochastic frontier approach, firms are assumed to not at all times be able to reach the maximum output, which thus allows for the measurement of inefficiencies, (Liu, 2010).

Cullinane & Song (2003) mention some reasons why frontier models have become increasingly popular. The frontier models are based on economic theory; the

deviations from the frontier can be naturally interpreted without complexity and policymakers can also use the results obtained.

Its ability to apply different functional forms gives the SFA the flexibility to be adapted to different situations. While this offers flexibility, it also is associated with one of the disadvantages of the model because an inappropriate functional specification would indicate inappropriate efficiency scores. The SFA can capture statistical noise which gives the models the ability to take into account factors that are exogenous to the model (environmental factors). This an advantage that some other methods like the DEA are incapable of (Cullinane & Song, 2003).

The SFA identifies the frontier through a regression method and with a composed error term. A frontier is to represent the *best practice* of the group of firms under examination. The developed frontier is then compared to efficiency (technical, scale, allocative, X, productive, dynamic, social, etc).

It is clear in theory that many functions can be applied to the SFA. The linear, the Cobb-Douglas, and the Constant Elasticity of Substitution (CES) belong to one group. The quadratic, translog and generalized Leontief functions are in the same group belonging to flexible forms. The extent to which the reality is reflected between these functions is different. In the selection of the function, one should thus select one closest to the reality of the sector or industry. Liu (2010) mentions two groups observed between the functions.

The purpose of the functions can be to measure production or costs. The production function can only handle a single output scenario and the cost functions can handle a multi-output scenario. The distance function usually is also applied to bridge the gap of the inability of the production function to cater for the multiple function scenario. Rodriguez-Alvarez, Tovar, and Trujillo (2007), Trujillo and Tovar, (2007), and Gonzalez and Trujillo (2008) consider the distance functions (Liu, 2010).

According to Merkel (2019), the Cobb-Douglas restricts the returns to scale to be constant, whereas the translog adds cross-product terms for each input. The difference between the translog and the Cobb-Douglas lies in the ability of the former to have less restrictive assumptions, which thus increases its flexibility.

According to Merkel (2018), the translog and the Cobb-Douglas are two methods with the ability to estimate the first and second-order unknown relationship. To assess the advantages that often exist between the translog compared to the Cobb-Douglas, the

likelihood ratio test can be applied. The choice is resolved by carrying both the functions and testing one which is more superior through the application of the LR test. The sufficiency of the Cobb-Douglas is tested across many SFA studies. (Trujillo and Tovar, 2007; Tongzon and Heng, 2005) found the Cobb- Douglas sufficient while (Coto-Millan et al.,2000; Estache et al.,2002) found the translog more superior.

The potential functions are not limited to these mentioned above, but these demonstrate the widely applied functions. Liu (2010) mentions other functions including – the generalized Box-Cox; Fourier; Laurent Barnett; McFadden; asymptotically ideal model; normalized quadratic; symmetric normalized quadratic; Hedonic.

3.3 Non- Parametric approach: The data envelopment analysis (DEA)

Discussions about frontier modelling were initiated by Farrell (1957). These approaches are based on assessing firms that are considered best performers or which can be said to have the best practices in the sample concerned, while regression analysis seeks to explain the average behaviour in the sample concerned. The DEA was first introduced by Charnes, Cooper, & Rhodes (1978), involving the use of linear programming methods to construct a frontier. The DEA as well as the SFA are two well-favoured efficiency analysis methods in literature across sectors (ports, shipping lines, health care, insurance, water) and have been applied in the aspect of both productions as well as on cost analysis. Wang & Cullinane (2015) mention that, although there is no agreement of which method is the 'best' in literature, the SFA and DEA results have shown a high correlation.

The term "DEA" arises due to the ability of the method to envelop the data as a frontier. DEA is categorized as a non-parametric technique (based on mathematical programming), while SFA (based on regression analysis) is a parametric technique. Both can be extended from basic to very complex analysis, depending on mostly the data availability as well as the problem at hand. The DEA can be extended to various types of models (CCR; BCC; window; additive; Malmquist) and can be evaluated in multiple stages; including stages with statistical analysis (to eliminate the disadvantage of the method lacking economic theory grounding).

The concept of efficiency which DEA boasts about was borrowed from engineering whereby a machine's efficiency is given by the output/input ≤ 1 , (Poitras et al., 1996).

The DEA creates a virtual output and a virtual input, based on this, the efficient firms are allocated along the frontier, thus the use of the “relative efficiency” concept. All other firms not along the frontier are said to be inefficient and expected to adjust relative to the firms along the frontier. It is impossible to judge whether the firms along the frontier are optimizing their inputs and outputs in actual reality or not.

There are various reasons why DEA is considered a powerful method in measuring efficiency and productivity. DEA gained popularity by having been able to improve the estimation techniques in econometrics. It is also able to measure various kinds of activities, such as performance, and can also assist in decision making. According to Tongzon (2001), the DEA requires no strict development of standards against measuring efficiency, even though these can be voluntarily included. According to Poitras, Tongzon, & Li (1996), DEA differs from the other regression analysis in that it focuses on the individual DMU instead of aggregates. DEA does not require a priori knowledge regarding the weights of the inputs and outputs. This is what makes the method to be considerably a more flexible method. The absence of the need to identify the relationships means that there is no need to know the production function in which the inputs are converted to the output.

Cullinane & Song (2006) mention several reasons for the wide popularity of frontier models. Among the reasons is that frontier models are strongly consistent with the theory of optimization. The structure of the frontier can also apply to many policies. DEA ensures that decision-makers are given a solution to improve efficiency by reducing input or output. The method also can identify the nature of returns and therefore the ability to minimize the average cost. The benchmarking also allows for the possibility to compare the counterparts and determine the firm that shows industry standards that apply to the other firms.

DEA can handle multiple inputs and outputs simultaneously, but without accounting for noise in the data (Paradi et al., 2018). DEA is thus very useful in making comparisons between firms (DMUs) that use similar input factors and operate in comparable environments. As mentioned by Cullinane, Wang, Song, & Ji (2006), it is of great significance that methods applied to the benchmarking of terminals are consistent, objective, and have the ability of applicability to all terminals.

Another powerful feature of DEA is that data can be used in its natural unit. In other words, there is no need to unify the units of measure and there is also no need to set rigid weights to the variables (Elferjani, 2015). One of the DEA assumptions is that all

the firms evaluated are operated to their best potential and thus from the sample, draws comparisons to identify the levels of efficiency; from this aspect, it can be said that the DEA offers a fair evaluation of DMUs. Paradi, Sherman, & Tam (2018) refer to this as the “benefit of the doubt”. Another value of the DEA is the provision of a “peer” reference which suggests an efficient firm(s) whose practices can be followed by an inefficient firm(s).

A unique feature by the DEA which is not presented by any other methods at present is the ability to determine the proportion of ‘wastage’ of resources, i.e input factors and (or) the additional output that can be achieved by the DMU concerned. These are given by ‘slacks’ and ‘target’ variables in the model.

As much as the DEA has so many advantages, it also has some limitations. Some factors that are disadvantageous to the DEA include that the DEA does not take into account the statistical inference procedures. According to Elferjani (2015), other limitations of the method are that selecting the input and output variables becomes very important as the method strongly relies on these variables to create weights. This, would therefore affect the accuracy of the efficiency scores produced. In accordance with Cook, Tone, & Zhu (2014), while DEA does not discriminate between raw and process data and can accept different dimensions of variables in one model, a misjudgment of the efficiency scores can occur when there are such differences. Although the author does not clarify the reasons behind taking such caution, it is highlighted as one of the pitfalls of the DEA.

The literature review of the DEA

The section above highlighted the different alternatives in measuring efficiency which include the DEA. Based on its advantages and appropriateness for the current analysis, the literature of the DEA is here reviewed further.

The need for firms to constantly remain mindful of their performance has given rise to the interest in the topic by researchers as well as managers among others (Taouab & Issor, 2019). Non-parametric efficiency estimation was initially used in the study of industrial-organizational performance in the 1970s (Chaouk et al., 2020). Later on, the DEA was developed and was used for the analysis of public entities across sectors. Banker et.al (1986) uses it for the analysis of hospitals; Ray (1991) in schools; Lewin et. Al (1982) in courts; Deprins et al. (1984) in post offices; Charnes et al (1985) in air force maintenance; Bourton and Phimister (1995) for Journal ranking; Banker and Johnston (1994) in airlines to name a few.

Although the literature shows that the DEA and SFA have proven themselves powerful across industries, studies that focus on just the literature review are very limited and include (Saeedi, Behdani, Wiegmanns, & Zuidwijk (2018); Paradi, Sherman, & Tam (2018); Mardani, Zavadskas, Streimikiene, Jusoh, & Khoshnoudi (2017); Valle (2015); Jarboui, Forget, & Boujelbene (2012); De Borger et al., 2002); Førsund & Sarafoglou (2002)) and specifically in the port sector (Chang & Talley, (2019); Langenus & Dooms (2015); Dutra, Ripoll-Feliu, Fillol, Ensslin, & Ensslin (2015); Park & De (2015); Dhingra, Prakash, & Raju (2014); Woo, Pettit, Kwak, & Beresford (2011); Panayides et al. (2009)). González & Trujillo (2009) and Lu & Wang (2016) specifically on container terminals.

Past studies making use of the DEA have shown to take on different directions and assumptions when formulating the models. Through snowballing, studies reviewed in this section will highlight the common understanding and the differences between the DEA studies in the container terminal and port sector internationally.

Table 3-1 specifically contains studies that analyze the performance of the container port or terminal sector. The comparison of the studies is based on the activity of involvement (port or terminal), the period over which the study is carried out, the location which the study analyses, the orientation assumption of the model as well as the variables (input and output) used for the analysis of performance.

Table 3-1: The DEA literature

AUTHOR	ACTIVITY AND PERIOD	REGION	ORIENTATION	MODEL TYPE	VARIABLES
Roll and Hayuth (1993)	20 Ports Hypothetical data		Both	DEA CCR	Input: Manpower, capital, cargo uniformity Output: Cargo throughput; level of service; user's satisfaction ship calls
Poitras et al. (1996)	23 cross-national Ports	5 Australian and 18 international	Not specified	DEA CCR	Input: Mix of 20-foot and 40-foot containers; Average delays in commencing stevedoring; difference between the berth time and gross working time number of containers lifted per quay crane hour; number of gantry cranes; frequency of ship calls Output: TEU berth hour; total number of containers handled per year
Martinez-Budria, et al (1999)	26 ports 1993-1997	Spanish	Not specified	DEA Banker, Charnes & Cooper (BCC)	Input: Labour, depreciation, other expenditures output: Total cargo revenue
Tongzon (2001)	16 Ports (1996)	4 Australian; 12 international	Input	DEA CCR Additive	inputs: Number of cranes, no. of container berths, number of tugs, terminal area, delay time, labour output: Cargo throughput and ship working time

AUTHOR	ACTIVITY AND PERIOD	REGION	ORIENTATION	MODEL TYPE	VARIABLES
Valentine and Gray (2001)	31 container ports	31 container ports out of the world's top 100 container ports	Not clear	DEA CCR	Input: total length of berth; total of investments Output: number of containers; total tons throughput
Itoh (2002)	8 ports	Japan	Not clear	DEA Window	Input: terminal area; number of berths, cranes and employees Output: TEUs handled
Barros (2003)	11 Ports	Portugal	Input	DEA	Inputs: labour (number of workers); capital (book value of the assets) Outputs: ships movement of freight; gross gauge; Break-bulk cargo containerized freight; solid bulk and liquid bulk Stage 1. Productivity: input (berthing capacity, cargo handling capacity) and output (cargo throughput, number of ship calls); Stage 2. Profitability: input (cargo throughput, number of ship calls) and output (revenue); Stage 3. Marketability: input (revenue) and output (customer satisfaction); and Stage 4. Overall efficiency: input (berthing capacity, cargo handling capacity) and output (customer satisfaction).
Park and De (2004)	11 Ports	Korea	Input	4-stage DEA	

AUTHOR	ACTIVITY AND PERIOD	REGION	ORIENTATION	MODEL TYPE	VARIABLES
Cullinane et al (2004)	25 Container ports 1992-1999	25 of 30 biggest terminals in the world	Output	DEA Window, CCR and BCC	Inputs: berth size; terminal area; number of berth cranes; number of yard cranes; number of straddle carriers Output: TEU handled
Turner et al (2004)	26 Ports & terminals	North American Ports	not specified	DEA (2stage)	Inputs: berth size; terminal area; number of cranes Output: TEU handled
Barros and Athanassiou (2004)	6 Ports	2 Greece and 4 Portugal	Input	DEA CCR and BCC	Inputs: Number of workers; book value of assets Output: ships movement of freight; total cargo handled; containers loaded and unloaded
Estache et al. (2004)	11 ports 1996-1999	Mexico	Input	TFP Malmquist Index DEA	Input: labour and capital (length of quay) Output: throughput
Min and Park (2005)	11 Container terminals 4 years	Korea	Not clear	DEA	Inputs: total length of quay; number of cranes; size of hard areas; size of labour force Output: cargo throughput
Cullinane, Song, Wang (2005)	57 Ports or terminals 1999	World	Input	DEA and FDH	Input: quay length; area, QC, yard gantries, SC Output: throughput
Barros (2006)	25 ports 2002 - 2003	Italy	Output	DEA CCR and BCC	Inputs: number of personnel; value of capital invested; size of operating costs Outputs: liquid bulk, solid bulk; number of ships; number of containers total receipts

AUTHOR	ACTIVITY AND PERIOD	REGION	ORIENTATION	MODEL TYPE	VARIABLES
Cullinane and Wang (2006)	69 Container terminals 2002	European	Output	DEA CCR and BCC	Inputs: terminal length, terminal area, equipment Output: container throughput
Cullinane, Wang, Song, Ji (2006)	30 Leading ports	International	Output	DEA BCC and SFA	Inputs: Length; area; number of cranes; number of straddle carriers Output: Throughput.
Rios and Maçada (2006)	23 Container Terminals 2002-2004	15 Brazilian, 6 Argentinean and 2 Uruguayan	Not specified	DEA BCC	Inputs: number of cranes, number of berths; number of employees; terminal area, amount of yard equipment Output: TEUs handled and average number of containers handled per hour/ship).
Al-Eraqi A, Adli M, Khader M, Barros CP (2008)	22 Ports 2000-2005	Middle East and Africa	Output	DEA standard & DEA window	Inputs: berth length, storage area, handling equipment Output: throughput
Wu, Goh (2010)	21 countries (ports) 2007	Cross country	Output	DEA CCR and BCC	Inputs: area, quay length; equipment Output: throughput
Hung, Lu and Wang (2010)	31 ports	Asia-Pacific region	Input	DEA CCR	Input: terminal area, QC; number of berths, terminals length of terminal, Output: throughput
Yip, Sun & Liu (2010)	141 container terminals 1997-2005	Cross continental	Output	DEA CCR 2 stage	Inputs: superstructure; infrastructure; yard equipment Output: throughput
Munisamy and Singh (2011)	69 Ports 2007	Asian region	Output	DEA CCR and BCC	Inputs: berth length, storage area, reefer points, total quay cranes, handling equipment Output: throughput

AUTHOR	ACTIVITY AND PERIOD	REGION	ORIENTATION	MODEL TYPE	VARIABLES
Bichou (2011)	10 Container terminals (2002-2008)	Global level	Input	Supply chain DEA CCR and BCC 2-stage	Input and output specified at gate, yard & quay and terminal level
Wanke et al. (2011)	25 Container terminals	Brazil	Input	DEA and SFA 2 stage	Input: number of berths, terminal area, parking lot Output: throughput and loaded shipments
Wanke 2013	27 Ports	Brazil	Not specified	DEA 2 stage	Inputs: no. of berths; area of warehouse area, yard area; intermediate input/output: solid bulk frequency; container frequency Outputs: solid bulk; throughput; Container throughput Contextual variables: private administration (yes/no)
Wang and Cullinane (2015)	104 Terminals	European	Output	DEA CCR and BCC	Input: length of terminal; area; equipment costs Output: throughput
Wanke and Barros (2016)	27 ports (2007-2011)	Brazil	Output	2 stage DEA BCC and bootstrapping technique	Input: quay length; maximal quay depth; warehousing area; yard area; channel width; channel depth Outputs: solid bulk loading hrs; container loading hrs; solid bulk; throughput; container throughput; solid bulk frequency; container frequency

AUTHOR	ACTIVITY AND PERIOD	REGION	ORIENTATION	MODEL TYPE	VARIABLES
Zheng and Park (2016)	30 Container terminals and ports 2014	Korean and China	Not clear	DEA CCR BCC	Inputs: berth length, yard area, number of quay cranes; number of yard cranes Output: total throughput
Luna, Mar-Ortiz, Gracia, & Morales-Ramírez, 2017)	152 containerships in a terminal	Mexico	Input	DEA 3 stage BCC	Input: number of quay cranes, number of trucks, number of yard cranes, service time Output: number of containers
Kammoun and Abdenadher (2018)	7 Ports (2007-2017)	Tunisia	Input	DEA SFA and	Inputs: Stevedoring equipment, area stores, Employees Output: Throughput (tons)

3.3.1.1 Orientation

In the DEA world, model orientation is one of the decisions that the analyst needs to make. The models can either be input- or output-oriented or assume no orientation. The orientation of the DEA determines what the model seeks to achieve concerning efficiency. An input-oriented model is assumed where the aim is to reduce 'excess' inputs while holding the output constant, whereas an output-oriented is assumed to 'expand' shortfalls in the output while holding constant the inputs. Non-oriented models are those assuming no orientation or considering both orientations (e.g. additive models), more of this is discussed in chapter 5. According to Coelli et al. (2005), the choice of orientation in the DEA is not as critical as in the case of regression analysis. The nature of the industry has high power to shape the orientation assumption. Input orientation is often the selected orientation in many fields due to the ability to control the input variables. In some industries, the input factors are fixed and the firms are to achieve the most output given those input factors.

Besides selecting the best model for efficiency, there is also an issue of ensuring that the data used is the best representative of the production function and that usable and actionable information for managers is captured. In 2018, Dyckhoff & Souren (2020) counted more than 1500 studies employing the DEA across sectors. The reliability of data being a common problem for the majority, therefore affecting the validity of the results.

In general, when selecting the input and output data, it is fundamental to consider that one set is minimized and the other is maximized (input and output, respectively). Another important aspect is evaluating the correlations of the variables and selecting variables that managers can easily interpret to improve the wastage of resources (Paradi et al., 2018). Generally, there is no limitation to the number of input and output variables but for reasons of degrees of freedom, too many data variables versus the number of DMU's can be hindering. According to Cooper et al. (2006), if the number of DMUs is less than the number of inputs and outputs, this will result in a large amount of DMU's being efficient when they actually may not be. Cooper et al. (2006) advise that first, a small number of variables is selected; when desired, these can be increased gradually, observing if any changes occur.

Another issue mentioned in Dyckhoff & Souren (2020) is the dependence of the weights of the DEA on non-negative data. Thus, the data cannot be negative and cannot be zero. For zero values, Huguenin (2012) suggests a substitution by a low value such as 0.01. This also means that in the DEA, the input and output variables must be equally available for all the DMUs to be analyzed.

The cornerstone of the direction of the results of the DEA is based on the input and output data, which is redirected to enhance performance. According to Paradi et al. (2018), this requirement has caused arguments between academia and industry where pure theory versus reality become entangled.

However, to overcome this problem, Cooper et al. (2006) suggest taking into account the degrees of freedom; to turn to algorithmic approaches, metric approaches, multiplier approaches, and statistical approaches.

Charnes, Cooper, & Rhodes (1978) propose a model that is input-oriented and some of the later studies to it assumed otherwise. In the port sector, Cullinane et al. (2004) associate input orientation with operational and managerial concerns and output orientation with planning and strategies. According to Cullinane et al. (2004), considering the long-term planning horizon and facilities of the port sector, input orientation can be a better representation as the output is expected to be fixed to the maximum capacity once the port is built. A port/terminal can under normal circumstances be able to predict its output at least for a year, especially given the quite stable client base. On the other hand, before making any further decisions for expansion, a port/terminal could use the knowledge of whether more output can be obtained from the present input factors. From that point of view, the output-oriented model can provide better insight.

In the literature sampled, it can be observed that there is a division between studies who assume output orientation, those who are lenient towards the input orientation, those who assume no orientation (in some cases it is not specified or not made clear), and others who assume both orientations.

If a terminal is to retain its profits or operate efficiently, then it is better to focus on minimizing its cost, which would require that the focus is directed to the input factors. According to Liu (2010), the infrastructure can hardly be adjusted, if not impossible, making it senseless to apply the input-oriented model. The input orientation option also allows for the possibility of the terminal to know precisely

where it stands regarding the sources of inefficiency within its yards. Input orientation aims to enable the terminals to plan, focusing on the identified sources of inefficiency. In some cases, e.g. regarding labour and yard equipment where the model identifies 'wastage' of the resources, it can be possible to use less by reducing the amount of used equipment, for instance and thus save on the running costs (the maintenance costs and labour associated with the machinery). Luna, Mar-Ortiz, Gracia, & Morales-Ramírez (2018) show an example of this. This could be done by adjusting accordingly depending on the number of containers to be handled. In the reviewed sample of DEA literature, some among those assuming an input-oriented model include (Barros, 2003; Barros & Athanassiou, 2004; Bichou, 2011; Cullinane, Song, & Wang, 2005; Estache, de la Fé, & Trujillo, 2004; Hung, Lu, & Wang, 2010; Kammoun & Abdenadhe, 2018; Luna et al., 2017; Park & De, 2004; Tongzon, 2001; Wanke, Barbastefano, & Hijjar, 2011). Barros (2003) assumes this with the assumption that seaports are concerned with cost control and difficulty controlling the cargo; Cullinane et al. (2005) is based on the fact that the output can be predicted in the short term and medium-term. Many of these studies propose the input orientation solely to minimize the outputs and costs.

According to (Liu, 2010), ports and terminals relate more to output orientation than input orientation. This is due to the port authorities' ability (to some extent) to influence production levels through policies and strategies and yet difficulties to change the input resources at least over a short period. Among others, (Al-Eraqi, Mustafa, Khader, & Barros, 2008; Barros, 2006; Cullinane & Wang, 2006; Cullinane et al., 2004; Cullinane, Wang, Song, & Ji, 2006; Munisamy & Singh, 2011; Wang & Cullinane, 2015; Wanke & Barros, 2016; Wu & Goh, 2010; Yip, Sun, & Liu, 2010) chose the output orientation in their DEA models. The output-oriented model in Cullinane et al. (2004) is applied purely from a pragmatic point of view and for ease of interpretation; Munisamy & Singh (2011) do so with the reasoning that ports can easily control the throughput; Wanke & Barros, 2016 recognize that orientation is not a consensual aspect of the DEA and assume the output orientation for the mere fact that ports and terminals will always attempt to maximize their output with the given level of output.

Other studies do not focus on this conception of orientation but rather focus on interpreting the results of efficiency scores by comparing the terminals or ports. Other studies have no explicit explanation regarding the direction of the model, (Itoh,

2002; Min & Park, 2005; Poitras et al., 1996; Rios & Maçada, 2006; Turner, Windle, & Dresner, 2004; Valentine & Gray, 2001; Wanke, 2013; Zheng & Park, 2016).

Only the case of Roll & Hayuth (1993) discussed the results from the perspective of both the orientations. Chapter 6 of this thesis will take on this approach where the perspective from both orientations of the model are explored.

3.3.1.2 Region

In an ideal situation, ports and terminals that are similar and which operate on similar environmental settings (such as the HLH ports and terminals) should give a good basis for comparison. Dyson et al. (2001) emphasize that factors should reflect the common activities among the analyzed decision-making units and that environmental changes should be unable to change the behaviour of the factors. Roll & Hayuth (1993) began discussing the DEA with a hypothetical situation. The rest of the studies that followed later would all be based on an analysis of some specific location making use of actual data. As can be observed in Figure 3-3, it is most popular to globally compare ports/terminals (Bichou, 2011; Cullinane et al., 2004, 2006; Valentine & Gray, 2001; Wu & Goh, 2010; Yip et al., 2010).

Studies making an evaluation of ports within the European region have also been popular (Barros, 2003, 2006; Barros & Athanassiou, 2004; Cullinane & Wang, 2006; Martínez-Budría, Díaz-Armas, Navarro-Ibañez, & Mesa, 1999; Wang & Cullinane, 2015). However, most of these studies are focused on a specific country rather than the continent itself e.g. Barros and Athanassiou (2004) comparing Greece and Portugal. To supplement this, Cullinane & Wang (2006) analyze 69 European container ports through the DEA BCC model taking into account the geographical influence as well as terminal size to evaluate and estimate efficiency. Merkel (2018) examines, in the context of European container ports, 77 large ports to find the relationship between technical efficiency and competition.

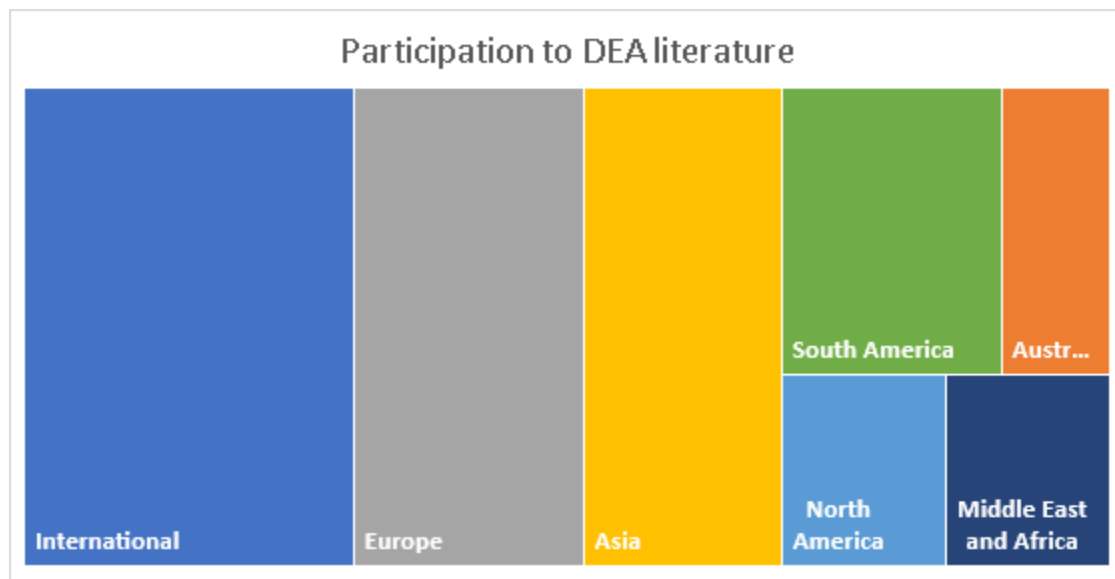
Studies representing Asia have given the same proportion as European studies. At a regional level, Munisamy and Singh (2011), apply the DEA CCR and BCC model to demonstrate the efficiency of 69 ports in Asia. Australia is represented in (Poitras et al., 1996; Tongzon, 2001). At an international level, Poitras et al. (1996) conduct a combination of the national category (Australia) with a comparison of 23 ports consisting of 18 international ports and five Australian ports by using a DEA CCR model. Tongzon (2001) attempts the same by also analyzing four Australian ports to 12 other international ports. (Valentine & Gray, 2001) select and analyze 31 ports from

the top 100 World container ports and apply the DEA CCR model to measure the port's efficiency. By examining ports and terminals in the same region, it can be assumed that the results are more meaningful than the global comparison, except for situations where the environmental variables and typologies have been accounted for. The port efficiency studies also have covered regions of closer distances. A representation of South America can be observed from (Rios & Maçada, 2006; Wanke & Barros, 2016; Wanke, 2013; Wanke et al., 2011). Rios and Maçada (2006) analyze the South-American (Mercosur) region's relative efficiency using the DEA BCC model. North America and the Middle East & African region are less represented in the DEA literature. Al-Eraqi, Mustafa, Khader, & Barros (2008) and Kammoun & Abdenadhe (2018) are among the few to contribute to the DEA inter-regional studies in the Middle East as well as Africa.

What is also noticeable with these studies is that there is a likelihood of domination by a specific author per region. The regional representation also shows a similar pattern that was discussed in the earlier section of this chapter in bibliometrics, except for Australia.

In chapter 1, it is mentioned that Asian and European ports are leading in terms of world container volumes, this representation is a mirror of that phenomenon. Therefore, it is not surprising that the regions also invest in research and development in the port sector. Data availability is also another factor affecting the research in some regions. Wu & Goh (2010) mention that due to data unavailability in Nigeria, the country's ports were eliminated in the sample. An issue of consideration highlighted in Coelli et al.(2005) when cross-country analysis is conducted is data quality. Ensuring that a uniform measure is applied and that the variables are estimated similarly is key. When monetary measures are taken into account, such as revenues and GDP, it is essential to convert to a uniform currency. Likewise, the labour sectors can be very diverse (discussed in chapters 1,4,5 and 6) from country to country, affecting the results of efficiency.

Figure 3-3: DEA literature by region



Source: Own compilation

3.3.1.3 Activity

As will be seen in Chapter 4, ports differ by many aspects. While there are a variety of agents such as port authorities, tug boats, terminal operators, etc., there can also be differences like operations carried out, the objectives, degree of competition, regulations (González & Trujillo, 2009). All these elements make it cumbersome to compare ports in a homogenous manner. Tongzon (2001) suggests that focusing on specific activities or ports is preferable in analysing comparisons.

While some among the studies are very careful in clearly identifying the element whose efficiency is being studied, some studies do not specify this element, causing confusion. Roll & Hayuth (1993) begin by generalizing their study and emphasized the complexity of the port sector where 20 ports were taken into account for the analysis. Poitras et al. (1996) make a follow-up on Roll & Hayuth (1993) mentions the significance of specifying the activity measured in the ports. The study makes use of the container activities in 23 ports across various nations (19 countries). In the literature reviewed here, many studies focus at the port level, and much fewer focus at the terminal level. While some of these studies have their interests specifically on ports, Cullinane, Song, Ji, & Wang (2004) specify that data unavailability is often the main reason for researchers to focus rather at the port level than at the terminal level, the authors themselves faced the same challenge. Among the studies whose focus is on container terminal activities are (Min & Park, 2005; Cullinane & Wang,

2006; Yip, Sun, & Liu, 2010; Wanke, Barbastefano, & Hijjar, 2011; Wang & Cullinane, 2015).

Although these studies may have a common study field, they do not necessarily have similar objectives in their efficiency analysis. The objectives may differ to a very large extent. Estache, González, & Trujillo (2002) for instance, focus on the promotion of the yardstick competition system through efficiency; Barros (2003) focuses on the role of efficiency on incentives issued by the Portuguese government; Barros & Athanassiou (2004) rely upon efficiency to make a comparison between the Portuguese and Greek port authorities; Luna, Mar-Ortiz, Gracia, & Morales-Ramírez, (2017) seek to compare Mexican container terminals and provide factors greatly affecting their terminals and Kammoun & Abdenadhe (2018) simply study efficiency for comparison of Tunisian efficiency scores.

Turner, Windle, & Dresner (2004) base their conclusions at both port and terminal levels. The sample of Cullinane & Wang (2006) is based on the analysis of either a container port or a container terminal within a port similar to that followed by Zheng & Park (2016). In the Asian region, Zheng and Park (2016) evaluate 30 container terminals in ports of Korea and China through a DEA CCR and BCC model intending to improve and manage terminals efficiently. The results conclude at a port level where efficiency levels for both country's ports are found equally efficient. This is different from previous studies where Chinese ports were usually found much more efficient.

One other issue regarding the activity specification is that some of these studies are titled 'ports' while they are focused on the terminal and vice versa. Twenty-five world Container terminals are evaluated by Tongzon and Heng (2005), although from the port perspective, the study uses container terminals as a measure of reference and indicates the presence of the private sector in the ports can improve operational efficiency. The Cobb-Douglas (CD) production function is applied to reach the findings. Cullinane and Song (2006) and Cullinane and Wang (2006) use the word "port" as a title to the papers and yet the measuring unit is the container terminal in the European region. This can mean that much more literature in the container terminal context exists but can be misinterpreted as port-level research due to the misconceptions. Some do not specify 'container' ports, giving the impression that the studies would be multi-output, while others are titled 'terminal', yet some of the

measures taken into account are port- rather than terminal-related. A reader thus needs to be conscious when interpreting this body of literature.

3.3.1.4 DEA Model types

The CCR model followed by the BCC model has been well received in the port and container terminal sector. Several other authors have made use of other varieties of the DEA models, such as the additive model in Tongzon (2001); the Window analysis (Itoh, 2002; Cullinane, Song, Ji, & Wang, 2004; Al-Eraqi, Mustafa, Khader, & Barros, 2008); Index DEA Martinez (2002) (Estache et al., 2004) making use of distance functions and decomposing productivity and efficiency; and FDH (Cullinane, Song, & Wang, 2005).

Various other authors make use of different models, which are not very likely applied in the DEA spheres. Park & De (2015) introduced the four-stage DEA method to measure seaport efficiency to overcome the weaknesses often associated with the ordinary DEA models. The two-stage models are usually carried out with a similar motive (Turner, Windle, & Dresner, 2004; Yip, Sun, & Liu, 2010; Bichou, 2011; Wanke, 2013; Wanke & Barros, 2016) of improving the weaknesses associated with the DEA. Yip, Sun, and Liu (2010) treat terminal operators as the decision-making units in evaluating port efficiency. The country and the port characteristics are taken into consideration using a two-stage DEA method that allows for the use of a parametric function. Luna et al. (2017) make use of the three-stage approach which is also an approach not very much visible in the port sector. Wanke and Barros (2016) evaluate the way in which different efficiencies are reached on 27 major Brazilian ports employing the non-parametric DEA. The study also investigates different conditions in which economies of scale are characterized highlighting a rare application of the DEA in a quantitative form, called the “multiplicative models” which helps to also extend the use of the DEA.

Chang et al. (2015) apply the new DEA model of Lozano and Villa (2004), which assumes a centralized structure of decision making (controlling all operating inputs) and also where aggregated outputs and inputs are more important than single units. In Chang et al. (2015), the new DEA is referred to as the centralized development envelopment analysis (CDEA), used to evaluate the resource allocation and performance of five container terminals supervised by shipping lines. Output is measured by throughput and output represented by four variables- number of quay gantry cranes, marshalling yards, labour, and hauling equipment. The advantage of

the approach used here is that it can show both the systematic and the centralized perspective of resource allocation by applying two different scenarios (by applying differing assumptions). On the other hand, the method treats all the terminals similarly without considering country differences. Different terminal/port importance levels and the hauling equipment may also be different and have differing capabilities.

For varying reasons, other authors apply both the DEA along with the SFA (Cullinane, Wang, Song, & Ji, 2006; Wanke, Barbastefano, & Hijjar, 2011; Kammoun & Abdenadhe, 2018). While Kammoun & Abdenadhe (2018) and Cullinane et al. (2006) utilize the two models to compare the efficiency scores on the same data set, others utilize the two models to highlight the weaknesses and strengths of each of the models (Erkoc, 2012). Specifically, Wanke et al., (2011) use the DEA solely to rank efficiency and make use of the SFA to determine if the inefficiencies are due to random noises or not. According to Cullinane & Song (2006), the results obtained from the SFA and DEA have been shown to have a very high correlation in several studies, e.g. Cullinane and Wang (2005) there was a 90% correlation between the scores.

3.3.1.5 Input and output variables

One of the advantages highlighted in chapter 5 regarding the DEA is its ability to capture multiple outputs and inputs. In the port sector, there is a broad consensus regarding the multi-output and -input nature of port activities, a feature so hardly captured in the port literature. As a DEA is strongly reliant on the input and output factors, it is important to evaluate the application of these variables in past literature. As mentioned by Paradi et al. (2018), one of the main challenges often faced by researchers is finding and including all the data that reflects the complete production function of a firm.

Dhingra et al. (2014) found a total of 133 variables used by researchers in the port sector. From the 133 variables, a majority (73) were used only once, 16 variables were used only twice and 25 variables were repeatedly used 3 to 5 times. Two variables were found consistently repeated in a majority of papers.

Measurement of output

In the container division, often the output is assessed through the total amount of handled containers ('throughput' in TEUs). Panayides et al., (2009) suggest that the reason behind this is due to the ease of collection of 'throughput' related data, especially at the port level. At the terminal level, the case is different. Another reason

is that 'throughput' is a good proxy to reflect the various facilities and activities. It is also the fact that it is the most common ground on which ports and terminals can be measured to assess the size, investment magnitude, activity levels, revenue generation, and the likes. Other studies combine 'throughput' and other variables to reflect their study objective better. For instance, Poitras et al. (1996) combine 'throughput' with TEU berth hour; Tongzon (2001) with ship working time; Valentine & Gray (2001) with total tons of containers; Barros & Athanassiou (2004) with ships movement of freight and containers loaded and unloaded; others doing the same include (Barros, 2006; Rios & Maçada, 2006; Wanke et al., 2011). Kammoun & Abdenadhe (2018) use tons of containers rather than the traditional number of TEUs. Another group of studies expresses the output in monetary terms (revenue) (Martínez-Budría, Díaz-Armas, Navarro-Ibañez, & Mesa, 1999), which is not very typical in the DEA body of literature.

Barros (2003) is a special case where more than the average number of output variables is used in the analysis. The study measures the output through ship movements of freight, gross gauge, break-bulk cargo, containerized freight, solid bulk, and liquid bulk. Similarly, Barros (2006) makes use of seven output variables. Another exceptional situation is of Park & De (2004), where various multiples of outputs show alternative approaches to measuring efficiency with the four-stage approach. In the study, the output alternates depending on measured productivity, profitability, commercialization, and global efficiency. Roll & Hayuth (1993) make use of four outputs, total cargo, level of service, customer satisfaction, and the number of ships, which is also above the average.

Measurement of input

The major equipment that operators invest in is the superstructure used to transfer the goods from ship to land or another ship, depending on the type of cargo handled and the procedures followed at the particular terminal (discussed in chapter 4). Dowd and Leschine (1990) and (Dyson et al., 2001) note that the production function of container terminals is determined by three main factors, i.e. labour, capital, and land. Panayides et al. (2009) mention that the factors of observation taken into consideration for the analysis should reflect the process of production. The inappropriateness of such a reflection can easily lead to biases. In the selection of the variables, omission of the important variables can lead to a misevaluation of efficiency.

Capital, labour, energy, materials and services (KELMS) are a core of discussion in performance measurement, but it is not the end of the story (Diewert, 2008). Among the crucial aspects in the inputs is the treatment of capital. Capital can be measured as an indicator of wealth and as a contribution of the assets to production (capital services) (Diewert & Schreyer, 2008). The complexity in the treatment of capital wealth is well covered in the literature (Diewert & Schreyer (2006); Diewert & Schreyer (2008); Diewert (2008)). The complexity of service provision considering fixed assets in technical efficiency and productivity is not well covered in the past container terminal literature. Terminals and ports have considered various strategies and solutions (chapters 5 and 6) to address the challenge.

The processes may differ in the container handling industry because the products handled are different, but the concept (of loading and unloading) is similar. Vanelslander (2005) comprehensively demonstrated a thorough cargo handling process. Ideas regarding influencing factors in the container handling industry are well documented. Details of activities (such as loading, unloading, storage, provision, maintenance, safety & security, and other sub-activities) are depicted and explained (also see chapter 4). The combination of activities varies even amongst actors fulfilling the same role, resulting in either the actor taking advantages or disadvantages of costs and therefore making profits or losses. Specifically, Vanelslander (2005) gives an overview of conditions that affect the container handling activities, which can impact the economies of scale/scope, amongst other things. These main factors are classified into policy (third largest), scope (fourth largest), chain (second largest), and terminal (largest) factors. These factors may or may not be fully controlled by the operator, whose discrepancy results in differences to the realization of costs and revenues of the container terminal operators. In assessing the operating costs, Vanelslander (2005) finds an important relation mainly between economies and terminal capacity, quay crane size, labor flexibility, and new and used quay cranes.

Accessibility to data plays a major role in the selection of the variables, as well as the sample size, explained further in chapters 5 and 6. Intermediate inputs such as water and electricity as well as labour are hardly ever considered. Capital, area, quay length, berth depth, quay cranes, yard equipment variables are consistently used in the DEA literature and other studies applying other efficiency methods. The consistency

suggests that these factors influence efficiency in the container port and terminal sector.

Some studies apply a considerably larger amount of variables, such as Poitras et al. (1996); Park & De (2004); Barros, (2006); Cullinane et al. (2006); Liu (2010); Zheng and Yin (2015). Others use only a few variables, for instance, Barros (2003) makes use of only two input variables, labour, and capital prices, while Estache, González, and Trujillo (2002) make use of only labour and length of dock as input variables. Although labour is a very significant input variable in production theory, it is often difficult to obtain. Demirel et al. (2012) discuss various circumstances where labour is often a politically sensitive situation. According to González and Trujillo (2009); Park and De (2004); Cullinane and Wang (2006); De Oliveira and Cariou (2011); Nguyen, Nguyen, and Chin (2011); yard handling equipment is an appropriate proxy for labour. Cullinane & Wang (2006) derive the labour variable from a predetermined relationship to terminal facilities. Tsunokuwa (1981) utilizes two input variables, quay length and the number of quay cranes, where labour is proxied by these two input variables. Many other container port and terminals studies that did not incorporate the labour variable include Cullinane et al. (2002), Cullinane et al. (2004), Cullinane et al. (2005a), Cullinane et al. (2005b), Cullinane and Song (2006), Cullinane et al. (2006), and Trujillo et al. (2013), Almawshaki and Shah (2015), (Liu, 2010), Sun, Yan, Liu (2006).

An expectation is that all container terminals will have a quay crane and a quay wall (or area), it would be expected then that all the studies would take these two variables into account, however, Barros (2006) and Wanke et al. (2011) show that this is not necessarily always the case. In Barros (2006), this is due to the nature and objectives of the study. The study is multi-output and is not solely focused on containers. There is no specific reason that can be pointed out in Wanke et al. (2011).

Except for a few studies (Martínez-Budría et al., 1999; Barros & Athanassiou, 2004; Barros, 2006; Wang & Cullinane, 2015), all the studies in the sample take into account solely the quantity data as opposed to financial data. Only Wanke (2013) considers the intermediate variables in this sample, which suggest that although these costs are necessary for production, they are hardly ever considered an effect to efficiency in the sector.

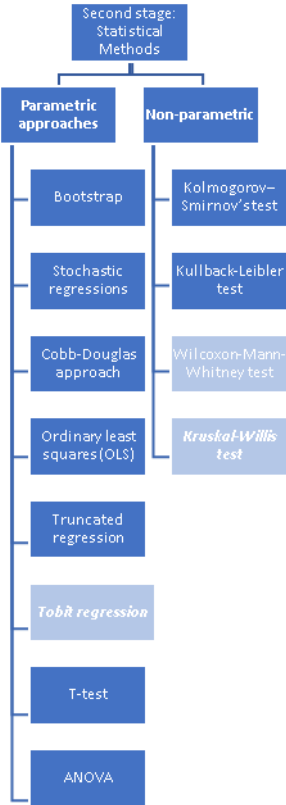
Panayides et al. (2009) thus strongly emphasize that future research should carefully distinguish between the different factors affecting ports before applying the DEA and making interpretations of the efficiency scores. In addition to the variables directly

affecting production, it should be remembered that there are also external influences to efficiency, as discussed in chapter 5. Panayides et al. (2009) warn that a waste of resources is a waste of resources regardless of the terminals' internal and external factors.

3.4 Statistical Methods for non-parametric approaches

A second stage of statistical analysis is alternatively incorporated with non-parametric methods such as the DEA. Alternatives exist between the parametric approaches (Bootstrap; Stochastic regressions; Cobb-Douglas approach; Ordinary least squares (OLS); Truncated regression; Tobit regression; T-test; ANOVA) and non-parametric approaches (Kolmogorov–Smirnov’s test; Kullback-Leibler test; Wilcoxon-Mann-Whitney test; Kruskal-Willis test).

Figure 3-4: statistical analysis methods



Since the first paper of the DEA (Charnes, Cooper, and Rhodes (1978)) (CCR) got published, there were many developments to the model, the BCC model being the most visible of the developments. Paradi et al. (2018) recognize five categories of model adaptations to the CCR model - the extension to evaluating contextual variables; model extensions; models with special data types; models of internal

structure examination and models which specifically measure environmental performance.

The two-stage evaluation where the DEA efficiency scores are later regressed against some contextual variables through statistical analysis has gained popularity. Although the DEA offers a great perspective on the efficiency and productivity of the container terminals, it does not always capture all the essential other factors to the overall efficiency. It becomes interesting then to find out whether the efficiency scores occur by chance or whether they are statistically significant. Often, a regression model is estimated on the second stage to the DEA, where the DEA scores are regressed with exogenous factors. These exogenous variables (sometimes referred to as environmental, external, contextual, surrogate non-discretionary) are simply factors that are not under the control of the management but which may likely affect efficiency and thus cannot be left unevaluated (Ramalho et al., 2010). According to Boussofiane et al. (1991), these factors can be measured directly or indirectly through a surrogate measure.

A statistical analysis comparing means, medians, and variances of the DEA efficiency scores can be carried out to find the effect of these external variables on the efficiency scores. When the differences are statistically significant, then the external variable is considered relevant in determining the container terminals' efficiency performance (Romano & Guerrini, 2011).

3.4.1 What can be benefited from the two-stage approach?

Coelli et al. (2005) discuss the advantages and disadvantages of the two-stage approach. The author mentions the bias that can be created when the variables in the first stage are highly correlated with the variables employed in the second stage. Barros, Chen, & Wanke (2015) discuss this endogeneity issue and suggest the Durbin-Wu-Hausman test as a potential solution. Another issue raised by Simar & Wilson (2007), which is still visible in many studies today, is that in many of the studies using the two-stage approach, the data-generating process for the second-stage variables is not described. There is often no clarity on the decision to consider the environmental variables rather than the input or the output in the production process.

On the other hand, the benefits of the two-stage approach lay in many reasons such as that multiple variables can be accommodated for; the method also allows for both, continuous and categorical variables (depending on model choice); the method offers

not only the possibility of determining the significance of the external variable but also the direction of influence; the methods are easy to incorporate with the DEA scores and they offer an opportunity for transparency in presenting the efficiency scores.

Fried, Lovell, Schmidt, and Yaisawarng (2002) emphasize that the two-stage DEA approaches are essential and deserve recognition for the ability to demonstrate that the variations in efficiency scores cannot solely be explained by managerial in/competence (BCC) and scale factors (SE).

Statistical treatment of the DEA score variations has been tackled in various ways, parametric and non-parametric (Wanke & Barros, 2016), as discussed below.

3.4.2 Parametric approaches

Cooper et al. (2007) acknowledge the growing literature combining DEA scores with multivariate data analysis in the second stage through regression analysis. Alternatives include methods such as the bootstrap, stochastic regressions, Cobb-Douglas approach, Ordinary least squares (OLS), Tobit regression. These methods are often justified to overcome the weaknesses of the DEA, in that it lacks statistical inference (Cooper et al. 2002). Approaches that explain the variations in the DEA efficiency scores by exogenous/external factors are discussed in Simar & Wilson (2007) and Wanke & Barros (2016). Specifically, Banker (1993) and Simar and Wilson (2007) show an evolution of such studies. A multivariate analysis is one of the methods employed in recent literature where DEA estimates are incorporated as a dependent variable.

The bootstrapping method has been identified to correct for biases in results and the ability to identify differences on different efficiency levels (sensitivity analysis). Bootstrapping provides amongst one of the best solutions to the drawbacks often highlighted about the DEA. Wanke and Barros (2016) study looks into how different efficiencies are reached on 27 major Brazilian ports employing the non-parametric DEA. The study also looks into different conditions characterised by economies of scale. The study also highlights a rare application of the DEA in a quantitative form, called the “multiplicative models” which assists in extending the use of the DEA. The study results can help assist the port in decision-making regarding the efficiency in future cargo growth and in setting priorities. The results are also valuable in assisting port authorities to use the information they have regarding the contextual variables

and the level to which the variables contribute to efficiency levels and therefore also can be able to prioritize within each group of ports.

According to Turner, Windle, and Dresner (2004), applying the OLS to explain differences in DEA efficiency scores is inappropriate. It will lead to inconsistent results owing to the dependent variable that is continuous but bounded between 0 and 1. Turner et al. (2004) thus recommend the Tobit model in such situations. Although the Tobit regression is similar to the OLS, it assumes a truncated normal distribution rather than a normal distribution, and it also employs a maximum likelihood-ratio estimation method.

As pointed out earlier, serial correlation is one of the disadvantages of the two-stage approach. Simar & Wilson (2007), one of the most influential studies in the two-stage literature, argue that a truncated regression combined with a bootstrapping re-sampling technique can overcome this problem. Before Simar & Wilson (2007), most of the studies applied the Tobit model (Chaouk et al., 2020). In their study, Simar & Wilson (2007) advocate strongly for the use of a bootstrapped model and further acknowledge the use of the OLS models (such as in Banker & Natarajan (2008)) for consistent estimation. However, mention is made about the OLS being possible only under peculiar and unusual assumptions to the data, which thus also limits the potential of the OLS to the DEA. Some of the studies (Byrnes et al., 1988; Nyman and Bricker, 1989; Ray, 1991) first transform the DEA scores by using log, logistic, or lognormal transformations before applying the OLS in the second stage. In some cases before the application of the OLS, an arbitrary constant is added or subtracted to avoid the division by zero or taking the log of zero.

Which method is superior to the other to conduct the second stage still lies unclear in theory, this is also debated in Chaouk, Pagliari, and Moxon (2020). The Tobit model will be applied to represent the parametric methods in Chapter 6 for this thesis.

3.4.2.1 Tobit regression

A Tobit regression (by Tobin, 1958) is similar to the OLS but different in that it assumes a truncated distribution compared to the OLS that assumes a normal distribution. The Tobit model falls under the category of censored regressions where the dependent variable is bounded from below or above, or in some cases both, (Chaouk et al., 2020). The Tobit also employs the maximum likelihood estimation applied in the context of the DEA by authors such as Windle and Dresner (1995); Windle and Dresner (2004); Latruffe et al. (2004); Bravo-Ureta et al. (2007); Banker &

Natarajan (2008) to name a few. Due to the bounded nature of 0 and 1 of the DEA efficiency scores, these authors strongly advocate using the Tobit model.

There seems to be an argument between especially the OLS and the Tobit model in the past literature. Many studies have applied the Tobit model, but there are still advocates of the OLS more than the Tobit model (Hoff, 2007 and McDonald, 2009). Given the type of data (truncated) from efficiency scores and the small sample size, the Tobit is still preferred.

3.4.3 Non-parametric

Similar to any statistical test, the non-parametric tests have also pros and cons. The strongest force of these tests is the independence on distribution assumptions like most of the parametric tests do. When the assumptions (of linearity, multicollinearity, normality of errors, independence, large sample size, continuous scale data) to carry out the parametric analysis are not met, it is not advisable to continue and interpret the results. These non-parametric methods then come in handy. Not only for these reasons mentioned above but also because the DEA makes no assumptions about the production function, thus pairing it with yet another non-parametric test seems more consistent. Another reason is bounded to the unknown distribution of the efficiency scores which makes it difficult to express it parametrically (Romano & Guerrini, 2011).

The Kolmogorov–Smirnov's test statistic, Kullback-Leibler test, Wilcoxon-Mann-Whitney test, and the Kruskal-Willis test are among other non-parametric tests often applied in conjunction with the DEA (Brockel & Golany, 1996). The disadvantage of the Kullback-Leibler test is that it has no power against the alternative hypothesis. In many cases, the Man-U test is most preferred (Banker and Natarajan (2004); (Cooper et al., 2006); (Nachar, 2008); Gomes et al. (2009); (Romano & Guerrini, 2011). An alternative to the Man-u test is the Kruskal-Willis test, whose advantage is that it can test more than two groups, compared to the Man-u.

3.4.3.1 *Wilcoxon-Mann-Whitney*

Cooper, Seiford, and Tone (2006) suggest the use of the rank-sum non-parametric (distribution-free) test developed by Wilcoxon-Mann-Whitney (Wilcoxon 1945, Mann and Whitney 1947). One of the strong points of the Mann-Whitney test (often used in comparison to the t-test) is that it is independent of the distribution of the dependent variable (Nachar, 2008).

By applying this test, conclusions can be reached on whether or not there are differences between two groups within the sample. Three assumptions of the Man-U test are that the two groups tested are randomly drawn from the target population; there must be independence within groups and mutual independence between groups. Lastly, the data measurement must be ordinal or continuous.

3.4.3.2 The Kruskal-Willis test

The Kruskal and Wallis (1952, 1953) test is a non-parametric equivalent to the Analysis Of the Variance (ANOVA) parametric test. The test is appropriate for three or more groups to compare and is an extension of the Man-U test (Akbari et al., 2020; Ostertagová et al., 2014). The Kruskal Wallis test is used to compare whether samples are from the same distribution or not. The test thus has the same assumptions as the Man-U test regarding normality, independence making use of rank data rather than raw data.

3.5 Conclusion

This chapter set the scene for the empirical analysis by providing detailed foundations of previous literature in the analysis of efficiency and the methods of measurements that will be applied in the empirical analysis of this thesis.

The literature review leads to several conclusions regarding efficiency analysis methods in container terminals. Efficiencies of various firms can be compared through various methods, the SFA and DEA being the most applied methods. To date, there is no consensus as to which model is surely superior to the other. The major deciding factor in applying either the SFA or the DEA is in the treatment of the external forces by the models. In SFA models, an inappropriate functional form can lead to erroneous results, for that, likelihood tests and Wald tests can be a source of correction. In addition to traditional models, the DEA can integrate statistical models to assess the impact of external factors on efficiency. Based on the strengths of the DEA and the relevance of the method to the case which will be investigated in the empirical chapter, the DEA will be employed for analysis.

Among the differences in the application of the DEA is the orientation of the models, which is very much shaped by the industry and the firm strategy (ie maximizing the output or minimizing the input). It, therefore, makes sense that there are different assumptions regarding this on different port and container terminal literature, as

discussed in chapter 2. A majority of literature bases their results on either one of the methods; however, for this thesis, the container terminals will be evaluated from both, input and output perspectives. Not only is Asia and Europe shown to have higher efficiency in many of the ports in the studies, but the locations are also found to have considerably higher research work in the field. However, in as much as Europe has a higher representation of studies, there is still no dominance of studies focusing on the HLH region and specifically on container efficiency. The HLH region is the choice region of analysis for this thesis.

A selection between the CCR and BCC models is made within the DEA models. The CCR is suitable in scenarios that embody constant returns to scale. At the same time, the BCC better illustrates situations where the returns to scale are changing and is an extension to the CCR. The DEA CCR and BCC have been found suitable among other approaches to analyze efficiency performance and benchmark the container terminals.

To improve one of the disadvantages of the DEA, the second stage can be incorporated into the analysis. The methods make it possible to consider the external effect to efficiency, using environmental factors not controlled by terminal managers. The Tobit regression has been chosen as the most suitable method to quantify the in/significance of the external factors to the efficiency scores parametrically. A Non-parametric approach that has been selected is the Kruskal Wallis test, in comparison to the Wilcoxon-Mann-Whitney. The test can handle and make a comparison of numerous groups of variables. There has long been a debate regarding which method the DEA is best linked within the second stage (i.e., parametric or non-parametric); applying methods from both perspectives offers greater reliability to the results. These methods shall be applied in the empirical analysis.

Chapter 4: Container terminal characteristics and efficiency factors

As shown in the literature review (chapter 3), container terminal operators seek to identify areas of improvement towards accomplishing efficiency, cost minimization, and competitiveness while remaining vigilant of the environmental impacts of their activities. To determine the crucial elements responsible for terminals' improved performance (efficiency and costs), it is paramount to recognize, understand, and identify the present-day variations between container terminals. This is in succession delineated by the terminal's processes, systems, layout, and activities. According to Kemme (2013) and Gharehgozli, Zaerpour, & de Koster (2020), albeit variations occur between container terminals in their manner of operation, comparability is feasible.

From a global view, variations of container terminals can be observed in terms of layout, state-of-the-art handling systems, processes, and activities. This directly indicates the differences present in the characteristics of operational efficiency and cost structures in these terminals. In chapter 3, the evaluation of technical efficiency is discussed, including the classic input and output factors of measurement. Through describing the sequential manner of the container handling process from (to) the “ship to shore sub-system” to (from) the “hinterland transportation”, this chapter demonstrates the characteristics and factors which influence the performance (efficiency) of container terminals. In this chapter, the classic factors of efficiency (as discussed in chapter 3) are identified and also the indirect and external factors are identified.

The selection of the identified efficiency affecting factors and characteristics depends on the container handling processes (sub-systems), past literature, and the observation of terminals. Terminal-specific factors are grouped into four categories (production, productivity, utilization, and service factors) while external factors are grouped into three (chain, policy, and scope). Some of the identified factors are mutually determined through various sub-sectors. Labour as one of the important factors to determining productivity is given a special elaborative discussion. Among

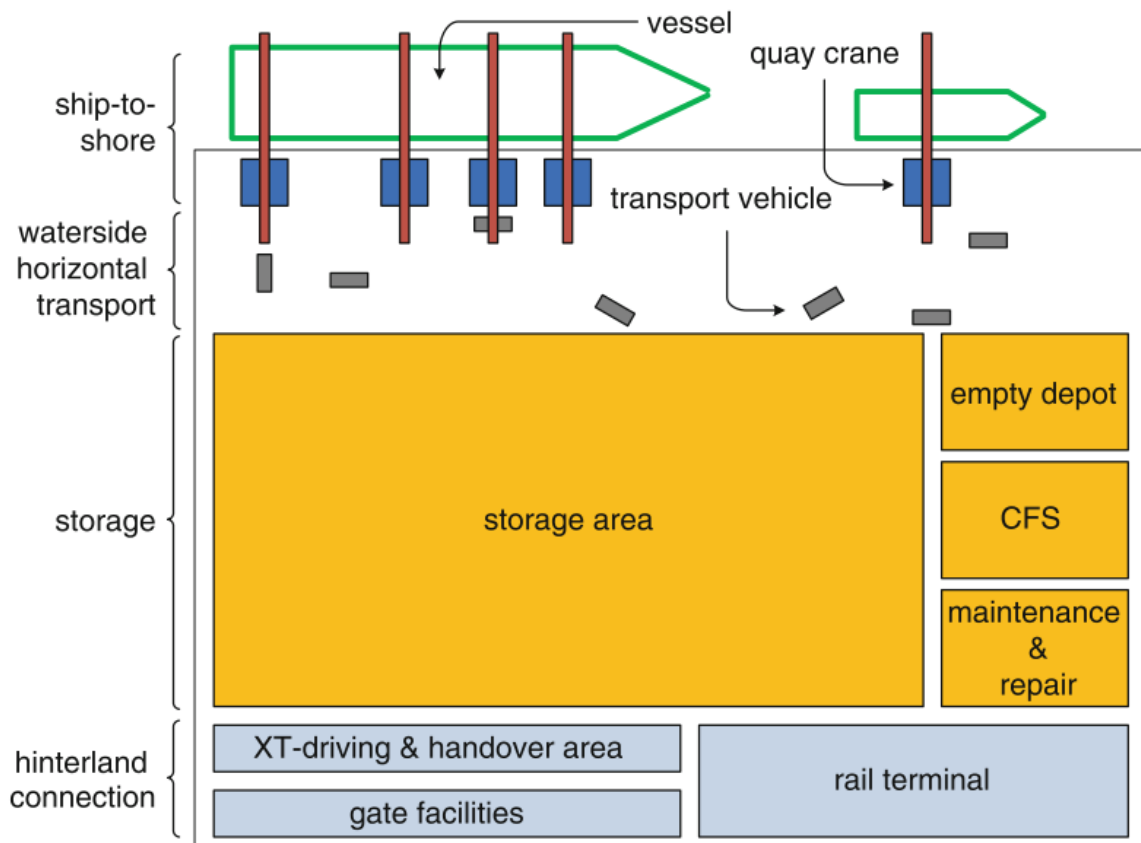
these factors, some are selected, quantified and their effect on efficiency in CTs is evaluated later on in chapters 5 and 6.

4.1 The logistics at a container terminal

A container terminal can be considered a complex system with numerous attributes, operations, and resources (Kemme, 2013). The main function of a container terminal is to serve as an interface between landside and seaside modes of transportation, transshipment of the containers from one mode of transportation to the other as well as for added value services. It is unlikely that the transshipment occurs directly from one mode to the other, thus, storage is often necessary for the containers waiting to be transferred. Added value services such as stripping, stuffing, repair, washing, maintenance, depot, in a terminal also make up the function. To simplify the complexities of a container terminal, the activities are divided by Steenken, Voß, & Stahlbock (2004); Günther & Kim, (2006); Kemme (2013) Stojaković & Twardy (2021) into several subsystems, i.e the ship-to-shore sub-system, the waterside horizontal-transport sub-system, the storage sub-system and the hinterland- connection sub-system. A diagram to show a general layout of a container terminal, as well as the sub-systems involved, is demonstrated in *Figure 4-1*

Generally, the process that takes place in *Figure 4-1* is that a vessel is moored, following that, gantry/quay cranes (QC) are located on the shore to load/unload. The QCs retrieve containers from (or to) the vessel to (or from) internal transportation, referred to as the yard horizontal transportation. The horizontal transportation vehicles then move the containers from (to) the quay area to (from) the yard. In the yard, the containers are temporarily stored in the storage area in wait to either be transferred to (from) external trucks or rails or transported to (from) another vessel or barge. Inland containers leave (or arrive) the terminal through terminal gates, after paperwork verifications and inspection, (Carlo et al., 2015).

Figure 4-1: Container terminal layout and sub-systems



Source: Kemme (2013)

The container terminal processes by configuration: Automated vs Non-automated

To provide cost-effective and efficient services to container cargo, terminal operators, shipping companies, port authorities continuously invest in new technologies, (Gharehgozli et al., 2020). The level of newer technologies can be assessed through the operating equipment of the terminals, which is different between automated and conventional terminals. Given the increase in throughput amount; land scarcity and decreasing costs of technologies and increase in the reliability of the newer technologies, container terminals continue to change in their development, (Gharehgozli et al., 2020).

From the very beginning of the globalization of container freight, the cornerstone has always been due to the efficiencies induced, given the ease of transfer from one mode of transport to the other. Till today, container terminals remain striving for higher efficiencies. As shipping lines augment, so do container terminals progressively

automate (Li & Lu, 2019). By virtue of the European container terminal (ECT), the home of container automation, the concept of automation continues to broaden. According to UNCTAD, since 2017, more than 50 CTs have been fitting under the category of 'automation or semi-automated CTs' worldwide, which amounts to 9.1% of all main CTs all around the world (Drewry, 2019). A survey conducted by Navis and Cargotec corporation to 60 Navis customers reveals that 83% of the survey respondents feel that decision making regarding artificial intelligence (AI) will be of utmost importance in the future innovation of terminals and 83% of respondents will increase investments towards AI over the next 1 to 3 years.

Automation can be translated into differing terms and it is closely related to "technology". Automation covers equipment and software. In its purest sense, automation is the replacement of human-operated activities by computer controls. Optical character recognition is another way in which humans can be replaced where for instance a checker is replaced by a handheld computer to record numbers or changing to the use of cameras at the gate compared to physically checking. The automation of equipment is more visible where human drivers have been replaced by robots. Automation in CTs can occur at vessel-quay; quay to stack; yard stacking and in-out gate, (Drewry, 2014).

As it is with the already existing automated terminals, an automated container terminal (ACT) is a terminal utilizing only automated systems in the yard as well as yard-dock movements. Crane-ship operations usually remain manually operated, while yard-gate operations remain assisted by remote controllers (major or total automation), (Martín-Soberón et al., 2014). In some terminals, gate activities are also automated (e.g the Maasvlakte II). In extreme cases, the quay cranes are also automated. In manual CTs, operations are all manually operated (all sub-systems). The flow of information, however, for nearly all CTs (automated or not) is centrally controlled, a very common system referred to as the terminal operating system (TOS) is used. An intermediate solution between these two extreme cases often referred to as partial or semi-automation is whereby some but not all of the major operations (yard-dock/yard) are not automated. Terminals where equipment is controlled remotely or where some functions are automated (or partially) are also referred to as the semi-ACT. Terminals will also have levels of automation depending on the combination between the minor and major automated equipment. In the Hamburg

Le-Havre, when evaluating the major container terminals⁹, a majority (68 percent) are lowly or not automated, while only a few are fully automated (26 percent) and even fewer (5 percent) have a combined feature between automated and non-automated.

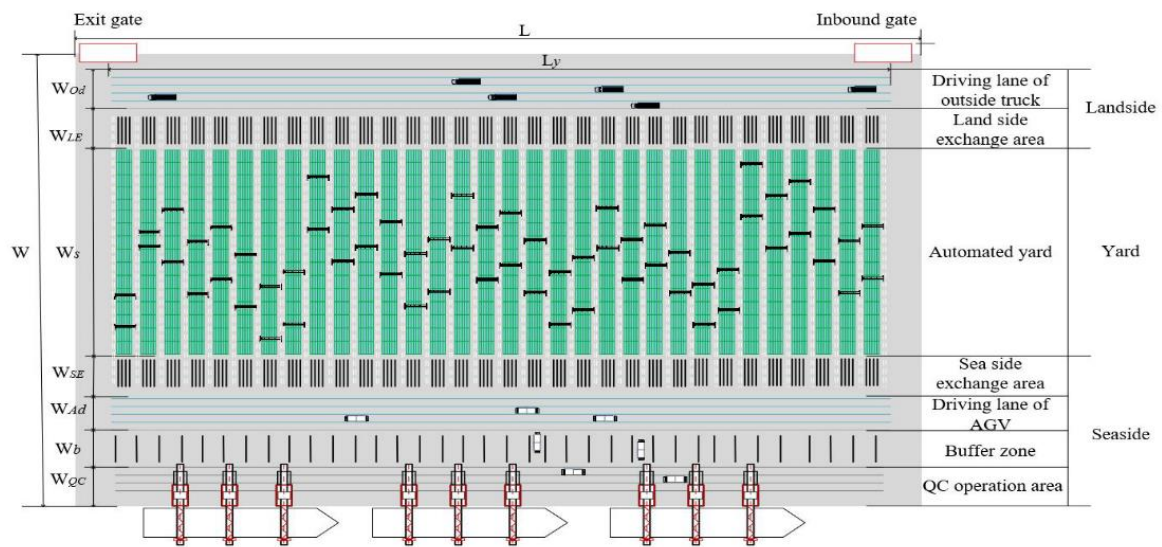
As can be expected, automating terminals already in operation (brownfield) is more complex compared to newly operating (greenfield) terminals. For the brownfield, a certain order ensuring coordination between operations and the operating system has to be followed. A drop in performance and efficiency is expected during this time, and in other cases, performance declines significantly longer. The Antwerp Gateway terminal is a good example of brownfield automation (Martín-Soberón et al., 2014). Some of the existing leading fully automated container terminals are located in the Port of Singapore, the Port of Hamburg, Thames port of England, and the TCB terminal of Nagoya Port in Japan.

The two extreme cases, a fully automated terminal (Wang, Chang, Shi, Yuan, & Gao (2019)) and a purely conventional terminal ((Young Kim & Hwan Kim, 1999)), are demonstrated in *Figure 4-2* and *Figure 4-3* respectively.

Given some advantages associated with improved technologies, to increase productivity and efficiency, among other reasons, CTOs have increased their share of automation of their equipment. According to Merkel (2018), when a port or terminal has modern equipment for handling the containers, this assists with reducing the user time to the consumer. Wang et al. (2019) associate automation with the advantage of manpower cost savings, capacity improvements, reduction of energy consumption, and the association of the terminal with an improved image to the general stakeholders. Drewry (2014) discusses the advantages of automation and adds to the list the safety of employees, that terminals gain longer working hours, and the reduction of human error.

⁹ Europe Terminal; Noordzee Terminal; Psa/Til Mpet Terminal; Gateway Terminal; Csp Zeebrugge; Eurogate Gmbh (Ctb); North Sea Terminal (Ntb); Msc Gate; Eurogate Gmbh (Cth); Burchardkai (Ctb); Tollerrort (Ctt); Altenwerder (Cta); Eurogate Gmbh (Ctw); Terminal Des Flandres; Terminal De France Gmp; Terminal Normandie Msc; Ect Delta; Euromax ; APM Terminals Maasvlakte Ii; World Gateway Rotterdam.

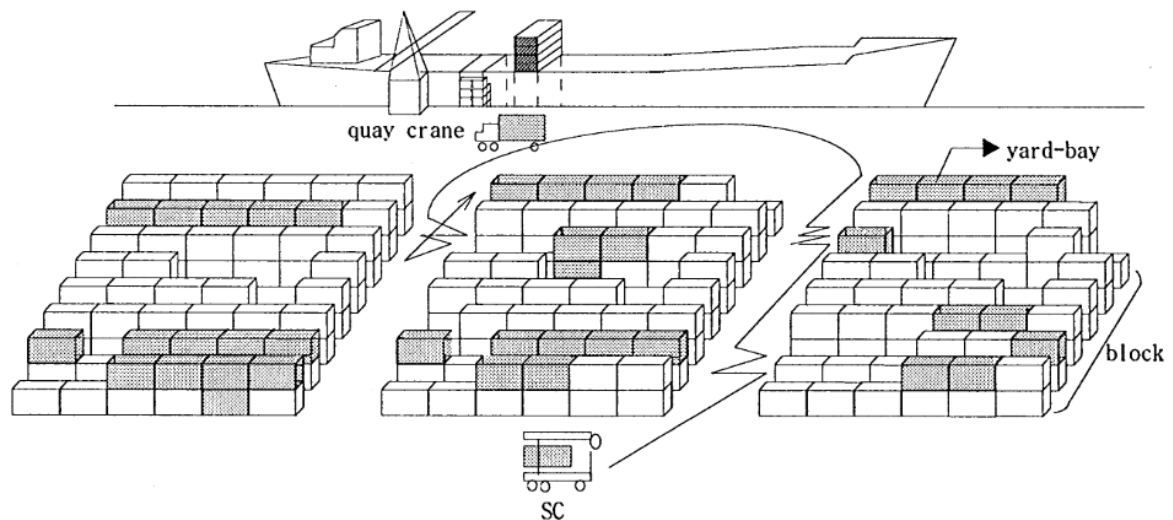
Figure 4-2: Automated container terminal



Source: Wang, Chang, Shi, Yuan, & Gao (2019)

In contrast to a fully automated container terminal, there is the manually-operated terminal. Figure 4-3 demonstrates such a terminal.

Figure 4-3: Traditional non-automated container terminal



Source: (Young Kim & Hwan Kim, 1999)

The layout of an automated terminal usually differs from that of a manual terminal. In non-automated terminals, storage yard blocks are usually parallel to the quay as demonstrated in Figure 4-3. This is also a typical setup of Asian CTs. Automated

terminals on the other hand have the yard blocks perpendicular to the quay, as demonstrated in Figure 4-2. This is a layout typical in European CTs. The layout affects the interaction of the QC and the yard equipment and therefore the speed and the efficiency over which activities in the yard are carried out.

The difference in the equipment employed in these terminals is shown in *Table 4-1*

Table 4-1: Container terminal equipment (automated vs manual terminals)

	Terminal equipment	
	Automated	Manual
Vessel to quay (STS)	Gantry cranes- Rail mounted gantry cranes (RMGCs); mobile harbour crane	Gantry cranes- Rubber-tyred gantry cranes (RTGCs); mobile harbour crane
Quay to stack (horizontal)	Automated guided vehicle (AGV); Automated straddles (AutoStrad) also known as automated lifting vehicles (ALVs)	Straddle carriers (SCs); trucks with trailers (Chassis) also known as tractor-trailer unit (TTU); Multi Trailer System trains (MTS); reachstacker; empty/loaded container handler; shuttle carrier (ShC); forklift; RTG; RMG
Yard stacking	Automated Stacking crane (ASC); Automated rail mounted gantries (ARMG); automated rubber- Tyred gantries (ARTG); Automated straddles (AutoStrad) also known as automated lifting vehicles (ALVs)	Straddle carriers (SCs), Forklifts, reachstacker; top lifter; Overhead bridge crane (OHBC); Chassis*;
In-out gate	Gantry cranes; Trucks; trucks with trailers; Multi Trailer System trains (MTS); reach stacker;	Forklifts; trucks with trailers; SC; Multi Trailer System trains (MTS); reachstacker; RTG; RMG

Sources: (Li & Lu, 2019); (Drewry, 2014); (Carlo et al., 2014); (Zehendner, 2013); (Brinkmann, 2011).

Typically, terminals select a combination of vehicles from the shortlisted horizontal transport in *Table 4-1*. The choice of equipment depends on space restrictions, history, economic reasons, hinterland transportation mode, the number of containers to be handled, and how a terminal is operated and results in the type of terminal storage layout.

In a study conducted by Wiese, Suhl, & Kliwer (2011), 67% of the terminals surveyed (out of 114) were using RTGCs also as the main stacking equipment. Automated RMGCs are also referred to as ASCs of whose use is more popular in European terminals. SCs, which are non-automated, are referred to as the most flexible equipment in the yard. SCs can perform both horizontal transport activities as well as yard stacking activities due to their feature of self-lifting and stacking. Newly developed SCs, called Autostrads, are common in Australian CTs, e.g. in Patrick Container terminals, Brisbane.

Both the automated and non-automated systems have pros and cons, highlighted in (Li & Lu, 2019). For instance, on one hand, is the straddle carrier system which proffers multi-functionality, flexibility, ease of balance, increased utilization rate, increased efficiencies, it also is associated with high initial investments, complicated structure, high wheel pressure, large body, and difficulty in stacking. On the other hand, the rail-mounted gantry crane system compensates for the ills of the former by allowing for simplicity, convenience, lower costs, reliability, electric drive, energy-saving, and ease. However, the system struggles with the unease of mobility to move boxes around the yard and in addition, also constitutes high initial investments. According to (Li & Lu, 2019), the middle playing field between full automation and conventional system offers better satisfaction.

In Gharehgozli et al. (2020), although higher efficiencies given the high investments are not guaranteed, it is predicted that container terminals of the future will be predominantly driven by vertical expansions. This indicates that the future layouts of future container terminals will have taller structures, which indicates that a change will occur in the used handling equipment and thus indicating a change in the levels of container terminal automation. This research field is howbeit in its infancy; thus, many possibilities are currently explored. In the Port 2060 project by Kalmar (part of Cargotec), the next-generation designs of container terminals mainly feature underground transportation systems, autonomous, smart containers, driven by data analytics.

4.2 The characteristics affecting efficiency: determined through the sub-systems

In this section, the four sub-systems (ship-to-shore; waterside horizontal transport; storage, and the hinterland connection subsystem) of a CT, with caution to the configuration, are discussed to identify the different characteristics existing between the world CTs and in so doing, highlight factors of performance.

Performance, which includes measuring productivity, efficiency, and effectiveness can be affected by various factors. The factors are grouped here into four categories, namely productivity, production, utilization, and service-related measures (Esmer, 2008).

4.2.1 Ship-to-shore sub-system

At this sub-system, loading and unloading of the container to the vessels or barges occur. It is an interface to one of the most important players in the terminal, the shipping lines. There is a difference in inventory depending on how a terminal is operated, i.e automated vs conventional seaport container terminals, as indicated in the previous section. Activities that occur in this sub-system, the differences between the terminals (where there exists any), and the performance affecting factors are discussed below:

4.2.1.1 Production factors

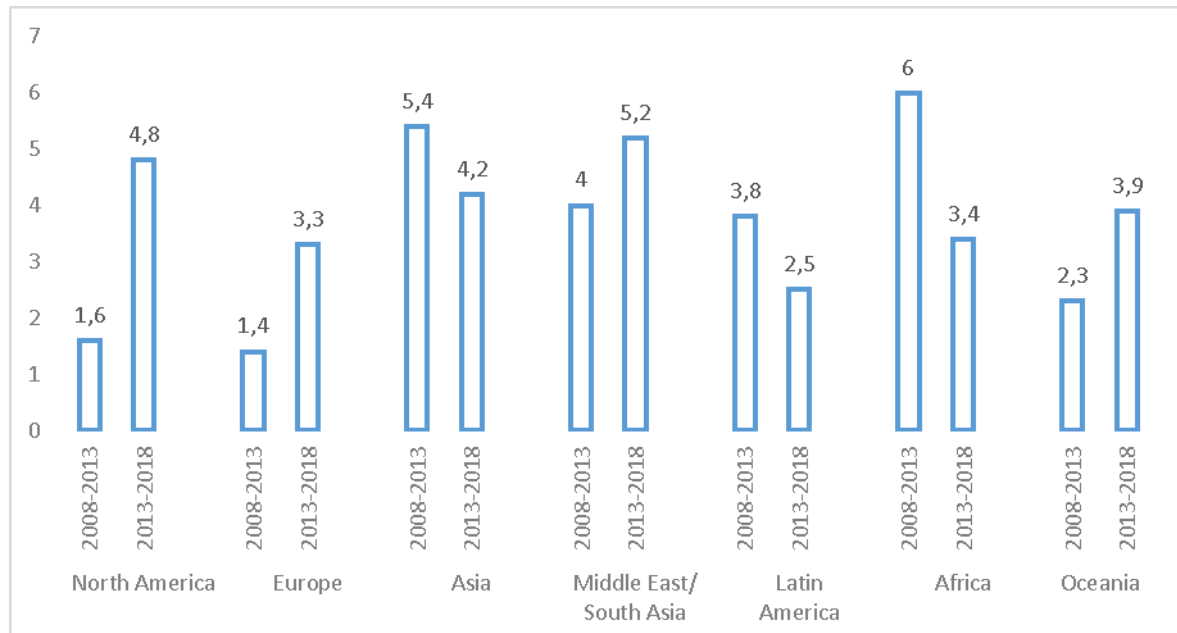
Production factors represent the level of business activity. To a large extent, this measure represents the quantity of cargo passing through a terminal (Esmer, 2008).

Ship throughput (tonnes or/and TEUs) and moves give the amount of container or the number of times a container is moved from (or to) the vessel. Unproductive moves occur when containers have to be unloaded and reloaded due to inappropriate storage such that containers that should be unloaded are located below other containers. The manner of reporting the output may differ given the location of where the CT is. These differences are especially visible between Asia and Europe. This may not necessarily affect the efficiency practically, but it may affect how efficiency is calculated and reported. *Figure 4-4* shows the average regional growth of throughput between 2008 and 2018.

Efficiently handling the containers (the output) circulating in a container yard requires that many other factors discussed in the sections that follow are well

integrated. To evaluate performance will require the knowledge of throughput and which with no doubt makes it necessary to understand the output factor of the container terminal at hand.

Figure 4-4: Average growth of port throughput (%)



Source: Own compilation based on Drewry (2014, 2019)

4.2.1.2 Productivity measures

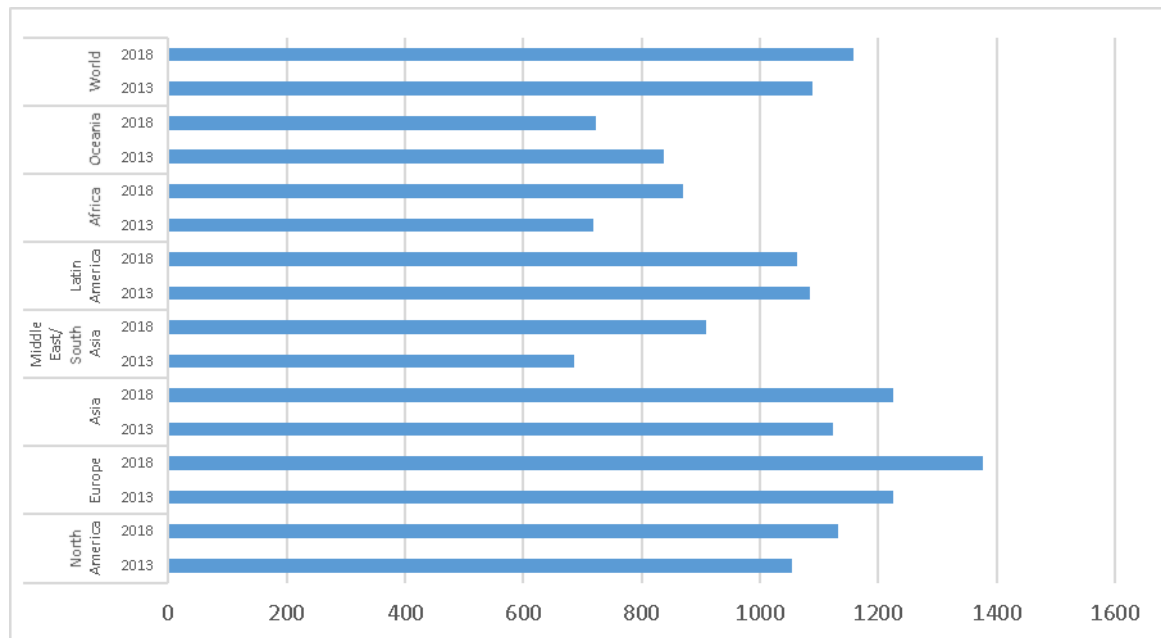
Productivity can be defined as the ratio between the volume produced and the resources employed to produce the output. Productivity can be measured through the analysis of all factors of production (total factor productivity) and/or through a single or a few factors (single/partial factor productivities) (Windle and Dresner 1992). The total factor productivity is preferable as it considers more details; however, it comes with more complexities in measurement and data.

In this sub-system, berth productivity is paramount, given the dependence of other sub-systems on it. The impact on efficiency is often indicated by the length of time which ships await at a terminal before being handled and the length of handling before the ship departs (total time at the terminal) (Stojaković & Twrđy, 2021).

The **berth length** defines the capacity of the feasible sum of vessels, which can be attended simultaneously. The berth length is one of the important characteristics influencing the technical operations and efficiency of container terminals. The length

of the quay wall will usually be natural or man-made. *Figure 4-5* indicates the average quay lengths by region in 2013 and 2018.

Figure 4-5: Quay length (m) by region

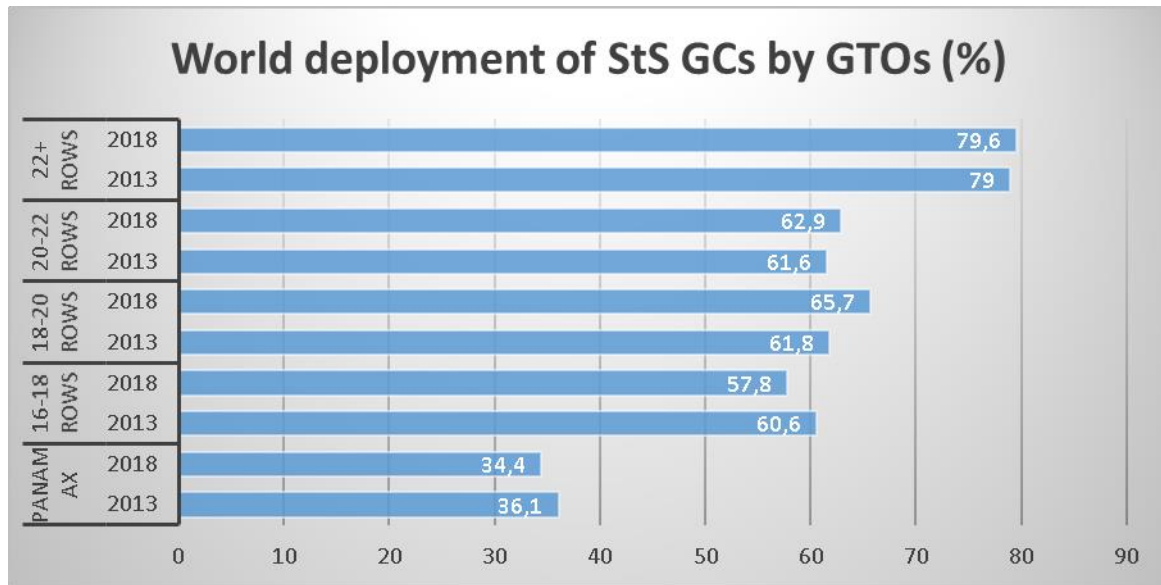


Source: own compilation based on Drewry (2014, 2019)

The inability of a terminal to cater for large vessels due to inadequate **water depth** is a deficiency likely to evoke harm to the terminal's efficiencies (Antoniou et al., 2019). Although continuous dredging (an extra service requiring the use of financial resources) is a possibility, it can also be restricted given environmental concerns of water life.

The emergence of high-efficiency quay/gantry cranes (G/GC) has greatly improved the service offer of terminal operators. To increase throughput, terminals may use multiple GCs, the passing, and the non-passing GCs. In the passing QC arrangement, the QC has an unequal length such that the smaller one can pass underneath the larger QC, common with the RMGCs. Container Terminal Altenwerder (CTA) in Hamburg, Germany has implemented this configuration. The non-passing arrangement is composed of two identical QCs serving one area of the yard, keeping a safe distance in between each other. This configuration is possible with both, the RMGCs and RTGCs and is implemented in the Euromax terminal in the port of Rotterdam, in the Netherlands. Container terminal Burchadkai (CTB) in the port of Hamburg, Germany implements a triple-crane strategy where there are two passing QCs and a larger passing QC (RMGCs), (Carlo et al., 2014).

Figure 4-6: Deployment of Ship to shore gantry cranes, outreach & ownership



Source: Drewry (2014, 2019)

4.2.1.3 Utilization measures

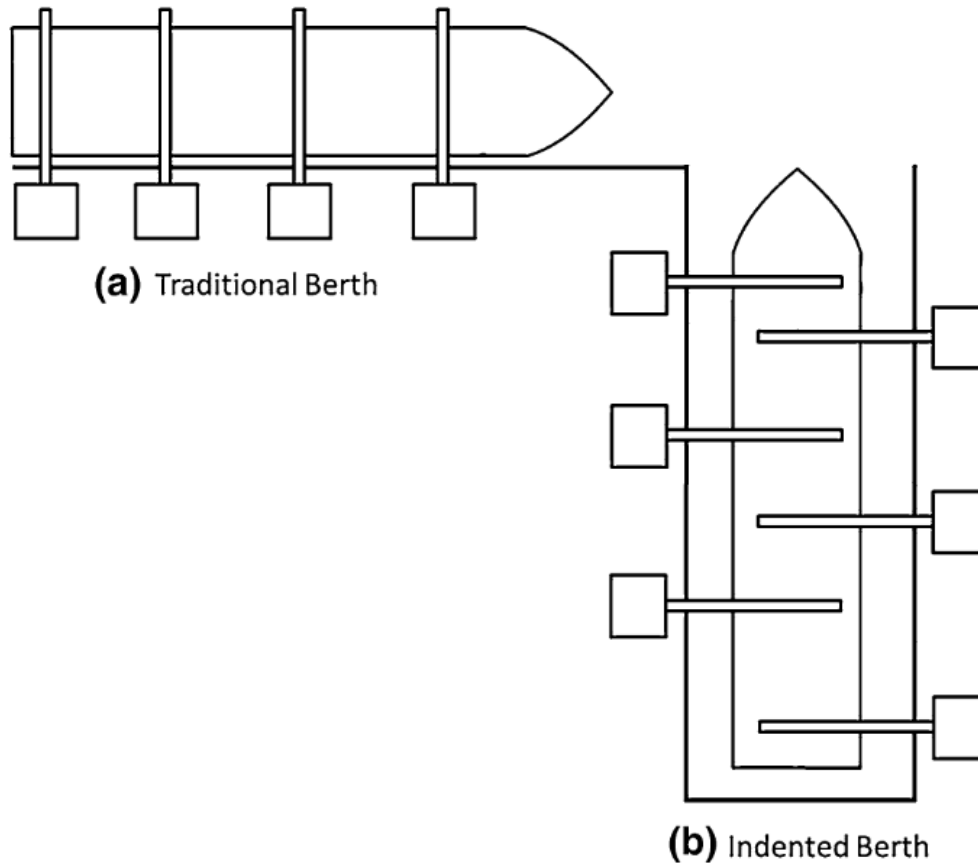
These factors contribute to the decision of how intensively the production resources are used (Esmer, 2008).

Before arrival, a vessel is assigned a berthing place (**berth allocation**). Schedules of large ships are known approximately a year before arrival and container allocations to that ship are ideally known approximately three weeks prior to the ship arrival. The schedule of trucks is almost unknown and thus plans related to trucks change almost all the times. In general, for this activity, to improve efficiency, the aim is to reduce the distances between yard and shore for all containers to be loaded and unloaded, (Steenken et al., 2004).

Traditionally, berths are parallel to the shoreline, referred to as the traditional berth or the marginal berth. A setup that has been associated with the limitation of the number of vessels which could be served at the same time, as well as the limitation of the number of vessels which a QC can serve, as only one side of the vessel can be operated at a time. To resolve this, indented berths can be a solution (although hardly available in practice at present), where a vessel is moored perpendicular to the shoreline. According to Carlo, Vis, & Roodbergen (2015), this type of berth is

associated with large investments to extend the quay wall. *Figure 4-7* illustrates both these quay walls.

Figure 4-7: (a) traditional and (b) indented berths



Source: (Carlo et al., 2015)

Another important element in this subsystem is the **quay crane allocation**. Prior to arrival, the vessels (or barges) are allocated specific quay cranes. As an interface of maritime, the quays of a terminal will have quay cranes for loading and offloading purposes. Moves per call determine the number of containers allocated to each of the quay cranes. Here, the expertise and skills of the crane driver can play a role, as well as the position of the vessel. A discrepancy between the discharging times of vessels has been observed in the past. It may occur therefore that while other cranes have already begun loading that other cranes are still discharging. These activities can therefore affect the efficiency levels of the terminal. Here, the objective is to increase the accessibility rate level of the crane while taking into account the technical data of ships and cranes, to minimize ship delay times, maximize ship performance, and for financial balancing.

In the **quay crane operation**, Kemme (2013); Steenken et al.(2004) & Wang et al. (2019), indicate the differences that can occur in QC allocation, QC cycle, and QC sequence. Traditionally, QCs operate in a single cycle mode, however implementing the dual-cycle mode reduced the QC empty travel and thus turns out to be more efficient. According to Carlo, Vis, & Roodbergen (2015), the dual cycle strategy has been well implemented in the port of Tacoma, Washington in the USA.

The competitiveness of large CTs relies greatly on the capabilities of the deployed QCs. Not all quay cranes can be operated on all ships. Ship length, crane length, and crane jib have to also be taken into consideration. In general, different container terminals have a combination of various types of cranes. Recently, container terminals compete to use the most advanced quay cranes available in the market.

Carlo, Vis, & Roodbergen (2015) discuss the latest developments in quay cranes, which include the multi-lift spreaders feature (e.g Mawan container terminal in Shenzhen implements a triple-spreader QC); **double trolley**, and **floating platforms**.

Some terminals may have fewer cranes by number and yet their cranes have dual capabilities, For instance, the Euromax terminal in Rotterdam has 12 deep-sea dual trolley container cranes. The Eurogate container terminal in Wilhelmshaven is also equipped with tandem lifting, handling four containers at one hoist, Drewry (2015). This will have an impact on efficiency.

When containers are unloaded and loaded, a plan that indicates the specific containers to be (un-)loaded, the vessel location of the containers, and the sequence of storing the containers is prepared, referred to as the **stowage planning**. The planning by the shipping line therefore also has to take into account all terminals of a vessel's rotation, (Steenken et al., 2004).

The objective here (from a shipper's point of view), is to plan in such a way that the number of moves during handling is minimized, that the utilization of the ship is maximized, that crane productivity is maximized, and like in all other activities and that there is cost minimization.

The plan has to be prepared in such a manner that the vessel's stability is maintained throughout the process and also it should be in such a manner that the sequence of the containers is properly identified as not doing so could lead to unproductive moves.

Although container sequence is usually known in advance, due to weather changes, tide, stability issues, a crane driver may decide to alter the pre-concluded sequence and this may cause changes to the schedules. To avoid problems, containers have to arrive in the quay at the right time, order, and in correspondence with horizontal transportation. Steenken et al. (2004) advise that online stowage planning will be the solution for the storage planning encountered problems. Reshuffling in terminals does occur but should be minimal. In manual terminals, the drivers will also have a higher impact on the performance than in automated terminals, however, even in automated terminals, transportation times cannot be exactly even, (Steenken et al., 2004).

4.2.1.4 Service measures

These measures are related to the satisfaction of the customers with the terminal (Esmer, 2008). This contributes to the effectiveness of the services provided by the terminal operator.

At the service level, the **turnaround time** is one important measure that affects the efficiency of a terminal. At this subsystem, minimization of the turnaround time is the major objective, (Covic, 2019). It turns out therefore that the change in the vessel sizes affects the activities in this sub-system and has changed some of the requirements to be met when carrying out activities in this sub-system.

For the ship to arrive safely at the shore, pilots and boatmen are hired to lead the ship safely. The **size of the ship** has a direct influence on the terminals which it can visit as large ships are not able to visit all terminals, given resources such as the water depth and length of quay walls.

Some terminals are equipped with locks that vessels have to pass through. At times, certain terminals are governed by certain regimes which may limit **access through the locks**, for instance in some terminals locks are only opened during certain time slots and therefore when the locks are closed, there is no access, this has an impact on efficiency.

For emergency reasons, vessels at times call at a port for **services** only. However, the presence of these services adds value to the terminal, which could assist in retaining already existing clients and attracting new clients, driving the cargo coming into the port up and thus affecting the terminal's efficiency.

4.2.2 Waterside horizontal transport sub-system

This sub-system acts as an interface between the ship-to-shore and the storage sub-system. From the quay crane, containers are moved to the yard (and vice versa) by the horizontal transport vehicles. The main objective of this sub-system is to move the containers between QC and the storage yard in a smooth, fast, and efficient way (Kemme, 2013). To achieve this, the selection of the horizontal transport vehicles has to be done carefully. The vehicles differ in several characteristics such as carrying capacity, flexibility, velocity, level of automation, etc. *Table 4-2* demonstrates the six most common systems by terminal operators.

Table 4-2: Horizontal transportation operations systems of CTs

Operations System	Required equipment per Quay Crane	Stacking Tiers [1-over-n-high]	Yard Capacity [TEU / ha]
(1) Reachstacker & TTU	3-4 Reachstackers +4-5 TTUs	3 4 5	350 500 950-1 000
(2) Pure SC	4-5	2 3	500 750
(3) RTG & TTU	2-3 RTGs 4-5 TTUs	4-5	1000
(4) RMG & TTU (blocks parallel to the quay)	2 RMGs 4-5 TTUs	4-5	1 000 (or more)
(5) RMG & ShC (blocks perpendicular to the quay)	2 RMGs 2-3 ShCs	4-5	1 000 (or more)
(6) RMG & AGV	5-6	4-5	1 000 (or more)

Source: Brinkmann (2011)

4.2.2.1 Production measures

Quay transfer throughput/moves express the amount of throughput moved between the quay crane and to the yard storage areas, a contributing factor in the analysis of efficiency.

4.2.2.2 Productivity measures

In a **reachstacker and TTU system**, a GC drops a container on a TTU, which transports the container to the stack area where a reachstacker stacks the boxes up to 5 rows high. Although no longer a common practice, but a forklift (fitted with a spreader) can also be used instead of the reachstackers. This system is best suitable for small and medium-size container terminals with a minimum of 3 reachstackers and 4 TTUs. In very small terminals where distances are short, the reachstackers can be used with no additional equipment. Although the system is favourable to investment

expenditure, operation costs, and flexibility, it may lead to lost time and additional moves due to the necessary handover procedures.

The **straddle carrier system** is one of the very much preferred systems in the non-automated systems as the SC can carry the containers in all yard activities (transport, stacking, (un)loading of trucks and rails). In this system, the GC places the container on the apron where the SC picks the container and stacks it into the yard. This system is best suitable for medium to large terminal sizes (approximately 500 TEUs per hectare) where flexibility is needed and it can stack up to 4 containers high. This system can also be assisted by container handlers and also by RMGs to assist in the rail yard. The system is relatively of lower labor costs compared to the TTU system but high investment and capital costs. When a terminal has longer distances to cover, SC are less efficient as they have limited speed compared to other vehicles.

In the **rubber-tired GC system**, an STS QC place the container in a TTUs which take the container to the yard, where the RTGC stack the containers up to 5 high. The RTGC can also service trucks and rail cars (landside operations). This system is suitable for medium to large CT and can deal with long travel distances in the yard. Investment in heavy concrete paving is necessary as the wheels are heavy. Besides flexibility, the system is associated with low space requirements and medium investment per piece of equipment. One of the disadvantages of the system is that two handovers are required and also trucks in the stacking area may cause disturbances to the TTUs.

The **RMGC system with TTUs** (with block parallel to the quay) is no different than the RTGC system above but instead of tyres, this crane is mounted on a railway. In comparison to the RTGs, RMGs stack higher, span wider, have moderate repair and maintenance costs, they are also easier to automate but make it difficult to change the yard layout.

In the **RMGC system with AGVs** (perpendicular to the quay), horizontal transportation is carried out by AGCs alternatively, the ShCs and stacking are carried out by the automated rail-mounted gantry cranes (ARMGCs). For safety requirements, truck handovers are always at the end of the stacking blocks. Low labor costs and high productivity are associated with this system, however, it requires high investment costs and well-trained labour.

4.2.3 Storage sub-system

The equipment employable is in *Table 4-2*. A storage yard takes up the majority of the terminal area and therefore, the layout has to be carefully planned. The storage function of a container terminal cannot be compared to a typical warehouse because the container terminal bases its model on transshipment as the aim is to have a short **dwelling time** and high throughput. Here containers are temporarily stored according to the needs of the shipper. The location is usually in the centre of the terminal and this is due to the decoupling of the waterside and landside activities. On the landside, the storage is connected to the hinterland by use of different modes of transport such as trucks and trains. Kemme (2013) refers to this as the most important sub-system which is an interface. It is arguably the core element of efficiency at a container terminal. For this sub-system to remain efficient, well planning is necessary through automated equipment.

In CTs, containers are stored either on chassis or stacked one on top of the other. Storage on chassis is very unlikely due to limited space and it is most common in North America. The rest of the World largely applies the stacking option. The main objective here is to create an efficient stacking execution by making strategic and operational decisions taking into account the sequence of the stacking equipment as well as the yard block layout. Important decision-making factors here include **yard capacity & utilization, planned throughput, expected dwell times, regulations from customs, environmental protection, and occupational safety** (Kemme, 2013).

4.2.3.1 Production measures

Similar to the other subsystems, container yard throughput measures the sum of container movements that take place in the yard. Increased traffic could mean increased revenues for terminals. When the traffic exceeds the handling capacity of the terminal, then congestion prevails, causing damage to the terminal in concern.

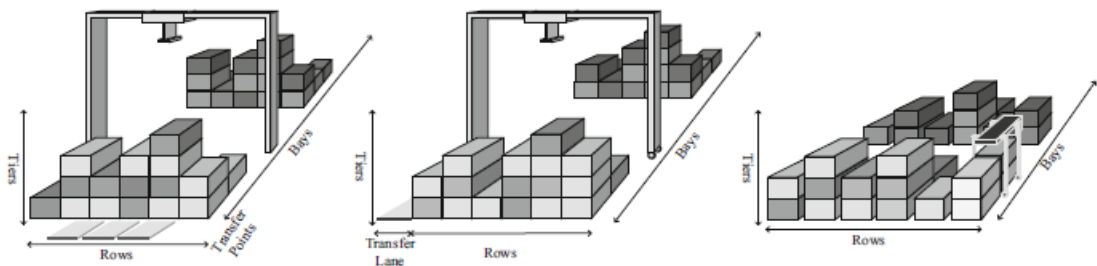
4.2.3.2 Productivity measures

A majority of terminals stack on the ground due to limitations (**yard and storage space**). A terminal layout and equipment can greatly affect the operations of the terminal. According to Meersmans (2002), stacking norms also can be associated with certain locations. The USA is commonly associated with containers on a chassis, while Europe and Asia, with generally smaller spaces, are generally associated with containers stacked one on top of the other in a certain fashion and pattern.

4.2.3.3 Utilization measures

The combination of YCs selected at a terminal affects the **layout of the yard** (contributes to calculating yard utilization). The layout is defined by the block structure, which is defined by the number of rows, defined by the amount of bays, the number of tiers, and by the space needed between rows and bays. Possible block structures due to the type of YC (given the features of handling) are illustrated in *Figure 4-8*. Noticeable differences are in the transfer possibilities from stack to means of transport. RTGs are associated with transfer lanes while RMGs are associated with both, the transfer lanes as well as the transfer points. A comprehensive overview of the storage yard sub-system can be found from Carlo et al.(2014).

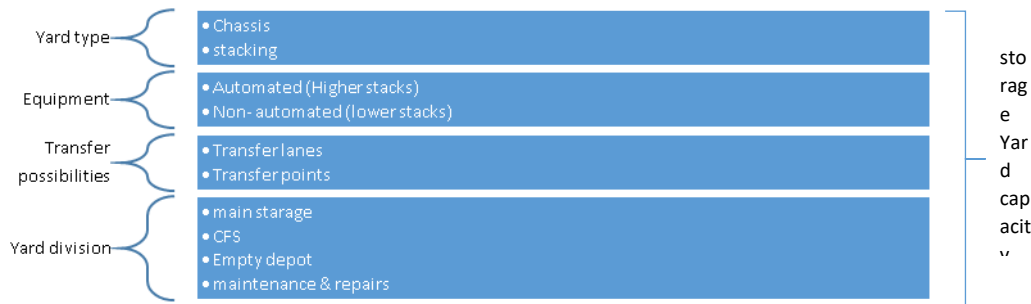
Figure 4-8: Block structures by RMG, RTG, and SC



Source: Wiese et al. (2011)

The handling capabilities of the terminal thus give rise to different stacking densities which can be calculated by Twenty-foot container ground slots (TGS) area and therefore the ability to define one of the important elements in identifying the operational performance of a container terminal, the total **storage area capacity** (determinant of yard utilization) for different yard sizes per annum. These various unique characteristics of the storage yard are depicted in *Figure 4-9*.

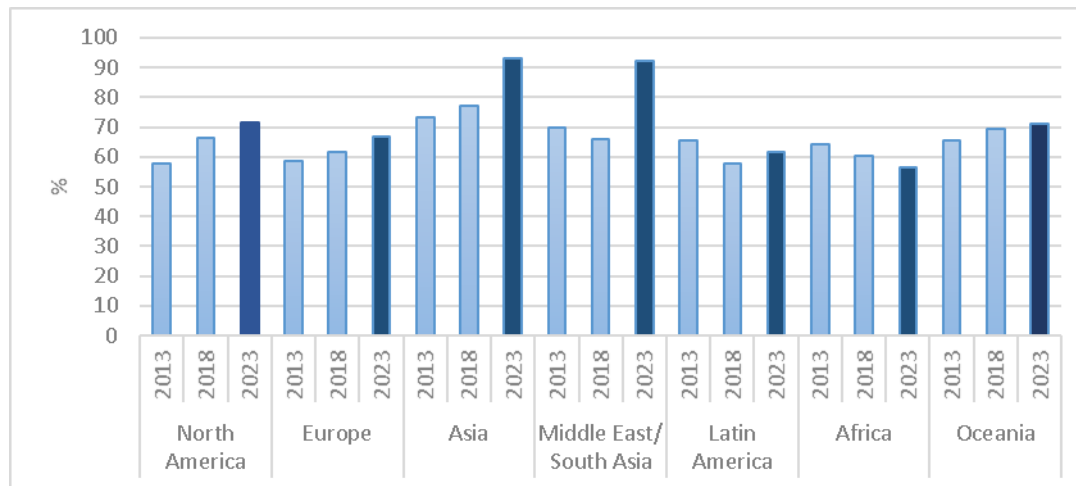
Figure 4-9: Characteristics of a storage system



Source: own compilation

The use of automated storage yards and retrieval systems (AS/RS) should be more targeted to improve the efficiency of storage yards (Wiese et al., 2011).

Figure 4-10: Terminal utilization and 2023 projections



Source: Own compilation based on Drewry (2014, 2019)

4.2.4 Hinterland connection sub-system

Hinterland connections facilitate the flow of imports and exports.

A container arrives at the terminal through various modes of transportation, either on the waterside or on the landside. Several modes of transport can serve this role. To be efficient, a terminal operator needs to manage these transport modes in consideration of the yard capacity. On the landside, containers can arrive/depart by external trucks (XTs) or rail. On the waterside, containers can arrive and depart by deep/shortsea vessels, feeders, or barges. The mode is defined by the flow of the container direction, i.e import, export, or transshipment. This is demonstrated in Table 4-3.

Table 4-3: Flow of good and mode of transport

		Mode of transport	
Flow of cargo		Arrival	Departure
	Transshipment		
	Export		
	Import		
	Land-Land		

Key:

Deep sea

short sea

feeder

Rail

Road

Feeder

Source: Own composition based on Kemme (2013)

4.2.4.1 Production measures

The differences in **intermodal delivery and receipt** lead to different cost structures in the terminals. This measures the activity relating to the delivery of outbound cargo and collection of inbound containers.

4.2.4.2 Utilization measures

Vanelslander (2005) mentions the impact of the **terminal gate opening hours** (time utilization) on the congestion of the terminal. How incoming and outgoing vehicles are processed is an important factor in terminal operation.

4.2.4.3 Service measures

The **time** taken to collect or deliver a container at a terminal (road/rail vehicle turnaround time) is important to trucking companies and thus this affects the efficiency of a terminal.

The **mode of transport** which the terminal can serve determines the containers able to reach and leave the terminal, affecting the amount/potential of handled containers.

Besides maritime accessibility and infrastructure, activities beyond the port such as **hinterland tariffs** can drive the traffic borne to the port (Vanelslander, 2005).

There are no clear boundaries concerning the **hinterland size** but it can be identified by the location which a port can reach with a competitive advantage compared to other ports, taking into account generalized transport costs. Hinterland by definition is blurred, it changes over time and cannot be precisely delimited.

An important determinant of port performance is **hinterland market size**. It can be expected and has been shown in previous research (Yuen et al., 2013; De Oliveira and Cariou, 2015), that ports serving larger local markets tend to be more efficient. To account for this effect, it is necessary to provide some measure of hinterland market size. One option is to use **population data** as a proxy for market size (De Oliveira and Cariou, 2015).

4.2.5 Summary: Sub-systems

The sub-systems and their characteristics are demonstrated in Appendix 4.1, varying between automated, semi-automated, and non-automated terminals. The typology is based on the physically observable activities in the container terminal. There are fewer observable differences at the ship to shore subsystem and the hinterland connection subsystems. There are significant observable differences between the horizontal transportation and the storage subsystems.

4.3 Characteristics determined indirectly to the sub-system

The factors discussed here are those which do not fall in a specific subsystem. Some are categorized as mutually determined factors and others that are exogenously determined.

4.3.1 Mutually determined internal factors

Some factors affect efficiency, not on a specific subsector but their impact can be visible across subsectors and can thus not be included in any subsector. These factors are discussed in this sub-section.

Port promotion strategies are likely to lead to increased revenues through increased traffic. However, a more than necessary amount of traffic leads to congestion, the effects of which can be detrimental.

To facilitate growth in the business and to expand their portfolios, CTOs often have to make **strategic decisions**. Some of the decisions could mean that the terminal may appear less efficient today but performance is intended to improve in the future. Ng and Pallis (2007) observe that different ports adopt different strategies and objectives, which are not only affected by external factors but also by internal factors

Liu (2010) identifies three classifications of container terminals, by **ownership**, by **geography**, or by **service** scale. The author recognizes the emphasis on ownership by operator type (local or global). The way of management between privately- and

publicly-owned firms can influence the drive of traffic towards the terminal and therefore affecting the efficiency.

The invention of the container is primarily for the facilitation of port handling efficiency (Liu, 2010). **The container size**, which refers to the length, width, and height dimensions of a container varies. Containers are all standardized according to the ISO 668 standard (Kemme, 2013). In general, a container is measured in the length of twenty-foot equivalent units (TEU) with a tare weight of 2,250 kgs, consequently, a forty-foot equivalent units (FEU) container is double the length of a TEU.

A container can also be classified according to the **basis of the cargo** carried, categorized into a dry container, tank container, open container, or reefer as the main types. The International Maritime Organization (IMO) also sets load standards by which a container can be categorized.

Differences in the type of a container mean a different implication to the costs/revenues. Poitras, Tongzon, & Li (1996) mention the importance of the type of container being handled to be one of the most important determinants of efficiency. Handling a 40ft container takes approximately the same time as handling a 20ft container. Larger terminals tend to handle a larger proportion of FEUs than smaller terminals. **Transshipment** of a container may contain more moves or fewer moves than an import container. A reefer **full/empty** container may require extra handling compared to a **full/empty** standard container. The type of container handled will likely lead to a different handling requirement for a terminal and therefore meaning a demand of different requirements of the handling machinery and handling technique.

Port Infrastructure refers to elements defining the accessibility of the ports; the factors affecting the layout; and the connections to the hinterlands.

Maintenance of infrastructure and superstructure has a direct impact on revenues and costs as the frequency of maintenance will lead to a different cost. The maintenance of activities such as dredging provides access to large vessels, therefore failure to do so may lead to loss of customers, create delays and technical defects can occur. Similarly, with the superstructure, irregular maintenance can lead to interruptions to the normal functioning of the machinery.

Regardless of the equipment installed in a terminal, a well cooperating **workforce** is imperative in achieving the desired outcomes.

Longer **leases** allow terminals to recover the costs of capital invested in the terminal. Due to longer **concession agreements** by the port authority to the terminals in the ports, the requirements by port authorities can be hard to be achieved by smaller terminals, giving the larger terminals higher chances of operating the terminals, outcompeting the smaller terminals, leading to takeovers and reduced competition.

Port management and operation are strongly reliant on the quality and speed of information flow between the players involved. The **Electronic Data Interchange (EDI)** enhances port efficiency, market outreach of their port, improved integration with the port community systems (PCS); a considerable decrease in costs and time spent during cargo handling, berth planning, gate control, and customs (Varbanova, 2018). The schedules from the shipping lines to the container terminal are transferred through the EDI enabling the terminal ship planner to do its job in an integrated manner such that the containers are allocated in an efficiency-enhancing manner. The process of container handling involves an aspect of documents and data exchange. Nowadays, the aim to achieve a fully paperless environment for administrative service and offer real-time based information is of importance. Heilig, Stahlbock, & Voß (2019), provide an overview of the transition from “paperless” procedures to the currently “smart” procedures of today. The first generation of the 1980s, when EDI was introduced, focused on achieving paperless procedures with the introduction of DAKOSY in the port of Hamburg in Germany. The first commercial terminal operations system (TOS), CITOS and Navis, were developed during this phase. The second generation, of the 1990s to 2000s was more focused on automation. This wave focused on drastically improving automation in container terminals, developing the TOS introducing technologies such as real-time locating systems (RTLS); global positioning systems (GPS); optical character recognition (OCR). The current generation, from 2010s till now, is the wave referred to as the “smart” procedures, which aims to facilitate real-time communication to further improve the services.

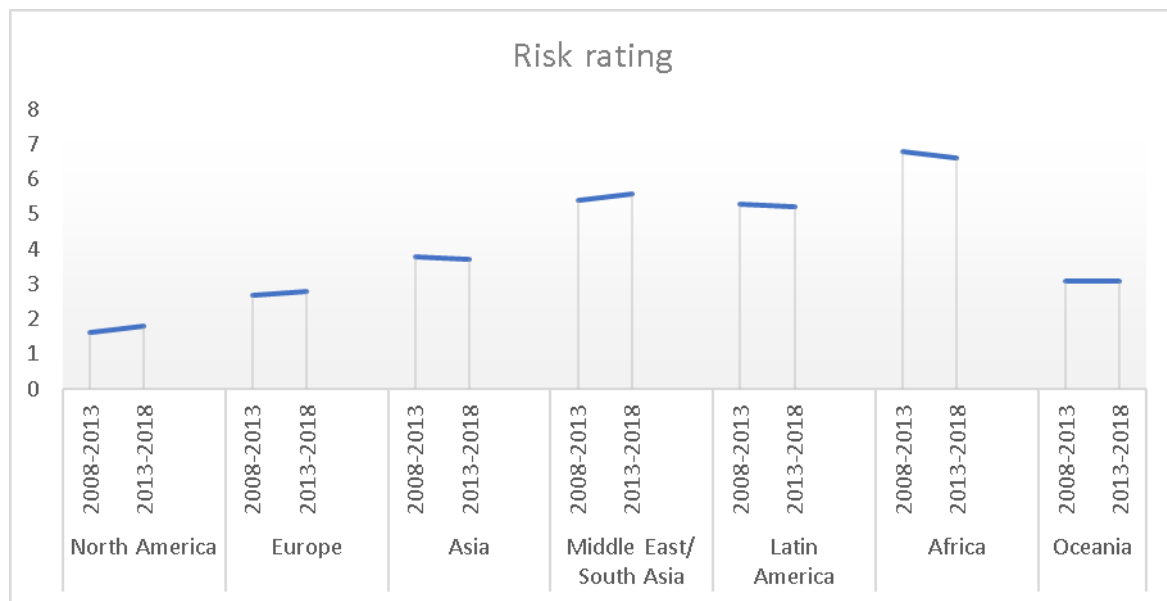
Terminals serve various stakeholders, including its customers. From the demand point of view, performance can be affected by the **perspective of the customers** and to the extent to which they are satisfied by the service they receive. Chang, Lee, & Tongzon (2008) show the qualities considered by shipping lines in deciding on the terminal they which to call.

Generally, the number of human injuries is low in ports (improved **safety**). Services offered to especially vessels entering the port such as pilotage services, towage, the buoys, lights, and other navigational aids can increase the service level of the terminal, and also when properly done, they can lead to increased attractiveness of the terminal. When not properly carried out, this can lead to increased time at the entry points. Ports also may tighten the procedures at their entry points, which could result in increased times and use of resources affecting the efficiency of the terminal services. Ports located in areas of harsh weather conditions may require tighter regulations, which may increase the time spent by the vessels at the terminal.

Contained within the metal boxes, are various types of products from sand to diamonds and high technologies, to exemplify a few value categories. This is one among other reasons that make container terminals a target by criminals. Solutions of protection against such activities thus have to be provided. Saavedra (2002) demonstrates the extent of such issues in South America. Terminals, therefore, need to dedicate resources (technologies and human capital) to control the **security** of their terminals.

The interruption caused by poor security can be detrimental to the terminal, in the short term and at times in the long run. A risk comparative relation conducted by Drewry (2014,2019) reveals differences in the risk profiles of container terminals at a regional level *Figure 4-11*. The rating is conducted through the OECD ratings (1 is low and 8 is high), taking into account political and economic factors to be considered when making investments in a region. For terminal operators, Drewry makes use of total capacity, as opposed to throughput for this association. Although the risk in North America is low, it has increased between the two periods, similarly in Europe and the Middle East. Although the trend in Africa has been negative over the period, the risk is the highest among the region. Overall, it is best to remember, following the economic theory that, 'profit is the risk return'.

Figure 4-11: Regional risk rating



Source: Own compilation based on Drewry (2014, 2019)

Several things can occur during handling, e.g damage during packing and unpacking; errors during loading, administrative errors, insecure packing, etc. Having these **services** at the terminal can increase the efficiency of the terminal.

4.3.2 Exogenous supply and demand factors

Other factors affect container terminal efficiency but occur outside the system. These factors are likely to affect the demand and supply of trade and thus affect how the terminal performs. Following Vanelslander (2005), these factors can be categorized into chain, policy, and scope factors. These are listed in this sub-section

4.3.2.1 Chain factors

The supply chain through which the containers flow incorporates within it some factors which can cause changes to performance. These factors are highlighted here.

According to Vanelslander (2006), cargo handling is made of a combination of the mentioned activities which highly affect costs and revenues. The profitability of **terminal co-operation** can be highly affected by the combination of activities which the companies involved decide to conduct and how.

Cooperation of handling companies can increase the market share of the resulting company, as well as set the conditions that allow for the economies of scale. An increased network between container terminals can prove beneficial for the

terminals involved. Medda & Liu (2013) suggest that specialization as opposed to multi-purpose at terminals lead to increased productivity.

Local terminal operators are terminals operating in a single world region and Global container terminal operators are terminals operating in more than one world region (**terminal market**). Due to their ability to connect at a global scale, know-how can be easily transferred between the different regions and this could be an added advantage compared to a local whose market is limited. This usually strengthens the terminal and trust is built to stakeholders and shareholders (Medda & Liu, 2013).

The presence of the **inland port** influence the cargo coming through the seaport if the radius of the seaport is in appropriate proximity to the inland port. The additional transportation such as barges from the seaport to the inland port could also influence handling prices at the seaport.

4.3.2.2 Policy factors

The decisions taken by governments and related institutions influence how supply and demand factors interact and thus influence the flow of trade. The factors affected in this case are discussed below.

Terminals are located in ports that are affected by distinguishable **environmental factors**. Concerns regarding the environment will have an impact on the terminals depending on where they are located. Environmental awareness is becoming a norm throughout the world across sectors. Protectionists of the environment have prevented port activities from expanding; terminals have had to invest in making use of technologies that help to reduce the environmental impact (example, low-emission terminal equipment). Both, the port and shipping industry has to adhere and comply with various rules and regulations due to the cost impact of their activities to especially climate change (Di Vaio & Varriale, 2018). Ports and shippers thus remain increasingly adhering to the pressures by the public institutions and the society to improve on their image and as part of social corporate responsibility.

Levinson (2016) mentions how the container made the world smaller and the world economy bigger. How the economy is performing (**economic growth**) will be mirrored by trade and vice versa.

4.3.2.3 Scope factors.

Terminal activity can be carried out in conjunction with other services. Some factors affecting efficiency in such strategies are taken into account here.

At times, seaports may have relations and links with inland ports (**terminal and inland port relation**). This can have various spill-over relations, which could lead to potential increases in the cargo of the affected terminals.

Services provided to containers, also by other companies at the port is an added advantage as it attracts containers that pass through the terminal.

Due to the international relations that occur in the movement of goods, ports and terminals do not operate in insulated environments but face **competition** in a variety of manner. This means that the terminal operators (and ports) frequently need to monitor their competitive position regarding rivalry, the threat of new entrants, global substitutes, and the market concentration of certain players. The impact of competition on efficiency remains inconclusive. A terminal may, due to increased competition, feel pressure and thereby increase its efficiency. On the other hand, a terminal faced with the competition may overinvest and which could lead to lower efficiencies, (Figueiredo De Oliveira & Cariou, 2015). To regulate competition, often port authorities issue long-term concession agreements. Concessions affect how power is shared between the port authority and the CTO. Often these agreements also specify the competition conditions during the concession period (Yip et al., 2014).

In an economy where production increases and demand for goods and services increases, there is bound to be an influence on port cargo (**industrial activity**). Sterlacchini (2001) discusses factors that determine the necessary conditions for firms to export/import their products and also their services.

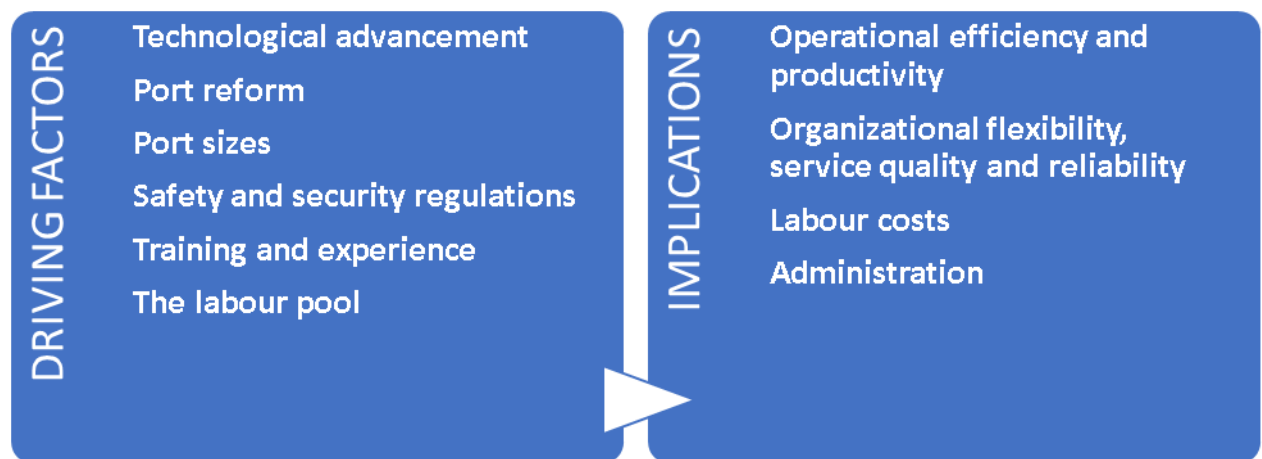
4.4 Port labour productivity

Admitting to the profound changes and shifts towards capital intensity over the years, the human factor remains one of the most valuable assets in achieving the goals and objectives of the port operations. Labour efficiency is an integral part of the overall performance of any business. From ship-to-shore to yard crane drivers and harbour pilots, port labour has proven to affect the port and terminal competitiveness (Satta et al. 2019; Turnbull and Wass 2007). Delays occurring during stevedoring can contribute largely to inefficiency (Tongzon, 1995). In the port sector, labour productivity is quite complex, especially given the diversity in the functionality of the yard handling equipment. The type of equipment used in the container yard

influences how labour performs, among many other factors (Notteboom 2010). In addition to the usefulness to performance at a company level, the OECD identifies labour productivity as a valuable tool to measure living standards, an important social element. That is important given the economic role of port labour in port cities and port countries. Based on the OECD (2021), high labour productivity growth can indicate a more effective use of capital; it can also mean a decrease in the employment of low-productivity workers or general efficiency gains and innovation.

Figure 4-12 demonstrates the changes in the port sector that have influenced port labour performance and, in turn, the performance of terminals from various observations.

Figure 4-12: Port labour changes and impact on terminal performance



Source: Own elaboration

The section that follows elaborates on the performance-related driving factors of change in the port sector, and the last section discusses the implications on terminal performance.

4.4.1 Drivers of change affecting port labour

Several factors of change have been shown to affect labour performance in the port sector. Those factors include changes in technologies utilised, port reform, firm strategies within the port and port sizes and they are discussed in this section.

4.4.1.1 Technological advancement

From the 1980s, ICT proved itself worthy by showing a significant rise in productivity. Automation and robotics were the first actions brought by ICT, followed by the digitalisation of communication processes and thereafter by the Internet of Things (IoT) (Esser et al. 2020). Given these developments, businesses intend to lower costs,

increase productivity and efficiency, improve flexibility, improve the environmental impact, and increase safety, among other things.

Some jobs are more prone to ICT developments in the terminal sector than others. New applications and technologies at the terminal affect both white-collar and blue-collar employees as it requires flexibility in adapting to the new skills required. The new skills often require less of the manual labour force and more of the skilled labour (Satta et al. 2019). At sea-side operations, the demand for changes in the superstructure, infrastructure, and operation increases, primarily due to the increasing container vessel sizes. Vessel traffic services (VTS), navigation support and collision avoidance have reduced the labour required in ancillary services. A similar phenomenon in labour reduction is also seen at the ship-to-shore and yard handling subsystems where handling equipment such as automatic crane controls (ACC), Automated Guided Vehicles (AGCs) and Automated Stacking Cranes (ASC) are used. In the hinterland connections, where a diverse mode of transportation is used, the number of workers and operational costs are reduced to become more efficient (Macharis and Bontekoning 2004). IT systems such as Electronic Data Interchange (EDI) and tracking systems require skilled labour, leaving out the lower-skilled portion of the labour sector. Although there are expected benefits from technological advancements, critical issues emerge from it affecting labour who become eliminated and those who remain.

4.4.1.2 Port reform changes

In the earlier days (up to and early 1960s), ports were characterised by a lack of job security, labour-intensive methods, poor working conditions, labour management, congestion, unpredictable cargo movements, contributing to inefficiencies (Haralambides 2021). Labour productivity grew approximately threefold, from 1 ton per person-hour. In such earlier days, inefficient ports were not viewed as such, given the availability of cargo volumes, till later on when ports were recognised as the bottlenecks causing disruptions in the entire supply chain. Such a recognition introduced increased privatisation, which is an action that removed many uncertainties in the sector and came along with further regulation of port labour (Haralambides 2021).

At the European Union level, several attempts have been put in place to liberalise port labour by regulating market access and providing port services. A section in chapter 2 of this thesis discusses a component of port reform to highlight the differences in

service provision across various port management types (private, public, tool and landlord ports). The introduction of the private sector became a norm in the industry, as it has in many cases been associated with increased competitiveness (Antoniou et al. 2019; Cullinane et al. 2006; Cullinane, Song, and Gray 2002; Hung, Lu, and Wang 2010; Meersman, van de Voorde, and Vanellander 2010; Tongzon and Heng 2005; van de Voorde, Eddy Meersman and Vanellander 2011. The reforms in the port sector, mainly due to the private sector's involvement, have fostered the reassessment of labour laws and practices.

4.4.1.3 Terminal sizes

The subject of firm size and productivity has attracted many economic studies. The observations claim that larger firms tend to be more productive, which would also be due to the labour force as means to productivity. In ports, controversies towards the efficiency and productivity benefits of size have been explored (Cullinane & Khanna, 1999; Cullinane & Wang, 2005; Cullinane et al., 2006; Martinez-Budría, Díaz-Armas, Navarro-Ibañez, and Ravelo-Mesa, 1999; Wang, Cullinane, and Song, 2005; Caves and Christensen, 1988; and De Neufville and Tsunokawa, 1981.

Terminals over the years have increased in size for various reasons such as mergers and acquisitions, integration, capacity building, greenfield growth, changes in commodity mix (containerisation), increased service provision and so on. Terminals with large volumes often invest in additional and improved infrastructure, in many cases ahead of demand (overcapacity). Infrastructural investments also impact on labour productivity as the condition of infrastructure affects how labour performs their work given a specific cargo size. Large terminals (especially due to larger concessions than in the past) also indicate more work; therefore, the gang required per call size will also determine productivity.

4.4.1.4 Safety and security regulations

Ports fall under the category of one of the riskiest places to work in. Chapter 1 of this thesis already addressed safety and security challenges at ports. Changes related to this element have led to increased improvements in many ports as work-related accidents and injuries have reduced over time. On one token, the downsizing of employees left behind a majority of skilled and highly experienced labours at ports. Employees also invest in improved equipment and take appropriate measures to improve safety and reduce work accidents, given the potential losses of productivity, which result in such hiccups. On the other token, the fewer labours place a more

significant portion of stress and responsibility per worker, which negatively contributes towards health.

4.4.15 Training and experience

Training and experience are linked with productivity and efficiency. Job qualification, career development, team organisation, education and training are critical to labour efficiency (Satta et al. 2019). Given the changes in operations, the skills required from labour have drastically changed over time. In many ports, the flow of progression or promotion techniques from one level to the next is particular. The rules around promotions are often linked with productivity; apart from salary increases, the dock labour enjoys a social status, an element that is often important in the circles. Another benefit from productivity is, of course, the port or the terminal itself (Bottalico 2017).

Given the continuous changes, there is a likelihood that the regulatory system might change at ports. The reduction of permanent port labour can be expected and an increase in non-permanent labour through the port labour systems, taking into account regulatory constraints, market requirements and ongoing processes of change (Bottalico 2017). In the Belgian case, there seems to be a trend towards a parallel system (non-unionised workforce outside the pool) provided by the new port reform, which addresses the needs of the shipping lines. Although there has been a recognition of the productivity and efficiency of the port labour in Antwerp, the European Commission has continued to initiate infringement procedures concerning the labour organisation.

The changes in the port environment also open a gap for terminals to be an accommodative environment with increased diversity in the recruitment processes. This means that terminals can now accommodate potential employees from various other backgrounds, such as IT, Engineering, openness to women, who were previously excluded; to young people who saw ports as unattractive environments with traditional processes. All these changes will have an influence on productivity and efficiency.

4.4.16 The labour pool

The labour pool systems were inherently designed to protect dock labour from fluctuating demand and supply scenarios at the port, which would also require less or more dock labour and thus meant a little guarantee of employability at a given time. A successful scheme would guarantee the operators a chance to obtain labour when

needed and guarantee labour work when it becomes available. To protect the interests of dockworkers, labour unions often want to avoid an oversupply of labour. Such may lead to labour shortages in handling vessels and is in itself one of the causes of disputes at the ports (Notteboom 2010). The productivity per vessel tonnage is rarely ever linked to an individual employee, but it is instead the whole gang, given the interdependence of the gang crew to each other. Labour pool management progressed to finding a proper balance, which, if not obtained, can increase costs (direct and indirect), leading to inefficiencies.

4.4.2 The Implications of the changes to ports and terminals

The transformations as mentioned above have proven themselves to have an impact on labour productivities and efficiencies. In turn, terminals faced with the changes can meet operational inefficiencies, lower flexibility, poor service quality, unreliable service, and increased labour costs.

4.4.2.1 Operational efficiency and productivity

The various changes affecting labour productivity can also be expected to have implications on the operational efficiency and productivity of the terminal itself. For instance, the labour reform introduced the participation of the private sector in ports. The private sector participation caused some changes in recruitment (for example, through private agencies), wages, and job security changes. The ability of the private firms to gain control of operations is associated in literature by essentially many benefits to performance, although not entirely. Private sector participation in the port sector is related to the adaptation of innovative managerial abilities, which positively contribute towards competitiveness and performance more than the case of the absence of private sector participation. On the other hand, increased privatisation is linked with a deterioration of dock workers' working conditions, downsizing, social tensions, and difficulties in managing the pool systems, among some of the challenges.

Another impact on terminal performance due to reform is through health, safety, and security changes. The private sector is associated with introducing innovative systems and techniques to improve performance directly and reduce workplace accidents, injuries and better organisation of safety procedures. The impact on this is that the costs associated with social payments reduce. Reducing absenteeism and unproductive time due to injuries and accidents leads to increased reliability and service quality, thus overall performance (Satta et al. 2019).

Although with many operational advantages (faster movement of containers, competitiveness, service quality, reliability), the increase in the application of ICT systems and automation at the port sector has been observed to create a mismatch between supply and demand of employment, skills and competencies (Esser et al. 2020). Due to automation, the middle-paid jobs are at the risk of disappearing, jobs on the floor will become assisted by robotics, and management jobs will become more complex, with a need for the ability to multi-task. The disappearance of jobs has caused labour unsettlement in many ports; it has negatively impacted on the mental state of the labour who remains and those who leave the labour force.

4.4.2.2 Organisational flexibility, service quality and reliability

The demand for stevedoring labour is highly variable, pending on the ships that arrive at the terminal. If terminals have little control in the labour supply, this creates less flexibility to adjust when necessary. Flexibility is essential as the peak times at the terminal are often irregular, and thus the number of available employees tend to affect productivity and efficiency.

The standard of training and qualifications of the labour force influences organisational flexibility. Based on (Bottalico, 2017), the possibility of producing such multi-skilled labour is only in certain limited professions. When labour is multi-skilled, it provides more flexibility to replace the workers without employing additional employees, contributing to labour efficiency and leading to multi-skilled labour.

The development of labour pools is to achieve flexibility, dependent on the labour regulations in the specific country. The possible adjustments provided by the labour pool system have benefited mostly the employers in having the possibility to mix and match the gang to the desired labour productivities per vessel and shift (Satta et al., 2019). In many cases, such benefits are often shared with employees through increased remuneration, defined by the trade unions. In some ports, such as the port of Antwerp, bonus systems are not allowed.

Automation also plays a role in flexibility by providing sophisticated and advanced systems that can be readapted. Automation is also associated with increased service quality and reliability. In Antwerp, automation is minimal due to the reduced flexibility of labour reorganisation. At the port of Rotterdam, where automation is higher, flexibility is not as high if the shift-loading plan has to suddenly change, for instance (Bottalico 2017).

4.4.2.3 Labour costs

The labour costs proportion on total costs varies by cargo type. While in dry bulk terminals, labour costs can be between 15 and 20 per cent, in container terminals, labour costs can be as high as 50 per cent of the total costs. Generally, labour costs in a port vary between 40 and 75 per cent (Notteboom 2010). Literature shows that high productivity and flexibility at terminals come at a price. The experienced changes in the labour sector productivities impact port costs, directly and indirectly, affecting the terminal cost structure.

Technological innovations and developments have a pronounced impact on labour productivity, as noted in the previous sections. While seeking innovation, many terminals also intend to reduce labour costs; in many cases, this indicates a reduction in the number of gangs. When labour costs do not decrease in line with the increase of innovation, this reduces the eagerness from the terminal operators managers to invest in such innovations, which may further influence inefficiencies (Notteboom 2010). Technologies are thus associated with reduced labour direct costs (except the expenses related to training) but with an increase in indirect costs. Among the indirect costs of technological advancement are the jobs lost due to automation. Automation can introduce an element of social tensions and conflicts given the role played by ports to employment. Too little labour left behind could indicate understaffing associated with high-stress levels, shortage strikes, accidents, traffic losses, inefficiencies, etc.

Implications due to labour unions and pools can affect labour management, affecting efficiency. Rigid and prescriptive work arrangements (e.g. the order of engagement, shift premiums) in the container stevedoring industry are likely to reduce productivity and reliability, increase processing times, and, ultimately, inflate labour costs (Satta et al., 2019). Specifically, in some cases, labour pools play a role in labour costs such that the cost element cannot be used as a basis for competitiveness by terminal operators. Antwerp labour is regulated through the port bible known as 'the CODEX', which would mean that the operators in the same port compete only on productivity, service level, etc., not on labour costs. At the port of Antwerp (where the cost per box is between 50 and 55 per cent, of which about 85 per cent consist of dock labour), dockworkers are represented by the labour pool. The payroll is provided by CEPA, where the labour is paid weekly and equally irrespective of the operator (Bottalico 2017).

4.4.2.4 Administration

Changes brought by port reform affected the processes at ports. As mentioned repeatedly, technology being one of the primary drivers of change was strengthened very much by the involvement of the private sector players. The introduction of community systems and common windows through improved ICT systems indicates more cooperation between port players. This collaboration demonstrates increased data availability collected between the port stakeholders. Although this leads to increased administrative work, the data collected can help find more solutions towards improving productivity and efficiencies. In addition, communication, IT and data-related skills will increase in demand at the ports as these become important.

Measuring the impact of labour on productivity and efficiency is very limited due to the inability to measure the actual labour force and obtain data. The inability to access data has been affected by the number of players in the ports. Different players offer services (terminal operators, labour pool organisation, port authorities, waterside nautical service providers, etc). The complexity in the collaborations between parties makes it challenging to collaborate on data with various entities with differing interests concerning labour.

4.4.3 Port labour policy

Changes in the sector also affected how port labour is described. How port labour is defined has changed over the years, the legal status of dock labour, labour pools functioning, labour arrangement, and labour definitions of qualifications. Considering dock labour definition by law, for instance, dockworkers reduced from 9 270 to 6 131 people between 1980 and 2015 (Bottalico 2017). The reduction is not solely a reduction but is also affected by the scope of the definition.

The inclusion of the private sector in the industry implicated the position of the terminal operators. For instance, the mergers, acquisitions and alliances of ship liners strengthened their bargaining power, limiting the freedom of terminal operators to demand more in contractual arrangements. M&As have to lead to an increase in some firm sizes, leading to increased power to negotiate. For instance, in Antwerp, port employers are obliged to be affiliated with CEPA Centrale des Employeurs au port d'Anvers), where the number of employees (salaries) managed by the company influences the possibility of a seat at the board of decision making, affecting the direction of policy.

The benefits of innovation to productivity and efficiency in comparison to the social costs (strikes, work accidents, mental health) associated with technologies have embedded difficulties in balancing policy compromise between innovation and quantity of dock labour.

The impact on port labour of the strategies (horizontal, vertical cooperation, mergers, acquisitions, etc.) applied by the port's stakeholders (multi(national) terminal operators, shipping lines, hinterland transportation, etc.) has hardly been explored in literature. The conditions of change in the labour pools, affected by many stakeholders, attract much of policymakers attention (such as the government and labour unions), who influence change in the port labour sector (Notteboom, 2010). A trend towards open and autonomous labour pools is evident across Europe. However, the pace of change has differed across the states and even within the same state (Bottalico 2017).

4.5 Conclusion

Faced with the rapidly changing environment, this chapter highlights the elements that terminal operators need to focus on to improve performance, some of which are internal and others externally controlled. How the yard is managed influences the competitiveness of the terminal. How the activities in the terminal are integrated is essential and assists in the cost minimization and improving efficiency at the terminal.

This chapter takes into account the characteristics of various container terminals based on the four sub-systems (ship-to-shore; waterside horizontal transport; storage and the hinterland connection subsystem). Terminals range from purely automated terminals, semi-automated and non-automated. This and the other factors identified to affect the layout used equipment and the manner of handling in the terminals. The waterside horizontal transport together with the storage sub-systems are where the terminals tend to have many visible differences by equipment, stacking, transfer lanes/points, etc. than the other sub-systems

This chapter lays a foundation for identifying efficiency factors for the empirical analysis in chapters 5 and 6. Based on previous literature (including Esmer (2008); (Vanelslander, 2005)), factors of efficiency are classified and categorized. Identifying

these factors is not solely important for the identification of technical efficiency but also cost effects to the CTs. Although all the factors mentioned are important, and although some affect some container terminals more or less than others, there can be a common ground.

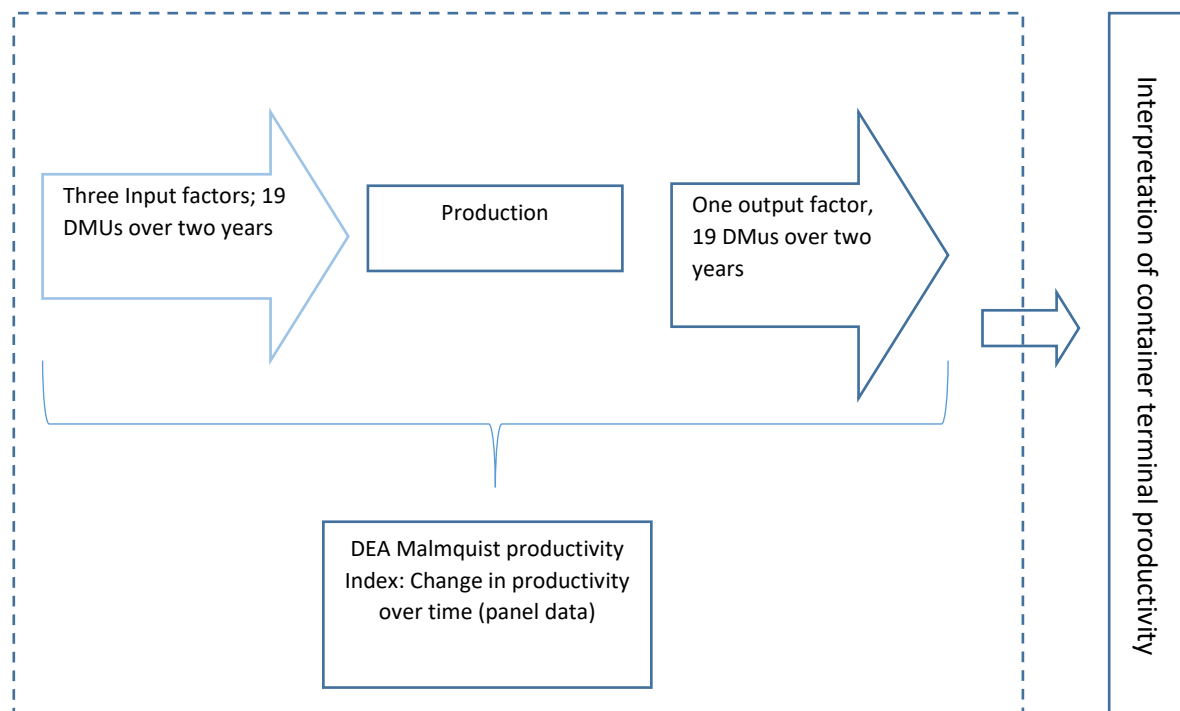
To facilitate performance on the waterside sub-system, the quay crane and the quay wall can be identified as the most important elements which are necessary for any container terminal. These elements can differ from the design point of view (e.g. quay wall design, quay crane features) to the plans implemented by the terminal managers. To facilitate the horizontal transportation and the storage yard, the horizontal transportation and the storage area become crucial in determining performance. The differences in these elements significantly influence the characterization of the level of automation of a container terminal. These two elements also affect the storage yard design (stacking and layout) and the overall maximum capacity of the terminal, which make the elements essential in the efficiency analysis. Labour is identified as a crucial element to supplement the functionality of the container yard, but due to data limitations, the variable is hardly ever included for analysis.

Although also important, the external factors are hardly or at times impossible to be determined by the managers of the terminal yet the factors influence how the terminal performs. Although terminal managers are not able to control the economic growth, it still plays a very significant role in demand and supply, which are factors driving throughput and thus the performance of a terminal. Several factors have been found in various research work to have an impact on performance, including the level of transshipment; the modal split options available in the hinterlands; the age of the terminal (linked to increased connections and client base); all these factors, although beyond the manager's control affect how the terminal will perform. These factors are taken into account in chapter 5 and six when the productivity and efficiency of the terminals respectively will be quantified. These factors are also valid when the cost structure of the terminal sector is redefined and calculated (Chapter 7)

Chapter 5: Container terminal total factor Malmquist productivity index at terminals in the Hamburg-Le Havre range

This chapter evaluates the total factor productivity (TFP) of deep-sea container terminals within the Hamburg- Le Havre range ports. The definition of performance terms including productivity and efficiency is in chapter 2. The measurement of TFP converts inputs to outputs over a certain period. The analysis positions the relative performance of the terminals within the sample and confers tentative solutions to improve the use of their critical resources. The selected method for the analysis is the Malmquist productivity index (MPI) which compares the overall productivity over two periods (2013 and 2018). Three input factors to measure against one output factor are used for 19 container terminals (Figure 5-1).

Figure 5-1: The DEA Malmquist total factor productivity analysis



Source: Own compilation

In this chapter first, the analysis method is described in the first section. Following that, a description of the selected sample is given. Section 3 describes the data.

Section 4 evaluates productivities generally and also through the Malmquist productivity index, where a detailed analysis of the results is discussed.

5.1 The method of analysis: The Malmquist total factor productivity index (MPI)

An index number is designed to measure comparison over time, space, or both (Coelli et al., 2005), (refer to Chapter 3 to read more details in this aspect). In the economic-theoretic approach, index numbers rely on well-defined properties, tests, and axioms, whereas in productivity measurement, the microeconomic theory of a firm is more relevant. The index approach involves constructing aggregates using a certain index formula. According to Liu (2010), the indices created are based on economic assumptions of production. The general index number problem can be specified as below:

$$V_{st} = \frac{\sum_{m=1}^M P_{mt} q_{mt}}{\sum_{m=1}^M P_{ms} q_{ms}} \quad 5.1$$

where the index V_{st} measures the changes in the value of quantities q_{st} and the prices P_{st} of M from the period s (base period) to period t (current period).

The Malmquist index (MI) approach demonstrates changes over time in TFP, technologies, technical efficiencies, and scale efficiencies. The model was introduced by Malmquist (1953) and then later developed into the DEA framework. As defined in Fare, Grosskopf, Norris, and Zhang (1994) and Caves, Christensen and Diewert (1982a, 1982b), the Malmquist TFP is decomposable into technical changes (shift of the frontier), indicating changes which are due to technological advances over time and technical efficiency changes (movement (catchup) along the frontier) which indicates the ability of how best a certain firm (or a Decision-Making Unit – a DMU) follows the best practices over time allowing it to operate on the frontier.

According to Coelli, Prasada Rao, O'Donnell, & Battese (2005), an output-oriented Malmquist TFP change index between period s and period t can be expressed as:

$$m_O^t(q_s, x_s, q_t, x_t) = \frac{d_O^t(q_t, x_t)}{d_O^t(q_s, x_s)} \quad 5.2$$

When s is the reference technology period, then the definition is:

$$m_o^s(q_s, x_s, q_t, x_t) = \frac{d_o^s(q_t, x_t)}{d_o^s(q_s, x_s)} \quad 5.3$$

where $d_o^s(q_t, x_t)$ is the distance from period t observation to the period s technology. A value of m_o greater than one indicates positive TFP growth between period s and t while a value less than one indicates a TFP decline.

The Malmquist index is defined by the geometric average of the two indices, at period t and s and represented as follows:

$$m_o(q_s, x_s, q_t, x_t) = [m_o^s(q_s, x_s, q_t, x_t) \cdot m_o^t(q_s, x_s, q_t, x_t)]^{0.5} \quad 5.4$$

Malmquist seeks to realize if the output has improved relative to the input factors over a certain time (Grosskopf 2002). The method constructs the so-called 'best-practice' frontier where all individual DMUs are compared to each other. The method can be applied in both the DEA and SFA, and it can either be input or output-oriented (Coelli, Prasada Rao, O'Donnell, & Battese, 2005). While some DMUs are considered to be technically efficient, others can be inefficient, which suggests improvements represented by 'efficiency change' and/or 'technical change'.

Where technical inefficiency is present, the formula above can be written as follows:

$$m_o(q_s, x_s, q_t, x_t) = \frac{d_o^t(q_t, x_t)}{d_o^s(q_s, x_s)} \left(\frac{d_o^s(q_t, x_t)}{d_o^t(q_t, x_t)} \times \frac{d_o^s(q_s, x_s)}{d_o^t(q_s, x_s)} \right)^{0.5} \quad 5.5$$

The ratio outside the large parentheses represents the change in the output-oriented technical efficiency and the ratio inside the bracket represents the geometric mean, capturing the shift in technology (technical change) between two periods.

The MI is not based on any specific assumption regarding the returns to scale properties of production technologies, thus both, constant and variable returns to scale (CRS and VRS) apply (Baran & Górecka, 2015). In the case of CRS, two sources of

productivity growth – efficiency change and technical change are possible. In the case of VRS, there exist additional possibilities, DMUs can exploit economies of scale and economies of scope achieved through the changes in the input and output mix. Scale efficiency changes (SEC) in the MI can be measured as follows:

$$SEC_o^{s,t}(q_s, x_s, q_t, x_t) = \left(SEC_o^s(x_s, q_s, x_t) \times SEC_o^t(x_s, q_t, x_t) \right)^{0.5} \quad 5.6$$

This measure is in line with the measures discussed earlier of technical change and technical efficiency change.

5.2 Selection of the terminals

The 19 global container terminals (GCTs) selected for the analysis are located in the HLH range, a region of economic importance to Europe and an area where at least four of the busiest container ports are located (Chen, Chou, & Hsieh, 2018). The ports along the region have been identified to be physically and economically similar (Meersman et al., 2014). The ports, and thus the terminals located along this region share a large portion of the hinterlands, therefore, a comparison between terminals with similar environmental factors can be expected to increase the reliability of the results. The 19 terminals are located in eight ports, as shown in *Table 5-1*¹⁰. The abbreviations of the terminals will be used in the discussion of the results.

¹⁰ A map detailing these container terminals is shown in chapter 1.

Table 5-1: Sample of container terminals under analysis

OPERATOR	CONTAINER TERMINAL	PORT	COUNTRY	ABBREVIATION
PSA	Europe terminal	Antwerp	Belgium	AntwpEuro term
PSA	Noordzee terminal	Antwerp		AntwNterminal
PSA	PSA/TIL MPET terminal	Antwerp		AntMPET
DPW	Gateway terminal	Antwerp		AntwGaty
CSP (COSCO)	CSP Zeebrugge	Zeebrugge		Zbrug(CSP)
EUROGATE	Eurogate GmbH (CTB)	Bremerhaven	Germany	BremGmbH(CTB)
EUROGATE	North Sea terminal (NTB)	Bremerhaven		BremNea(NTB)
EUROGATE	MSC gate	Bremerhaven		BremMSC gate
EUROGATE	Eurogate GmbH (CTH)	Hamburg		Ham(CTH)
HHLA	Burchardkai (CTB)	Hamburg		HamBurch(CTB)
HHLA	Tollerrort (CTT)	Hamburg		HamTol(CTT)
HHLA	Altenwerder (CTA)	Hamburg		HamAlt(CTA)
EUROGATE	Eurogate GmbH (CTW)	Wilhemshaven		WilmGmbH(CTW)
CMA CGM	Terminal des Flandres	Dunkirk	France	DunkDesFlan
DPW	Terminal de France GMP	Le Havre		LHTdeFrance
TERMINAL INVESTMENT LIMITED (TIL)	Terminal Normandie MSC	Le Havre		LHTNorm
HP	ECT Delta	Rotterdam	Netherlands	RottECTDelt
HP	Euromax	Rotterdam		RottEuro
APM	APM terminal	Rotterdam		RotAPM

5.3 Description of data

One of the historical impediments in resolving efficiency and productivity concerns lies in selecting the combination of the input and output variables (discussed in chapter 3). In the boom of efficiency analysis, Farrell (1957) identified that theory neglecton primarily causes problems concerning variable selection. Chen et al.

(2018) outline the significance of identifying essential factors in the analysis of port (and terminal) efficiency, both input, output, and environmental factors.

Container terminals worldwide are financed in various ways, depending on the form of ownership, which could be private or public. In either one of these cases, there needs to be a return for the assets invested. There is also a relationship between the assets/resources injected into the container terminal and the output achieved. Furthermore, the investments poured into the container terminal directly impact the operational costs and, therefore, on the handling charges, *ceteris paribus*. Ferrari, Puliafito, and Tei (2018) and Merk (2020) explain how concession agreements can enhance terminals' performance by introducing specific goals following the concession duration. To maximize handling activities, terminal operators often need to invest in particular infrastructure and superstructure, including quay cranes, yard handling equipment, and at other times also the buildings. The current analysis will consider this capital (discussed in chapter 3) as part of the input factors of the analysis.

The data was collected through the container terminals' websites (quay length, quay cranes, area) for the year 2018. For the year 2013, the same input data was collected from the Dynamar reports. For confirmation of the data where clarity lacks, the terminals were contacted directly through e-mails, telephonically, and in some cases physical visits. For throughput data, the Drewry annual review and forecast reports as well as Dynamar reports were consulted.

A significant amount of literature has shown that some diverse factors and forces influence determining efficiency and productivity in a DMU, as described in chapter 3. What is of significance is that the selected input and output variables can represent the production function of the concerned DMU. The terminal processes must be well described, from the quayside to horizontal movements, stacking, distribution to hinterlands (Han, 2014). The selected factors should cover the main resources used to achieve the output. Ill-defined and erroneous variables will lead to wrong conclusions. The factors should be common among the DMUs and it is also necessary that the environmental changes are unable to change the behaviour of these factors (Dyson et al., 2001). The environmental (external) variables will also be taken into account in chapter 6 of this thesis

Terminals have characteristics that make them unique; for instance, an automated versus a non-automated terminal will have different assumptions to labour.

Cullinane and Wang (2006) emphasize that the assumption should be taken with caution as terminals with different labour arrangements and equipment may translate into different results. Efforts were made to obtain reliable labour input data for this analysis which led to trial and error as a result of the unavailability of data across the CTOs. In this analysis, labour is excluded but a pre-determined relationship assumption will be held to proxy for labour. In addition to this, Cullinane and Song (2006) indicate that there exists a very high correlation between physical quantities of capital and labour. Although this is a reliable assumption specifically for fully manual terminals, this is not the case with automated and partially automated terminals. In this analysis, therefore, efficiency is measured without taking into account this factor.

A vast number of combinations of especially the input factors can be hypothesized as influencing factors to terminal efficiency and productivity. Besides the typical practicality challenges of data collection, the issue of multicollinearity¹¹ becomes paramount and therefore also limits the number of variables that may be employable in one model. On one hand, the inclusion of too many factors can lead to a reduction in the discriminatory power of the technique and thus create incorrect efficiency levels. On the other hand, the exclusion of some input factors will lead to efficiency scores that do not capture the entire process. The key is to include at least the key variables to represent the entire handling process from the container terminals operators' point of view (in this case). Annual physical information has been collected on the quay length (meters) and total yard area (hectares) as a proxy for land; the number of gantry cranes (quantity) and the total number of yard handling equipment (quantity), a proxy for capital as input factors. The data is for a two-year window period (panel data) for the Malmquist model. Throughput (TEUs per annum) is used as an output measure. The data is in quantities and excludes costs due to the unavailability of the required data across all the terminals.

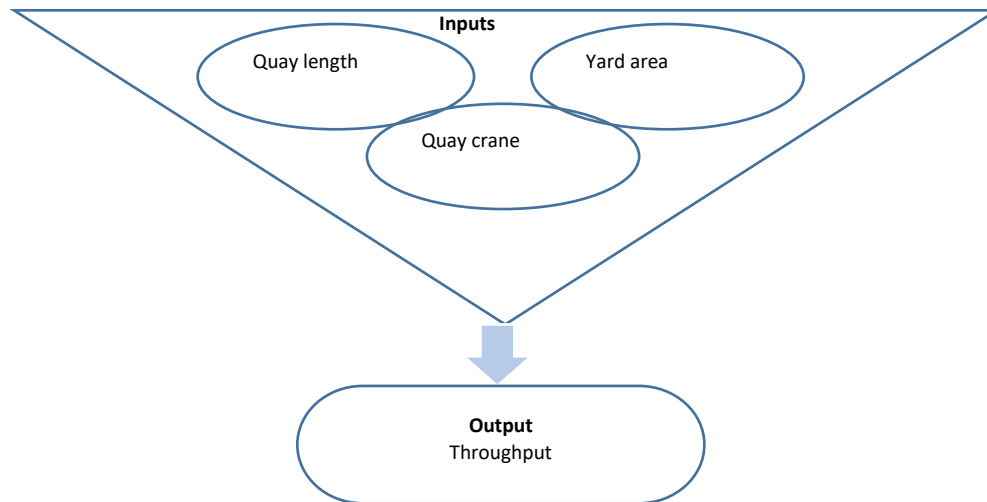
Chapter 3 attributes various studies in container ports and terminals and how the output is measured. The studies reveal that in ports and container terminal efficiency (and productivity) literature, the consensus is augmented towards the use of

¹¹ Multicollinearity is the occurrence of high intercorrelations among two or more independent variables in a multiple regression model (Shrestha, 2020).

throughput in TEUs handled (production function) and turnover (cost functions). In this analysis, throughput measured in TEUs will be made use of.

Figure 5-2 visualizes the input and the output variables which are defined.

Figure 5-2: The input and output variables



Source: Author's compilation

5.3.1 Output measures

5.3.1.1 Throughput:

While physical capacity (which is important mainly for the terminal operator) is the maximum volume that can potentially be handled by the terminal given the various resources, throughput is the actual handled volume at a given terminal given the resources available. It is typically measured by the number of twenty-foot Equivalent Units (TEUs) per annum that are moved across the quay face of a particular terminal. Throughput relates very closely to the cargo handling facility activities and services. Throughput also serves as a reliable basis of comparison when terminals or port size, investments, revenues, and activity are being measured, (Cullinane & Song, 2006).

- *Data:*

For the analysis, throughput is measured by the number of containers handled in TEUs per annum. Data is collected for two years: 2013 and 2018. The throughput data is collected for a total of 19 terminals. The throughput data was obtained through Drewry reports (2016, 2017, 2018, and 2019) as well as the Dynamar report (2015). The throughput used is only of the terminal's main handling activity and excludes other operations that the terminal may have.

5.3.2 Input measures

5.3.2.1 Quay length:

Berth allocation involves assigning vessels to a berth along the quay wall, making berth characteristics such as quay length (QL) and water depth important (Carlo et. al, 2015). To cater to the ever-growing vessels, the quay length is important as it can limit the number of vessels served at a particular time. The quay wall plays a major role in the ship-to-shore subsystem, a compulsory stage to container handling. As explained in chapter 4, there can be various factors of consideration at berth, such as the quay wall design, berth allocation, quay crane assignment, scheduling as well as maintenance cost where applicable. According to Imai, Nishimura, and Papadimitriou (2001), low usage of berth coupled with low volumes of cargo leads to lower efficiency and productivity as it increases costs per container handled. On the other hand, Issam et al. (2019) explain how congestion can be the main factor disturbing performance in a container terminal. A rule of thumb suggested by Drewry (2014) is that operators should allow for a QC for every 100m of the quay. Notwithstanding the importance of falling under the category of 'land' in the production function, the quay length distinctly contributes to determining the levels of efficiency and productivity of a terminal.

- *Data*

The quay length (in meters) data is also reported for the years 2013 and 2018. The quay length data is collected for a total of 19 container terminals. The QL data was obtained through the terminal websites as well as the Dynamar report (2015).

5.3.2.2 Quay cranes:

To cater for the ever-growing vessels, terminal operators have had to invest in mega cranes with the potential of handling larger vessels than there currently are in the market. Quay cranes play a very significant role in any terminal. Carlo, Vis, and Roodbergen (2015) refer to them as the bottleneck of the entire operation due to:

- The dependence of the other activities on them,
- The complexity of their task,
- The importance of their speed,
- The importance of the number of QCs in a terminal
- The safety concerns often needed to abide by the CTO.

The latest innovations of quay cranes give the ability to load/unload four TEUs in tandem and twin; others may have double trollies or floating platforms. Taking these differences into account, similar to Bichou (2011), can fully capture the capacity of the quay crane which has not been done in this current analysis due to lack of publicly available information across all the 19 container terminals. This variable assists in explaining efficiency in the ship-to-shore subsystem and forms part of the 'capital' category of the production function.

- *Data*

The total quantity of quay cranes in each of the 19 CT is collected for the study period from the container terminal's websites and Dynamar. Where the information was not online, the terminal operators were contacted directly for confirmation.

5.3.2.3 Yard area:

To transport containers around the yard, from (to) the vessel to (from) the yard and from (to) the yard to (from) the hinterland transportation, container terminals require handling space (elaborated in chapter 4). A larger area contributes to increased capacity, which is so because a bigger yard is associated with increased yard stacking slots. A large portion of the total space within the container yard is taken by the storage yard, (Wang et al., 2019). In the storage yard, the allocation of resources is determined, the selection of technologies as well as the optimization of blocs within the yard. Specifics of these characteristics are explained in chapter 4. The change in the capability of handling equipment has meant that operators can stack containers up to 6 or 7 high (when empty) which has therefore changed the maximum capacity per hectare of the terminal (Wang et al., 2019). The amount of land required has now reduced compared to the past for the same quantity of containers. In the production function, this variable represents 'land' and it is part of the storage sub-system in the container terminal operation.

- *Data*

The total terminal area gives the total number of containers (TEUs) handled per hectare per terminal per annum. Measurements of the yard area for each of the 19 CTs are collected for the study period from the container terminal's websites, Drewry, Dynamar. Where the information is not online, the terminal operators have been contacted directly for confirmation.

5.4 Descriptive statistics

The descriptive statistics of the Malmquist model are demonstrated in *Table 5-2* for a sample of 19 container terminals. The model will compare two years: 2013 and 2018. The descriptive statistics indicate the average values of the period. The minimum and the maximum values of the variables indicate a terminal with the average lowest and highest throughput amount (measured in TEUs), quay length (measured in meters), area (measured in hectares), quay cranes (measured in quantity), between the two years. The mean indicates the average sizes of the same measures (input and output) over the period. The standard deviation (square root of variances) indicates how closer to the average the general measures are while the sum statistics show the abundance of output, land, and capital that the terminals in the region hold in conjunction.

5.5 Correlation

One of the fundamental attributes of the DEA is the dependence of the results on input and output variables. As mentioned earlier, there is an abundance of literature employing the DEA model in the container port and terminal field. A majority utilises a similar nature of variables, but most of which ignore the potential impact of correlation between the input and output variables. Interactions between the DEA attributes are hardly ever explained or modelled in past literature. López, Ho, and Ruiz-Torres (2016) conducted a study dedicated to the effect of the degree of correlation between input and output variables on DEA efficiency scores. The findings show that, on average, the high correlation between output and input variables tends to lead to increased efficiency scores and when the input/output correlations are low, the results tend to be erratic. The degree of correlation between input variables tends not to affect the results when the correlation between input and output is high; however, when the correlation between input and output variables is low, the degree of correlation between input variables often leads to low scores (López, Ho, and Ruiz-Torres, 2016).

Table 5-2: Descriptive statistics for the Malmquist for the period (2013 and 2018)

DESCRIPTIVE STATISTICS					
FUNCTION	Description Averages per period	Variable	N	Sum Statistic	Mean
OUTPUT	Total throughput (000' TEUs/period)	TEUs_Av	19	34 295	3015.5
INPUTS	Quay length (m/period)	QL_Av	19	31707	1674.1
	Total yard area in hectares (h)/period.	Area_Av	19	1 898	48.9
	Number of quay cranes	QC_Av	19	285	6.0

In 1997, Smith (1997) studied the impact of correlation in the DEA and found that correlation between input and output variables has a significant impact on efficiency scores. Lee and Choi (2010) proved that the non-negative weighted sum of output and the non-negative weighted sum of inputs ('perfect' correlation) is not relevant for a DEA study, it does not affect efficiency. In assuming erroneous results due to serial correlation, Adler and Berechman (2001) use the principal components analysis to resolve the correlation in the DEA and thus also to eliminate some variables. Dyson et al. (2003) prove that this cannot be a solution as it can take away elements providing important information to efficiency. According to Wang et al (2011), a correlation is only meaningful when isotonicity is satisfied, which means that output should increase when inputs increase. Correlation analysis indicates the significant positive relationships between the inputs and the outputs, which are, therefore, isotonic and justly included in the model (Wang et al., 2011). Also, the DEA is a non-parametric method that does not impose a functional form or shape on the production set and thus is not bounded by the assumption that often holds for parametric measurements (Romano & Guerrini, 2011).

Pearson correlations between each of the variables for this analysis are displayed in Table 5-3 for the Malmquist model.

Table 5-3: Correlation between inputs and output

	TEUs	QL	Area	QC
TEUs	1	.89**	.84**	.93**
QL		1	.90**	.92**
Area			1	.89**
QC				1

**Correlation is significant at the 0.01 level (2-tailed).

The input variables are reasonably and positively correlated to the output. All the inputs have a correlation exceeding 80% to the output in both samples, which indicates their importance in explaining the throughput. The highest correlation to throughput is that of the quay cranes (QCs), which indicates the importance of quay cranes to a container terminal. Following QCs is the quay length, followed by the area. Comparatively, the difference in the correlation scores is very little, in agreement with the high significance levels at 0.01 on a two-tailed test. The slope for all the independent variables is positive, which confirms past literature that the more resources a unit (terminal) holds, the more potential there is to create more output (containers handled). This confirms to Grosskopf (2002) states that output growth is

expected to follow input growth, based on Solow-Swan of the neoclassical school of thought.

5.6 Estimation of the results

5.6.1 Global container terminal productivities

This section presents an overview of productivity to indicate the intensity of resource usage in the CTs compared to the HLH region concerning the world average CTOs.

Figure 5-3 to Figure 5-6 provided below indicate the intensity of usage of assets in the handling of TEUs (throughput quantity) per annum. The quay length (meters), area (hectares), the number of quay cranes deployed for the year 2013 and 2018 are examined and compared. The performance evaluation is from a global perspective (Asia, World, the Middle East/South Asia, Africa, Europe, Latin America, North America, Oceania); from a regional perspective (Hamburg-Le Havre region), and also at a terminal level. These variables reflect the basic assets that terminals require for the handling of containers, which is relevant for performance and risk and return.

Drewry (2019) composes average metrics for operational performance based on a sample of 350 container terminals, internationally operating and all of which handle a minimum of 200 000 TEUs for the year 2018. The sample represents approximately 60% of the global container throughput and therefore can be used as a reference of comparison. Based on the metrics, worldwide, container terminals have on average a length of 1 130 meters of a quay, handling approximately 1 150 TEUs per meter based on the 2018 data. The average world area is 52 hectares, handling approximately 25 000 TEUs per hectare in 2018. Container terminals worldwide also own on average 10 quay cranes and handle approximately 130 000 TEUs per quay crane in 2018.

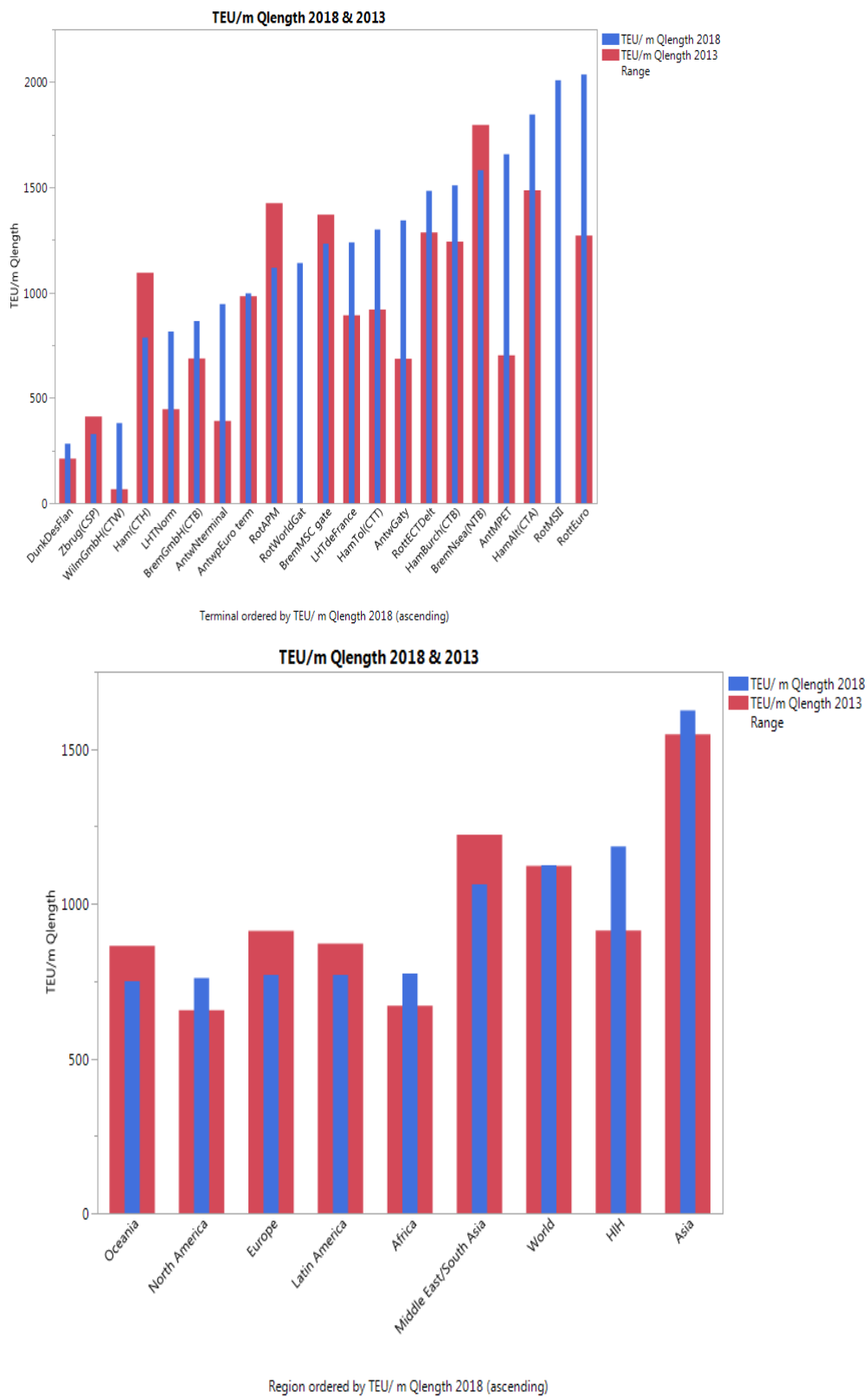
At the Hamburg-Le Havre region collectively, the terminals deploy an average quay wall of 1 682 meters, a length greater than that of the world average quay length, handling about 1 186 TEUs per meter. The region's area averaging 107 hectares, about double the size of the world's average terminal area, and handling 18 900 TEUs per hectare which is less than that of the world averages. The terminals use on average 15 quay cranes per terminal which handle approximately 133 969 TEUs per quay crane. These observations are demonstrated in the section that follows.

Under normal circumstances in a container terminal, throughput expands from one year to the next. If the resources in that particular terminal remain the same, then

this would indicate that the assets are better utilized, when comparing the previous year to the next. However, due to changes in various factors such as ownership, terminal contractual expansion, contractions in terminal physical sizes, etc, this can lead to deviations from those expectations. For instance, throughput in the HLH range sample of terminals has collectively been inclining by 17% between 2013 and 2018. Similarly, there has been a positive average percentage change with the quay length (4%), area (13%), and quay cranes (9%) increase of the assets from 2013 to 2018 for the entire group. However, at the individual terminal level, the scenario is much different. There has been a total of 11 terminals that had encountered changes (increase and decrease) in the length of the quay, 11 terminals with changes in the area, and 10 terminals with changes in the amount of owned quay cranes over the period.

From a global point of view, there has been a decline in the use of assets in most of the regions. There has been a decline in the use of quay wall by meter (given throughput) in all regions between 2013 and 2018, except for Asia and North America. This can be an indication of larger infrastructural investments across the regions. With regards to the utilization of quay cranes, only the Middle-East, Africa, and North America are an exception to the decline. Regarding the use of land (area), only the Middle East and North America employed their terminal areas better than in 2013 given the jointly generated throughput.

Figure 5-3: Throughput (TEUs) per quay length in meters (terminal vs region)



Source: Author's compilation

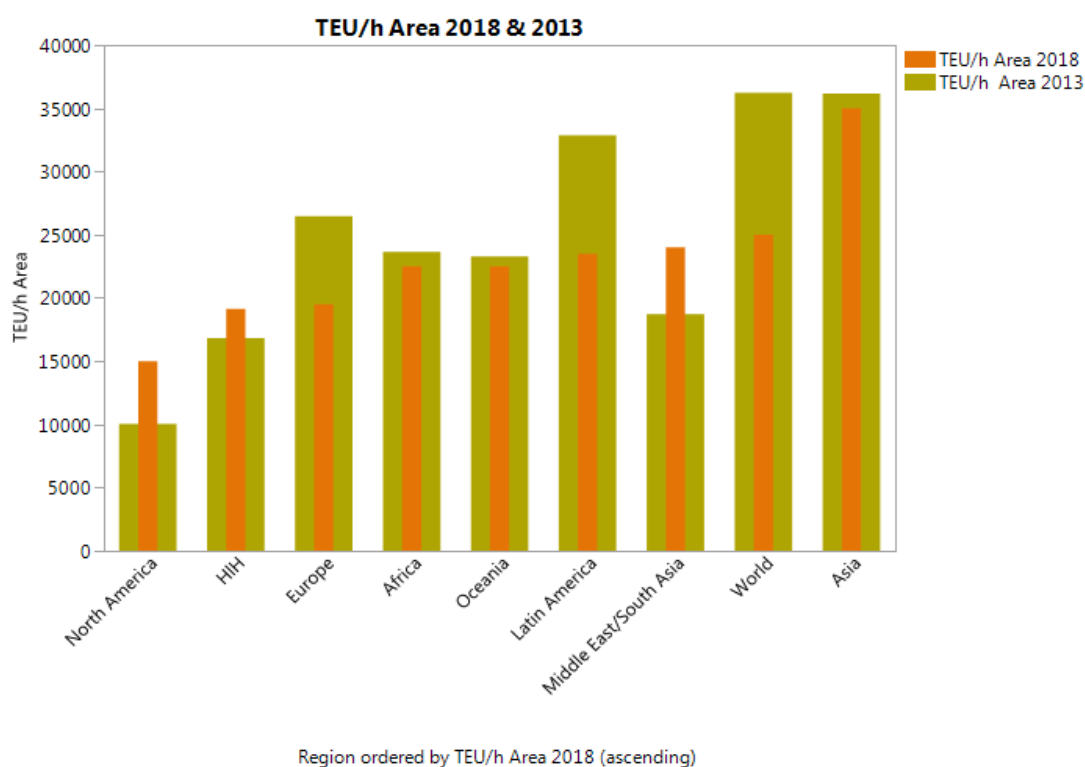
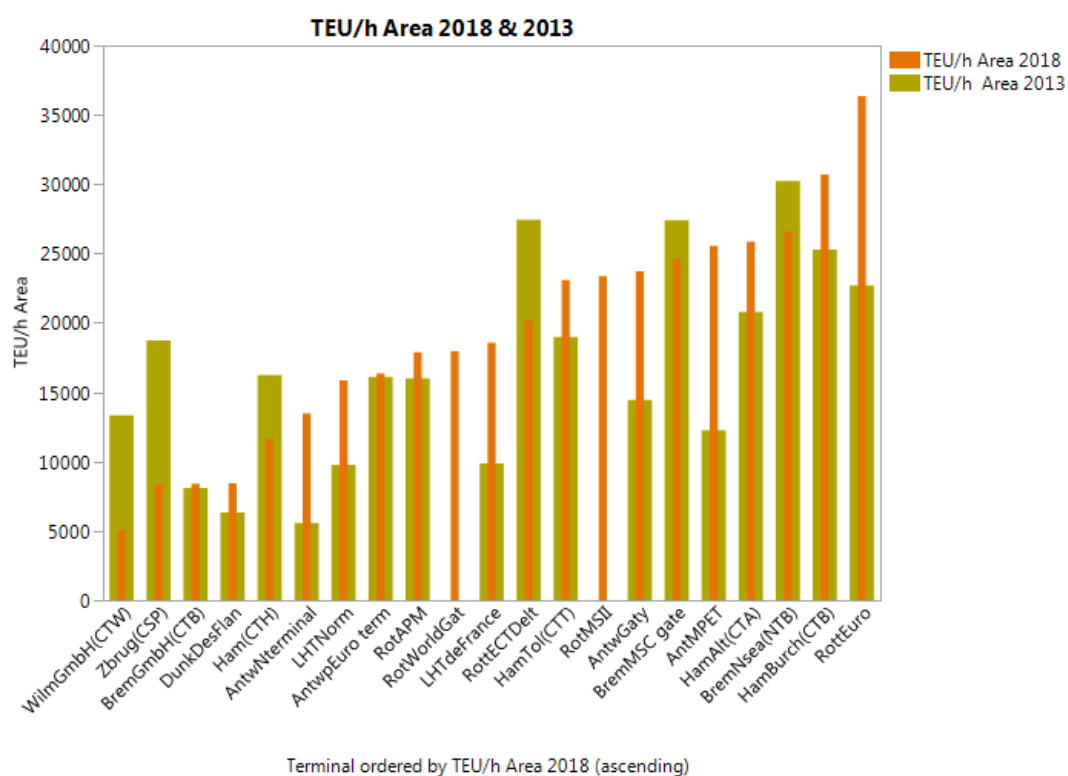
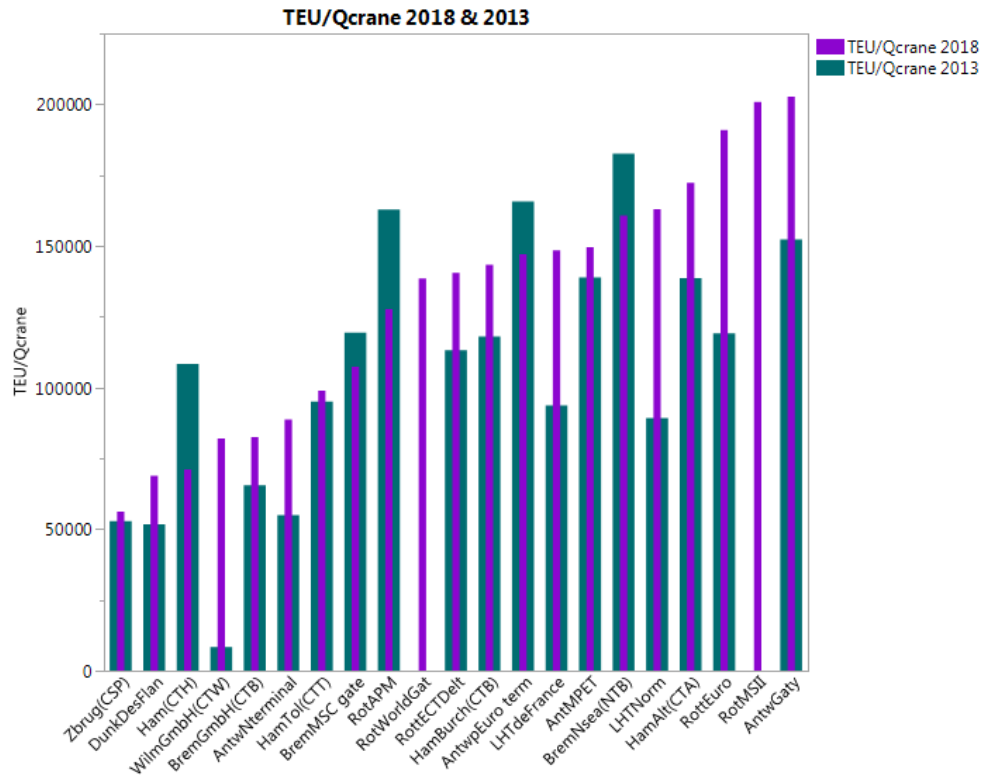


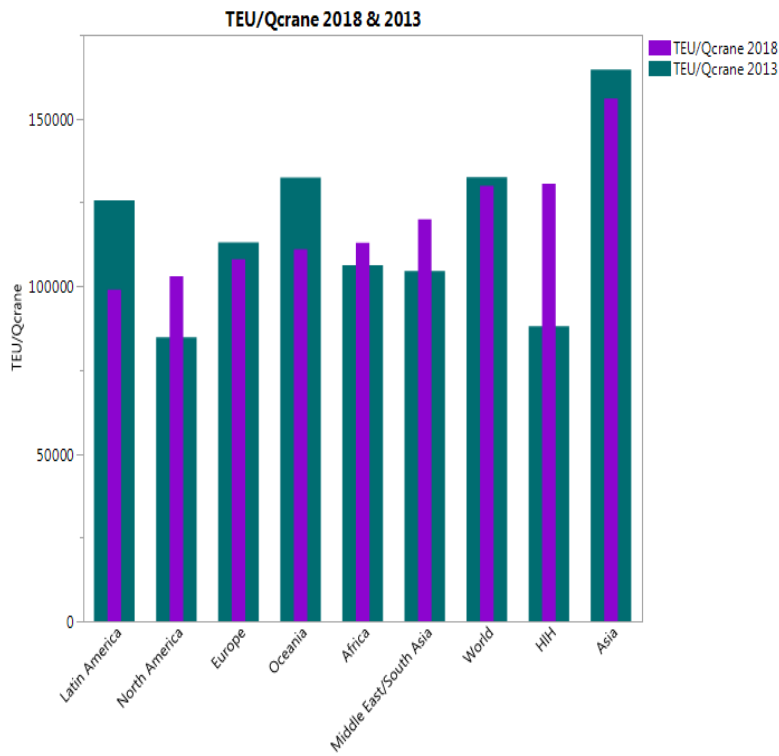
Figure 5-4: Throughput (TEUs) per area in hectares (terminal vs region)

Source: Author's compilation

Figure 5-5: Throughput (TEUs) per quay crane (terminal vs region)



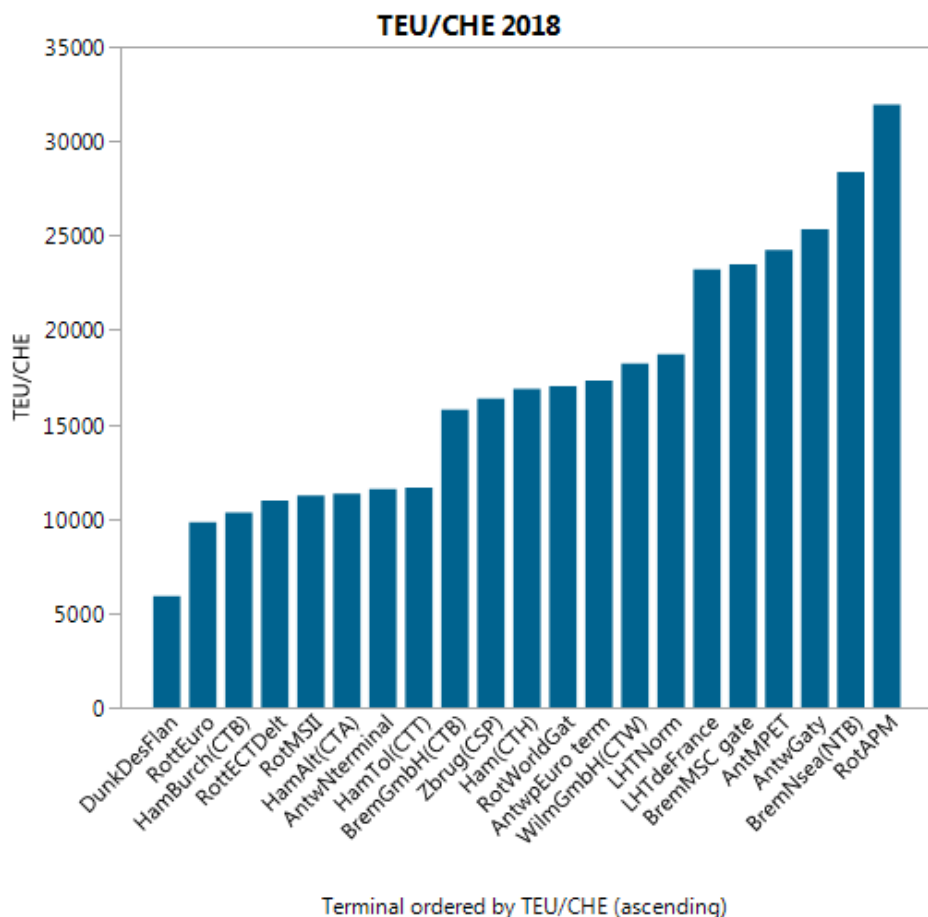
Terminal ordered by TEU/Qcrane 2018 (ascending)



Region ordered by TEU/Qcrane 2018 (ascending)

Source: Author's compilation

Figure 5-6: Throughput (TEUs) per yard handling equipment (terminal)



Source: Author's compilation

As part of the world's international players, the performance of the Hamburg Le Havre region's container terminals with regards to the throughput (TEU) per meter of the quay and per quay crane is comparatively at a better rate than the performance of the world combined container terminals, concerning data from Drewry (2019). Throughput per meter of the quay at HLH is 5.45% more than the world rate and 54.08% more than of the European region. Similarly, the performance of throughput per gantry crane is 3.05% higher than the world average and 24% higher than the European rate. This indicates that the terminals make better use of the quay wall as well as of the quay crane. On the other hand, the throughput per hectare is about 24% less than the world measures and about 3% less than the general performance of the European region (see Figure 5-3, Figure 5-4, Figure 5-5, Figure 5-6). Theoretically, this indicates that the terminals generally are not efficiently utilizing the area compared

to the other global container terminals. At the global scale, these differences in the use of the area can be due to layout differences between the regions of the world, i.e. container terminals in Asia tend to stack higher than terminals in Europe.

This analysis illustrates a helicopter overview of productivities for the region. The analysis above however indicates the potential sources of unproductiveness and inefficiencies. It further also exhibits the variations between the world, regional, and terminal performance values.

Applying the Malmquist model will reveal the relative productivities and efficiencies, considering all input variables and comparing all the terminals.

5.6.2 The DEA Malmquist productivity index (MPI)

Efficiency and productivity are cumbersome concepts to distinguish. It can be distinguished that while efficiency is concerned with how best to utilize resources for production, productivity is concerned about obtaining more output from the input sources.

One of the elements of performance is productivity. The distance function-based Malmquist productivity index (MPI), an alternative to the window analysis is the chosen method here. When the analysis is carried out over cross-sectional data, specifically over a very short period, it means that the comparison of the DMUs is based on the production of that current period, and the role of time is ignored. According to Cullinane and Wang (2010), this can be rather misleading as it also will make excessive use of the resources during the analyzed time when in fact the resources are intended to give beneficial results in the future periods. Measurement that is based solely on the current period is relevant when planning to compare counterparts during the point of interest, but it fails to indicate the direction overall, which is where measurement of panel data, such as the Malmquist productivity index comes in. They compare counterparts with each other and over a certain length of time and are thus more informative. Needless to mention, in today's world, terminals need to be efficient if they are to survive in the competitive environment (Wang and Cullinane, 2015).

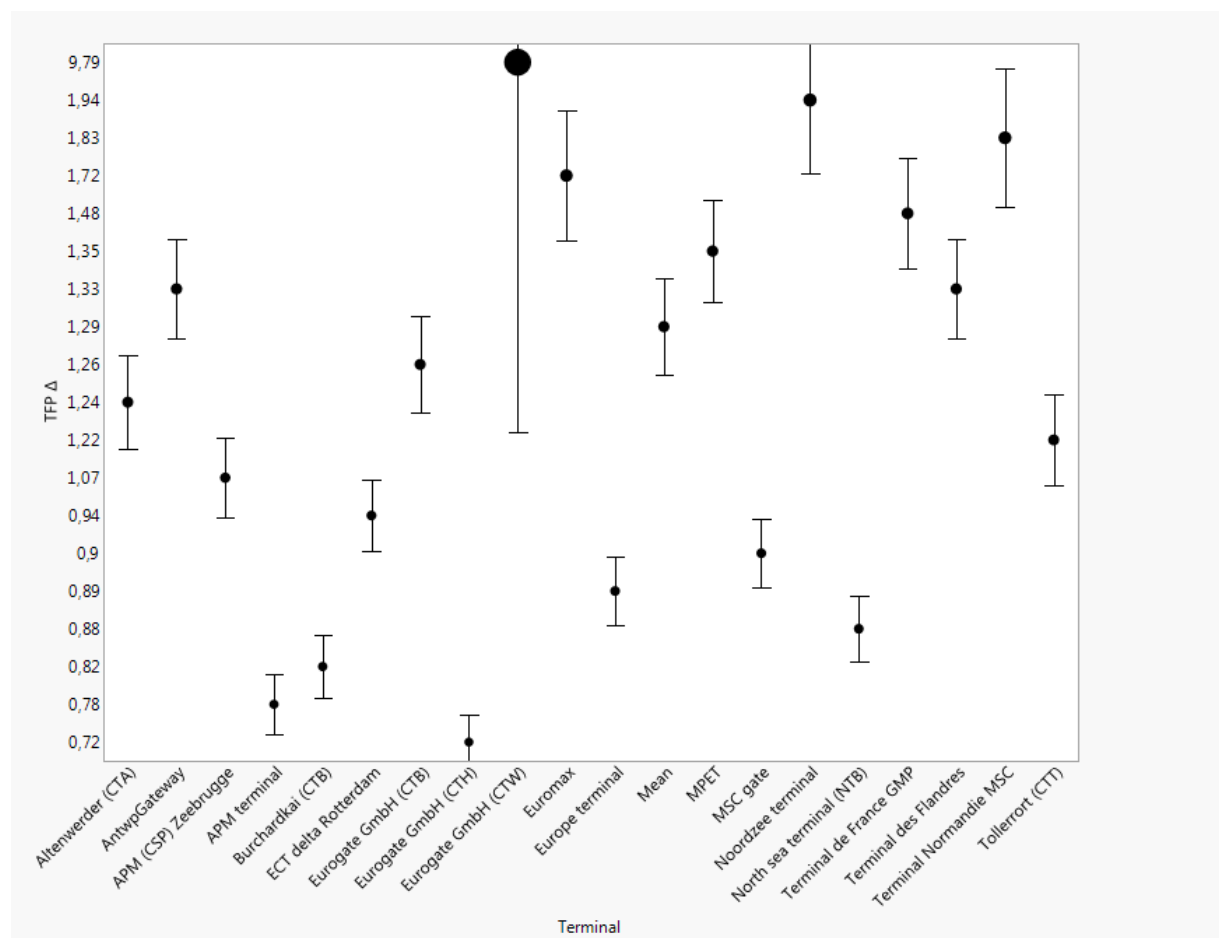
The DEA MPI model assumes variable returns to scale and is output-oriented. According to Coelli (1996), it is only appropriate to consider CRS, when the DMUs analyzed operate at an optimal scale. Where imperfect competition and financial constraints exist, the firms are likely to operate below optimal scale and therefore

assuming variable returns to scale is most appropriate. As explained earlier in the data description section, the analysis will consider the years 2013 and 2018 to give an overview of the changes in productivity in the 19 container terminals in the HLH region. Input is measured using the quay length (m), quay cranes (number), area (h), and output measured in throughput (TEU).

The results in *Table 5-4* indicate that over the years, the container terminals in the HLH range experienced on average a positive change in TFP indicated by the index of 1.29 (29%¹²). The TFP changes for the terminals are depicted in *Figure 5-7*. The scores are also expressed in error intervals to demonstrate the distance of the TFP change score from the mean score. The Wilhelmshaven CTW terminal displays the largest interval from the mean, explained by the much largest TFP score than the rest of the terminals, explained later in this section. The TFP is largely due to technical change (26%) compared to (3%) efficiency change during the entire period. The efficiency changes are further decomposed into pure efficiency change (-2%) and scale efficiency change of (5%). These results are to indicate the group of terminals collectively from the region but do not suggest homogeneity in the determination of TFP. At the terminal level, the productivity levels of the terminals vary from the group average.

¹² Please note that the indices have been multiplied by 100, and 100 deducted i.e. $((TFP*100)-100)$ to interpret the proportion of change in the index.

Figure 5-7: Total Factor productivities (TFP) changes of the container terminals



Source: Author's compilation

The realized changes in productivity can largely be due to developments in the terminals, such as new investments in new technologies, increases in terminal sizes, ownership changes, etc. Broadly, the terminals in the region underwent some alterations regarding their yard sizes during the 2013 to 2018 period. Nine (out of 19) of the terminals under consideration deviated in size through the period, of which two reduced in size while the rest expanded. There also follows a trend of alterations in the quay lengths of the terminals. Eight of the terminals made some changes in the length of their quays of which three reduced in length. Eight out of the 19 terminals reconsidered the number of quay cranes distributed along the quay walls of the terminals from 2013 to 2018. From these terminals, only the ECT Delta terminal shows a reduction by 2 GCs. Recently, in 2019 and 2020, many more terminals are investing in additional, more advanced quay cranes to improve efficiency. For instance, the Eurogate North sea terminal in Bremerhaven (BremNea(NTB)) added 4 QCs in 2020, following new order of 6 QCs in 2018 (Eurogate, 2020). DPW gateway in

Rotterdam (RotWorldGat) also welcomed 2 new QC (and 25 new Automated Guided Vehicles (AGVs)) in 2020 (Rotterdam world gateway, 2020). HHLA Burchardkai in Hamburg (HamBurch(CTB)) is also following the same trend with two new orders of quay cranes in 2020.

Table 5-4: Malmquist index summary of annual means

Terminal	Efficiency change	Technical change	Pure efficiency changes	Scale efficiency changes	Total factor productivity changes
Europe terminal Antwerp	0.64	1.39	1.00	0.64	0.89
Noordzee terminal	1.54	1.26	1.16	1.33	1.94
MPET terminal Antwerp	1.07	1.26	1.21	0.89	1.35
DPW Gateway terminal Antwerp	0.96	1.39	1.00	0.96	1.33
Eurogate GmbH (CTB)	1.11	1.13	1.11	1.00	1.26
North sea terminal (NTB)	0.78	1.13	0.80	0.97	0.88
MSC gate Bremerhaven	0.75	1.21	1.00	0.75	0.90
Terminal des Flandres	0.96	1.39	1.00	0.96	1.33
Eurogate GmbH (CTH)	0.63	1.13	0.84	0.76	0.72
Burchardkai (CTB)	0.68	1.21	0.68	1.00	0.82
Tollerrort (CTT)	1.01	1.21	1.08	0.94	1.22
Altenwerder (CTA)	1.10	1.13	1.01	1.08	1.24
Terminal de France GMP	1.18	1.25	1.01	1.18	1.48
Terminal Normandie MSC	1.31	1.39	1.08	1.21	1.83
ECT delta Rotterdam	0.80	1.17	0.93	0.86	0.94
Euromax Rotterdam	1.33	1.30	1.14	1.17	1.72
APM terminal Rotterdam	0.62	1.28	0.83	0.74	0.78
Eurogate GmbH (CTW)	7.02	1.39	0.90	7.79	9.79
APM (CSP) Zeebrugge	0.79	1.35	1.00	0.79	1.07
Mean	1.03	1.26	0.98	1.05	1.29

On average, over the period, positive total factor productivity changes (29 percent) were experienced by the HLH terminals. From the analysis, it is visible that 12 from the 19 container terminals have experienced an increase in TFP, indicating that a majority of the HLH terminals have been managing their resources well from the technical point of view. With regards to the sources of productivity, the impression given by efficiency changes is that there is an average positive change of 3%. At an individual terminal level, 10 of the 19 terminals (53 percent) experienced negative efficiency changes while the other nine experienced positive changes. The sources of TFP are a result of efficiency changes or/and technical changes.

The efficiency changes are further decomposed into pure efficiency and scale efficiency. A larger part of the HLH inefficiency is due to pure efficiency (0.98) compared to the positive contribution of scale efficiency (1.05), indicating 2 percent of pure inefficiency and 0.5 percent of scale efficiency. Conforming to Chan et al. (2018), pure efficiency suggests the manner of resource management of the CTO. Despite being less efficient from a management perspective (pure efficiency), the results indicate that at an individual terminal level, many of the terminals (42 percent) contributed positively towards pure efficiency, in comparison to the 32 percent of those who contributed negatively and 26% with neutrality (i.e full efficient at 100 percent). It can be suggested that the positive trend in the pure efficient terminals in the HLH region can well rest in experience. In global supply chains, the North European region is known for its expertise in the container sector (second after Asia) where most of the best performing terminals are located, as discussed in chapters 1 and 2 of this thesis. Several factors (including competition) can lead to the terminals in this region embodying comparable industry practices. As Cullinane and Wang (2015) supported, container terminals in Northern Europe face fiercer competition than other container terminals in the rest of the world. Due to the proximity of competitor ports, European terminals tend to enhance similar behaviour. Although the practices in the region can be much similar among the terminals, it is also interesting to note that the changes for pure efficiency are not always the same of terminals operated by the same operator. For instance, Eurogate operates 5 terminals which are located in three regions in this sample, Bremerhaven, Hamburg, and Wilhelmshaven. The terminals display different changes over the period: one of the terminals experienced positive pure efficiency changes, three of the terminals experienced negative changes and one terminal experienced no changes relative to pure efficiency.

On the other hand, all the three PSA container terminals experienced positive changes in pure efficiency. This indicates that the management of the resources and the application of industry norms is much different when comparing the terminals and can be attributable to other endogenous and exogenous factors. The use of advanced information systems is a phenomenon well invested in at the HLH range for managing terminal operations. For the same efficiency gains, Eurogate's North sea terminal (NTB) transitioned its IT system to Navis N4 in 2017. This, together with a proper balance between labour and scale economies, can prove beneficial for the

CTOs, thus in agreement with previous findings, Wanke et.al (2018), Ross and Droge (2004), and Wanke & Barros (2016).

Scale efficiency is a measure indicating the intensity to which the input factors (assets) are converted to the output (throughput). This indicator has positively (0.5 percent) contributed to TFP through the period, nevertheless, a majority (58 percent) of the CTOs during the period were less scale efficient; 11 percent were neutral and 32 percent have shown a positive scale efficiency contribution to TFP. The direction of scale, either increasing or decreasing, is not specified by the model; this issue will be dealt with later on in the BCC model (chapter 6). The suggestion is that most HLH terminals can benefit from improving on their production scale for enhanced throughput. Given the investments by the CTOs in acquiring more assets for the terminals, with throughput growing at a steady rate of 8%, (UNCTAD, 2019), it can be reasonable to justify why many of these terminals are experiencing inefficient scales. By 2018, on average, according to Drewry (2019), the HLH terminals have a yard capacity utilization rate of 71%. The Rotterdam HP-operated Euromax terminal is an exception as the terminal has a utilization rate of 120%. On the other end, the Zeebrugge CSP terminal, Hamburg's Eurogate (CTH), both have in 2018 a yard capacity utilization rate of 40%. This justifies that terminals are likely to invest in excess capacity for future cargo security, causing short-term inefficiencies that are expected to be beneficial in the future. The technical change measurement better explains this asset acquisition.

Regarding technical changes (an indicator of technological advancement), all the terminals have experienced positive changes (average of 26 percent), which demonstrates the extent to which many of the terminals acquired more technologies to gain increased efficiencies and productivity. This is an important measure for the HLH CTOs as this is the greatest gear to productivity compared to efficiency, indicating that the investments into the assets are hitherto paying off.

A container terminal that experienced the largest increase in TFP during the entire period is the Eurogate terminal in Wilhelmshaven, Germany (WilmGmbH (CTW)) (9.97). The Eurogate terminal in Wilhelmshaven was inaugurated on 2012 September: it is among the newest terminals in the region in this sample. The JadeWeserPort terminal (WilmGmbH (CTW)) was implanted based on its advantage to deep waters as well as for the handling largely transshipment. Not only did the terminal experience

low volumes¹³ at the beginning of operations solely due to the mere fact of being a new terminal but also it was due to the unfavourable economic environment during the time. There was an expansion with the area and the quay wall, accompanied by an increase in volumes. In 2018, while its sister terminals in Hamburg and Bremerhaven experienced a decline in volumes, the terminal had an 18% incline in volumes, although the terminal remained underutilized (48% utilization rate). It can be expected that with increased years of operation in the CT sector that the handling techniques and strategies should also change. Wang & Cullinane (2015) mention that in the long term terminals need to ensure sustainable development, be able to cover their costs, and be productively efficient. According to the Malmquist model, it is suggested that the TFP changes of the terminal are largely driven by efficiencies (7.02) relative to the 1.39 changes due to technologies.

Other terminals which on average showed higher TFP over the period than the other CTOs include PSA Noordzee terminal in Antwerp (AntwNterminal) (94%); TIL Terminal de Normandie in Le Havre (LHTNorm) (83%); HP Euromax Rotterdam (Rott Euro) (72%).

The relative productivity change of 94 percent in the PSA Noordzee terminal is mainly driven by efficiency (54%) rather than by technical change (26%). With regards to efficiency, the terminal experienced positive changes in pure efficiency (management) (16%) and with scale efficiency (33%). In 2016, PSA invested in both the Europa and Noordzee terminals in Antwerp to cater for the temporarily displaced customers during the big move of the MPET terminal (Drewry, 2017). From 2015 to 2016, throughput at the Noordzee terminal decreased by about 10% before increasing again in 2017 and excessively in 2018. As a result, 2018 is the period marked by the greatest productivity for this terminal, very largely induced by scale efficiencies rather than by technical efficiencies.

In the case of AntMPET, where an increase of 35% TFP occurred during the observed period, the change is largely due to technical change (26%) compared to efficiency change (7%). At 4.5 million TEU, the PSA's home terminal capacity in Antwerp was fully utilized in 2013 and it was time for the biggest terminal move in history. The Antwerp MPET terminal faced a large change in ownership structure and operations from the period of 2013, which had a major impact on the change in throughput, as

¹³ Although discounts incentives were granted for port dues to attract traffic. A strategy to assist attract cargo especially at the early stages of operation, (Wang & Cullinane, 2015).

well as on the production properties of the terminal. The home terminal in Delwaidedok became saturated and in May 2014, MSC would consolidate its terminal handling activities to the current (2020) Deurganckdok terminal. The move involved 21 (out of 24) quay cranes, a project that was completed in 2016. These factors could have resulted in these major technological changes over the period. As demonstrated in the productivity changes, the greatest disruptions occurred in 2015 (252%), the second most productive terminal following the Eurogate's Wilhelmshaven terminal during 2015. The productivities are very clearly driven by production technologies as the terminal expanded in quay length, area, and quay crane, while throughput had also increased. This also affected the productivity of DP World's Antwerp terminal (AntwGaty), which lost some space with a 7-year concession from the Antwerp port Authority. This meant rescaling in the quay length and area of the terminal. On average, DPW Gateway terminal in Antwerp experienced positive TFP (33%) but experienced negative efficiency changes (-4%) and 39% technical change.

TIL Terminal de Normandie in Le Havre (LHTNorm) experienced an overall increase of 83% in TFP, predominantly driven by technical advancements (39%) and efficiency (31%). Between 2013 and 2018, there were no major technical changes in the terminal. Throughput however declined slightly between 2017 and 2018 (8%), this is in contrast to the group's throughput which increased significantly (17%) between 2016 and 2017 and 8.5% to 2018.

The Rotterdam ECT Delta terminal (RottECTDelt) is one among the container terminals that face a decline in TFP (-6%), which is a result of inefficiency (-20%). In 2016, there was a 2% decline in throughput across the HP terminals worldwide. The throughput was however boosted the following period due to the acquisition of 20% of PSA's stake, (Drewry, 2017). In 2015, the group experienced a 3% decline in volumes. Specifically, in the European ports, the throughput of HP's terminals reduced by 6%, predominantly as a result of the Rotterdam terminals who are the largest contributors for the operator. The declines confirm the share impact of the Maasvlakte new terminals had become apparent.

Although also with a slight increase in productivity over the period, CSP Zeebrugge has been stable as well over the period with an average of 1.07 of total average productivity. The inefficiency of the terminal was solely due to scale efficiency. The major change in the terminal over the period is a strategic change where CSP took over the terminal's ownership and sold 10% to CMA CGM.

Three terminals that mostly negatively contributed to the HLH range TFP include the Eurogate container terminal in Hamburg (Ham(CTH)); APM terminal in Rotterdam (RotAPM) and HHLA Burchardkai in Hamburg (HamBurch (CTB)); with average productivity growth of -28%, -22% and -18% respectively. For this group of terminals, the negative productivity growth is due to efficiency as they all experienced negative efficiency growth while technical changes were all positive. The Hamburg (CTH) terminal and the APM terminal faced some decline in throughput in some years during the period due to strategic decisions. For instance, the APM terminal in Rotterdam faced a significant decline in volumes between 2015 and 2016 due to the MSII beginning operations and thus shifted some cargo to the MSII terminals. The APM terminal generally faced an issue of excess capacity during a time of slow growth in volumes in its Northern European facilities. Other strategic influences such as the changes by shipping lines (Maersk) also influenced the volumes in this terminal during this period, negatively affecting the pure efficiency levels and productivity levels of the terminal. At Hamburg's Eurogate container terminal, the decline in productivity is largely driven by the same factor, efficiency change, specifically by scale efficiency rather than pure economies. According to Drewry (2015, 2016, and 2017), the terminal had been struggling with volumes especially in 2013 and 2014. In 2017, the terminal was greatly affected by the mergers between UASC and Hapag-Lloyd and between China Shipping and Cosco. In addition to that, there was the insolvency of Hanjin in 2016. Further shipping alliance decisions greatly influenced volumes at the terminal, affecting its performance. During the same period, the terminals invested in two quay cranes.

5.7 Conclusion

With the expansion of global trade, terminals need to frequently review their capacity to ensure that their service provision is competitive enough while limited resources are utilized economically.

Among others, the advantage of the Malmquist approach is its ability to measure not only productivity changes but also the ability of DMUs to catch up on the frontier and also the ability of the DMU's innovation advancements. This separation ability between the sources of productivity allows the terminals to divert focus on the relevant areas requiring more attention.

The positive average TFP at the HLH indicates positive progress of performance of the container terminals in this region collectively. Ten terminals can be well-referred from the sample as 'the catchers' given their vigorous alignment towards the frontier (positive efficiency changes). The HLH terminals can be collectively referred to as the 'innovators', as they wholly managed to shift the production frontier positively.

Contributing to the HLH region's strong points is that the container terminals have collectively managed to grow positively (TFP changes), efficiently handle containers (efficiency changes) and positively advance their technologies (technical changes). Nonetheless, the results indicate that more can still be done to better manage the resources in the terminals (negative pure efficiency). The terminals collectively experienced positive scale efficiencies (5%), however, a majority (58%) compared to the 32% that experienced positive scale efficiencies, of the terminals have indicated scale inefficiency.

Underutilization of capacity in terminals has long been identified as one of the challenges. It is important to note that one of the reasons for the underutilization well lies in the typical investment strategy of terminals. Terminals are likely to expand their capacities, considering future throughput, which in this model will indicate lower efficiencies, given the lower rates of utilization. This is not an absolute indication of poor productivity and efficiencies; however, it indicates the rate and level of the misuse of the resources, regardless.

Therefore, it is highly significant that the terminals are aware of their relative productivity changes to observe and align their strategies with the resources held, thereby reducing wastage and allowing innovation. Knowing productivity and efficiency can enable the terminals to know when and where change is necessary and when new goals need to be set. When efficiency is achieved, so can the resource utilization be improved.

Chapter 6: Container terminal Efficiency analysis at the Hamburg-Le Havre range

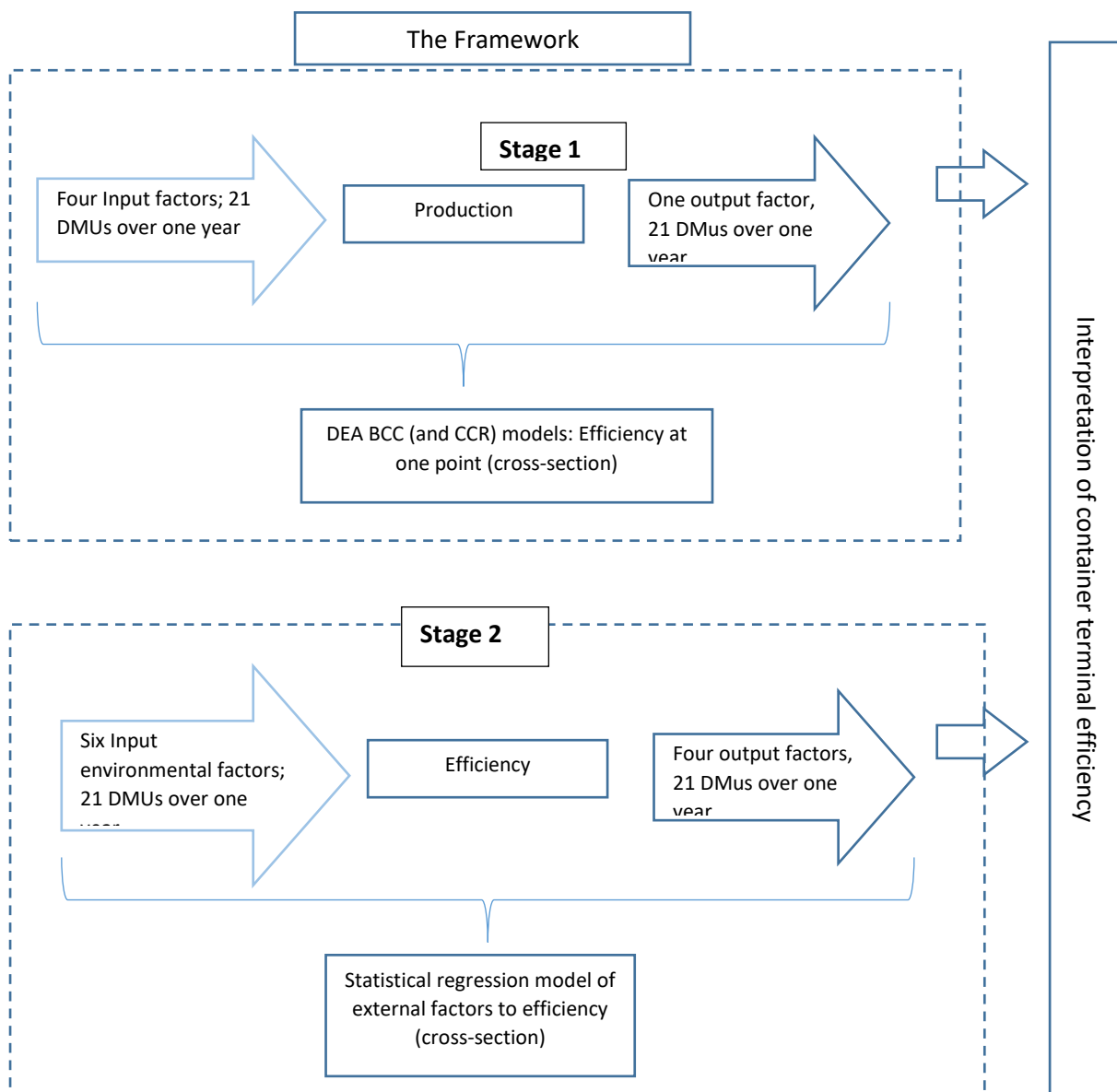
The fundamentals of efficiency models, including the data envelopment analysis (DEA), were discussed in chapter 3. Here the applied DEA models are defined and applied. The current study measures and quantifies efficiency, contributing to an in-depth analysis of the Hamburg Le-Havre (HLH) container terminals for 2018. The BCC and CCR models in this analysis incorporate additional factors omitted in the Malmquist analysis (chapter 5) and thus provide new-found insight to the analysis. The DEA analysis is supplemented by statistical analysis, applying a Kruskal – Wallis test and a Tobit test to project the impact of the external factors on efficiency. Subsequently, in this chapter, a two-stage analysis is undertaken, where:

- In the first stage, the DEA Banker, Charnes and Cooper (BCC) (and Charnes, Cooper, and Rhodes (CCR)) models are carried out to analyze 21 container terminals along the Hamburg- Le Havre (HLH) region for one year (2018).
- In the second stage, the efficiency scores from the first stage are regressed against exogenous variables identified.

The framework in which efficiency is analyzed in this chapter is outlined in Figure 6-1.

The first section specifies the models experimented within the two stages of the analysis. The section that follows discusses the selection of the sample and the variables for the DEA approach. Following that, the descriptive statistics and correlations of the DEA data are demonstrated and discussed. A section that follows expands on the results of the DEA (BCC and CCR models). Following that, the second stage statistical analysis is conducted. In a similar fashion to the DEA analysis, the section is processed by describing the variables used for the analyses, preceded by the key assumptions for parametric testing. The final section discusses the results from the statistical test, supplemented by the analysis of efficiency-affecting contextual variables which are indirectly observed. Subsequently, a conclusion for the entire chapter is provided.

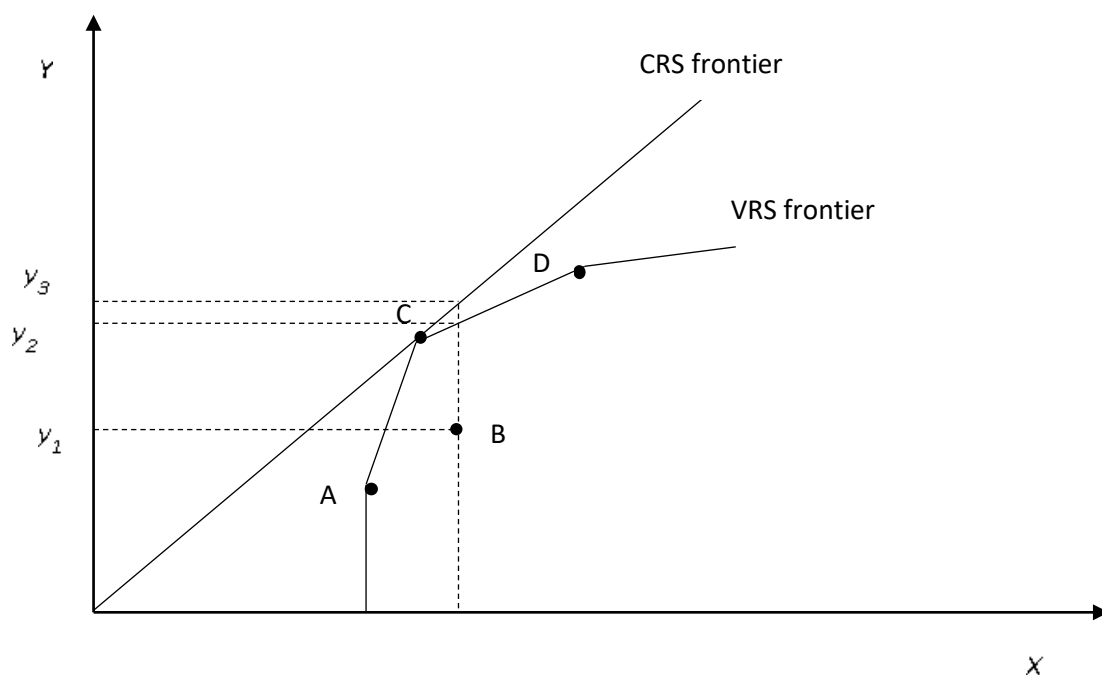
Figure 6-1:: The methodology framework



6.1 Model specification: Data envelopment analysis (DEA)

The DEA Charnes, Cooper, and Rhodes (CCR) and the Banker, Charnes, and Cooper (BCC) models are specified, tested and the results are discussed in this chapter. *Figure 6-2* demonstrates a DEA where the constant return to scales (CRS) and variable returns to scale (VRS) assumption is held. While in the VRS, the increasing and decreasing returns to scale, the CRS reflects that output will change by the same proportion as the change in inputs. In *Figure 6-2* four data points (A, B, C, and D) are used to estimate the efficient frontier and the level of capacity utilization, assuming fixed inputs (X) for the full capacity output (Y). Under the CRS assumption, point C determines the most efficient point. Under the VRS assumption, points A, C, and D define the efficient frontier. In both cases, point B represents inefficiency and thus underutilization of capacity.

Figure 6-2: The conceptual graph related to the Data envelopment analysis (DEA)



Source: Adapted from (Cooper et al., 2006; Dyckhoff & Souren, 2020)

The analysis involves identifying the sources of technical inefficiency. Technical efficiency can be measured by the Charnes, Cooper, and Rhodes (CCR) (1978) model and the DEA Banker, Charnes, and Cooper (1984) (BCC) model. The ability to increase output while increasing efficiency (output orientation) and without any absorption of complementary resources (input orientation) is key for any industry. In this analysis, both the input and output orientation perspectives will be taken into account. The

results of the efficiency scores will provide information that suggests how the respective container terminals can adjust and improve their performance, given the resources of the container terminal (inputs), and vice versa, how to reduce the use of resources (inputs), while obtaining the best maximum output (throughput).

6.1.1 DEA Charnes, Cooper, and Rhodes (CCR) model

A starting point for efficiency analysis is always Farrell (1957) (further details in chapter 3), where efficiency is given by taking the weighted sum of input(s) responsible for the weighted sum of output(s) (Cubbin & Tzanidakis, 1998). In 1978, Charnes et al. proposed a model that assumes constant returns to scale (CRS). The CRS model, also known as the CCR model (an acronym of the creators' family names), is appropriate when all DMUs operate at an optimal scale (Huguenin, 2012). The ratio of efficiency as described by (Charnes et al., 1978b) is:

$$TE_k = \frac{\text{weighted sum of outputs}}{\text{weighted sum of inputs}} = \frac{\sum_{r=1}^s u_r y_{rk}}{\sum_{i=1}^m v_i x_{ik}} \quad 6.1$$

where TE_k describes the technical efficiency (ratio) of a firm k using m inputs to produce s outputs, y_{rk} is the quantity of output produced by firm k , x_{ik} is the quantity of input i consumed by firm k , u_r is the weight of output r , v_i is the weight of input i , n is the number of firms to be evaluated, s is the number of outputs, and m is the number of inputs

The goal is to find weights that maximise the efficiency scores with a constraint that the DMUs cannot obtain scores greater than 1 (Cubbin & Tzanidakis, 1998). This linear programming problem requires maximizing the output and minimizing the inputs such that the scores obtained are neither less than 0, nor greater than 1 (determined by the weights¹⁴). Holding inputs constant (output-oriented) and holding outputs constant (input-oriented), the dual form of these equations, as demonstrated by Johnes & Johnes (2004) for the models become:

Output-oriented dual (CRS)

Maximize ϕ_k

¹⁴ The weights are assumed to be strictly positive and universal.

Subject to

$$\theta_k y_{rk} - \sum_{j=1}^n \lambda_j y_{rj} \leq 0 \quad (r=1, \dots, s) \quad 6.2$$

$$\begin{aligned} x_{ik} - \sum_{j=1}^n \lambda_j x_{ij} &\geq 0 \quad (i=1, \dots, m) \\ \lambda_j &\geq 0 \quad (v_j) \end{aligned} \quad 6.3$$

Input-oriented dual (CRS)

Minimize θ_k

Subject to:

$$y_{rk} - \sum_{j=1}^n \lambda_j y_{rj} \leq 0 \quad (r=1, \dots, s) \quad 6.4$$

$$\begin{aligned} \theta_k x_{ik} - \sum_{j=1}^n \lambda_j x_{ij} &\geq 0 \quad (i=1, \dots, m) \\ \lambda_j &\geq 0 \quad (v_j) \end{aligned} \quad 6.5$$

where $1/\theta_k$ and θ_k represent the technical efficiency of firm k , and λ_j represents the weights of outputs and inputs for firm j .

6.1.2 DEA Banker, Charnes, and Cooper (BCC) model

In 1984, Banker et al. proposed an assumption of variable returns to scale (VRS) and relaxed the CRS assumption of the CCR model proposed above. The VRS model is one of the extensions to the pure DEA model proposed by Banker, Charnes, and Cooper (BCC) (1984). This model is appropriate when DMUs are not operating in an optimal situation, in other words, in a situation where competition is imperfect. According to

Johnes & Johnes (2004), a DMU will have the same efficiency status (efficient/inefficient) regardless of the orientation in the BCC model, however, the score can differ by no large amount. This is different from the CCR model, where the scores are similar regardless of orientation.

Its output and input orientation models (dual equations) are demonstrated below:

Output-oriented dual (VRS)

Maximize ϕ_k

Subject to

$$\phi_k y_{rk} - \sum_{j=1}^n \lambda_j y_{rj} \leq 0 \quad (r=1, \dots, s) \quad 6.6$$

$$x_{ik} - \sum_{j=1}^n \lambda_j x_{ij} \geq 0 \quad (i=1, \dots, m) \quad 6.7$$

$$\begin{aligned} \sum_{j=1}^n \lambda_j &= 1 \\ \lambda_j &\geq 0 \quad (v_j) \end{aligned} \quad 6.8$$

Input-oriented dual (VRS)

Minimize θ_k

Subject to

$$y_{rk} - \sum_{j=1}^n \lambda_j y_{rj} \leq 0 \quad (r=1, \dots, s) \quad 6.9$$

$$\theta_k x_{ik} - \sum_{j=1}^n \lambda_j x_{ij} \geq 0 \quad (i=1, \dots, m) \quad 6.10$$

$$\sum_{j=1}^n \lambda_j = 1$$

$$\lambda_j \geq 0 \quad (\forall_j)$$

6.11

The CRS assumption is relaxed in this situation and a convexity constraint is added:

$$\sum_{j=1}^n \lambda_j = 1 \quad 6.12$$

Scale efficiency

The VRS DEA frontier is based on the laws of return (Coelli, 1996). In previous container port and terminal studies ((Almawsheki & Shah, 2015), Wanke & Barros, 2016), a note is made of the potential benefits of large-scale production in the sector. While the CRS implies proportional change for input and output variables, the VRS implies no proportional changes between input and output variables (can be increasing and decreasing).

The ratio for calculating scale efficiency is represented below:

$$SE_k = \frac{TE_k, CRS}{TE_k, VRS} \quad 6.13$$

Scale efficiency (*SE*) calculated in this manner denotes the degree to which the object is efficient concerning the optimum output enabling the maximal use of inputs (Baran & Górecka, 2015). When both, the CRS and VRS models are constructed, technical efficiency (*TE*) is decomposed into two components, scale inefficiency, and pure technical efficiencies. If there is any difference between CRS and VRS, this indicates a presence of scale inefficiency (Coelli et al., 2005).

A non-increasing returns to scale technical efficiency (NIRS *TE*) problem is run in conjunction with the *SE* model to determine the area of the firm's operation. The NIRS *TE* and the VRS *TE* scores are compared to determine the direction of scale

efficiencies. When output changes at the same levels as input, constant returns to scales are achieved. When input levels change beyond the output levels, then decreasing returns to scale are experienced and on the contrary, when inputs change at a level lower than the output, then increasing returns to scales are experienced, Zheng & Park (2016).

6.1.3 Other DEA extensions

There are three types of DEA models identified in Cooper, Seiford, & Tone (2006): The non-additive (the radial) and the additive (the slack-based models (SBM)).

The main difference between the radial and additive models is that while the radial models require an orientation assumption, the additive models combine both orientations in one model. The additive models also address an essential issue of a 'trade-off' between the input variables, i.e. allocative efficiency. The models focus on the maximum slacks which the DMUs can adjust to be efficient. While the radial models easily express efficiency utilizing proportions, it is not straightforward to do so using the additive models (Paradi et al., 2018).

Radial models observe their input factors in a line, radially from the point of origin. The CCR and BCC are two radial models that have been dealt with already.

SBM's augment the additive model (Cooper, Seiford, & Tone, 2006). This model aims to overcome the additive model weaknesses, that is – it should be able to measure different inputs and yield an efficiency score regardless of the metric, whether it's in kilometres or miles, a property referred to as 'units invariant' or 'dimension free'. Another improvement is to have the ability to provide a standard efficiency score. This model was introduced by Tone (1997, 2001), according to Paradi et al. (2018).

In addition to these two traditional types of DEA models, Cooper, Seiford, & Tone (2006) bring to light the Russell measure models, which improve the slacks-based measure (SBM) models. These are under the foundations of Fare and Lovell (1978), criticized for the difficulty of computation. On the other hand, the model can capture all possible efficiencies as opposed to the radial models, which cover only some input and output efficiencies.

Cooper, Seiford, & Tone (2006) provide a summary of the various alternatives of the DEA models and important characteristics of the models represented in *Table 6-1*.

Table 6-1: Summary of DEA models

Model		CCR-I	CCR-O	BCC-I	BCC-O	ADD	SBM
Data	X	Semi-p	Semi-p	Semi-p	Free	Free	Semi-p
	y	Free	Free	Free	Semi-p	Free	Free
Trans	X	No	No	No	Yes	Yes ^α	No
Invariance	Y	No	No	Yes	No	Yes ^α	No
Units Invariance		Yes	Yes	Yes	Yes	No	Yes
θ^*		(0,1)	(0,1)	(0,1)	(0,1)	No	(0,1)
Tech. Or Mix		Tech	Tech	Tech	Tech	Mix	Mix
RTS		CRS	CRS	VRS	VRS	C(V)RS ^b	C(V)RS

^α: The additive model is a translation (Trans) invariant only when the convexity constraint is added.

^b: C(v)RS means constant or variable returns to scale according to whether or not the convexity constraint is included.

In Table 6-1 ‘Semi-P’ means semi-positive, indicating that the data is non-negative with at least one positive element for the DMUs and ‘Free’ indicates that the data can be negative, positive, or zero. θ^* represents the efficiency score which can range between zero (0) and one (1), except for the additive model (ADD). The additive model is therefore coordinate-free and translation-invariant. The model evaluates a DMU’s efficiency by t_1 -metric distance from the efficient frontier. Translation invariance indicates that the models are invariant with regard to the translation coordinate system. According to Cooper, Seiford, & Tone (2006;p93), “Given any problem, a DEA model is said to be translation-invariant if translating the original input and/or output values results in a new problem that has the same optimal solution for the envelopment form as the old one”.

Tech or mix indicates whether the model measures technical efficiency or mix efficiency. RTS shows the returns to scale. In the BCC-I (o) model, variables are free due to the translation invariance theorem. CCR/BCC-I indicates the input-oriented model; CCR/BCC-O the output-oriented model and SBM the slack-based model.

6.2 Model specification: Second stage

In this section, the statistical models are outlined.

6.2.1 The Tobit model

The Tobit model is specified as follows:

$$y^* = X'\beta + \varepsilon, \text{ with } \varepsilon | X \sim N(0; \sigma^2) \text{ with } y = y^* \text{ if } y^* > 0, \text{ and } y = 0 \text{ otherwise} \quad 6.14$$

where y is the observed variable of interest and y^* is the latent variable. In this equation, the expected effect of X on y^* is monotonic; the residuals follow a normal distribution and the dependent variable (efficiency scores) is censored (Chaouk et al., 2020 and Amore & Murtinu, 2019).

6.2.2 The Kruskal- Wallis test

Based on Ostertagová, Ostertag, & Kováč (2014), the Kruskal- Wallis test statistic measures the degree to which the actual observed mean ranks $\bar{R}_i$ differs from their expected value $(N+1)/2$. If the difference is large, the null hypothesis H_0 is rejected.

When no two observations are equal, then the test statistics is tested as follows:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i (\bar{R}_i - \bar{R})^2 = \frac{12}{N(N+1)} \sum_{i=1}^k n_i \left(\frac{R_i}{n_i} - \frac{N+1}{2} \right)^2 \quad 6.15$$

The coefficient $12/N(N+1)$ is a suitable normalization factor.

The distribution of the test statistic H is well approximated by the chi-square distribution with $k-1$ degrees of freedom. The null hypothesis is rejected on significance level α , when:

$$H > \chi^2_{1-\alpha, k-1}$$

Where $\chi^2_{1-\alpha, k-1}$ is $(1- \alpha)$ a quantile of the chi-square distribution with $k-1$ degrees of freedom.

6.3 Selection of the terminals

Due to its non-parametric nature, the DEA is well suited with small sample sizes; however, as Tongzon (2001) proposed, a larger sample should be used than a small sample to reduce biases. The sample size of the DEA is much dependent on the

number of input variables to be included in the sample, for homogeneity reasons, in many other studies. For accuracy, according to Zheng & Park (2016), the selected amount of container terminals (DMUs for a DEA) should always be greater than the sum of input and output variables. For the same reason, Banker et al. (1984) state that the number of container terminals should at least exceed the total sum of the number of inputs and outputs by three times, while Golany and Roll (1989) indicate that for homogeneity reasons, the DMUs should be at least twice the total number of outputs and inputs. Boussofiane et al. (1991) mention that the sample of the terminals selected should at least be greater than the product of inputs and outputs. Due to the flexibility in the choice weights to inputs and outputs, according to Boussofiane et al. (1991), having fewer DMUs than the sum of inputs and outputs can affect the discriminatory power of the DEA. This would result in some DMUs being considered efficient when they could be inefficient.

Based on the guidelines from the authors above (Zheng & Park (2016), Banker et al. (1984), Golany and Roll (1989), and Boussofiane et al. (1991)) and taking into account the total input and output variables (four inputs and one output) of the current analysis, it can be concluded that the adequate amount of DMUs (decision-making units) for the current analysis can be either - $DMU \geq 5$; $DMU \geq 15$; $DMU \geq 10$; $DMU \geq 4$, respectively. This, therefore, suggests that a minimum of four container terminals can be enough to carry out the analysis, following the stipulation of Boussofiane et al. (1991).

For the importance of relevant comparison, the analysis is limited to the container terminal level rather than the port level. This is in line with studies supporting that for container terminal efficiency, comparison can only be attained at container terminal level, rather than at port level, Wang Cullinane (2015); Goss (1990); Heaver (1995); Alderton (1999); Heaver et al (2000); Heaver, Meersman and van de Voorde (2001). The selection of the container terminals is first based on the ports along the HLH range. Initially, 30 container terminals were taken into consideration along the HLH range. The terminals should handle not solely short sea but also deep-sea going vessels. The considered container terminals should measure their output in throughput (TEUs) and must employ at least four input factors (quay, quay crane, yard area, and yard equipment) to achieve the output produced. Due to the conditions mentioned above and data

availability, the sample includes a total of 21 container terminals which are carefully selected such that the HLH region is well represented. In comparison to the Malmquist productivity model (chapter 5), the current analysis takes into account an additional two container terminals, the APM Maasvlakte II in Rotterdam as well as the DPW World gateway terminal in Rotterdam (highlighted in *Table 6-2*), given their establishment post the year 2013.

Table 6-2: Sample of container terminals under analysis

OPERATOR	CONTAINER TERMINAL	PORT	COUNTRY	ABBREVIATION
PSA	Europe terminal	Antwerp	Belgium	AntwpEuro term
PSA	Noordzee terminal	Antwerp		AntwNterminal
PSA	PSA/TIL MPET terminal	Antwerp		AntMPET
DPW	Gateway terminal	Antwerp		AntwGaty
CSP (COSCO)	CSP Zeebrugge	Zeebrugge		Zbrug(CSP)
EUROGATE	Eurogate GmbH (CTB)	Bremerhaven	Germany	BremGmbH(CTB)
EUROGATE	North sea terminal (NTB)	Bremerhaven		BremNea(NTB)
EUROGATE	MSC gate	Bremerhaven		BremMSC gate
EUROGATE	Eurogate GmbH (CTH)	Hamburg		Ham(CTH)
HHLA	Burchardkai (CTB)	Hamburg		HamBurch(CTB)
HHLA	Tollerrort (CTT)	Hamburg		HamTol(CTT)
HHLA	Altenwerder (CTA)	Hamburg		HamAlt(CTA)
EUROGATE	Eurogate GmbH (CTW)	Wilhemshaven		WilmGmbH(CTW)
CMA CGM	Terminal des Flandres	Dunkirk	France	DunkDesFlan
DPW	Terminal de France GMP	Le Havre		LHTdeFrance
TERMINAL INVESTMENT LIMITED (TIL)	Terminal Normandie MSC	Le Havre		LHTNorm
HP	ECT Delta	Rotterdam	Netherlands	RottECTDelt
HP	Euromax	Rotterdam		RottEuro
APM	APM terminal	Rotterdam		RotAPM
APM	APM terminal Maasvlakte II	Rotterdam		RotMSII
DPW	World Gateway	Rotterdam		RotWorldGat

6.4 Definition of the variables

In chapter 5, section three, the data is described. The throughput (in TEUs per annum) is defined as an input factor. In this analysis, throughput is considered only for the year 2018.

Four input factors are considered for the DEA analysis: the quay, quay crane, yard area¹⁵, and yard equipment. The former three inputs are similar to those considered in chapter 5 but only the year 2018 is used for this analysis.

The additional variable in this chapter, the container yard handling equipment (CHE)¹⁶, captures the differences within the terminal yard. Through the CHE, the level of automation for a container yard is determined. This variable captures the transportation of containers from/to the quay crane to/from the stacking yard and to/from the inland transportation. As described in chapter 4, different types of machines are deployed and vary across the container terminals. Wiśnicki, Chybowski, and Czarnecki (2017) emphasize the importance of selecting the CHE and its role in productivity and efficiency. The organization of each of the processes within the yard, the interrelations, and synergies will impact efficiency. These capital measures play a role in the 'horizontal-transportation', 'yard storage', and 'hinterland transportation' sub-systems within the container yard.

The yard handling equipment for the container terminals is specified, allowing for the division to 'automated', 'semi-automated', and 'non-automated' conventional terminals. The yard handling equipment is added per container terminal.

The total amount of the yard handling equipment units from the horizontal transportation to hinterland transportation is added and represented as one variable for the analysis. Unlike the other variables, the CHE data is only available for 2018.

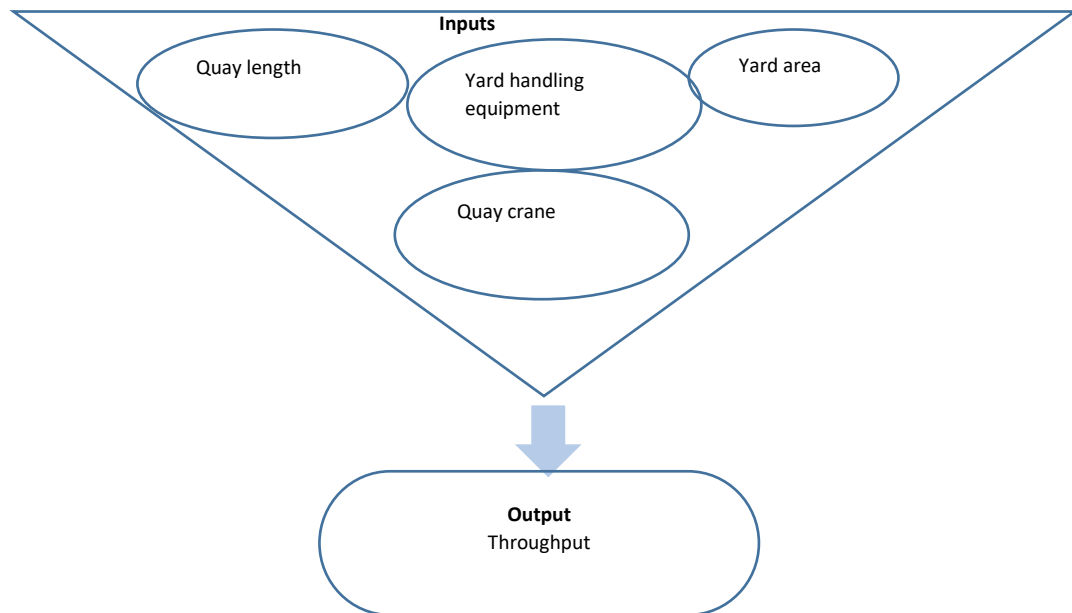
The CHE quantity data is obtained from the websites of the terminals; the CTs were contacted through e-mail and telephonically for the confirmation or further clarity of the data. Wang and Cullinane (2015) use the yard equipment costs rather than quantity. Cost data is arguably a better representation. However, obtaining all the necessary and reliable information (such as the year of purchase, initial price,

¹⁵ These input factors are all described in chapter 5.

¹⁶ The CHE variable is not included in chapter 5 due to the unavailability of the data.

interest rates, depreciation & amortization, manufacturer, etc.) across all CTs would be cumbersome as it is not publicly available for all terminals. Figure 6-3 visualizes the input and the output variables for the DEA analysis.

Figure 6-3: The input and output variables



6.5 Descriptive statistics

The descriptive statistics of the variables of the 21 container terminals are demonstrated in *Table 6-3*. The sample size in this analysis is larger than that of the Malmquist model in chapter 5 due to the newly developed terminals post-2013 (APM terminal Maasvlakte II in Rotterdam (RotMSII) and DPW Gateway in Rotterdam (RotWorldGat), which are discarded in the Malmquist analysis.

The descriptive statistics demonstrate the throughput, quay length, area, quay cranes, and yard handling equipment for the year 2018. This dataset thus includes the container yard equipment (CHE), which is not included in the Malmquist model due to data unavailability over the years studied. The largest terminal in the region handled approximately 6 million TEUs in 2018, which is 16 times larger than the smallest terminal. The high variance value shows the potential in the variability of the terminal sizes. The variance also indicates the outliers in the dataset, which is not a necessary concern given the non-parametric nature of the DEA.

Table 6-3: Descriptive statistics for the CCR and BCC

Descriptive Statistics					
Function	Description	Variable	N	Minimum	Maximum
Output	Total throughput (000'TEUs/Ann.)	TEUs	21	393	6 130
Inputs	Quay length (m/Ann.)	QL	21	1 000	3 700
	Total yard area in hectares (h)/Ann.	Area	21	47	265
	Number of quay cranes	QC	21	6	41
	Number of container yard handling equipment (CHE) /Ann.	CHE	21	24	487

6.6 Correlation

One of the fundamental attributes of the DEA is the dependence of efficiency on inputs and outputs. As mentioned earlier, there is an abundance of literature employing the DEA analysis in the container port and terminal field. A majority

employs a similar nature of variables, but most of which ignore the potential impact of correlation between the input and output variables. Interactions between the DEA attributes are hardly ever explained or modelled in past literature. López, Ho, and Ruiz-Torres (2016) conducted a study dedicated to the effect of the degree of correlation between input and output variables on DEA efficiency scores. The findings show that, on average, the high correlation between output and input variables tends to lead to high-efficiency scores, and when the input/output correlations are low, the results tend to be erratic. The degree of correlation between input variables tends not to affect efficiency scores when the correlation between input and output is high. However, when the correlation between input and output variables is low, the degree of correlation between input variables often leads to low-efficiency scores (López, Ho, and Ruiz-Torres, 2016).

In 1997, Smith (1997) studied the impact of correlation in the DEA and found that correlation between input and output variables has a significant effect on efficiency scores. Lee and Choi (2010) proved that the non-negative weighted sum of output and the non-negative weighted sum of inputs ('perfect' correlation) is not relevant for a DEA study, it does not affect efficiency. In assuming erroneous results due to serial correlation, Adler and Berechman (2001) use the principal components analysis to resolve the correlation in the DEA and thus also to eliminate some variables. Dyson et al. (2003) prove that this cannot be a solution as it can take away elements providing critical information to efficiency. According to Wang et al. (2011), a correlation is only meaningful when isotonicity is satisfied, which means that output should increase when inputs increase. Correlation analysis indicates the significant positive relationships between the inputs and the outputs, which are, therefore, isotonic and justly included in the model (Wang et al., 2011). Also to consider is that the DEA is a non-parametric method that does not impose a functional form or shape on the production set and thus not bounded by the assumption that often holds for parametric measurements (Romano & Guerrini, 2011).

Pearson correlations between the variables are presented in *Table 6-4*.

Table 6-4: Correlation between inputs and output (BCC data)

	TEUs	QL	Area	QC	CHE
TEUs	1	.84 **	.84 **	.88 **	.82 **
QL		1	.91 **	.93 **	.70 **
Area			1	.89 **	.69 **
QC				1	.72 **

The input parameters are reasonably and positively correlated to the output. All the inputs have a correlation exceeding 80% to the output in both samples, which indicates their importance in explaining the throughput. The highest correlation to throughput is that of the quay cranes (QCs), which indicates the importance of quay cranes to a container terminal. Following QCs is the area, followed by the quay length. Last is the handling equipment with a correlation of 88%. Comparatively, the difference in the correlation scores is very little, in agreement with the significance levels. Significance levels indicate that the variables are significant at the level of 0.01 on a two-tailed test. The slope for all the independent variables is positive which is a confirmation of literature, that the more resources a unit (terminal) holds, the more potential there is to create more output (containers handled), as Grosskopf (2002) states that output growth is expected to follow input growth, based on Solow-Swan of the neoclassical school of thought.

6.7 Discussion of the first stage results: Application of the CCR and BCC DEA models

The results of the DEA, including the evaluation criteria, can be easily adjusted and reviewed by the user following the specific operating environment of the DMUs to reflect certain characteristics. According to Paradi et al., (2018), as much as the DEA offers this flexibility, it also requires that one understands how the nature of such constraints affects the results of the DEA. While the last section evaluates the changes in productivity and efficiency over the years, this section will evaluate the current state of efficiency of the container terminals of the HLH. Cullinane et al. (2004) discourage the reliance on cross-sectional studies as it can be misleading in measuring performance. Although panel data and time series measures are more informative, cross-sectional analyses are also useful in measuring the current scenario and useful in situations with limitations to data.

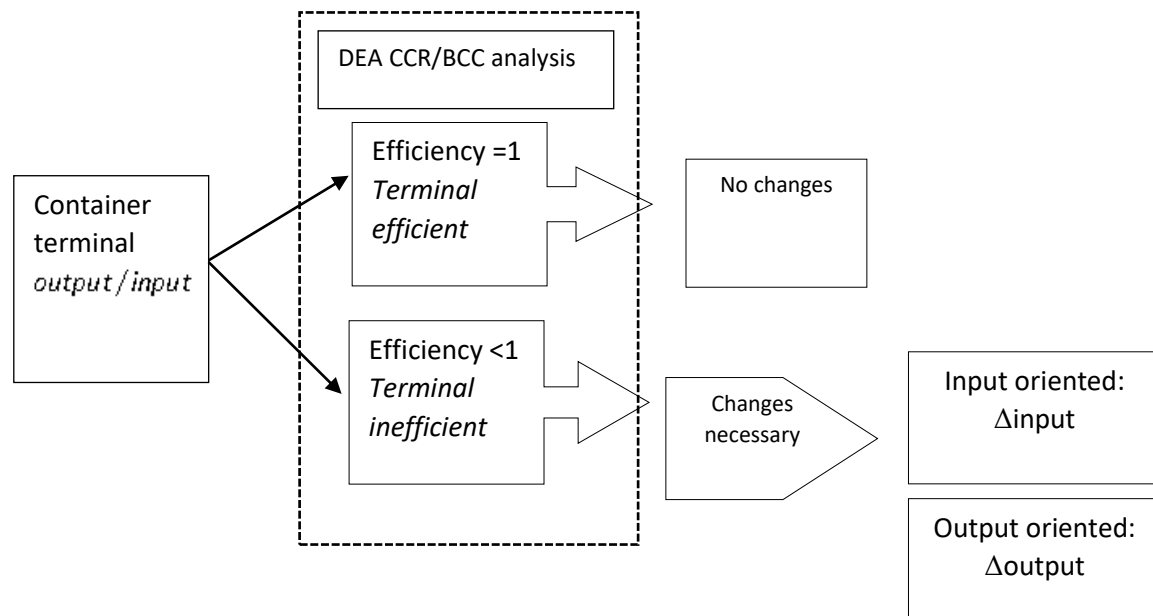
While frontier techniques do not provide the exact cause of inefficiency, they do provide the basis for analyzing relationships between the efficiency estimates and other structural characteristics. The purpose of the DEA is to construct a non-parametric envelopment frontier over the data points such that all observed points lie on or below the production frontier.

The advantages of the DEA are dealt with in-depth in chapter 3. The ability of the DEA to make an accurate comparison of similar container terminals is one of its

advantages. The DEA is based on the economic theory of production, which is consistent with the production processes of the container terminal. The freedom of choice between output-maximization or input-minimization confers the container terminals' ability to use the model following their long-term objectives and strategies. Moreover, the power of the DEA to adapt other assumptions (other DEA model types) is useful for the suitability of differing underpinnings and rationales. For instance, the BCC allows that the container terminals are compared to each other in respect of their sizes, whereas the CCR model compares the container terminals as if they all operate in the same scale.

The DEA strictly bases its outcome on values provided for the inputs and outputs for all decision-making units (DMUs), giving a score limited to 1. DMUs with an efficiency score of less than 1 indicate inefficiency and, therefore, a potential to improve by making the necessary changes from either an input or output perspective. The input factors taken into account in the analysis assist in identifying the specific activities which the terminals can focus on, represented by the quay length, quay crane, area, and yard handling equipment. These can guide the terminal operator to focus on improving quayside activities, horizontal activities, stacking yard-related activities, or a combination of all. Differences in efficiency could be due to a variety of factors related to economic factors (Miyashita, 2005); management competency, organizational style/ culture, and the ability to access certain major trade routes (Tongzon, 2005); or it just could be due to competition between the terminals in a certain region (Cullinane, 2000; Cullinane, Wang, and Cullinane, 2004). DMUs with a score equal to 1, forming the efficient frontier, are efficient and can presumably not make any further changes to improve their efficiency given their current input factors. This is depicted in Figure 6-4.

Figure 6-4: DEA CCR and BCC process



Source: own compilation

To avoid misrepresentation, the terminals are divided into two, automated and non-automated. The two groups are analyzed separately, and the DEA outcome scores are represented in combination. The current model is based on the assumption of variable returns to scale (VRS), the Banker, Charnes, and Cooper (1984) (BCC) model of the DEA where a container terminal is technically efficient if the technologies of the DMU are combined in a way that places the DMU along the feasible production frontier. There are various models estimated within the DEA model. TE (total efficiency) measures are derived under the assumption of constant returns to scale, and thus referred to as CRSTE (constant returns to scale technical efficiency) and represented by the Charnes, Cooper, and Rhodes (1978) (CCR) model of the DEA. PTE (pure technical efficiency) is derived under the assumption of variable returns to scale and thus referred to as the VRSTE (variable returns to scale technical efficiency) and represented by the BCC model of the DEA (an extension to the CCR model). The VRSTE model is often praised for its ability to consider market competition, applying a convexity constraint, while the CRS assumes perfect competition which is not the likely situation in the real sector, especially in the container terminal market. Demirei et al. (2012) and Bichou (2012) argue that port and terminal operation follow VRS while Wang (2002) argues that both the CCR and VRS model should equally matter unless there is economic proof to indicate that CCR is unnecessary otherwise, both models

should be presented. SEs allow firms to analyze the efficiencies that are due to size and thus make the necessary changes to the scale of production (Kumar and Gulati, 2008). According to Huguenin (2012), technical inefficiency can be due to two sources; poor management and operation of a firm (observed from the VRSTE model) and secondly it can be due to firm operation at an inappropriate scale (observed from the SE model). This analysis will apply the variable returns to scale assumption.

The orientation of the model is another fundamental guideline in the application of the DEA which entails specifying whether the intention is to increase output while holding constant all the inputs (output orientation) or reducing the inputs (input orientation) for the given amount of output. Input orientation is associated with resolving operational as well as managerial issues while output orientation is much related to the resolution of planning and macroeconomic issue, (Baran and Górecka 2015; Cullinane et al. 2004), both these targets are vital in the container sector. Coelli & Perelman (1996) mention that the values of efficiency in a CRS are similar on either input or output orientation models, whereas the efficiency values change in a VRS model when orientation changes. The decision to output or input orientation should be decided upon depending on which variables the decision-maker has more control over. In this study, the objective of the terminals is assumed to be input minimizing and output maximizing. In this analysis, the results are reported for both, input as well as for output-oriented techniques. This is to give both perspectives as terminal's strategies and needs may differ.

The two-stage DEA is associated with maximization in the sum of slacks, rather than minimization. In this analysis, the multi-stage DEA is applied which compiles a sequence of radial movements leading to identifying the most efficient points along the frontier. This avoids the necessity to maximize on slacks.

The estimations from the models are demonstrated in

Table 6-5: Input & output-oriented DEA CCR BCC model

	Output-oriented				Input-oriented			
Container terminal	Ocrste	Ovrste	Oscale	ODirection	Icrste	Ivrste	Iscale	IDirection
Europe terminal Antwerp	0.91	1.00	0.91	irs	0.91	1.00	0.91	irs
Noordzee terminal	0.58	0.78	0.74	irs	0.58	0.98	0.59	irs
MPET terminal Antwerp	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
DPW Gateway terminal Antwrp	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
Eurogate GmbH (CTB)	0.55	0.68	0.82	irs	0.55	0.94	0.59	irs
North Sea terminal (NTB)	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
MSC gate Bremerhaven	0.93	1.00	0.93	irs	0.93	1.00	0.93	irs
Terminal des Flandres	0.43	1.00	0.43	irs	0.43	1.00	0.43	irs
Eurogate GmbH (CTH)	0.57	0.59	0.97	drs	0.57	0.65	0.88	irs
Burchardkai (CTB)	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
Tollerrort (CTT)	0.87	1.00	0.87	irs	0.87	1.00	0.87	irs
Altenwerder (CTA)	0.98	0.99	0.99	drs	0.98	0.99	0.99	drs
Terminal de France GMP	0.92	1.00	0.92	irs	0.92	1.00	0.92	irs
Terminal Normandie MSC	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
ECT Delta Rotterdam	0.84	1.00	0.84	drs	0.84	1.00	0.84	drs
Euromax Rotterdam	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
APM terminal Rotterdam	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
APM terminal Maasvlakte II	1.00	1.00	1.00	crs	1.00	1.00	1.00	crs
World Gateway Rotterdam	0.81	0.81	0.99	drs	0.81	0.91	0.89	irs
Eurogate GmbH (CTW)	0.60	0.88	0.67	irs	0.60	0.95	0.63	irs
APM (CSP) Zeebrugge	0.51	1.00	0.51	irs	0.51	1.00	0.51	irs
Mean	0.83	0.94	0.88		0.83	0.97	0.86	

The estimates presented here are based on both, the input and output orientation and draw reference to the CCR model (output/input-oriented constant returns to scale technical efficiency (O/Icrste) and to the BCC model (output/input-oriented variable returns to scale technical efficiency (O/Ivrste) and output/input-oriented scale efficiency (O/Iscale), where #1= 'efficient' and <1='inefficient'

The efficiencies recognized by the CCR (CRSTE) model are always equal or lower to the efficiencies recognized by the other two models (VRSTE and SE) because the model is stricter to the DMUs by not adjusting to the differences in size but rather assuming that all the DMUs are the same, which is arguably not a reflection of reality. The CCR model makes no distinction between the sources of inefficiency while the BCC distinguishes between in/efficiencies to management and scale. The BCC accounts for these differences and thus relaxes the strict assumption of the CCR model. To obtain scale efficiencies, the efficiency scores from the CCR model are divided by the efficiency scores of the BCC model.

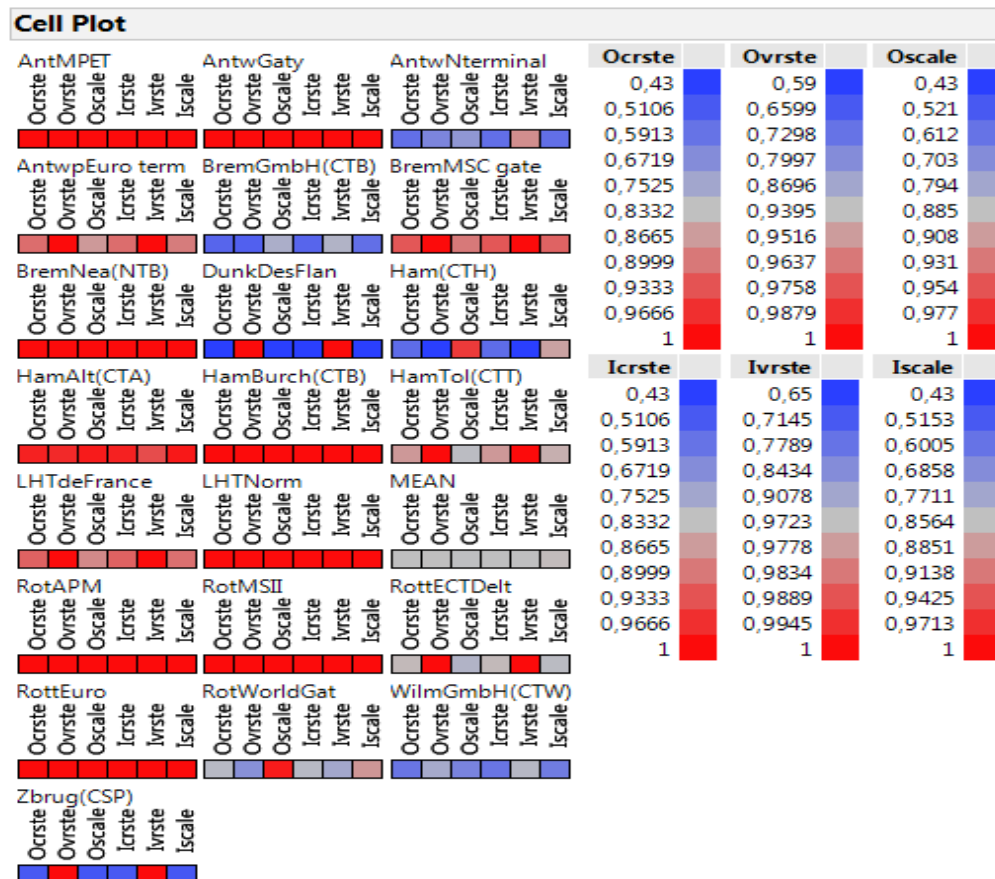
According to Coelli (1996), the orientation of the DEA model should not create a major influence on the results obtained. Between the input and output-oriented models, a Pearson model indicates a 100% correlation constant returns to scale technical efficiency models (CRSTE); 82% correlation in the efficiency scores as reported in the variable returns to scale technical efficiency model and 94% correlation in the scale efficiency model. This thus indicates that the level of efficiency for these container terminals largely does not change following whether a decision to maximize output or to minimize input is being made, therefore leaving the decision of orientation to the particular terminal concerning the objective and goals of the terminals.

The CCR model, which indicates that all DMUs are operating under similar conditions, shows that 38% (8) of the container terminals are fully efficient in both the input- and output-oriented models. This model does not indicate the sources of efficiencies but takes both scale and pure efficiencies as equals. According to the CCR, the container terminals of the HLH are 83% efficient.

The BCC model reveals more terminals to be considered efficient compared to the CCR model. The BCC differentiates between the efficiencies sources and does not assume that the container terminals operate under similar conditions. In both, the input and output orientation, 71% (15) of the container terminals are fully efficient in the VRSTE model and 38% (8) are fully efficient when the scale efficiencies are evaluated. Although not by great amounts, in the BCC model, the efficiency levels of the container terminals vary under each model (with the exclusion of the fully efficient DMUs). In the output-oriented perspective, the container terminals are 97% efficient and 94% efficient in the input-oriented perspective. In the scale efficiency model, the container terminals are collectively efficient by 86% and 88% in the output and input-oriented models, respectively.

The cell plot (Figure 6-5) indicates the different scores of the container terminals concerning the six models, clearly illustrating the various container terminals' performance across the estimated models, indicating efficiency from low (0) to high (1).

Figure 6-5: Cell plot of container terminal efficiencies across models



Source: Owner's compilation

The sections that follow elaborate on the models.

6.7.1 CCR model of the DEA: Constant returns to scale technical efficiency (CRSTE)

Whereas the efficiency values may change concerning the orientation in a VRS model, according to Coelli & Perelman (1996), the efficiency values in a CRS are similar on either input- or output-orientation models. According to this model, the HLH region can collectively improve the use of resources by 17% in both orientation perspectives. This indicates the extent of wastage in the container terminals collectively. In the output-oriented model, theoretically, this indicates that the HLH container terminals can increase their level of output to 1.2 (=1/0.83) times as much as

the current level of throughput while utilizing the same amount of input. The input-oriented model indicates that input factors can be reduced 1.2 times as much as the current level of inputs for the same amount of throughput currently produced.

The conduct of the output-oriented model can be easier to understand. Analyzing the input-oriented model can be cumbersome in the context of a container terminal as the input factors such as the quay length, yard area, the quay cranes and yard equipment are fixed (to some extent). The input-oriented model diverts focus on the resource-wasting inputs. 'Improving' on such inputs is highly subjective from terminal to terminal. Maximizing the output given the existing resources indicates the most potential handlings for the container terminal. However, achieving that target could mean applying various techniques and strategies for the terminal.

The specific 8 CTOs who achieved efficiency and thus considered optimal according to this model include PSA/TIL MPET terminal in Antwerp (AntMPET); DPW Gateway terminal in Antwerp (AntwGaty); Eurogate North sea terminal in Bremerhaven (BremNea(NTB)); HHLA Burchardkai in Hamburg (HamBurch(CTB)); TIL Terminal de Normandie in Le Havre (LHTNorm); HP Euromax terminal in Rotterdam (RottEuro); APM terminal in Rotterdam (RotAPM) and APM terminal Maasvlakte II in Rotterdam (RotMSII).

In comparison to the Malmquist 'efficiency changes' model examined earlier in chapter 5, these terminal's reputation of efficiency was positive for 3 CTOs (MPET terminal Antwerp, Terminal Normandie MSC, and Euromax) and negative for Gateway terminal Antwerp, BremNea(NTB), Burchardkai (CTB) and APM terminal Rotterdam. The APM terminal Maasvlakte II cannot be compared as it is still relatively a new terminal. Although the Malmquist efficiency model offers a background check, it is not completely comparable to this CCR model due to the missing 'yard handling equipment' variable in the Malmquist model versus the current CCR (and BCC) model. The potential reasons behind the levels of efficiency could be explained by resource management (VRSTE) and scale efficiency (SE) models which are explained below. These container terminals operate along the frontier and are thus part of the container terminals which could be used as a reference point to the industry practices when holding the assumption that all the container terminals operate under a similar business environment (constant returns to scale). A situation that deviates from this assumption is represented in the section that follows.

6.7.2 The BCC model of the DEA

The BCC model is assessed through the variable returns to scale technical efficiency (VRSTE) model scale efficiency (SE) model.

6.7.2.1 The variable returns to scale technical efficiency (VRSTE) model

The BCC further reveals the sources of in/efficiency, either from the management (through the VRSTE model) or from scale advantages or disadvantages (scale efficiency model).

The pure technical efficiency of the BCC model can be obtained through the following equation:

$$\text{Pure technical efficiency} = \frac{\text{technical efficiency}}{\text{scale efficiency}} \quad 6.16$$

As expected and observed from the cell plots, the VRSTE model reflects that more terminals (15) are theoretically fully efficient compared to the CRSTE model. Not only are there more terminals that are efficient in the vrste model but also the efficiency levels of the container terminals are generally higher (97% in the input-oriented model and 94% in the output-oriented model). This indicates that the efficiencies of the HLH range can collectively improve by 3% and 6%, respectively. Because the VRSTE model indicates the level of management of resources, the implication is that in the output-oriented model, the HLH container terminals can increase their level of output to 1.06 (=1/0.94) times as much as the current level of throughput while utilizing the same amount of input. In the input-oriented model, this indicates that input factors can be reduced 1.02 (=1/0.97) times as much as the current level of inputs for the same amount of throughput currently handled. At the HLH range, efficiency is thus derived more from the management of resources than from scale efficiencies as the scale efficiency score is lower in both orientations. This is also a confirmation of the Malmquist findings that over time, the HLH container terminals derived efficiency more from pure efficiencies than from scale efficiencies. Scale efficiencies were negative over the 5-year observed period.

As much as this indicates a strong factor for the HLH container terminals, it also indicates that there is little that most terminals will benefit from improving resource management for efficiency purposes, *ceteris paribus*; however, more can still be done.

The implication on this indicates that the container terminals need to divert focus more on scale efficiencies.

Container terminals that appeared inefficient under the CRSTE model but which are considered efficient under the VRSTE model include the PSA Europa terminal in Antwerp (Europe terminal); Eurogate MSC gate terminal in Bremerhaven (BremMSC gate); CMA CGM Terminal des Flandres in Dunkirk (DunkDesFlan); HHLA Tollerrort in Hamburg (HamTol(CTT)); DPW terminal de France in Le Havre (LHTdeFrance); HP ECT Delta in Rotterdam (RottECTDelt), and CSP terminal in Zeebrugge (Zbrug(CSP)). The reason behind these terminals being considered efficient under the VRSTE in comparison to the crste model is that as mentioned earlier, the CRSTE model makes no distinction between the sources of efficiency, which makes it impossible to identify the reason for inefficiency from the model itself. Therefore, these terminals being efficient under the VRSTE model indicates that the sources of efficiency are derived from scale inefficiency which will be discussed later on in this chapter.

The DEA Malmquist explains that the average container terminal in the HLH region is more productive due to technologies rather than efficiency, although they are both important. This means that efficiency in itself will not be the only solution to ensuring improved productivity of the container terminals. CTOs which according to the analysis have shown to derive some inefficiencies from resource management include PSA Noordzee terminal in Antwerp (Noordzee terminal Antwerp); Eurogate Container Terminal Bremerhaven (BremGmbH(CTB)); Eurogate container terminals in Hamburg (Ham(CTH)); HHLA Altenwerder terminal in Hamburg (HamAlt(CTA)); DPW gateway in Rotterdam (RotWorldGat), and Eurogate terminal in Wilhelmshaven (WilmGmbH(CTW)).

For all these terminals, the efficiency levels according to the model are above 90% with the exception of Eurogate container terminals in Hamburg (Ham(CTH)) whose theoretical efficiency is 65%. This is the only container terminal whose inefficiencies are derived largely from resource management as compared to scale efficiencies. The Malmquist model also proved this over the period of observation where the terminal's efficiency was largely (and negatively) affected by pure efficiencies in comparison to scale efficiencies with the potential reasoning to this owed to low volumes in relation to capacity in the container terminal. The terminal's capacity utilization for the year 2018 is 39.90%, which is very low in comparison to the rule of 70-80% and it is also generally lower than the average HLH group of terminals. It would be interesting to

observe the impact on the terminal's efficiency given by a change in resource allocation.

As the DEA is a benchmarking method, it also suggests that Eurogate GmbH (CTH) Hamburg will benefit from adapting to practices of Eurogate North sea terminal in Bremerhaven (BremNea(NTB)) among other efficient terminals referred to as best practice or peer referents by Almawshaki & Shah (2015). Coelli et al. (2005) suggested that the multi-stage DEA solves for a sequence of radial and slack movements. The radial movements indicate the movement that an inefficient firm has to make to locate itself on the frontier. The slack movements indicate the further movement needed for a firm located on the frontier to move to an efficient frontier. According to the DEA model's presented slack and radial movements, the container terminal can theoretically improve its performance (measured by the output) by 70% utilizing the same input resources. The input factors are traditionally in the DEA translated such that they indicate the proportion by which they can be reduced so that the DMU in concern is efficient (Dyckhoff & Souren, 2020). As this introduces implications for translation for the container terminal context, it can be translated in the sense of future possible production possibilities. This would thus stand alongside the assumption that terminals are likely to invest considering current volumes and future volumes. For instance, the Eurogate GmbH (CTH) Hamburg terminal would benefit from 'reducing' the quay length, area, quay cranes, and container yard handling equipment by 13%, 22%, 21.7 %, and 0% for the input factors, respectively. Therefore, this means that, at the present moment, these input factors are in excess by these proportions which should still be made use of, taking into account the throughput that can potentially be handled till efficiency is achieved.

Furthermore, this also means that a greater focus should be diverted to yard space distribution and quayside activities and less attention to horizontal transportation and to stacking activities to improve. The container terminal would be considered efficient with these movements, *ceteris paribus*. According to Coelli (1996), the proportional changes are applied following the returns to scale laws. The absolute comparison of the Eurogate GmbH (CTH) Hamburg terminal with regards to the performance of throughput, it can be observed that it handled lower throughput per quay length, area, quay cranes, and container yard handling equipment for the year 2018 relative to the HLH average, which can also serve as an indication of resource wastage in comparison to the 21 HLH container terminals.

On average, the HLH container terminals have proven to be highly efficient (97% and 94%) from the resource management perspective. While a majority of the terminals (71%) were fully efficient according to this model, those CTOs experiencing inefficiency need to take into account the returns to scale model and come with strategies that could assist in ensuring improved efficiencies. Also, important to take into account is that, this analysis makes conclusions only based on the sample, could there be a different combination of terminals in the sample, the situation could have been different. As the current sample considers the important terminals in the region (according to volumes and activity), this should not be a major concern, especially to the container terminals forming part of the frontier.

6.7.2.2 The scale efficiency (SE) model

The scale model is discussed in chapter 5 and it is generally expressed as follows:

$$\text{Scale efficiency} = \frac{\text{Technical efficiency}}{\text{pure technical efficiency}} \quad 6.17$$

In the scale efficiency model, there is a greater variation in the treatment of the terminals. Terminals in the VRSTE model were all treated the same either in the output- or in the input-oriented model. In the SE, a few terminals have obtained different efficiency scores between the output and the input orientation models. This indicates that the container terminals need to be mindful with regards to the proportional allocation of the resources referring to the orientation of their desires (maximizing output or minimizing the input).

Compared to the VRSTE model, there are fewer container terminals considered theoretically efficient. According to the scale efficiency model, the level of efficiency is 86% and 88% (input and output orientation, respectively). This indicates a resource wastage or potential improvement of 14% and 12% in scale efficiency, this is higher than the 3% and 6% required in the VRSTE model of pure efficiencies. Therefore, this suggests that, to improve and become more efficient, the HLH group of terminals has to divert more focus on improving scale economies.

When the model is oriented to output, the HLH container terminals can increase their level of output to 1.13 (=1/0.88) times as much as the current level of throughput while utilizing the same amount of input. The input-oriented model indicates that input factors can be reduced 1.17 (=1/0.86) times as much as the current level of inputs for the same amount of throughput currently handled. Of course, these dramatic

adjustments would be possible when appropriate production strategies are implemented as well as if production occurs at the appropriate scale.

The DEA Malmquist model indicated a negative impact on productivity flowing from scale efficiencies compared to pure technical efficiency, total efficiency, and technical change throughout the period observed.

CTOs considered efficient in this SE model have also been considered efficient in the CRSTE and VRSTE models and are namely PSA/TIL MPET terminal in Antwerp (AntMPET); DPW Gateway terminal in Antwerp (AntwGaty); Eurogate North sea terminal in Bremerhaven (BremNea(NTB)); HHLA Burchardkai in Hamburg (HamBurch(CTB)); TiL terminal de Normandie in Le Havre (LHTNorm); HP Euromax terminal in Rotterdam (RottEuro); APM terminal in Rotterdam (RotAPM) and APM terminal Maasvlakte II in Rotterdam (RotMSII). These terminals making up 38% of the sample form the efficient frontier and can be referred to by other inefficient terminals as the best practice or peer referents terminals as they make strategies of improving their terminal efficiencies.

All the terminals in this group (38%) relatively handle throughput by a meter of quay, area, quay crane, and container yard handling equipment (CHE) above the HLH average (1 185 TEUs per meter of quay; 19 144 per hectare; 130 538 TEUs per quay crane and 17 357 TEUs per CHE based on 2018 measurements) except for Terminal Normandie MSC and APM terminal Rotterdam with regards to quay length, area, and quay crane throughput. Similarly, these two terminals generally handled lower throughput than the average terminal. According to Cullinane, Wang, Song, & Ji, (2006), larger ports (by throughput) are associated with high scale efficiency. This cannot be entirely confirmed in this sample as these two terminals above are among the efficient terminals, although they handled a lower number of containers than the average terminal in the sample.

At the same time, the HHLA Altenwerder terminal in Hamburg (HamAlt(CTA)) (99% scale efficient) and the HP ECT Delta in Rotterdam (RottECTDelt) (84% scale efficient) handle throughput above the average terminal in the HLH sample but they are considered inefficient. In the specification of Cullinane, Wang, Song, & Ji, (2006), a cutoff point between large and small terminals is 1 million TEUs per annum. In this case, some large terminals would still be excluded in the 'efficient' group, but on the other end, all the small terminals in the sample are indeed considered scale-inefficient. What is also similar with these terminals is that they are all facing

constant returns to scale in production, which indicates that the output changes at the same proportion to input factors.

Cullinane, Wang, Song, & Ji, (2006) mention that large ports usually face constant returns to scale (CRS) or decreasing returns to scale (DRS) due to their size limitations to expansion. Having reached a certain level of operational scale, growth becomes limited, given the need for high investments. Resource limitations such as land unavailability can also be an issue, as in the case of PSA home terminal in Antwerp (later MPET). It is also very likely that these considerably 'large' terminals already have invested in expensive, new equipment.

This group of terminals consists of high and semi-automated CTs such as Gateway terminal Antwerp; Burchardkai (CTB); Euromax, and APM terminal Maasvlakte II. On the other hand, smaller terminals are not very likely to struggle regarding the potential to expand, although they also face challenges associated with for instance capital for further expansions, as compared to their larger counterparts. Gateway terminal Antwerp for instance faces the challenge of inability to expand its terminal further by land. Losing part of the concession period of seven years led to the terminal losing some of its quay wall and area, having an impact on the performance and growth potential of the terminal.

Another group of terminals under the SE model are the terminals whose inefficiency is only due to scale. This group consists of 7/21 (33.33%) terminals. These include the PSA Europa terminal in Antwerp (Europe terminal Antwerp); Eurogate MSC gate terminal in Bremerhaven (BremMSC gate); CMA CGM Terminal des Flandres in Dunkirk (DunkDesFlan); HHLA Tollerrort in Hamburg (HamTol(CTT)); DPW terminal de France in Le Havre (LHTdeFrance); HP ECT delta in Rotterdam (RottECTDelt) and CSP terminal in Zeebrugge (Zbrug(CSP)). All these container terminals have experienced increasing returns to scale with the exception of RottECTDelt, which experienced decreasing returns to scale. The terminals experiencing IRS are at an advantage because they can theoretically still very much benefit from increasing production by utilizing the same resources. The terminals all can be regarded as 'small terminals' as they all (except the ECT Delta), have handlings less than the average of the HLH group of terminals which indicates room for potential improvement in handlings if business conditions allow it.

With regards to yard performance, these terminals do not all handle more than the average TEUs per meter of quay length, hectare, quay crane, and CHes. Terminal des

Flandres and CSP Zeebrugge achieved less than the average TEUs across all these measures, similarly to the Europa terminal, which only achieved just above the average TEUs per quay crane. Terminal des Flandres and CSP Zeebrugge also achieved the lowest efficiency levels compared to the entire HLH group and both inefficiencies were sourced from scale inefficiency. These two terminals also handle the lowest per annum throughput compared to the entire HLH group. The fact that these terminals handle less throughput with their given resources than the average other HLH container terminals signifies the reason behind the scale inefficiencies and therefore the wastage of resources.

Over the observed period in the Malmquist model, ECT Delta began experiencing positive technical changes in 2018 (13%) compared to the previous years. Faced with local competition and the desire to handle large vessels efficiently, there were huge investments in technologies at the terminal. There was an expansion on the quay wall, yard, and the deployed quay cranes. HP also closed down the home (city) terminal in Rotterdam in 2015 which may have significantly affected the volumes and strategic decisions of the Rotterdam HP group. The new terminals in MSII (DPW World gateway) introduced previously unforeseen effects to the terminal (especially concerning shipping lines). This would result in excess capacity versus lower volumes, which means higher fixed costs and lower prices. In 2016, the new developments of terminals at the Maasvlakte led to a decline of 6% in throughput. These could have had an influence on the scale inefficiencies at the terminal.

The rest of the terminals in the SE model are the container terminals which are considered inefficient in both the VRSTE and SE models. This indicates that the terminals are not only facing challenges related to size issues but also resource management-related issues. These are inclusive of the PSA Noordzee terminal in Antwerp (AntwNterminal); Eurogate Container terminal Bremerhaven (BremGmbH(CTB)); Eurogate container terminals in Hamburg (Ham(CTH)) ;HHLA Altenwerder terminal in Hamburg (HamAlt(CTA)); DPW gateway in Rotterdam (RotWorldGat) and the Eurogate terminal in Wilhemshaven (WilmGmbH(CTW)).

The situation at the Eurogate GmbH (CTH) Hamburg terminal is unique in comparison to the rest as it is the only terminal in the HLH group whose inefficiency is largely due to resource management compared to scale inefficiency. This unique situation occurs in both, the input and output-oriented models, however, the returns to scale are different on the models. The container terminal experiences increasing

returns to scale in the input-oriented model and faces decreasing returns in the output-oriented model. What can explain this behaviour at the terminal could be well linked to the overall available capacity and its underutilization. The underutilization of the terminal's resources explains much of the results in the VRSTE model and the distribution of the resources to achieve the output explains more the scale efficiencies. From the output perspective, this entails that the maximization of output occurs at a rate greater than the increase in inputs on the other end, from the input perspective, entails that the terminal is not likely to achieve much improvement on the output from reducing the inputs.

Another terminal that experienced the same regarding the returns to scale is the World Gateway Rotterdam. This terminal experienced IRS in the input-oriented model and DRS in the output-oriented model. In the input-oriented model, inefficiency is derived from scale production, while in the output-oriented model, the sources of inefficiency are largely due to resource management.

The inefficiencies from the Eurogate GmbH (CTB) terminal are split between management and scale. The scale inefficiencies in this terminal are only on the input-oriented model. In the output-oriented model, the inefficiencies are derived from the VRSTE. The Malmquist also had indicated this trend in the terminal over the 5-year observed period where efficiency and technologies were positive but scale efficiency was negative. There were no major technological changes in the terminal over the period, only the area increased and throughput increased by 21% between 2013 and 2018. The insolvency of the Hanjin shipping line affected the volumes in the terminal, including its sister company, North Sea terminal (NTB), which could have affected efficiency. Generally, the Eurogate terminals suffered a decline in throughput in 2017 and 2018 (2% decline), Drewry (2019). The same year the Eurogate Container Terminal Bremerhaven (BremGmbH (CTB)) gained Hamburg Sud as a new customer, and there were also new mergers between shipping lines which significantly affected the throughput of the terminal.

The differences in the scale economies seem to have a relationship with the scale of production or so-called throughput. If reference is made to the cutoff point of 1 million¹⁷, as proposed by Cullinane, Wang, Song, & Ji, (2006) to differentiate between big and small terminals, then it can be concluded based on this HLH sample that,

¹⁷ Ham(CTH) and World Gateway Rotterdam are large according to the 1 million TEUs cutoff but in 2018 their throughput is lower than the HLH average.

concerning the degrees and the direction of scale economies, there seems to exist a certain trend. Decreasing returns to scale in production are experienced only by the large container terminals. Although these terminals face decreasing returns to scale, they have achieved higher scale efficiencies than most of the other terminals which are considered smaller. On the other hand, all the small terminals (Terminal des Flandres; Eurogate GmbH (CTW); CSP Zeebrugge) have experienced increasing returns to scale and have also experienced among the lowest scale efficiencies compared to the entire sample. The Noordzee terminal Antwerp can also be included in the list as it handles just a little above 1 million TEUs and also falls under the terminals with among the lowest scale efficiencies in the sample.

As can be observed from the BCC model, in the HLH region, greater efficiencies are largely due to management (VRSTE) and lesser efficiencies by scale. The inefficiencies observed in the region, indicate that collectively, more can still be done to improve on the utilization of the HLH region's scarce resource. To achieve overall efficiency in terminals, Bichou (2011) suggests that the input resources should be minimized per sub-system, rather than the entire terminal. This would enable a terminal operator to for instance focus on the yard dwelling time where inefficiency is realized at the yard-quay sub-system. In the gate-yard subsystem, the queuing headways for inbound trucks could be minimized. Additionally, the efficiencies between the sub-systems can be incorporated to achieve improved overall efficiency in the container yard. More attention should be paid to inefficient variables and more resources allocated to improve these (Chen et al., 2018). Slack movements of the DEA have indicated differences between automated and non-automated terminals. In the case of automated terminals, the model suggests no necessary adjustments in the quay length and very minimal adjustments in the number of quay cranes. This indicates that based on the existing technologies, there are not many adjustments to suggest at the waterside activities while for the non-automated terminals, the proportion of adjustments to these activities is a little more. Proportionally, in the HLH, both automated and non-automated terminals are to largely focus on improving the utilization of yard space, although the yard handling equipment has given no suggestions for adjustment for both types of terminals.

However, one should bear in mind that there could also be some other external factors that exert significant influence on the analyzed container terminals when it comes to efficiency analysis. Panayides et al., (2009) thus strongly emphasize that

future research should carefully distinguish between the different factors affecting ports before applying the DEA. The authors warn that a waste of resources is a waste of resources regardless of the terminals' external factors.

6.8 Second stage: Statistical analysis

The consciousness of efficiency affecting factors (internally and externally derived factors) is necessary for container terminal operators to survive the competitive environment. This analysis explores external factors which influence efficiency. These factors are introduced in chapter 4. Only a selected few are discussed and analyzed using the Tobit model and the Kruskal-Wallis. First, the variables taken into account are described. For statistical analysis, various assumptions should be met to determine whether or not conclusions can be drawn from the analysis; these are discussed in the previous section. Following that, the results of the tests are reported. Lastly, supplementary elements affecting efficiency are discussed as well as the expected relation to efficiency.

6.8.1 Description of the variables

There are considerably numerous sources of inefficiency in container terminals as indicated by the characteristics in the typology chapter (4). Here, a selection of those exogenous factors are employed and their significance to efficiency is statistically tested. The set of variables considered for the analysis are here described and demonstrated in Table 6-6. The table illustrates the variable's name, the description and measure of the variable, the form in which the data is represented on each model, i.e., the Kruskal Wallis model and the Tobit model. In the Kruskal Wallis model, the variables are evaluated using a binary form, while in the Tobit model continuous data is used. The last column indicates the source where the data has been collected.

6.8.1.1 Automation

Bichou (2012) indicates how cargo configuration may affect a terminal's efficiency. It is mentioned that configurations and operating systems influence the manner of management in a terminal, affecting efficiency. Automation has in past literature been associated with increased technical efficiency when coupled with appropriate container terminal design and synchronization (Watfa and Karmadi, 2019 and Chen et al., 2020). As a result of the potential promises from automation, a growing amount of container terminals are considering automation as a solution to many of their obstacles.

For this analysis, to determine whether the terminal is automated or not automated, the equipment utilized for handling the containers is evaluated (specific details on automation are in chapter 4). The details of the equipment that define the level of automation of a particular terminal are discussed in chapter 4 (e.g. use of rail mounted gantry cranes (RMGCs) vs use of rubber-tyred gantry cranes (RTGCs)). Terminals with no existing automation are regarded here as the non-automated terminals. Terminals that are partially automated and those which are fully automated are referred to as the 'automated' terminals. The semi-automated terminal (DP world gateway terminal in Antwerp) is considered an automated terminal in this analysis. Non-automated terminals (dummy 0) are the majority in the HLH range making up a total of 67% compared to 33% of automated terminals (dummy 1).

6.8.1.2 Transshipment

Transshipment occurs when traffic moves vessel-port-vessel, while gateway traffic moves from vessel to hinterland. The mixture of cargo in a container terminal determines the function with which the particular terminal is associated. This function is also very reliant on the vessels calling at the terminal which could be in a manner of transshipment to (from) short-sea vessels from (to) a deep-sea vessel or/and from (to) barge to (from) deep sea.

According to Lu (2014), gateway terminals mainly handle exports and imports and have a high connection to the hinterlands while transshipment terminals largely redistribute cargo to a final destination through other means of transportation, e.g. through a hub and spoke network. Other container terminals can be categorized as a blend of the two extreme cases. While some Asian CTs, e.g in Singapore, handle higher volumes of transshipment, the CTs at the HLH range are a good example of terminals handling a low proportion of transshipment (in proportion to their total volumes), with the exception of the Bremerhaven terminals. In the case of the HLH range, the analysis aims to compare and contrast the levels of transshipment between the CTs, not divide between the strict distinction of 'gateway' and 'transshipment'. As mentioned by Zeng, Feng, and Chen (2017), transshipment has become an important element for indicating the competitiveness of a port (terminal).

Yard capacity is determined by dwell time, among other factors (refer to chapter 4). Transshipment containers have a low dwell time (Cochrane, 2008) and thus positively contributing to capacity and efficiency. Transshipment terminals tend to

store containers by the quayside to facilitate re-export on the other hand, gateway terminals tend to store the containers in the yard for longer, which has an impact on efficiency. According to Lu (2014) and Zeng, Feng, and Chen (2017), a high amount of transshipment is associated with higher efficiencies in container terminals. Cullinane et al. (2006) mention that gateway ports are likely to exhibit lower technical and scale efficiency scores than transshipment ports. This is attributable to the footloose nature of transshipment versus gateway cargo, resulting in increased efficiency. As a consequence of fast-changing and overlapping hinterlands, cargo assurance in gateway ports and terminals is reduced and potentially along with efficiency.

By definition, a terminal is said to be a transshipment terminal when the measured transshipment is greater than 50%. A majority of the terminals in the Hamburg Le Havre region exhibit transshipment incidents lower than 50%. For this study, transshipment is used as a means of comparison between the container terminals in discovering whether, in the HLH region, the level of transshipment is sufficient in explaining the levels of efficiency. A threshold of 25% is logically used to discriminate between the levels of transshipment between the container terminals. Percentage amounts exceeding 25% indicate higher levels of transshipment while lower than 25% indicate levels of lower transshipment. As indicated in Table 6-6, the average transshipment for all the terminals in the region is 37%, with the lowest levels of 8% and the highest being 55% for the year 2018. In the Kruskal-Wallis analysis, terminals with higher transshipment are represented with a dummy variable 1 versus dummy 0 for lower transshipment.

6.8.1.3 Inland navigation

Cochrane (2008) shows how transshipment can be related to inland navigation. Generally, higher gateway containers can indicate a potential of higher barge cargo, depending on the modal split. A shift towards inland waterways is considered a global environmentally-friendly action and thus its importance has grown more than ever as authorities advocate for change. Rogerson et al. (2020) demonstrate that the barriers to increased modal split differ from a perspective of many factors.

Terminals with inland waterways require differing and additional equipment as well as a unique layout different from the traditional terminals without inland waterway transport. According to Wiegman and Witte (2017), this has a major impact on efficiency. Whether or not barges call at the deep-sea going side or not is another

factor to take into account. This can influence the technical efficiency of a terminal. The direction of the impact to overall technical efficiency depends on the utilization rate of the inland waterways service for that particular terminal, which in turn also depends on many factors, as identified in Rogerson et al. (2020). The modal split share dedicated to container cargo carried through inland waterways is used as a measure for inland navigation. Table 6-6 illustrates that the average terminal in the HLH region handles inland water transportation of 20%, while the lowest terminal handles 2% and the largest 36% for the year 2018. For the Kruskal Wallis model, the dummy variables 0 and 1 are used to represent lower levels and higher levels of inland navigation respectively. For this analysis, a threshold of 25% is used to indicate whether or not the particular terminal is regarded as a terminal with a considerably high amount of barge handling or not.

6.8.1.4 Age of the terminal

The age of the terminal can determine the extent of connection to the supply chain as well as the reputation in the industry. Aronietis, Van de Voorde, and Vanelslander (2010) identify port reputation as one of the important characteristics of competitiveness. Reputation goes hand in hand with the quality of service delivery which is also directly linked to the infrastructure and superstructure of the port. This will thus have an influence on efficiency in that particular terminal.

For the analysis, the age of the terminal is determined by the years over which the terminal (not specifically the operator), has been in existence. Table 6-6 indicates that the youngest age of the terminals is 3 years and the oldest age of 52 years in 2018. For the Kruskal Wallis model, dummy variables 0, 1, 2, 3, and 4 represent very young, young, old, and very old categories respectively.

Table 6-6: Description of the external variables

External variable	Description	Measure	Data form	Source
Automation	Total yard handling equipment in 2018	The total amount of vessel to quay equipment, quay to stacking yard, from stacking, and yard to gate equipment	Kruskal Wallis Non-automated- 0 Automated-2 $\Sigma=14$ (67%) Automated-1 $\Sigma=7$ (33%) Tobit Non-automated- 0 Automated-1	Terminal websites and contacts.
Transshipment	Proportion of containerized transshipment cargo in 2018	Total percentage of transshipment at port level applied to terminal level.	Kruskal Wallis Lower transshipment - 0 Higher transshipment - 1 Tobit data $\mu=0,37$ $\leq 0,08$ $\geq 0,55$	Port contacts and websites.
Inland Navigation	Proportion of containerized inland navigation cargo in 2018	Total percentage of containers transported through inland waterways at port level applied to terminal level	Kruskal Wallis Lower Inland Navigation - 0 Higher Inland Navigation - 1 Tobit $\mu=0,20$ $\leq 0,02$ $\geq 0,36$	Port contacts and websites.
Age of terminal	The age in 2018 from when the terminal began operations	The difference in years between 2018 and the initial date of terminal operation.	Kruskal Wallis Very young- 0 Young- 1 Old- 2 Very old-3 Tobit $\mu=19$ ≤ 3 ≥ 52	Terminal websites.

6.8.2 Key assumptions for parametric testing

Table 6-7 illustrates a summary of the basic assumptions required (indicated by the tick) for parametric tests. Parametric tests are predicated upon fundamental assumptions for the results of the regression to be valid. The diagnostic tests for a Tobit are demonstrated in Appendix and discussed below.

Table 6-7: Assumptions of a parametric test

Key assumptions	Type of regression			
	General linear	T-stat	OLS	Tobit
Linearity	✓		✓	✓
No Multicollinearity	✓		✓	✓
Normality of errors	✓		✓	
Independence	✓	✓	✓	✓
Large sample size	✓	✓	✓	✓
Truncated regression				✓
Continuous/ordinal scale		✓		

6.8.2.1 Linearity

The assumption of linearity proposes that the dependent variable is linearly related to the independent variables (and other influencing factors). The scatterplot in Appendix A.1 indicates the linearity (with minor deviations) among the variables (efficiency scores versus the external variables).

6.8.2.2 No autocorrelation and multicollinearity

For the ability to carry out a Tobit test, there must be little or no correlation (≤ 0.70) between the explanatory variables, and that there must be a high correlation (≤ 0.30) between the explained and the explanatory variables. Autocorrelation is the degree of correlation between the values of the same series demonstrated in Appendix A.2. Correlations can also assist in observing multicollinearity, a situation that arises when the predictor variables are highly correlated among themselves, (Kutner et al., 1997).

The correlation between the explanatory variables and the explained variable is violated between dependent variables by almost all the variables (there is a low correlation) with the exception of transshipment as well as automation versus the efficiency scores. Automation violates the rule only concerning OVRSTE and IVRSTE. This suggests that an increase in the level of automation, as well as an increase in transshipment, can be expected to tend to increase efficiency.

There is not much evidence of autocorrelation, only transshipment is strictly and highly related to automation.

6.8.2.3 Independence of errors and Homoscedasticity

This assumption states that there should be independence of the errors, this indicates that the residuals should not be highly correlated. To test for this, a scatterplot is demonstrated in Appendix 6.3. The rule of thumb is that the predicted values, as well as the standardized residuals, are between the values of 3 and -3. As can be seen in Appendix 6.3, this assumption is met.

When the assumption of independence is not satisfied, then heteroscedasticity (absence homoscedasticity) is said to be present. On the other hand, when the assumption of independence is not satisfied, that indicates the presence of autocorrelation, (Poole & O'Farrell, 1971).

6.8.2.4 Non-normality

The test of normality examines whether the dependent variables are statistically significantly different from a normal distribution. The assumption of normality also applies to the error terms which is important in indicating the omitted factors. It is evident on all tests of normality in Appendix 6.1- Appendix 6.4 that the dependent variables are not normally distributed, (Kutner et al., 1997). The non-normality of the dependent variables is as a result of the bounded nature (min,0 and max,1) of the efficiency scores. The normality tests indicate that the dependent variables are statistically significant, P-values $0.00 \leq 0.05$ on both, the Kolmogorov-Smirnov and in the Shapiro-Wilk test, suggesting that the null hypothesis was rejected.

6.8.3 Discussion of statistical results

The external variables are reported from two perspectives, first are the external variables directly observed in this study and secondly reported from indirect observation, through past literature.

6.8.3.1 External variables: Direct observation

The Tobit model in parallel with the Kruskal-Wallis test were selected for this analysis and the results are represented in *Table 6-8*. The fundamentals of the models and the reason for choosing the tests were provided in chapter 4. The main reason for the appropriateness of the Tobit model is the ability to cater to the nature of the DEA

scores which are bounded between 0 and 1. This boundedness of the efficiency scores also leads to the non-normality of the data distribution of the dependent variables. The inability to meet this assumption makes it impossible to carry out a majority of parametric analysis, the Tobit thus comes as a rescue for such encounters (Chaouk et al., 2020 and Banker & Natarajan, 2008). On the other hand, is the non-parametric statistical method, the Wilcoxon-Mann-Whitney test, and the Kruskal-Willis test. The advantage of the Kruskal-Willis over the other is its capability to cater to more than two groups of variables.

The results in *Table 6-8* indicate the significance levels of the variables in explaining the efficiency levels determined by the DEA models. The alpha value (p-value, shown on the table) is compared with five percent (0.05) significance level. The null hypothesis is rejected if the p-value is less than the 5 percent significance levels. Rejecting the null is an indication that the variations in the independent variable are associated with the changes in the dependent variable. On the other hand, a p-value greater than five percent indicates insufficient evidence that there is a non-zero correlation.

Table 6-8 demonstrates the results of both, the Kruskal – Wallis models, and the Tobit model.

Table 6-8: Significance levels of the external variables

DEA Model		Ocrste		Ovrste		Oscale		Icrste		Ivrste		Iscale	
	N	21		21		21		21		21		21	
	Mean	0,83		0,94		0,88		0,83		0,97		0,86	
	Min	0,43		0,59		0,43		0,43		0,65		0,43	
	Max	1		1		1		1		1		1	
Statistical model		Tobit test	Kruskal-Wallis	Tobit test	Kruskal-Wallis	Tobit test	Kruskal-Wallis	Tobit test	Kruskal-Wallis	Tobit test	Kruskal-Wallis	Tobit test	Kruskal-Wallis
External variables	Automation	0,05**	0,22	0,38	0,21	0,07	0,16	0,05**	0,22	0,58	0,82	0,06	0,22
	Transshipment	0,5	0,8	0,31	0,26	0,82	0,64	0,52	0,8	0,7	0,26	0,54	0,8
	Inland_Nav	0,2	0,27	0,29	0,18	0,36	0,31	0,19	0,27	0,3	0,18	0,29	0,32
	Capacity utilization	0,01**	0,05**	0,02**	0,06**	0,12	0,08	0,011**	0,05**	0,05**	0,08	0,04**	0,06
	Age	0,8	0,86	0,45	0,57	0,7	0,83	0,17	0,86	0,4	0,66	0,95	0,83
	GDP	0,2	0,29	0,54	0,61	0,25	0,16	0,23	0,29	0,71	0,76	0,24	0,29

Where ** represents asymptotic statistical significance at the levels of 5%

Output/input-oriented constant returns to scale technical efficiency (O/ICRSTE)

Output/input-oriented variable returns to scale technical efficiency (O/IVRSTE)

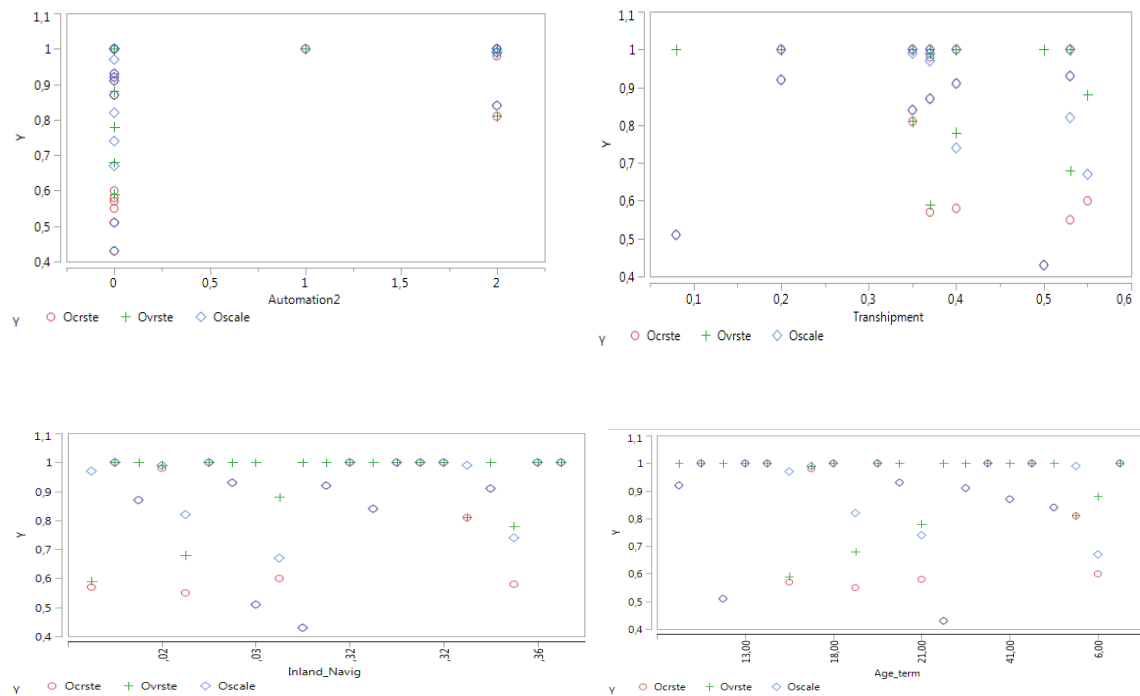
Output/input-oriented scale efficiency (O/ISCALE)

A simple regression is implemented to the efficiency scores of the HLH region. The findings indicate that automation is the only external variable that significantly explains the efficiency levels in the HLH container terminals. The other external variables have demonstrated statistical insignificance to explain the differences in efficiency scores in the HLH group.

Automation shows statistical significance in the CRSTE model ($p\text{-value} > 0.05$). In the scale model, the $p\text{-value}$ is ($=0.07$) and (0.06) in the output and input scale models respectively, which can also be said to hold some level of statistical significance as this is close to the critical value of 0.05 . This indicates the importance of automation for container terminals in general (CRSTE model) and in agreement with past literature. Indicated in *Figure 6-6* are the efficiency scores (only the output-oriented) in relation to the external variables. In the CRSTE model, it can be seen that automated terminals are likely to achieve high efficiency while on the other hand, the lower efficiency scores are to be found in the non-automated container terminals. The statistical significance of automation in the scale efficiency model indicates the importance of size and the role of automation in assisting larger container terminals to reach increased efficiency levels (explained later).

As already proven in past literature, for the general container terminal (CT) view, all these external variables have shown to affect efficiency. The statistically insignificant results in the rest of the external variables are therefore surprising, however, because these CTs operate in a very similar environment, showing differences by characteristics will not always be clear but can however show some evidence of some relation. In *Figure 6-6*, transshipment, inland navigation and age are not following any clear pattern, which indicates the reason for the lack of statistical significance. With a greater sample, from a diverse environment, these would be clearer to observe. In the HLH context, transshipment and inland navigation are all similarly experienced by all the container terminals operating in the same port (transshipment and inland navigation). This is the main reason for the difficulty to observe any differences in this sample. This, however, indicates that container terminals that operate under a similar environment can indeed be compared without the influence of external factors, which justifies that the efficiency scores obtained from this analysis are usable and reflective for benchmarking.

Figure 6-6: Efficiency vs external variables



Source: Owner's compilation

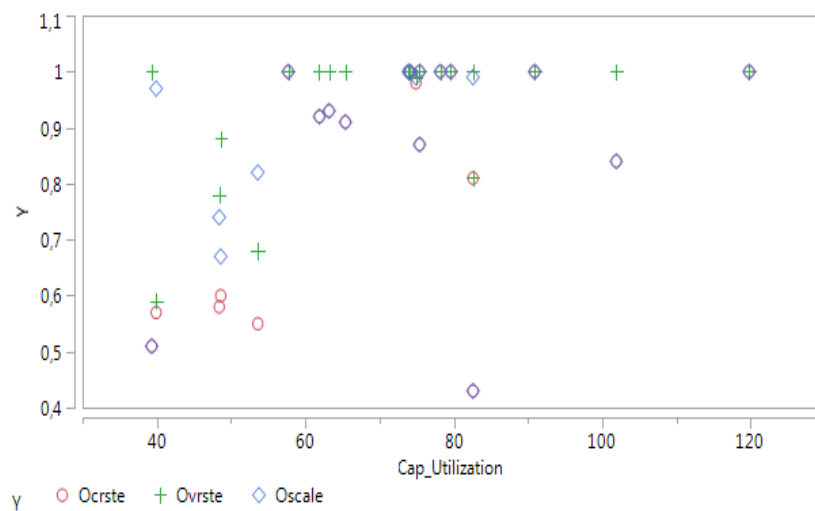
6.8.3.2 External variables (indirect observation)

Drawing from the literature of container port and terminal efficiency, numerous factors have been proven to stimulate specific effects on efficiency. Some of those factors are cumbersome to quantify and even when they are quantifiable, measuring the relationship with efficiency may require extensive research. To find the relationship of some further factors to efficiency, this study uses pre-existing literature and suggests a relationship between the factors and efficiency as a general rule.

6.8.3.2.1 Capacity utilization

Capacity utilization is inherently reflected in the efficiency scores and thus not examined statistically but graphically to observe the patterns relating to the efficiency scores. In Figure 6-7, it can be observed that indeed, lower levels of efficiency are visible in the lowly utilized container terminals and not in the container terminals with high utilization rates. The evidence illustrates that the CT with the lowest efficiency score is indicated by the scale efficiency model and it is also the terminal with the lowest capacity utilization rate.

Figure 6-7: Capacity utilization



Source: Owner's compilation

6.8.3.2.2 Competition

Competition is expected to hold a positive relationship with efficiency (van den Broeck et al. (1994). On one hand, Cullinane and Wang (2010) counter-intuitively suggest that *ceteris paribus* strong local competition of ports can reduce efficiency. The sample of evaluated ports included Rotterdam, Hamburg, Antwerp, Bremen/Bremerhaven, Los Angeles, and Long Beach. On the other hand, ports facing less competition are found more efficient. This is because during the time of the analysis, these ports heavily invested port facilities and infrastructure, in response to competition, and given that reason, efficiency then has been reduced theoretically. Van den Broeck et al. (1994) however associate this aspect with the short run.

6.8.3.2.3 Ownership

The reduced internal competition when it comes to public service ports on one extreme and private service ports on the other extreme is one of the contributors to reduced efficiency. According to Cullinane et al. (2006), managerial incompetence is rather the main cause of the association with lower efficiency than the merit of just being public or private. Privatization has been widely accepted when a port is to increase competitiveness, (Antoniou et al., 2019). Specifically concerning port efficiency, Lu (2014); Cullinane, Song, and Gray (2002); Cullinane et al. (2006) and Tongzon and Hang (2005) identify a positive relationship between efficiency and privatization. Cullinane et al. (2006) specify a positive relation between technical and

scale efficiency and private-sector participation. On the other hand, Tongzon (2002) finds that full privatization hinders efficiency but argues that partial participation of the private sector is more preferable. In addition to this, Lu (2014) suggests that the operator of the container terminal also has an influence as the global operators have different management practices.

6.8.3.2.4 Concession agreements

Port authorities create and grant to the private sector long term concession contracts (usually ranging between 25 to 40 years) where the interests of the port are prioritized and which include higher performance and efficiency (Antoniou et al., 2019). Ferrari, Puliafito, and Tei (2019) show a model where both the terminal operator and the port authority benefit from concession agreements based on performance and thus from increased efficiencies.

6.8.3.2.5 Shipping cycles

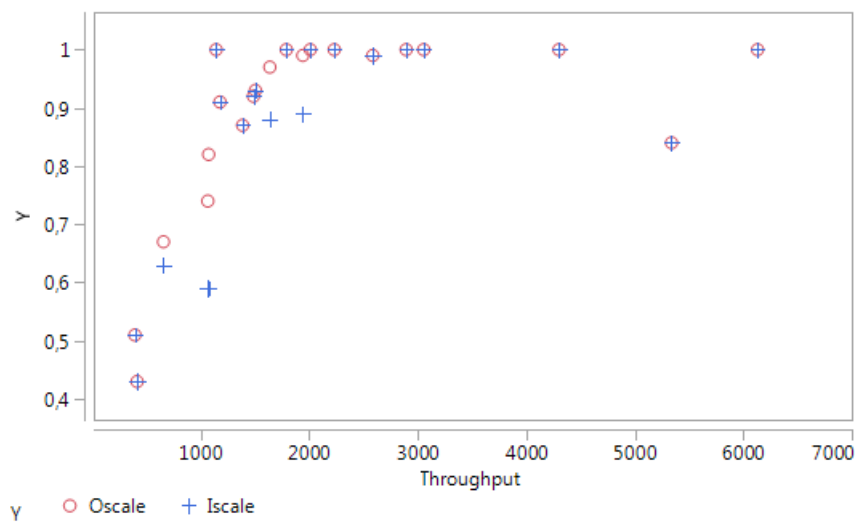
Shipping cycles determine the length of time an average ship spends at a terminal. Any terminal aims to shorten the time spent at their terminals and thereby improving efficiency (Bichou, 2011).

6.8.3.2.6 Terminal size (throughput)

There is a leniency that suggests that larger (by throughput) ports have more efficiency than their small counterparts due to larger-scale production (Cullinane & Khanna, 1999; Cullinane & Wang, 2005, (Cullinane et al., 2006) Martinez-Budría, Díaz-Armas, Navarro-Ibañez, and Ravelo-Mesa (1999) and Wang, Cullinane, and Song (2005); Caves and Christensen (1988) and De Neufville and Tsunokawa (1981).

Figure 6-8 indicates the relationship between throughput and scale efficiency of the HLH container terminals, which confirms the claims of the authors above that there is a positive association between throughput and efficiency.

Figure 6-8: Scale efficiency vs throughput (TEU 000's)

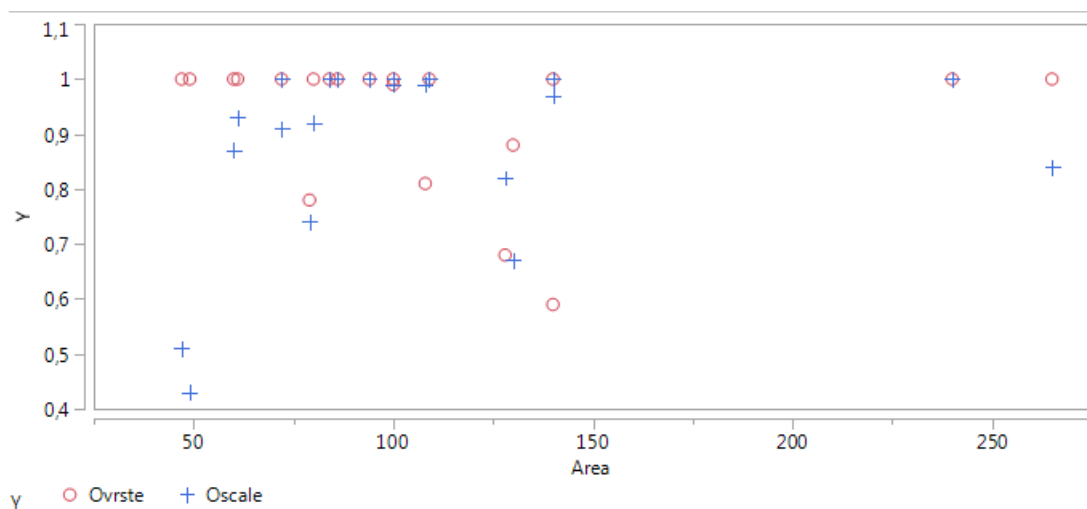


Source: Owner's compilation

6.8.3.2.7 Terminal size (area)

Prior findings associate efficiency with the container yard area. Turner et al. (2004), Tongzon and Heng (2005), Niavis and Tsekeris (2012) indicate that larger terminal sizes tend to operate more efficiently than smaller terminal sizes. According to Lu (2014), this is because big scale ports often use large, specialized equipment than smaller ports (and terminals), enabling the terminals to handle large vessels arriving at their terminals in the best possible manner. Lu (2014) also mentions how terminals are heavily invested in extra capacity which may temporarily lead to lower levels of efficiency but a situation that can be expected to change with time as the terminal adjusts. According to Figure 6-9, there is no clear relation, it can be witnessed that both, small and large terminals can be fully efficient. However, it can also be confirmed that none of the large terminals is on the low end of efficiencies.

Figure 6-9: Efficiency vs Area (hectares)



Source: Owner's compilation

6.8.3.2.8 Water depth

Another efficiency determining factor often taken into consideration in ports and terminal studies, as shown in chapters 2 & 5 is the water's depth. This indicates the sizes of vessels able to visit the terminal. Chen et al. (2018) state how ports can now build artificial water depth. An indication that location becomes less of a concern regarding water depth. The author finds that deep waters have a positive effect on efficiency.

For the HLH container terminals, the water depth is almost similar for all the container terminals taken into account and thus, it was not taken into account for the analysis.

6.8.3.2.9 Container sizes

Chapter 4 makes a distinction between different container sizes. Throughput is usually measured in TEUs, thus when a forty-feet equivalent unit (FEU) is moved across the quay face, it is counted as two. This would indicate that a transshipped FEU container is counted as four TEUs (Cochrane, 2008). In container terminals where the proportion of TEUs is higher, this indicates that the number of lifts and handling is higher, thus imposing a negative effect on efficiency compared to FEU container types.

6.8.3.2.10 Operating hours

Operating hours are mentioned in past literature to influence the level of capacity and throughput. Wiegman and Witte (2017) compare container terminals 24/7 and those that operate from Monday to Friday from 08.00 to 17h00. The findings indicate that operating for fewer hours will negatively affect the efficiency of a terminal. If efficiency is measured taking into account the terminal's capacity versus throughput then a terminal operating 24/7 will achieve higher throughput for the same amount of input resources.

6.8.3.2.11 Other factors

Conclusions on the relation to the efficiency of other factors such as the container mix, work practices, crane efficiency, vessel size, and cargo exchange are based on Tongzon (1995), where the determinants of the container terminal efficiency and performance are analyzed.

Based on these findings, some preliminary suggestions can be given regarding the extent of the effect by certain variables to the levels of efficiency, demonstrated in Appendix 6.6.

6.9 Conclusions

In this chapter, the methodology to estimate container terminal technical and scale efficiency is specified and estimations are compiled. This chapter reflects all the previous chapters where the aim is to evaluate terminals' efficiency, taking into account the changes that have occurred in the sector. The continuous changes in the industry show the importance of resilience in ensuring business survival. The ability of a container terminal to efficiently serve its customers is important, especially in a highly competitive environment like the HLH. The importance of capacity investment and its role in efficiency is discussed in detail. In addition to capacity building, the vitality of resource management and the continuous observation of the external factor are extensively highlighted. The analysis shows that, although terminals can be looked at under one roof, all of them are unique. Therefore, as much as efficiency evaluation offers some relative insights, the ultimate decision always needs to consider the terminal at hand.

The derivation of the analysis from terminals of the same region justifies the reliability of the efficiency scores as it translates to a stronger standardization of environmental factors. Making use of both, the CCR and BCC models in the container terminal context help provide detailed results to delineate possible scenarios. On the one hand, a scenario where there are limited business economic forces, such as competition and so no distinction made regarding the sources of efficiency, is a situation depicted by the CCR model. On the other hand, where a distinction between efficiency sources is accounted for, the BCC can well demonstrate the possibilities for the container terminal operators. In addition to this, the models also substantiate scenarios when the terminals are assumed to be output maximizers (output-oriented) compared to input minimizers (input-oriented). Assuming that the terminals have different strategies and that the focus may deviate pending a situation at hand, it adds value to clarify both perspectives.

The CCR and BCC models indicate that high-efficiency levels in the HLH are primarily a consequence of the desirable management of resources. The efficiency of the terminals is negatively affected by scale, which is similar to what is observed in the Malmquist model, particularly at the individual terminal level.

Automated terminals are observed to have lower 'wastage' of resources than non-automated terminals at the waterside sub-system. Both, automated and non-automated terminals are examined to improve yard utilization. Regardless of the level of automation, what should be kept in mind is that the terminal resources (input factors) may vary from one terminal to the other (differences are not catered for in this analysis). For instance, the quay cranes may have different features regarding capacity (tandem lifting potential, trolley features), quay walls vary in the outline, the yard area may have a different outline, it may also be used for different other activities in other terminals, yard handling equipment choices are different as shown in chapter 4. All these factors will make the recommendation to 'adjust' subjective to the terminal combination of factors.

Although 'wastage is wastage', the other reality is that there are some factors that cannot be controlled through terminal management but which nonetheless affect efficiency. The Kruskal Wallis and the Tobit models are used to test the significance of four environmental variables. Naturally, given that these terminal operators are located in the same regions, they are affected by a similar business environment, it nevertheless remains interesting to evaluate within the region, the existence of any

external factors which could be significant enough to lead to a variance in the performance of these terminals. All the identified external variables have been in the past literature proven to affect terminal efficiency to some degree, however, in this analysis, only automation is significant enough to explain the variations in efficiency levels. This suggests that apart from the efforts of managing the container terminal resources and scale, it is also beneficial to explore opportunities brought forward by investing in new developments and technologies so as improve performance and resilient infrastructures in the sector.

It should be noted that the aim of measuring efficiency is not to instruct terminals on achieving high-efficiency scores but it is a way of indicating a way to sustainable competitive advantages. Thus, if the strategic goals of a terminal are met, then there remains no obligation to consider low-efficiency scores as a hazard.

Chapter 7: The analysis of the cost structure at a container terminal: Antwerp case

Recent literature has emphasized the importance of transport costs and infrastructure costs on trade (Park & An, 2020). Container terminal costs are one of the most important costs in the maritime supply chain. Disruptions at container terminals have been proven to lead to inefficiencies. Inefficiencies have been proven costly to the terminal operator and the rest of the supply chain. According to Clark, Dollar, & Micco (2002), shippers who improved their efficiency from the 25th to the 75th percentile reduced costs by 12%. The study found that inefficient ports tended to increase handling costs. The aim of this chapter is to evaluate the cost structure of two terminals identified to be highly efficient in chapter 6.

The HLH region is characterized by the shared hinterlands between the ports, which indicates the high competition for the players located in the region, making the costs important among other factors. Port players have different objectives and strategies but the players with a large customer base and who are influential (e.g shipping lines and terminal operators) have a high potential to shape the port. The concentration of global container terminal operators (CTOs) such as PSA, HPH, DPW, etc has shaped how vertical integration occurs with the shipping lines (SLs). The alliances and mergers in this sector have made a balance for competition even harder not only for SLs and CTOs but also for port authorities (PA) and the hinterland players. All these factors will have an impact on how the cost structure is shaped and eventually on the pricing of the terminal, a scenario represented in Figure 7-1.

Figure 7-1: Container cost identification



Source: Own compilation

In cost theory, a firm uses a set of inputs to achieve the output, a similar scenario to what was observed in the HLH case. In the previous chapters, the typologies of various CTOs are generalized (chapter 4). Later on, the efficiency and total factor productivity levels of specifically the HLH range of container terminals were evaluated in-depth (Chapter 5 and 6).

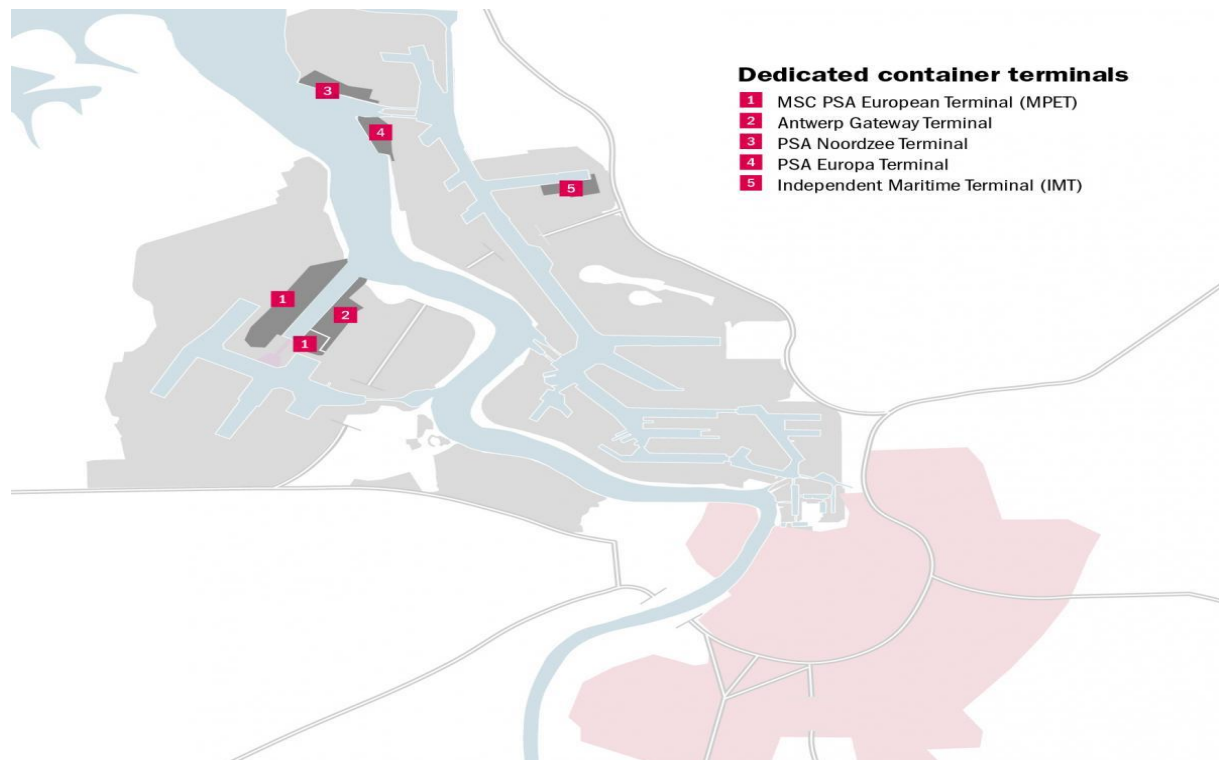
In this chapter, a generic cost model for handling containers is demonstrated and calculated to where cost elements of a container terminal are illustrated and the cost for the containers assessed from the perspective of throughput (TEUs) and also in terms of total moves. The port of Antwerp is one of the port leaders, not only regionally but also globally. It is interesting to therefore evaluate the container handling costs from the perspective of this competitive player in the port sector. First, the chapter highlights the competitive advantage at the port of Antwerp. Following that, the cost structure of a generic terminal is outlined to build a background for the cost calculation. Section 7.3 specifies the methodology and the model used to calculate the cost elements for this analysis; then the method of analysis follows where the model which is adapted is specified. The section also discusses the sample selection and variables description. Following, the results and discussion from the analysis are discussed then finally the impact of costs on prices before the conclusion is reached.

7.1 The competitive advantage of the port of Antwerp

The Port of Antwerp (PoA) is home to five specialized deep-sea container terminals demonstrated in *Figure 7-2*. Port activity is often complex and involves various players and activities which are integrated differently in different ports. Apart from throughput, a port's competitive position can depend on how the service is provided. Although operating in usually very competitive and rapidly changing environments, global ports are often regarded as the drivers of economic growth. Many factors driving global competition affect ports, including mega vessels, market entry of integrators, network linkages between port players, unitization of general cargo, hinterland transport networks, etc. Haezendonck, van den Broeck, & Jans (2011) realize that Northwestern European ports cannot expect to compete based on only low costs but should rather develop higher-order sources (non-cost factors) of competitiveness. By prioritizing, in addition to costs also a wider range of service, modern infrastructure, facilities, equipment, and highly-skilled labour, the PoA seeks

to remain competitive (Haezendonck & Langenus, 2019). The port is run through the landlord port model where the state owns the port infrastructure and the services are offered by private operators through usually long-term lease agreements (concessions); more of this aspect is discussed in chapter 2.

Figure 7-2: Antwerp specialized container terminals



Source: Port of Antwerp (2020)

In academic literature, the importance of performance – in financial and non-financial terms – has been the main focus. Financial parameters have been highlighted as means of thriving through the competitive environment. Haezendonck, van den Broeck, & Jans (2011) mention that cost-related elements which benefit various actors in a port cluster are important determinants of performance in port competition. Similarly, the findings of Rezaei, van Wulfften Palthe, Tavasszy, Wiegmans, & van der Laan, (2019) emphasize the importance of transport costs and time in determining competitiveness. Roll & Hayuth (1993) mention the importance of also non-cost factors such as service quality, labour regulations, etc to performance. Haezendonck, Pison, Rousseeuw, Struyf, & Verbeke (2016) confirm this phenomenon in the port of Antwerp where qualitative elements, as well as non-cost elements, are identified to explain more of the competitive advantage of the port cluster for general cargo and containers. The strengths

common to the firms within a port become instrumental in what becomes considered 'port specific' advantages.

Given the high productivity and the labour pool flexibility, Haezendonck et al., (2016) identify warehousing and transshipment as one of the core competency factors of the port of Antwerp. One of the unique features of the PoA is the flexible organization of the port labour system (allows firms within the cluster to share dock labour) which makes a distinction between permanent (usually white-collar labour) and non-permanent (usually the dock labour) (Bottalico, 2017). This becomes an interesting phenomenon to observe as it affects productivity and efficiency, and in turn costs. The second factor identified in the study of Haezendonck et al., (2016) is the hinterland connection through inland navigation which is owed to the geographical location of the PoA as well as its network. The PoA's location (further inland) allows that the customers of the port are closer, reducing especially transportation costs. Port hinterland connections are one element of increasing performance and competitiveness, however, according to Haezendonck & Langenus (2019), the PoA's hinterland extensions are still premature and have not yet resulted in a source of competitive advantage. Value-added activities in the PoA add to the competencies of the port, characterized by experienced operators. Specifically related to the terminal operators, the strength of the port of Antwerp lies in the manner in which cooperation between operators occurs and the support from the Flemish regional government. Lack of adaptability and uniformity of port logistics systems is however highlighted to hinder the optimization of this relation. The inter-sectoral competence among port operators is also evident in the port, evidenced by how the operators can cooperate not only with local clients but also external clients. These factors regarding the PoA are also confirmed in Haezendonck & Langenus (2019) where it is stressed that port integration although important does not necessarily automatically and immediately lead to increased competitiveness.

There is a need for PAs and operators to focus not just locally, but further on to other areas and globally to remain competitive. Terminal operator's strategies should also not solely focus on financials such as profit maximization and cost reduction but should also focus on non-financial factors such as labour skills (human capital and development), synergies (connectivity and integration), environmental costs (corporate social responsibility), etc.

7.2 The general cost structure of a container terminal

Given that the supply sector operates in a chain, resolving pressure on one link can easily transfer the problem to another point along the chain. For instance, increasing capacity by the carrier may spread costs over several containers, but this may result in increased costs for a terminal operator who needs to have the handling capacity for increased cargo. These supposedly increased costs can easily be transferred to the hinterlands and the final consumer, *ceteris paribus* (Meersman et al., 2010).

The cost structure is determined by endogenous factors and exogenous factors (discussed in chapter 4), including the advancement of technology and trade performance (Meersman et al., 2010) on the one hand. On the other hand, how a port operates will shape how costs are incurred and how the port players can interact. Liu (2010) outlines port type's basic administration and operation activities, as demonstrated in *Table 7-1*. The manner in which ports operate is closely related to the manner of financing. The table shows the administrative and operational roles played by both the private and the public sectors, which can thus be translated into costs.

Table 7-1: Basic port administration/operation modes

Labour				
Equipment				
Infrastructure Superstructure				
	Public service port	Tool port	Landlord port	Private service port

Port authority costs

Stevedore costs

Source: based on Liu (2010)

In deciding on which port assets should be at the hands of the private sector and which ones should be under the control of the state, port planners have to evaluate some inherent risks and potential returns. Although costly to maintain, some long-

lived assets such as breakwaters, channels, turning basins, are very likely to remain in the hands of port authorities due to the marginal benefit derived. Provision of these services by the private sector would also likely result in monopoly profits and reduced use than economically desirable. Long-lived and high-cost infrastructures such as terminals and quay walls are however very likely to be leased out to the private sector due to the ability to balance out the marginal cost, the marginal use, and benefit of the assets. As indicated in *Table 7-1*, the private sector can provide all these services, depending on the port. The private sector also evaluates the risk and return on what service/s they can provide, taking into account the extent to which regulations are instilled relative to price setting.

In developing a cost structure for intermodal rail terminals, Wiegman & Behdani (2017) make note of seven characteristics of infrastructure investments, which form part of the cost structure of terminals. First is that the economic life of the investments, as well as the payback period, is typically long, 15 to 30 years, which generally means that these private operators would need to generate profits over a short period. Secondly, the initial investments are usually very large and lead to immediate costs, while sales are very little or not yet generated. Thirdly is that the nature of the business environment involves many stakeholders, which thus becomes political, causing delays that often cause cost implications for projects. Fourth is that the projects usually cannot be reversed once started. Fifth is that the construction projects are usually very long, incurring costs before there are sales. Sixth is the uniqueness of the project, although processes are similar in ports but the yard layout is usually different and thus giving little comparability possibilities and finally, the variable costs are usually low, and thus setting prices based on marginal costs usually is not satisfactory. There are various ways in which the charges are regardless reverted to the private sector when the PA has decided to remain to provide certain services. According to The World Bank (2007), the balance between the two service providers greatly contributes toward the competitiveness of the port.

Cost drivers in CTs are related to assets such as infrastructure, superstructure, and area of the terminal. These assets are likely to contribute towards fixed costs. The cost structure of CTs largely typically consists of fixed costs and lower variable costs, (Marek, 2019). Wiegman et al. (1999) propose a general cost structure for a terminal. The cost structure is evaluated through company size making use of personnel costs, depreciation, housing & handling equipment costs, external costs, and other costs.

The cost structure confirms the existence of scale by finding that, when the size of a terminal increases, the financial results reduce. When the size of the terminal reduces, 'other costs' and 'external costs' decrease. On the other hand, when the size of the terminal increases, there is even a larger increase in costs related to housing and handling equipment. There is a mixed relationship regarding depreciation and personnel costs to the terminal size (Wiegmans et al., 1999).

Container terminal strategies can shape the cost structure. Costs in terminals are attached to service provision and for the benefit of especially the customers of the terminal. An under-evaluation of costs may lead to severe repercussions in the short and long-run (UNCTAD, 1975). Terminal operators use cost centres to classify cost services, facilities and to address different issues. A favourable location along with the service provision (through vertical or horizontal direction) can lead to a specific cost centre. Some of the costs can be directed to the time factor, which is incurred when the service is performed. A cost centre can also be defined based on the user of the services and facilities. These costs are important in also determining the pricing structure (UNCTAD, 1975).

7.3 Method of analysis

For the estimation of costs in the transport sector, the application of cost functions dominate. On the one hand, academia uses cost functions to provide insights on scale economies and thus provide recommendations for regulation, competition, and productivity growth. On the other hand, policymakers trust cost functions for making comparisons and for decision making (Blauwens et al., 2020).

Econometric cost functions based on linear relations, quadratic, Cobb-Douglas, translog, and hedonic functions have been used to estimate costs in cargo handling. In recent literature, the cost functions are taken into account in various forms. Oum, Tretheway, & Zhang(1991) explore various literature where costs are estimated. The emphasis of the study is on the method of Caves et al. (1981) where a variable cost function, as opposed to a total cost function, is proposed for reasons related to inputs that cannot be fully optimally adjusted (quasi-fixed) over a while. Wiegmans et al. (1999) use Activity-based Costing (ABC) to assess and allocate costs to freight terminals. Tovar (2003) makes use of a multiproduct cost function. Jara-Diaz, Tovar, & Trujillo (2011) estimate costs using a long-run quadratic function. Seedah, Harrison, Boske, & Kruse(2013) propose an analysis toolkit for cargo handling costs. To reduce

costs incurred in a port in Indonesia Rahman, Sarno, & Effendi (2018) use the goal programming approach. Zhang & Liu (2020) base their cost estimation on interval and principal component analyses.

Among the methods of cost calculations is the queuing model. Zenzerović, Vilke, & Antonini (2013), use the queuing theory to find an optimal capacity of a terminal operator through estimating a total cost function. A queuing theory is a mathematical study of waiting lines to demonstrate a scenario of congestion, formation, and queues (Anokye et al., 2013). Wherever competition exists for limited resources, queueing is likely to result. The simplicity and ease of defining queues through probabilistic distributions often make this model better preferred than other models. In identifying the optimal capacity of a container terminal, the model of Zenzerović et al. (2013) takes into account the impact of the service rate on the total ship stay at a terminal, the impact of the number of berths on the total ship stay at a terminal and also the total costs of a container terminal. Thus, in this current analysis, the adaptation of the total costs formulas are adjusted to fit the current situation. Additional cost elements are taken into account to depict more of the terminal operator's activities.

The models mentioned above, made deductions about the marginal costs and cost minimizations/optimization, which requires time series data. An engineering generalized cost applied in this analysis is a basic but powerful method of cost estimation. It can be used to fit in any terminal given any structure. This thus makes the model relatable to a wider range of terminals, even outside the container sector, and offers a solid starting point for the cost calculation and cost structure analysis.

The productive factors necessary for port operations were discussed extensively in the previous chapters. According to cost theory, production costs comprise fixed and variable costs. Use will here be made of seven input variables that capture the costs incurred for production. Labor, yard area, quay wall, quay cranes, straddle carriers, and/or automated stacking cranes (ASC), material costs, and other operating costs are to represent the input factors to production (costs). For the output (service or product), the total number of handlings (throughput), as well as the total number of container moves, are used to represent the production at the container terminals.

The analysis is based on data of container terminals in the port of Antwerp. Data covers two CTO's in the port of Antwerp. The estimations are based on a year (2018)

on information obtained from the terminal operators and financial statements obtained from Bel-first.

7.3.1 Model specification

The cost estimation in this chapter is adapted from the model of Zenzerović, Vilke, & Antonini (2013), where the optimal capacity planning for a port container terminal is determined. The model is based on the queuing theory to achieve the optimal functioning of the terminal. The total cost function of the model by Zenzerović et al. (2013) is an expansion of a cost model by Schonfield and Frank (1984). The total cost (C) model in Zenzerović et al. (2013) is presented as a function of berth costs (C_{ql}), cost of container cranes (C_q), costs of transport- handling equipment i.e Straddle carrier/ Automated stacking cranes (ASC) (C_s/C_{as}), Cost of labour (C_l), stacking yard area costs (C_{yrd}). The cost function thus represented as:

$$C = C_{ql} + C_q + C_{s/as} + C_l + C_{yrd} \quad 7.1$$

The model in this chapter adapts Zenzerović et al. (2013)'s model and, in addition, takes into account material costs C_m and also other operating costs C_{op} . The connotations of all the total costs are in the section that follows.

Energy and concession costs are among the crucial elements in the operation of a terminal. The elements should be included in economic analyses, primarily when strictly examining fixed and variable costs. The model of Zenzerović et al. (2013) does not consider these variables for example, which include important costs elements such as insurance, consultancy fees, marketing, indirect fees, etc. In this analysis, these cost elements are considered crucial in determining the total costs of operating a container terminal and are included.

Thus, adding these two elements express the total cost function as:

$$TC = C_{ql} + C_q + C_{s/as} + C_l + C_{yrd} + C_m + C_{op} \quad 7.2$$

The single costs are expressed in the formulas:

$$C_{ql} = Ql_m \cdot M_q \quad 7.3$$

where:

C_{qt} - quay wall total costs/annum

Ql_m - quay wall (meters)

M_q - Annual maintenance cost of the quay wall

To calculate quay crane costs, information received from the terminal operator is used.

$$\text{Cost of container cranes/month} = n_q * c_q$$

n_q - number of quay cranes

$$C_q = \left[D_0 \cdot \frac{i(1+i)^{Nd}}{(1+i)^{Nd} - 1} + M_q \right] \quad 7.4$$

where:

C_q - cost per crane/annum

D_0 - initial price of the crane

M_q - annual maintenance cost per crane

N_d - economic life of the QC

i - interest rate

The same procedure as of the QC is followed for the straddle carrier (SC).

$$\text{Cost of Straddle carrier/annum} = n_s * c_s$$

n_s - number of straddle carriers

$$C_s = \left[D_0 \cdot \frac{i(1+i)^{Nd}}{(1+i)^{Nd} - 1} + M_s \right] \quad 7.5$$

where:

C_s - cost per SC/annum

D_0 – initial price of the SC

M_s – annual maintenance cost per SC

N_d – economic life of the SC

i - interest rate

M_s – Annual maintenance cost per SC

The annual maintenance costs are :

$$M_s = ((R \cdot t) * t_m) * t_y \quad 7.6$$

where R is the hourly rate, t is the number of hours per shift (7.52), t_m is the monthly measure and t_y is the annual measure.

The same procedure as of the QC is followed for the ASC.

$$C_{as} = n_{as} * m_{as} \quad 7.7$$

where:

n_{as} - number of ASCs

m_{as} - annual maintenance cost

$$TC_l = \sum(C_l, Q_l) * EC \quad 7.8$$

$$C_l = [Sh_l, f_l] + fp_l \quad 7.9$$

where:

TC_l – Total cost of labour/annum

C_l – Cost of labour/ annum

Q_l - Quantity of labour

EC – Employer's costs paid to the regulator

sh_l – Shift rate per designation according to Codex

f_l – Average weighted shift factor

fp_l – Fixed premium

$$C_{yrd} = Yrd_c \cdot K_{yrd} \quad 7.10$$

where:

C_{yrd} - yard area total costs/ annum

Yrd_c – Yard capacity

K_{yrd} – Yard costs per unit, with formula

$$K_{yrd} = \left[D_0 \cdot \frac{i(1+i)^{N_d}}{(1+i)^{N_d} - 1} + M_q \right] \quad 7.11$$

where:

K_{yrd} – Yard costs per unit/ annum

D_0 – Initial price of yard

M_q – Annual maintenance cost

N_d – Economic life of the yard

i - interest rate

$$C_m = D_m \quad 7.12$$

The cost of material C_m is given by direct costs D_m incurred for providing the service.

$$C_{op} = IC \quad 7.13$$

Operating costs C_{op} is given by indirect operating costs IC .

The output is measured from two perspectives, the throughput by TEUs and also through total moves.

The throughput is expressed as follows:

$$\sum TEU = \sum H \quad 7.14$$

The sum of TEUs $\sum TEU$ is given by the total of handled containers $\sum H$ per annum.

$$\sum MV = \sum (M_{vs} + M_{bg} + M_{rl} + M_{tr} + M_{sc}) \quad 7.15$$

The total number of container moves takes into account the productive and unproductive moves of the containers. It takes into account the annual total moves related to vessel moves M_{vs} , barge moves M_{bg} , rail moves M_{rl} , truck moves M_{tr} , straddle carrier moves M_{sc} .

Taking into account the model of Zenzerović et al.(2013), the total cost ($TC(x)$) calculation in the current model is delineated into fixed costs (FC) and variable costs ($VC(x)$), where x is the quantity of containers handled (throughput):

$$TC(x) = FC + VC(x) \quad 7.16$$

where

$$FC = C_{yrd} + C_{ql} + C_q + C_{s/as} \quad 7.17$$

$$VC = TC_l + C_m + C_{op} \quad 7.18$$

Therefore, the average costs per container AC_{teu} expressed as:

$$AC_{teu} = \frac{\sum FC}{\sum UCAP_x} + \frac{\sum VC}{\sum UCAP} \quad 7.19$$

The average cost per container is given by the fixed costs in relation to capacity utilization $\sum UCAP_x$ given the throughput handled (x) and the variable cost proportion given the total throughput handled (or the total capacity used) $\sum UCAP$.

And the average costs per move AC_{mv} expressed as:

$$AC_{mv} = \frac{\sum FC}{\sum mv_x} + \frac{\sum VC}{\sum mv} \quad 7.20$$

Another alternative for measuring the output is expressed through the number of total moves of the containers. The fixed costs demonstrate the costs necessary and unavoidable regardless of the number of containers handled while the variable costs illustrate the costs associated with the changes to the throughput handled in the CTs. AC_{tcu} indicates the average cost per container given by the proportion of the total costs in relation to U_{cap} , capacity utilization (throughput given the maximum capacity) and the variable cost proportion of the throughput. The average costs are also expressed through the total number of moves AC_{mv} which indicates the average cost per move.

7.3.2 Sample size selection

At the port of Antwerp, two container terminals referred to, for this analysis as container terminal 1 (CTO1) and container terminal 2 (CTO2) are selected to represent the Antwerp container ports.

7.3.3 Description of the data

The overall summary of the cost elements is illustrated in *Table 7-2*.

7.3.3.1 Yard area costs

In terminal operations, a yard plays a vital role in acting as a nerve centre in operations (Chen, 1999). The yard costs represent the 'land' related costs and taking into consideration the costs incurred due to the yard area. Yard area total costs are calculated by taking into account the loan repayments for the yard (including the interest rates), the amount for yard improvements or replacements, and the maintenance costs. The cost elements to calculate the total cost of yard area costs are obtained from the container terminals. The data is available for the year 2018.

7.3.3.2 Quay wall

Investments to the quay wall are made by the port authority. The maintenance contributed by the terminal operator to the quay walls as part of the concession fee.

The cost is based on the maintenance costs which are based on the quay length. The cost elements are obtained from the container terminals for the year 2018.

7.3.3.3 Quay crane

The cost of the cranes is calculated on the basis of the initial price of the crane, the economic life cycle, and the annual maintenance cost per crane. Depreciation is not taken into account. The elements to calculate the quay crane costs are obtained from the container terminal operators for the year 2018.

7.3.3.4 Straddle carrier

Similar to the quay cranes, the costs for the straddle carriers are calculated based on the initial price of the straddle carrier, the economic life cycle, and the annual maintenance cost per straddle carrier. Depreciation is not taken into account. The elements to calculate the straddle carrier costs are obtained from the container terminal operators for the year 2018.

7.3.3.5 Automated stacking cranes (ASC)

Similar to the quay cranes, the costs of ASCs are calculated based on the initial price, the economic life cycle, and the annual maintenance cost. Depreciation is not taken into account. The elements to calculate the ASC costs are obtained from the container terminal operators for the year 2018.

7.3.3.6 Labour

The labour costs are taken based on the quantity of labour demanded each month; the price of labour differs according to the designation of the labour, which is calculated based on the CODEX rates regulated by CEPA at the PoA. The cost of labour is also further adjusted to take into account the additional margins paid by the TOs to the regulator (CEPA) for the service rendering this is captured by the employer's cost component. The average weighted shift factor is to account for the average of all shifts, to neutralize the shift times as this affects the labour costs. The costs also take into account a fixed premium per day, a cost unavoidable regardless of the employee's job function. The cost of labour includes both the full-time and dock labour. The cost elements were contributed by the container terminals and also a confirmation of the rates from CEPA for the year 2018.

7.3.3.7 Material costs

The cost of material is obtained from the financial statements of the TOs and includes costs that can be directly linked to the rendering of the service. These are costs related to energy; transportation; overhead costs (consultancy, IT costs, Marketing costs, facility costs incl taxes & insurance, etc) and concession costs. The cost element is obtained for the year 2018.

7.3.3.8 Other operating costs

These are indirect operating costs that exclude the costs directly related to the cost of handling the containers and also excludes the cost of labour. This element is obtained from the financial statements of the container terminals.

7.3.3.9 Throughput

The throughput is measured in TEUs to capture the total amount of containers handled in 2018.

7.3.3.10 Total container moves

The total moves are taken into account, productive and unproductive, related to vessel moves, barge moves, rail moves, truck moves and straddle carrier moves. This output measure is important in that it captures the activities of all the subsystems of the terminal.

Table 7-2: Cost elements

	Variable (EUR/annum)	Symbol	Description	Sample size	Source of data
Cost measures	Yard area	C_{yrd}	Total costs (€) of the yard area (hectors) per annum (2018)	2	Container terminals
	Quay wall	C_{qt}	Total costs (€) of the quay walls (meters) per annum (2018)	2	
	Labour	TC_t	Total costs (€) of labour (number) per annum (2018)	2	
	Quay crane	C_q	Total costs (€) of the quay cranes (number) per annum (2018)	2	
	Straddle carrier/	C_s/C_{as}	Total costs (€) of the straddle carrier or the	2	

	Automated stacking cranes (ASC)		automated stacking crane (number) per annum (2018)		
	Material costs	C_m	Total costs (€) of material per annum (2018)	2	
	Other operating costs	C_{op}	Total costs (€) of operating per annum (2018)	2	
Output measure	Throughput	$\sum_{teu}$	Total number of containers handled in TEUs per annum (2018)	2	
	Total moves	$\sum_{mv}$	Total number of moves per annum (2018)	2	

7.4 Results and discussion

To make business decisions for the present and future moment, it is vital for any business to evaluate the cost incurred and understand the business's cost structure.

7.4.1 The total average costs

Generally, as described in chapter 4, the technical process for handling containers involves three sub-systems, whose service requires inputs that can be described aggregately through labour, space, facilities, and equipment. All these factors are necessary to achieve the throughput (moves) of the container terminals. Clark et al (2002) claim that a rise in transportation costs by 10 percent leads to a reduction in volumes by more than 20 percent.

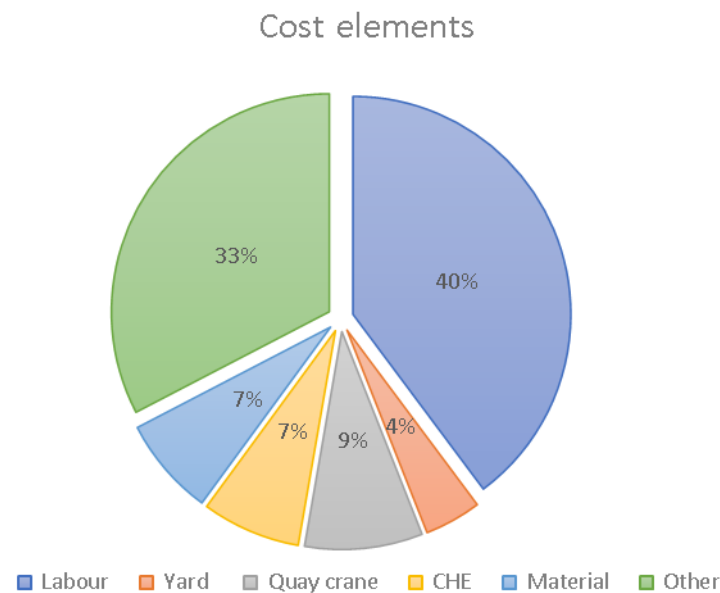
Wiegman et al. (1999) observed a decline in costs as the total of handlings increase. A small terminal with an average of 2.500-5.000 handlings incurs an average total cost of €30-€40 per handling while a larger terminal with about 20.000 handlings incurs about €15-€20 per handling. This indicates that an increase to about 75% of terminal size can lead to a reduction in costs of about 63%. In Vanelslander(2005), operating costs for terminals with varying sizes were measured where the findings show that a terminal in the port of Antwerp with a throughput of 1 655 341 TEUs incurred €22.36 per container and a terminal with a throughput size of 80 3103 TEUs incurred €29.47 per container, confirmation of the former regarding the size and the costs.

The cost of the quay crane and yard handling equipment is calculated based on the loan repayments of the initial price and the maintenance cost. Costs related to space

(land) are calculated from yard area costs as well as costs incurred from the quay wall. These costs are also further coupled with other costs incurred related to labour, material costs as well as other costs incurred. All this information is received from the terminal operators.

Figure 7-3 shows the cost elements by proportion to total costs while Figure 7-4 demonstrates the proportions of the number of moves per activity in the terminal.

Figure 7-3: Cost elements to container handling



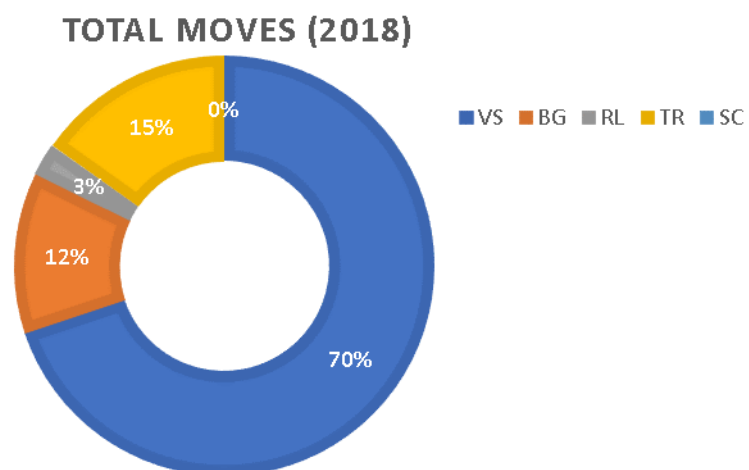
Source: Own compilation based on the Antwerp container terminals (2018)

In Wiegmans et al. (1999), an expectation (cost structure) of the relation of the cost variables to total costs is given. It is mentioned that the importance of “other costs” tends to decrease with the increase of the output. Handling equipment costs are said to be positively related to output, thus with more output, these costs increase the average cost. The direction of personnel costs is found to be somewhat mixed in relation to the output. In the terminals in the port of Antwerp, labour costs are the largest (40%) contributor to container handling costs, the yard costs being the least contributor (4%). In chapter 6, it was confirmed that the performance of the HLH region CTOs is relatively higher compared to the world average where throughput per meter of quay and throughput per quay crane are 5.45% and 3.05% (respectively) higher. The performance of the yard area utilization was however lower in comparison to the global standards. This indicates that although the costs related to

yard space are proportionally lower than other costs for the terminal operator, they are still significant to determine performance. Quay crane and CHE costs are significant cost contributors, which goes to show the effort put in place by the terminal operators in improving technologies in their terminals to improve efficiency and general performance.

A great portion of the moves in the terminals happen along the quayside where the vessel moves are the largest at 70%, followed by truck moves (15%), barge moves (12%), rail moves (3%), and lastly straddle carrier moves representing the lowest towards the total moves within these container terminal yards. First, this says much regarding the modal split share of the port, secondly the resource consumption of activities in the terminal and thus the areas of the terminal where a majority of the costs are dedicated to.

Figure 7-4: Container terminal total moves

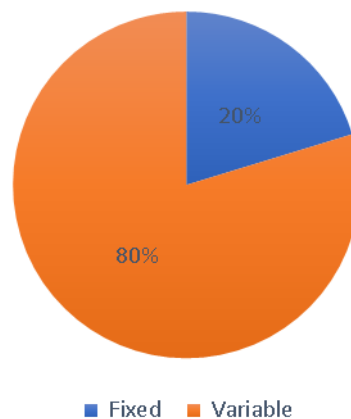


Source: Own compilation based on the Antwerp container terminals (2018)

7.4.2 The average variable and fixed costs

Figure 7-5 demonstrates the contribution of the variable and fixed costs in a terminal based on the total costs of the terminals.

Figure 7-5: Fixed vs variable costs



Source: Own compilation based on the Antwerp container terminals (2018)

There is a grey line in determining which costs are completely fixed and completely variable in the case of a CTO. This is because while some costs such as the investments in quay cranes, yard equipment, and yard space are generally fixed and are of long term, some costs incurred due to these investments such as interest repayments and at times maintenance can be regarded as variable and short-term in nature. The fixed costs collectively represent 20% of the total costs per container cost per annum, in comparison to the 80% proportion represented by the variable costs. This observation is in contradiction with the cost structure often observable, e.g in Marek (2019). This is likely due to the treatment of the labour costs in the port who are not accounted for in the payroll of the company and thus only hired in line with demand (provision of flexibility discussed in chapter 4), as opposed to having full-time labour and thus invariable with demand.

While the proportion of fixed costs is significantly lower than that of the variable costs in determining the average container cost, fixed costs signify an important aspect. First, because these costs cannot be avoided by the terminal operator regardless of the amount of containers handled. Second because they determine a proportion that is strongly tied to the physical assets of the terminal, which is necessary to achieve the goals and objectives of the terminal. Economies of scale are recognised if the long-term average cost curve decreases as output increases (Blauwens et al., 2020). Economies of scale are affected by fixed costs and less by variable costs (Cullinane & Khanna, 1999). The lower fixed costs proportion in these container terminals indicates a potential of lower economies of scale. While this is a

disadvantage, it benefits terminals by providing flexibility in adjusting the variable cost elements when it is beneficial.

Ports in the HLH region have long also been associated with technological advancements. As indicated in chapter 6, the CTO's in the HLH range have average positive technical changes over a prolonged period of the analyzed five years, which shows the investments made in the terminals and thus the importance of the fixed element to the total costs. The HLH ports are also associated with underutilization of the yard area, perhaps the smaller proportion of the yard costs could be a driver to the lower focus on the yard utilization. In chapter 6 the investment to excess capacity, due to future supply and demand expectations, was also identified as a driver of the utilization levels to some of the terminals. However, in the case of Antwerp, most terminals have shown high yard utilization.

The terminals in the PoA are an interesting point of observation, making a comparison between some important factors such as automation, which shows the advancement in technologies of the terminal, and perhaps the investments in more costly technologies in the yard. The terminals vary from low automation to high automation. This can help to make a cost comparison between the expenditure of terminals in equipment.

Variable costs are important in determining the majority of the total cost and play a large role in determining the terminal's short (and long) term success. The variable costs are usually different pending on the decision taken, from day to day (Blauwens et al., 2020). These costs can be adjustable by the terminal operator and therefore, some of these costs, for example, the labour, material costs, and other costs, can be adjustable based on the management and strategy of the CTO. From this perspective, the management of the terminal can be easily observed through the analysis of the terminal. This corresponds with the analysis in chapter 6 where the container terminals in the HLH range are considered well-managed and therefore the resources are generally efficiently utilized.

The variable cost element (not excluding the fixed) helps managers to align CTO costs along with areas that require a change within the terminal, ensuring that the terminal reduces wastage and takes advantage of cost-effective decision-making. Chapter 6 also revealed the returns to scale of the HLH container terminals. This is an element observable through the variable costs as well. Oum et al., (1991) highlight two aspects (among others) regarding a cost function, first that calculating a cost function

based on total costs and not taking into considerations the quasi-fixed costs which are neither variable nor fixed creates an error to the results and secondly, the possibility of calculating returns to scale from the variable cost function for firms not operating from an equilibrium point with regards to some inputs. Also, given the capital intensity of the sector, returns on the investments at the CT can be realized from managing the variable inputs and handling as much containers as possible given the capacity of the terminal.

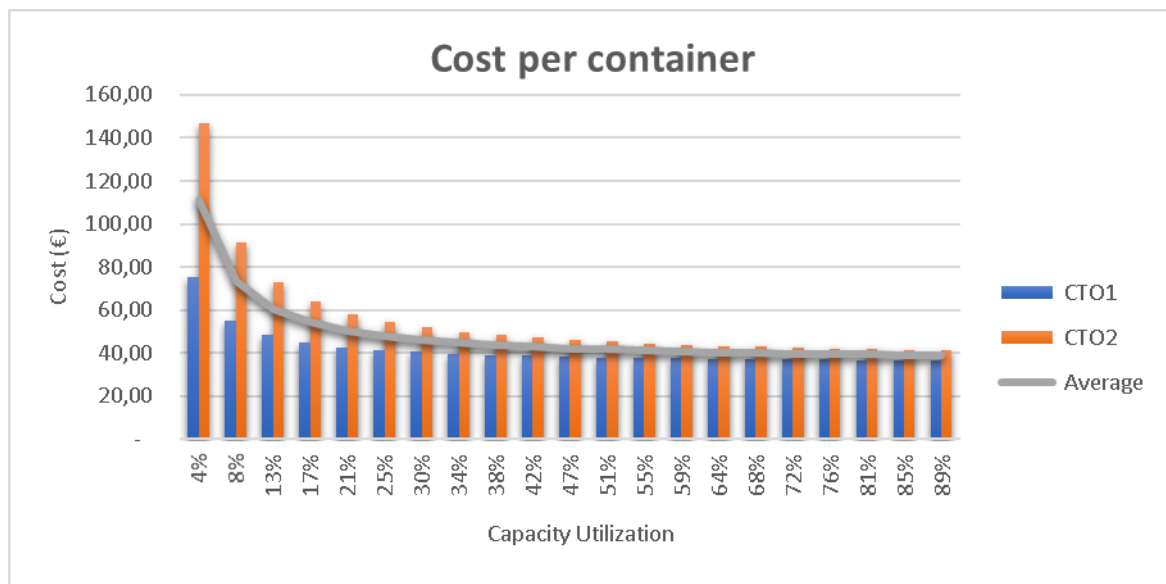
Concession fees affect the pricing strategy of a port authority and affect the costs of a terminal operator (Meersman et al., 2014). It is important that there is value created for the port authority and also for the port users, inclusive of the terminal operators. Ferrari et al., (2019) highlight the impact of concession fees on performance. It is highlighted that variable concession fees can lead to the optimization of port facilities.

7.4.3 The average cost per container and per move

Figure 7-6 demonstrates how much the costs would change if *ceteris paribus*, the throughput handled would change, represented by capacity utilization. On the other hand, *Figure 7-7* demonstrates the average container move costs.

As the amount of handled containers expands, the cost per container reduces. Based on the 2018 data, the cost per container on average is €44.47, at which point the total containers handled is approximately a representation of a utilization rate of 71% of the total capacity of 11.8 million TEUs. If an average of 11.8 million TEUs (full capacity of the terminals) was handled, then the average price would be €38.66. Theoretically, this would very likely lead to congestion (from about 70% of capacity utilization) and thus increase costs, therefore if practically this would be the case, it would be interesting to observe as the container terminals are not very far from reaching full capacity. The rule of thumb for the maximum occupancy rate for container terminals is 80%, at which point the cost would be approximately €39 per container handling based on this analysis. Although the existence of scale efficiency was determined in chapter 6, the HLH range terminals were collectively considered to have a weakness in scale efficiency in comparison to technical efficiency. This cost analysis, therefore, shows the potential advantage of large-scale production in comparison to small-scale production. This would be better demonstrated with a larger data set.

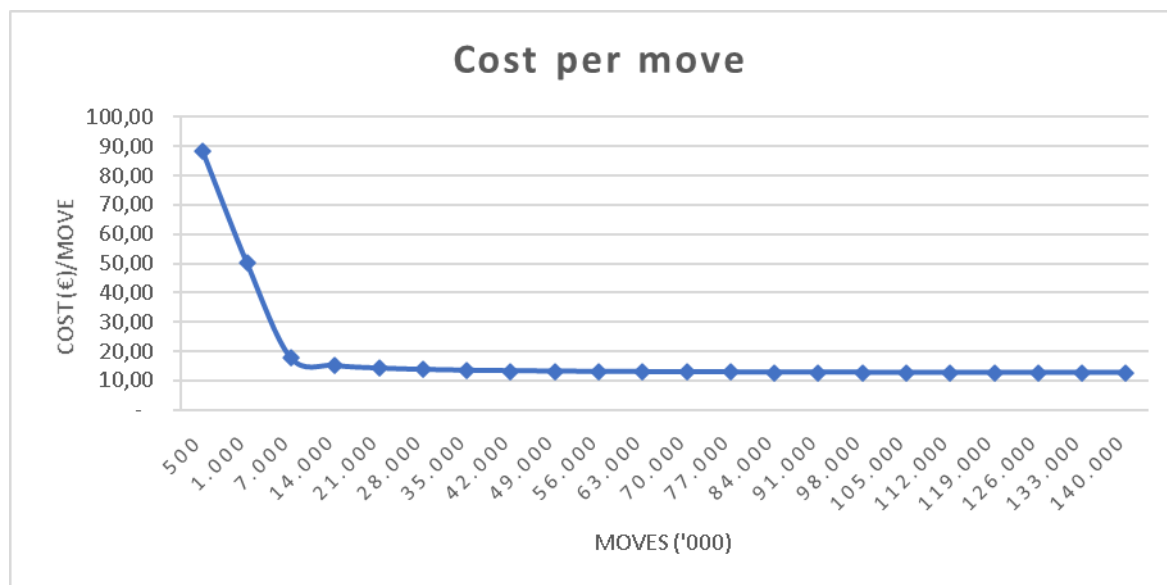
Figure 7-6: Average cost per container



Source: Own compilation

The costs per move indicate that at an average of 140 million moves per annum, the containers incur approximately €12.60 per move. The number of moves is more informative compared to throughput in that they properly demonstrate all the processes of the yard across all subsystems. Container terminals charge their clients only for certain moves (e.g. charge & discharge moves, yard or adjacent moves, and restow moves), all other moves including shifting and shuffling are not included in the price but remain a cost to the terminal operator. In practice, most containers are not moved immediately between the transport modes. They are usually in the storage yard earlier to provide enough convenience for the terminal operator to have access to them when the time to move them arrives, but of course, also not too early as this would result in many unproductive moves. Such unproductive moves are what contribute to reduced productivity and efficiency. According to Clark, Dollar, & Micco (2002), higher inefficiencies lead to higher costs and higher charges in ports. The factors of efficiency and productivity are discussed in chapter 4 and analyzed further in chapter 6. Although many of these factors have been in past literature identified to affect efficiency, in the specific case of the HLH range, automation and capacity utilization have been proven to play a significant role in efficiency. This explains one of the reasons behind terminals improving their technologies with the hope to expand capacity so as to reach higher productivities and efficiency, and likely lead to reduced costs and higher competitive edge.

Figure 7-7: Average cost per move



Source: Own compilation

The key to resolving problems related to the number of moves is developing and maintaining good planning procedures (Chen, 1999). Unproductive moves can also be due to a shortage of storage capacity where containers are stacked in a mixed position due to lack of space and thus causing more shuffling and shifting. How information is transferred in the yard also influences the organization of the containers. It is thus through improving on the organization of these elements that unproductive movements can be avoided and so that operating costs can be reduced (Chen, 1999).

7.4.4 Other costs

The excessive growth of transport-related services is a universal occurrence, linked to industrial development. At a local level, some costs are linked to the service provision but often these costs are not internalized and are referred to as 'external costs'. According to Blauwens et al.,(2020) such costs are costs that the service provider causes to third parties and which they do not pay for, for example, air pollution, noise pollution, safety risks. To cut down on such problems, these costs have to be passed down to the users of the services. In some instances, these costs are accounted for through levies which the user of the service has to pay for. Generally, in the transport sector, a choice would need to be made between making use of either a barge, truck, rail. The user would (through levies) be required to pay the difference of social costs between these modes of transport, logically, the user will only pay a

higher price for the other option when they feel that it is worth it (opportunity cost). Therefore, deciding to transport some part of the distance by sea or road or rail will change the total cost of handling that container. That decision is also affected by other costs, for instance, in the container terminal, the cost of handling a container to a barge is much higher than handling a truck container. This is due to the number of moves involved between these two modes.

Governments all over the world try through taxes (the costs can be priced through marginal costs) to reduce the social burden linked to services that cause external costs as much as possible. The European Union strives for sustainable mobility in the transport system (for both goods and passengers). The emphasis is more on the modal split where roads account for 45.6 %, railway 10.5%, inland waterways 3.3%, oil and pipelines 3.2% of the demand, (Grzelakowski, 2020). Although many regulations have been imposed, the modal shift to the more environment-friendly modes remains a problem. An estimation of the external costs comparing the three modes of inland transport (rail, road, inland waterways) is extensively demonstrated in (Hofbauer & Putz, 2020); a network optimization model is solved in Ambrosino, Ferrari, Sciomachen, & Tei (2018) aimed at highlighting the effects of increasing capacity of the rail share in the planned interventions. In Ambrosino, Ferrari, Sciomachen, & Tei (2016) the external costs are taken into account as means of decision making for locating inland ports. In Merchan, Léonard, Limbourg, & Mostert (2019) external costs for inland freight transport modes are demonstrated and compared using a lifecycle assessment methodology.

7.4.5 The container terminal charges

It is not the intention in this section to analyze the price of the container terminals but this element cannot be ignored as costs are one of the elements affecting price. Port pricing remains a complicated issue that lacks transparency. With improved structured port charges, there is a potential of enhancing port efficiency alongside trade (Meersman et al., 2014). Given that one of the objectives of terminal operators is to maximize profits, this generally indicates that the manner of price setting will be affected by the cost structure. The pricing structure also determines the role of the costs to the pricing, such as cost-based pricing, cost-recovery pricing, congestion pricing, strategic pricing, or quality pricing, to name some (Vanellander & Sys, 2014).

Terminal users will use the terminal if there are benefits to tap; through those benefits, terminal operators can create some revenues through those provided

benefits. Three elements are essential for port pricing: the flow of costs, benefits, and revenues (UNCTAD, 1975). The pricing of the terminal operator should induce efficiency, such that the price encourages the economic utilization of the terminal assets.

The supply chain operates in a manner that the price of one party is the cost of the other party. The diversity in the service offer contributes to the price determination, especially the vessel operators and cargo owners are faced with. According to Meersman, Strandenæs, & Van de Voorde (2014), prices also reflect whether the service suppliers are private or public entities. Given that the resulting prices affect the port choice decisions by vessel operators and cargo owners, port competition softens.

Blauwens et al.(2020) give an overview of considerations that could cause significant impact, especially to transportation firms when setting prices. The price may differ pending on whether the long-term or the short-term decisions are considered, which affects either fixed or variable costs consecutively. Another dimension that can shape pricing is the quantity adjustment where the operator can adjust production (or service) level relative to demand or set the price relative to the market. Another assessment is to consider the balance between the actual demand and the potential demand, both affected by the sensitivity to time and by capacity availability. Whether or not the operator offers a single or joint product (or service) is another crucial consideration to note, as this will affect the interdependency when calculating the price.

Port pricing structures are dissimilar across different ports, reflecting the level of traffic, among other factors, and also reflecting the level of revenues, (Martinez-Budria et al., 2001). From the perspective of the port operators, the aim is to obtain efficient infrastructure prices that cover costs. Seedah, Harrison, Boske, & Kruse (2013) show charges imposed on a shipping line by various CTOs in different ports. It is shown that on average, shipping lines were charged \$25.00 for loading and unloading full containers and \$2.00 for empty containers, based on the port of Brownsville. Seedah et al., (2013) show a total cost model for a vessel. It is confirmed that these costs can greatly differ from port to port as some ports may allocate lower or higher costs associated with navigational services berth services, and cargo operations. This impacts the competitive position in which the port and the related firms are located. By using a game-theoretic pricing model, to show both cooperative

and competitive games, Zhou & Kim (2020) determine the prices of container handling companies. The findings indicate a price of \$60.3 per TEU for a container terminal handling 269 908 TEUs and \$94.3 per TEU for a CTO handling 290 143 TEUs with profits of \$15.5 and \$7.8 million, respectively. The comparison is made to indicate the influence of size on the price. It is shown that larger operators tend to incur less costs and are thereby able to charge lower prices and thus maintain a better competitive position. Whether or not the general prices charged by operators are high or low remains unclear.

7.5 Conclusion

Terminal efficiency has been proven now and again through past literature to be an important determinant and driver of costs. Terminals that are considered highly efficient have been shown to incur lower costs and can charge less for their service and consequently maintain their competitive position. Given the impact of efficiency on costs and the importance of terminal costs in the maritime chain, it becomes interesting to identify the costs of handling a container.

The cost structure in the container terminal is found to be very reliant on the port's operation type, which determines which costs are covered by the terminal and in what manner. Labour is found to be the highest cost contributor to the total costs and the yard-related cost to be the least contributors to the total costs. In contrast to the norms, the variable costs (80%) here are found to dominate the fixed costs (20%), which are largely owed to the labour costs.

A generic cost model in a total cost function is calculated to indicate the basic model of a container terminal. The current cost of €44.47 per container is 34% higher than in 2005 (without adjusting for inflation) based on Vanelslander (2005). At the current calculated cost of €44.47 cost, the yard is 71% occupied and at 100% the cost would reduce to €38.66, this is at a rate however that any terminal would prefer but also not aim for as it would lead to high congestion. One of the solutions in reducing operational costs is through a good organization of the yard as this would limit the amount of unproductive moves, which affects efficiency as well as costs consequently. Container terminal costs are not solely about the physical and operational costs but there exists also the external cost which play a part in the total costs and should be incorporated into the cost calculation, however here these costs are not taken into account, but this does not to any extent indicate lack of importance.

The direct costs and external costs of the terminal operator make up its social costs which is a keystone for price charging in the transport sector (Grzelakowski, 2020). To achieve a more efficient method of infrastructure pricing than the current one, the EU transport policymakers would benefit by finding a way to incorporate these two costs. According to Grzelakowski (2020), this would reflect the true cost incurred, it would give the right and optimal price signal, it would take into account the real needs of providers and users, and it would take cognizance of scarce resources. From the European Commission (EC) point of view, the internalization of external costs will send signals to transport users on how to change their market behaviour (Schroten et al., 2019).

Chapter 8: Conclusions

This chapter reflects how the aims and objectives of the thesis were achieved. First, the main conclusion highlights the findings for each chapter. The second section delineates the contribution of this research. In the third section, several policies are recommended to terminal operators, port authorities, and researchers. The fourth section brings to light the limitations encountered when carrying out this research and specifies the research work's scope. For further developments in the knowledge of this research domain, recommendations with brief guidelines of future research are given in the fifth section.

8.1 Main conclusion

Supply chains are highly dynamic environments and are quickly changing, a trend that has been vastly accelerating over the past years. The changes (governance, regulations, vessel sizes, technology, and so on) in the supply chains have been shown to impact the operation of terminals significantly; they shape the strategies applied by terminal operators, the management of resources and the costs. The role played by the terminal operators is vital, given the interface position they hold along the supply chain.

Efficiency and productivity have been proven to express the container terminal's strength in terms of its capability to serve its clients effectively, using the available resources with minimal waste. While inefficiency can show itself in low volumes handled, it can also breed congestion. Lack of capacity and insufficient investments can lead to terminal congestion. Inefficiently managed terminals can result in the slowing down of economic activities, causing reduced inventory in stores, vessel delays, road congestions, and overall, a disruption to the supply chain. Recent relatable events where most of the public witnessed the impact are through the COVID-19 pandemic and a glimpse through the 'Ever Given' mega vessel, which spent days blocking the Suez Canal. In parallel with the disruption to 'business as usual' and 'the everyday life', inefficiency in the form of congestion instigates additional costs to terminal's customers, such as shipping lines, truckers, rail operators, barge

operators, forwarders, hauliers, exporters and importers, passing the costs to the end consumer.

Overcapacity and overinvestments can also cause inefficiency, often observed through low volumes (opposite congestion) in a terminal. While highly congested terminals directly negatively affect the entire industry, overcapacity is directly a terminal concern. Additional costs in this scenario are observable through resource management.

Relative efficiency is especially vital to evaluate in a business environment where competition is high. Failure to offer competitive service delivery can easily result in a change in terminal choice by customers. The HLH terminals operate in a similar business environment, with shared hinterlands, and thus staying ahead becomes key to remaining relevant to customers.

The internal management of resources is vital in determining efficiency, especially for the benefit of a terminal operator. Although the external factors are cumbersome to measure, it can be concluded that they also have an immense contribution to efficiency. Summing up, it can be supposed that inefficiencies are costly and should be eliminated as much as possible.

Summing up, inspired by the recent major transitions in the sector, in this thesis the following overarching research questions are addressed:

What are the measured technical efficiencies and productivities in container terminals? What can be observed in the container terminal cost structures?

The thesis untangles the main question by answering the questions that follow.

What is the level of play between competition or cooperation in the Hamburg-Le Havre container terminals? (chapter 2)

Container terminals in the HLH are unique in that their hinterlands are close to each other. Thus, remaining relevant and competitive means that the terminals need to offer a certain level of service to keep customers satisfied as switching costs and efforts are low.

This chapter shows how governance has evolved in the port sector from a global point of view and also highlights the trends at the HLH. Resource management (efficiency) can reflect the applied strategies by an entity. The impact of privatization on

efficiency is one of the much-discussed topics in port competition research. Gaining power along the maritime supply chain (horizontally and/or vertically) is the aim of many entities in the sector as it allows increased control and higher flexibility. The driving factors to cooperation and potential impact on efficiency are highlighted in this chapter.

Cooperation in many forms has manifested in the industry, across the globe, through horizontal and vertical structures, changing how the firms identify themselves. In the HLH, while shipping lines are visibly getting powerful, terminals are also observed growing horizontally. For instance, much earlier in the 1990s, the mergers between Hessenatie and Noordnatie in Antwerp were already observed; the recent cooperation between ports (port of Antwerp and the port of Zeebrugge). Such events highlight how regulation and strategy influences competitiveness and performance.

Competition and cooperation have taken stride and changed the shape that was observable in the past for the sector. Global carriers have formed alliances with each other and have formed various other relations with terminal operators and hinterland transportation. All this is to expand their service offers; to increase efficiency, convenience, and market power (among many other drivers). These types of relations are expected to continue. Mergers between CTOs and carriers are found to be more suitable when they are based on an alliance. CT horizontal relations are also seen to continue increasing with the long-term collaborative projects currently in the plan.

What container terminal efficiency and productivity methods have been applied previously? (Chapter 3)

Chapter 3 depicts the developments in the literature of efficiency and productivity in the container port and terminal sector. The chapter shows the suitable methods for the HLH's efficiency measurements, taking into account its business environment.

Although there are many alternatives to analyzing efficiency, the DEA and the SFA are popular in the maritime and ports sector. The suitability of the DEA for container terminal representation is primarily due to the appropriateness of the DEA models in dealing with a single (and multiple-output) scenario while taking into account the input resources.

The DEA is based on the theory of Pareto-efficiency, which abides by the laws of economics. Such a theoretical foundation helps explain the port sector, which is strongly connected to economic activity.

The importance of management and strategy is mentioned repeatedly in this research; the DEA offers terminals an opportunity to explore various aspects of the relative performance of their entity in comparison to its competitors. The model offers an opportunity for the terminals to measure relative (in-)efficiencies, an opportunity to identify the exact elements leading to inefficiencies, suggestions of improvements, and suggestions to industry peers whom the concerned entity may observe to improve. The flexibility in the model allows terminal managers to align their actions well with their strategies (for example, the choice between input orientation versus output orientation).

The cumbersomeness of measuring, especially the external factors affecting terminal efficiency (and also some internal factors), is a cause of concern. The DEA can be combined with either statistical and non-statistical methods to resolve such limitations. Furthermore, potential variables of interest may have data limitations due to the unavailability of large datasets. Unlike the parametric approaches such as the SFA, the DEA can handle small datasets. Such an advantage is especially vital in the container terminal sector, where there is minimal data sharing.

The importance and popularity of the DEA continue to increase now more than ever before. It is revealed that apart from the internationally-focused port efficiency research work, Europe and Asia also contribute a significant proportion to the body of literature. This indicates that the extrapolation of the model is widely accepted and that the results observed in one location can also be replicated in another location. In as much as Europe is highly represented in the DEA (and performance in general) literature, there is low extrapolation of the model to the HLH region, and especially to a single terminal level. It becomes exciting to thus explore an analysis in this region, at the terminal level, using the DEA.

What are the various container terminal characteristics (performance affecting) that can be identified along the region? (Chapter 4)

The measurement of performance, specifically productivity and efficiency, has relied on methods summarized in chapter 3, using some variables to capture the observed relationships. Vital efficiency factors are identified through a comprehensive

description of the terminal operation process in this chapter, distinguishing between an automated and non-automated terminal.

In this chapter, it is confirmed that across container terminals, there are many similarities, as well as many differences. Across terminal operators, there are comparable variables, some measurable and others not. The factors identified in this chapter become a practical application in the analysis of this research work.

In as much as the infra/superstructure are fundamental to production and consequently to efficiency, labour is highlighted as an essential factor in production. In the urge to improve on technologies, attention is turning towards automation. About 9.1% of the CTs worldwide are considered automatically operated. If the social costs from automation are not resolved, it could lead to other underlying issues such as labour unrest, loss of jobs, mental health, to name some.

It is confirmed in this chapter that the level of automation creates major differences in how a terminal is operated, which creates a divergence in the operation of the container terminals. Regardless of the internally identified factors of production, efficiency and productivity are also affected by external factors, where no/limited amount of control exists.

The distinction between the internal and the external variables is important as it indicates factors that terminal managers can control when making decisions about their terminals. There is an indication that there are many external factors that CT managers cannot control. However, operators need to remain resilient and prepare to withstand external shocks where possible.

Critical variables for container terminal handling identified in this chapter are used as input in productivity and efficiency analysis in this thesis.

How has container terminal productivity changed in the HLH range over the previous years? (Chapter 5)

The extent and significance of change in the maritime and port industry were highlighted. It was shown that the changing sector requires adaptability and continuous performance improvement. The HLH, as a region with high competition and also where the port serves a significant economic role, the evolution of productivity against the evolution of time is measured. The HLH is characterised by heavily investing in terminals, with some terminals considered to have excess capacity. On the other hand, some terminals are approaching full capacity, with a few

exceeding their technical capacities. From that light, the chapter measures productivity to determine if more or less investments are needed in the HLH terminals' infrastructure to serve their customers efficiently.

The Malmquist model is found appropriate for the analysis in this chapter, where total factor productivity (TFP) is measured to indicate the change in productivity over time in the HLH region. It is confirmed in this chapter that the HLH range terminals are found to use their resources better than the world average (TEUs per quay, gantry crane) but found to underutilize land (hectares). This is an indication that the available resources over time have been efficiently used in serving the customers.

In the two years observed (2013 and 2018), the HLH container terminals indicate positive progress in TFP collectively. This chapter demonstrates the impact of heavy investments in the container terminal sector, an approach used to ensure adequate capacity for not just the current cargo volumes handled, but also for the future. A comparison between the two periods also illustrates how excess capacity can lead to short term inefficiencies and the implications in the long run. This serves as an investment decision indicator, not only to terminal operators but also to port authorities and influential policymakers.

The scale inefficiencies in a majority of the region's terminals place a greater emphasis on the prioritization of investment decisions on capacity building by terminal operators and port authorities. The positive growth in productivity over time in the region should be taken as a positive sign that the resources accumulated and available have been well managed in the region. Hence, the players are identified as the 'catchers' given the positive efficiency changes and identified as 'innovators' given the positive shift of the production frontier.

What are the technical and scale efficiency scores associated with the container terminals? (Chapter 6)

In chapter 6, the Banker, Charnes and Cooper (BCC) model and Charnes, Cooper and Rhodes CCR models of the data envelopment analysis (DEA) are used to illustrate the technical and scale efficiencies of the container terminals in detail.

One significant change is observed in terminals' employed technologies, infrastructure, and superstructure. The yard equipment used in a container terminal defines the level of automation or the extent of advancing the terminal's technologies. In comparison to chapter 5, this chapter supplements by considering

the container yard equipment but for only a single period. The observation helps realise differences in handling containers between automated and non-automated terminals. The question of overcapacity and capacity shortage is revisited as a critical element. Not only is that beneficial for investment decisions but it is also vital for resource allocation, saving terminals operators and their customers from additional costs that can occur due to inefficiencies.

The analysis indicates that resource management of the HLH CTs has resulted in high levels of efficiency. However, the terminals experience negative scale efficiencies resulting in a decline in overall efficiency, confirming the results of the Malmquist model used in the previous chapter. Automated terminals are observed to have lower 'wastage' of resources than non-automated terminals at the waterside sub-system. Yard utilization is an issue that both automated and non-automated terminals need to improve. Disregarding automation, there exist as many different terminals as there are in this sample, indicating that, as much as they are said to be aggregately similar, they are also very different, and thus, the goal of each terminal would be to reach its objectives as much as possible rather than trying to score a higher rate of efficiency.

Crucial factors to container handling are used in this chapter, with an emphasis on the exclusion of the labour variable. In addition to the internal, measurable factors, the chapter shows how significant the external factors can impact the efficiency of a terminal on the one hand. On the other hand, the external factors are also shown to have a negligible impact when the business environment of operation is similar. Managers must ensure that resources are optimised as much as possible to reach the best possible output; not doing so will lead to inefficiency. This calls into action also the applied strategies for current and future plans, as this influences performance. This chapter provided evidence that regardless of the internal managerial efforts of the terminal, there are also factors beyond control. Although such external factors cannot be controlled, measures to remain resilient within the terminals should be implemented.

Although results derived from the DEA offer crucial insights on optimum production, the results should always be interpreted with caution. First of all, to reach interpretable conclusions, the researcher needs to have good knowledge of the terminals in specific details. For instance, the measurement of the quay lengths in some terminals is distributed in total amounts, including barge quay walls, whereas

other terminals only distribute the deep-sea quay wall information on their websites. Comparing such a measure without such knowledge could thus lead to fallacious interpretations. This is the same with the quay cranes; they have varying features which can cause handling capacity variations.

What is the observed cost structure of the CTs? (chapter 7)

Given the importance of costs in determining performance, chapter 7 gives an in-depth analysis of specific terminals of a single port (Port of Antwerp) taking into account the cost factors of the terminals. The handling costs are calculated per container and per move. It can be observed that labour costs are the largest contributor to the final costs while yard (land) related costs contribute the least to the average cost. This high cost of labour is one of the leading factors to increased automation among terminals. The labour element is hardly ever publicly available from terminals, making analysis for benchmarking techniques difficult to complete and fully resemble the processes.

Given that inefficiency is costly, it is concluded from the analysis that the cost of handling one container is approximately €44.47 and is also €12.60 for each move when taking into account 2018 data. The cost is expected to reduce when more containers are handled per period. Considering external costs is what policymakers need to target when setting competition and/or cooperation policies. This also represents the costs to the society and environment and thus indicates the true cost of terminal activity.

8.2 Research contribution

This research serves to contribute in various spheres.

The research focuses on a single geographical area in the north of Europe. Evaluating container terminals within the same proximity has numerous benefits. The focus on one region gives the opportunity of assessing business units that are faced to a larger extent by a similar business environment, policies and governance traditions. The regional focus strengthens the interpretation of the results, which adds value when extrapolating the results as comparison can be linked to specific regional characteristics. There is also insufficient research conducted in the HLH from the container terminal perspective.

The Data Envelopment Analysis (DEA) applied in this research is very widely applied in the port sector. This research offers an opportunity for a more comprehensive understanding of the HLH performance, which is easier to compare with research conducted elsewhere. The specific details covered in the model's analyses (such as observation from various orientations and statistical models) contribute to the field. This is an advantage to the body of research and policymakers who wish to make critical decisions for the economies concerned.

Four percent of the Connecting Europe Facility (CEF) budget is dedicated to ports. Part of the port budget is towards the improvement of the infrastructure. This research contributes to the dialogue about terminals' capacity building, which many terminals continue to improve.

Automation is taking over in the sector. Given the continuous plans and projects worldwide towards automation, it is valuable to contribute towards the knowledge of the link of automation to efficiency, especially as the movement has a significant impact on labour. The implications for labour are not only for the social aspect, but it also questions the formulations and assumptions of production theories, which are the cornerstone to many of the performance methods applied today.

Analyzing individual terminal operators at an 'operator' level requires confidential data. This is the limitation of a majority of previous studies attempting 'operator' level analysis. This research collects data from the terminals under study, making use of specialized reports that are not easily accessible to the general public. Direct communication with the terminal operators strengthens the dependability of the results.

8.3 Policy recommendations and managerial implications

Based on this research work, it can be concluded that measuring relative performance is a complex issue that requires an extensive evaluation of the terminal sector and the external environment. The importance of the port environment to economic development requires that influential groups in the supply chain take precautions for several matters affecting performance. The recommended policies are observable from three viewpoints, i.e. from the perspective of the terminal operator and port authorities; the port stakeholders & local communities; and for academia and research field, as elaborated below.

Implications to container terminal operators and ports

- The drivers of performance (efficiencies) should be observed by paying close attention to resource management (pure technical efficiencies), port size (scale efficiencies) and external factors.

Policymakers should put in place more observations that will improve the major regional concerns (geographical patterns and location challenges) to improve efficiency. This can save ports and terminals high costs. The savings should be channelled towards benefiting the economies on which these entities are operating.

Implications to the port stakeholders and local communities

- The social and internal impact of automation should be observed.

The costs and benefits (long and short term) of automation should be a consensus reached by various stakeholders to consider port and terminal performance and consider the social effects of automation (such as labour and carbon emissions). There should be a concrete understanding of the port or terminal's need for performance improvement and the external impact.

- The need for further investments in infrastructure should be well examined.

While economies of scale are relevant in the port sector, they are also known not equally to benefit all the stakeholders. The impact of port expansions should prioritise the needs of the local communities. Cost-benefit analyses should be conducted to assess the impact on local livelihoods. Solutions should consider technical gains for the ports and port operators and benefit the communities of interest.

- The incentives to encourage data (standardised data) sharing should be implemented.

The IMO has made it mandatory for all its member countries to exchange key data electronically primarily for the harmonisation of vessel arrival, stay and departure. The Facilitation of International Maritime Traffic (FAL) Convention provides governments with recommendations to improve communication between parties, transitioning to full-fledged maritime single windows (MSWs). To facilitate and promote the MSWs and port community systems (PCS), multi-stakeholders and the legal frameworks need to cooperate and aim towards increased data sharing.

Authorities should find solutions for improving data sharing in a manner that offers incentives and recognition for data sharing. Trust is one of the components hindering further sharing of information. Ports should ensure that trust can be built between stakeholders by installing controls allowing for improved trust relations by introducing laws to enforce the agreements established.

Implications to academia and research

- Stimulate collaborations of the industry with academia

Through cooperating, there is a chance for firms to acquire external knowledge and supplement internal knowledge. On the other hand, the analysis methods for academia also improve and adjust to represent reality-enhancing innovation. Information unavailability often leads to methods guided by a lack of data rather than theory. Platforms of engagement such as conferences, seminars, and webinars, where academia and industry share views and challenges, should be encouraged while maintaining independence. The initiative will enrich and incentivise the consensus between the two fields.

8.4 Limitations and scope

The limitations experienced when carrying out this research were mainly due to data unavailability, methodological constraints and physical barriers, elaborated below.

Several variations of container terminals can exist: there can be marine terminals, barge service terminals, empty depots, container repair, rail terminals, trimodal terminals, etc. Multi-purpose and ro-ro container terminals were excluded from the sample.

Terminals are selected within the Global top 50, and those outside this scope are automatically excluded, regardless of the size of the output.

Having selected container terminal operation as a broad base of study; the thesis focused explicitly on analysing one of the regions of economic importance in the ports of Europe and the world - the Hamburg-Le Havre range. Terminal operators outside this region are not taken into account in this thesis.

One major drawback of this type of analysis is its inability to capture intended future demand. Some terminals may have current assets in anticipation of future demand,

which will not be captured in the analysis but which would likely deem the terminal inefficient.

The data for these cases has been gathered by contacting the terminals in person and through the websites and specialized documents. Where information is unavailable across all terminals (such as labour and financial data), such variables, or the terminal itself, were eliminated. This influenced the sample size.

Given the unavailability of data, it is difficult to reflect the entire production function of the terminals.

The variable 'container handling equipment' is only available for one year and hence not included in the Malmquist model.

The sample size between the two stages is not equal due to some terminals which did not exist throughout the five-years.

8.5 Recommendations and further research

Without some of the highlighted limitations and with the insights gathered through the research, more answers in this field of research can be provided.

One of the pressing concerns in the ports sector is the disruption to the environment, especially carbon emissions. One of the DEA advantages is the ability to cater for multiple outputs, regardless of the unit of measure. Future research should take into consideration environmental sustainability as the pressure stands to reach zero emissions by 2050. Such a measure can also be combined with other environment-friendly measures, such as those for port accessibility. Such research would consequently measure the rationale behind infrastructural investments. As an additional measure of output, the number of moves should also be considered in giving in-depth details in this type of research.

There is an indication in supply chain research that efficiencies are costly. The costs from inefficiencies do not only affect the terminal or port (for example, demurrage, detention, storage, social costs). It would add value for future research to outline the relationship between inefficiencies and costs to the terminal and the port stakeholders. The models can also consider different scenarios, e.g. where there is a large amount of transshipment and where there is little or no transshipment, and where there are variations to the automation levels.

The SFA as a competing method to the DEA could be interestingly applied in the HLH context to evaluate the differences in the results of the two methods. Further, the research could also attempt to combine the DEA and the SFA by taking the favourable components from each method to develop a new method with less of the disadvantages often highlighted from each method.

There are various characteristics of the terminals which were not tested in this analysis. More of the characteristics listed in chapter 4 should be observed.

A multi-regional approach has its challenges, but it would allow for the possibility to test various other external variables that are impossible to test in the same region as the results would likely show insignificance in explaining efficiency.

In specifying the variables, further research can take more caution in developing especially the input variables. For instance, measuring quay cranes in terms of their capacity and potential would be more valuable than measuring just the quantity. The yard equipment could also be represented in value rather than quantity. Future research work should be careful in the assessment of capital as this variable can present complexities when the value is considered.

Furthermore, the strategic developments taking place horizontally and vertically have an impact on the decisions of many stakeholders. Future research should look into the implications of such strategic changes on efficiency.

Bibliography

- Aamlid, C. (2019). *Threats to Container Ports and Preventative Security Measures* (Issue May 2019). University of South-Eastern Norway.
- Acciaro, M., Ferrari, C., Lam, J. S. L., Macario, R., Roumboutsos, A., Sys, C., Tei, A., & Vanelander, T. (2018). Are the innovation processes in seaport terminal operations successful? *Maritime Policy and Management*, 45(6), 787–802. <https://doi.org/10.1080/03088839.2018.1466062>
- Acciaro, M., & McKinnon, A. (2015). *Efficient hinterland transport infrastructure and services for large container ports* (pp. 75–101). <https://doi.org/10.1787/9789282107850-5-en>
- Afriat, S. N. (1972). Efficiency Estimation of Production Functions. *International Economic Review*, 13(3), 568–598.
- Aigner, D., Lovell, C. A. K., & Schmidt, P. (1977). Formulation and estimation of stochastic frontier production function models. *Journal of Econometrics*, 6(1), 21–37. [https://doi.org/10.1016/0304-4076\(77\)90052-5](https://doi.org/10.1016/0304-4076(77)90052-5)
- Akbari, N., Jones, D., & Treloar, R. (2020). A cross-European efficiency assessment of offshore wind farms: A DEA approach. *Renewable Energy*, 151, 1186–1195. <https://doi.org/10.1016/j.renene.2019.11.130>
- Al-Eraqi, A. S., Mustafa, A., Khader, A. T., & Barros, C. P. (2008). Efficiency of middle eastern and East African seaports: Application of DEA using window analysis. *European Journal of Scientific Research*, 23(4), 597–612.
- Almawshaki, E. S., & Shah, M. Z. (2015). Technical Efficiency Analysis of Container Terminals in the Middle Eastern Region. *The Asian Journal of Shipping and Logistics*, 31(4), 477–486. <https://doi.org/10.1016/j.ajsl.2016.01.006>
- Almawshaki, S. E., & Shah, Z. M. (2015). Technical Efficiency Analysis of Container Terminals in the Middle Eastern Region. *The Asian Journal of Shipping and Logistics*, 31(4), 477–486. <https://doi.org/10.1016/j.ajsl.2015.08.008>
- Ambrosino, D., Ferrari, C., Sciomachen, A., & Tei, A. (2016). Intermodal nodes and external costs: Re-thinking the current network organization. *Research in Transportation Business and Management*, 19, 106–117. <https://doi.org/10.1016/j.rtbm.2016.05.001>
- Ambrosino, D., Ferrari, C., Sciomachen, A., & Tei, A. (2018). Ports, external costs, and Northern Italian transport network design: effects for the planned transformation. *Maritime Policy and Management*, 45(6), 803–818. <https://doi.org/10.1080/03088839.2018.1448475>
- Amore, M. D., & Murtinu, S. (2019). Tobit models in strategy research: Critical issues and applications. *Global Strategy Journal*, October, 1–25. <https://doi.org/10.1002/gsj.1363>
- Anokye, M., Annin, K., & Oduro, F. T. (2013). Application of Queuing Theory to Vehicular Traffic at Signalized Intersection in Kumasi-Ashanti Region, Ghana. *American International Journal of Contemporary Research*, 3(7), 23–29.
- Antoniou, C., Roh, S., Seo, Y.-J., & Kwak, D.-W. (2019). The impact of service

- privatization-The case of a container terminal at the Port of Limassol. *KMI International Journal of Maritime Affairs and Fisheries*, 11(1), 1–20.
- Aronietis, R., Van de Voorde, E., & Vanelslender, T. (2010). Port Competitiveness Determinants of Selected European. *Association for European Transport and Contributors*, 1985, 1–17.
- Banker, R. D., & Natarajan, R. (2008). Evaluating contextual variables affecting productivity using data envelopment analysis. *Operations Research*, 56(1), 48–58. <https://doi.org/10.1287/opre.1070.0460>
- Baran, J., & Górecka, A. (2015). Seaport efficiency and productivity based on Data Envelopment Analysis and Malmquist Productivity Index. *Logistics & Sustainable Transport*, 6(1), 25–33. <https://doi.org/10.1515/jlst-2015-0008>
- Barros, C. P. (2003). Incentive regulation and efficiency of Portuguese Port authorities. *Maritime Economics and Logistics*, 5(1), 55–69. <https://doi.org/10.1057/palgrave.mel.9100060>
- Barros, C. P. (2005). Decomposing growth in Portuguese seaports: A frontier cost approach. *Maritime Economics and Logistics*, 7(4), 297–315. <https://doi.org/10.1057/palgrave.mel.9100140>
- Barros, C. P. (2006). A Benchmark Analysis of Italian Seaports Using Data Envelopment Analysis. *Maritime Economics and Logistics*, 8(4), 347–365. <https://doi.org/10.1057/palgrave.mel.9100163>
- Barros, C. P., & Athanassiou, M. (2004). Efficiency in European seaports with DEA: Evidence from Greece and Portugal. *Maritime Economics and Logistics*, 6(2), 122–140. <https://doi.org/10.1057/palgrave.mel.9100099>
- Barros, C. P., Chen, Z., & Wanke, P. (2015). Efficiency in Chinese seaports: 2002–2012. *Maritime Economics & Logistics*, 18(3), 295–316. <https://doi.org/10.1057/mel.2015.4>
- Basarici, A. S., & Satir, T. (2019). Empty container movements beyond the effect of trade imbalance: Turkish terminals. *International Journal of Logistics Systems and Management*, 33(1), 1. <https://doi.org/10.1504/ijlsm.2019.10014706>
- Bearzotti, L., Gonzalez, R., & Miranda, P. (2013). The event management problem in a container terminal. *Journal of Applied Research and Technology*, 11(1), 95–102. [https://doi.org/10.1016/S1665-6423\(13\)71518-9](https://doi.org/10.1016/S1665-6423(13)71518-9)
- Belfkih, A., Duvallet, C., & Sadeg, B. (2017). The Internet Of Things For Smart Ports Application To The Port Of Le Havre. *Proceedings of IPaSPort 2017*, May.
- Benacchio, M., & Ferrari, C. (2002). Recent Trends in the Market Structure of Container Terminal Services: Which Way To Integration? *Pomorski Zbornik*, 40(1), 153–176.
- Bichou, K. (2011). A two-stage supply chain DEA model for measuring container-terminal efficiency. *International Journal of Shipping and Transport Logistics*, 3(1), 6–26. <https://doi.org/10.1504/IJSTL.2011.037817>
- Blauwens, G., De Baere, P., & Van de Voorde, E. (2020). *Transport Economics* (Seventh Ed). De Boeck Publishing Company.
- Böse, J. W. (2020). General Considerations on Terminal Planning, Innovations and Challenges. In *Handbook of Terminal Planning* (Second Edi, pp. 3–27). Springer

Nature Switzerland AG.

- Bottalico, A. (2017). *Across the chain . Dock labour systems in the European ports : A comparative analysis on two container*. UNIVERSITÀ DEGLI STUDI DI MILANO.
- Brinkmann, B. (2011). Operations systems of container terminals: a compendious overview. In J. Böse (Ed.), *Handbook of Terminal Planning* (pp. 25–39). Springer.
- Brocket, P. L., & Golany, B. (1996). Using rank statistics for determining programmatic efficiency differences in data envelopment analysis. *Management Science*, 42(3), 466–472. <https://doi.org/10.1287/mnsc.42.3.466>
- Burmeister, H.-C. (2020). Ensuring Navigational Safety and Mitigate Maritime Traffic Risks While Designing Port Approaches and Ship Maneuvering Areas. In *Handbook of Terminal Planning* (p. 269). Springer Nature Switzerland AG.
- Carlan, V., Sys, C., Calatayud, A., & Vanelander, T. (2018). *Digital Innovation in Maritime Supply Chains: Experiences from North Western Europe* (Issue April). <https://doi.org/10.18235/0001070>
- Carlan, V., Sys, C., & Vanelander, T. (2016). How port community systems can contribute to port competitiveness: Developing a cost-benefit framework. *Research in Transportation Business and Management*, 19, 51–64. <https://doi.org/10.1016/j.rtbm.2016.03.009>
- Carlo, H. J., Vis, I. F. A., & Roodbergen, K. J. (2014). Storage yard operations in container terminals: Literature overview, trends, and research directions. *European Journal of Operational Research*, 235(2), 412–430. <https://doi.org/10.1016/j.ejor.2013.10.054>
- Carlo, H. J., Vis, I. F. A., & Roodbergen, K. J. (2015). Seaside operations in container terminals: literature overview, trends, and research directions. *Flexible Services and Manufacturing Journal*, 27(2–3), 224–262. <https://doi.org/10.1007/s10696-013-9178-3>
- Caserta, M., Schwarze, S., & Voß, S. (2020). Container Rehandling at Maritime Container Terminals: A Literature Update. In *Handbook of Terminal Planning* (Second Edi, p. 343). Springer Nature Switzerland AG.
- Castelein, R. B., Geerlings, H., & van Duin, J. H. R. (2019). Divergent effects of container port choice incentives on users' behavior. *Transport Policy*, 84(April), 82–93. <https://doi.org/10.1016/j.tranpol.2019.04.010>
- Caves, D. W., Christensen, L. R., & Swanson, J. A. (1981). Productivity growth, scale economies, and capacity utilization in U.S railroads, 1955–74. *The American Economic Review*, 71(5).
- Caves, D. W., Christiansen, L. R., & Trethway, M. W. (1984). Economies of destiny versus economies of scale: Why trunk and local service airline costs differ. *RAND Journal of Economics*, 15(4), 471–489. <https://doi.org/10.2307/2555519>
- Chang, S.-M., Wang, J.-S., Yu, M. application of centralized data envelopment analysis in resource allocation in container terminal operations A. application of centralized data envelopment analysis in resource allocation in container terminal operationsng-M., Shang, K.-C., Lin, S.-H., & Hsiao, B. (2015). An application of centralized data envelopment analysis in resource allocation in container terminal operations. *Maritime Policy & Management*, 42(8), 776–788. <https://doi.org/10.1080/03088839.2015.1037373>

- Chang, Y.-T., & Talley, W. K. (2019). Port Competitiveness, Efficiency, and Supply Chains: A Literature Review. *Transportation Journal*, 58(1), 1–20. <http://www.jstor.org/stable/10.5325/transportationj.58.1.0001>
- Chang, Y. T., Lee, S. Y., & Tongzon, J. L. (2008). Port selection factors by shipping lines: Different perspectives between trunk liners and feeder service providers. *Marine Policy*, 32(6), 877–885. <https://doi.org/10.1016/j.marpol.2008.01.003>
- Chang, Y. T., Park, H. (Kevin), Lee, S., & Kim, E. (2018). Have Emission Control Areas (ECAs) harmed port efficiency in Europe? *Transportation Research Part D: Transport and Environment*, 58(November 2017), 39–53. <https://doi.org/10.1016/j.trd.2017.10.018>
- Chaouk, M., Pagliari, D. R., & Moxon, R. (2020). The impact of national macro-environment exogenous variables on airport efficiency. *Journal of Air Transport Management*, 82(May 2019), 101740. <https://doi.org/10.1016/j.jairtraman.2019.101740>
- Charnes, A., Cooper, W., Golany, B., Seiford, L., & Stutz, J. (1985). *Foundations of Data Envelopment Analysis for Pareto-Koopmans Efficient Empirical Productions Functions*.
- Charnes, A., Cooper, W. W., & Rhodes, E. (1978a). Measuring the efficiency of decision making units. *European Journal of Operational Research*, 2, 429–444. <https://www.utdallas.edu/~ryoung/phdseminar/CCR1978.pdf>
- Charnes, A., Cooper, W. W., & Rhodes, E. (1978b). Measuring the efficiency of decision making units. *European Journal of Operational Research*, 2, 429–444. <https://www.utdallas.edu/~ryoung/phdseminar/CCR1978.pdf>
- Chen, H. K., Chou, H. W., & Hsieh, C. C. (2018). Operational and disaggregate input efficiencies of international container ports: An application of stochastic frontier analysis. *International Journal of Shipping and Transport Logistics*, 10(2), 113–159. <https://doi.org/10.1504/IJSTL.2018.090092>
- Chen, T. (1999). Yard operations in the container terminal-a study in the 'unproductive moves. *Maritime Policy and Management*, 26(1), 27–38. <https://doi.org/10.1080/030888399287041>
- Chen, X., He, S., Zhang, Y., Tong, L. (Carol), Shang, P., & Zhou, X. (2020). Yard crane and AGV scheduling in automated container terminal: A multi-robot task allocation framework. *Transportation Research Part C: Emerging Technologies*, 114(September 2019), 241–271. <https://doi.org/10.1016/j.trc.2020.02.012>
- Clark, X., Dollar, D., & Micco, A. jandro. (2002). Maritime Transport Costs and Port Efficiency. In *wp52781* (No. 2781; Development Research Group Macroeconomics and Growth, Issue February). <https://doi.org/10.2139/ssrn.898468>
- Cochrane, R. A. (2008). The effects of market differences on the throughput of large container terminals with similar levels of efficiency. *Maritime Economics and Logistics*, 10(1–2), 35–52. <https://doi.org/10.1057/palgrave.mel.9100190>
- Coelli, T. (1996). A Guide to DEAP Version 2.1: A Data Envelopment Analysis (Computer) Program. *CEPA Working Paper 96/08*, 1994, 1–49.
- Coelli, T. J., Prasada Rao, D. S., O'Donnell, C. J., & Battese, G. E. (2005). *An introduction to efficiency and productivity analysis* (Second edi). Springer Science and business media. <https://doi.org/10.2307/2531310>

- Coelli, T., & Perelman, S. (1996). *EFFICIENCY MEASUREMENT, MULTIPLE-OUTPUT TECHNOLOGIES AND DISTANCE FUNCTIONS: WITH APPLICATION TO EUROPEAN RAILWAYS*. [http://orbi.ulg.be/bitstream/2268/35665/1/105.CREPP 9605 Coelli-Perelman.pdf](http://orbi.ulg.be/bitstream/2268/35665/1/105.CREPP%209605%20Coelli-Perelman.pdf)
- Cook, W. D., Tone, K., & Zhu, J. (2014). Data envelopment analysis: Prior to choosing a model. *Omega (United Kingdom)*. <https://doi.org/10.1016/j.omega.2013.09.004>
- Cooper, W. W., Seiford, L. M., & Tone, K. (2002). *Data Envelopment Analysis: A Comprehensive Text with Models, Applications, References and DEA-Solver Software*, Springer, Kluwer Academic Publishers. <https://doi.org/10.1017/CBO9781107415324.004>
- Cooper, W. W., Seiford, L. M., & Tone, K. (2006). *Introduction to data envelopment analysis and its uses*. Springer Science+Business Media, Inc.
- Covic, F. (2019). *Container Handling in Automated Yard Blocks : an Integrative Approach Based on Time Information*. Springer. <https://www.worldcat.org/title/container-handling-in-automated-yard-blocks-an-integrative-approach-based-on-time-information/oclc/1083641762>
- Crotti, D., Ferrari, C., & Tei, A. (2019). Merger waves and alliance stability in container shipping. *Maritime Economics and Logistics*, 0123456789. <https://doi.org/10.1057/s41278-019-00118-6>
- Cubbin, J., & Tzanidakis, G. (1998). Regression versus data envelopment analysis for efficiency measurement: an application to the England and Wales regulated water industry. *Utilities Policy*, 7(2), 75–85. [https://doi.org/10.1016/S0957-1787\(98\)00007-1](https://doi.org/10.1016/S0957-1787(98)00007-1)
- Cullinane, K., & Khanna, M. (1999). Economies of scale in large container ships. *Journal of Transport Economics and Policy*, 33(2), 185–208. http://www.bath.ac.uk/e-journals/jtep/pdf/Volume_33_Part_2_185-207.pdf
- Cullinane, K. P. B., & Wang, T.-F. (2006). The efficiency of European container ports: A cross-sectional data envelopment analysis. *International Journal of Logistics Research and Applications*, 9(1), 19–31. <https://doi.org/10.1080/13675560500322417>
- Cullinane, K., & Song, D.-W. (2006). ESTIMATING THE RELATIVE EFFICIENCY OF EUROPEAN CONTAINER PORTS: A STOCHASTIC FRONTIER ANALYSIS. *Port Economics Research in Transportation Economics*, 16, 85–115. [https://doi.org/10.1016/S0739-8859\(06\)16005-9](https://doi.org/10.1016/S0739-8859(06)16005-9)
- Cullinane, K., Song, D.-W., Ji, P., & Wang, T.-F. (2004). An Application of DEA Windows Analysis to Container Port Production Efficiency. *Review of Network Economics*, 3(2), 184–206. <https://doi.org/10.2202/1446-9022.1050>
- Cullinane, K., & Song, D. W. (2003). A stochastic frontier model of the productive efficiency of Korean container terminals. *Applied Economics*, 35(3), 251–267. <https://doi.org/10.1080/00036840210139355>
- Cullinane, K., Song, D. W., & Wang, T. (2005). The application of mathematical programming approaches to estimating container port production efficiency. *Journal of Productivity Analysis*, 24(1), 73–92. <https://doi.org/10.1007/s11123-005-3041-9>
- Cullinane, K., Wang, T.-F., Song, D.-W., & Ji, P. (2006). The technical efficiency of

- container ports: Comparing data envelopment analysis and stochastic frontier analysis. *Transportation Research Part A*, 40, 354–374.
<https://doi.org/10.1016/j.tra.2005.07.003>
- Cullinane, K., & Wang, T. (2010). The efficiency analysis of container port production using DEA panel data approaches. *OR Spectrum*, 32(3), 717–738.
<https://doi.org/10.1007/s00291-010-0202-7>
- De Borger, B., Kerstens, K., & Costas, Á. (2002). Public transit performance: What does one learn from frontier studies? *Transport Reviews*, 22(1), 1–38.
<https://doi.org/10.1080/01441640010020313>
- Debreu, G. (1951). The Coefficient of Resource. *Econometrica*, 19(3), 273–292.
- Dhingra, T., Prakash, N. B., & Raju, T. B. (2014). The known and unknown facets of port efficiency measurement : A systematic review of literature . *International Conference on Frontiers of Infrastructure Finance*At: Vinod Gupta School of Management, IIT Kharagpur, April 2016.
- Di Vaio, A., & Varriale, L. (2018). Management innovation for environmental sustainability in seaports: Managerial accounting instruments and training for competitive green ports beyond the regulations. *Sustainability (Switzerland)*, 10(3), 1–35. <https://doi.org/10.3390/su10030783>
- Diewert, E. (2008). What Is To Be Done for Better Productivity Measurement. *International Productivity Monitor*, 16(2001), 40–52.
- Diewert, W. E., & Schreyer, P. (2006). *The measurement of capital The Measurement of Capital The Measurement of Capital*.
<https://www.researchgate.net/publication/228395231>
- Drewry. (2014). *Container terminal capacity and performance benchmarks: Terminals automation*.
- Drewry. (2017). *Drewry: Container Port Operators Eye Mergers and Acquisitions | World Maritime News*. <http://worldmaritimenews.com/archives/198481/drewry-container-port-operators-eye-mergers-and-acquisitions/>
- Drewry. (2019a). *Global Container Terminal Operators The Global ContainerTerminal Industry*.
- Drewry. (2019b). *Global terminal operators: Annual review & forecast*.
- Drucker, P. (1963). *Managing for business effectiveness*. Harvard Business Review.
- Dutra, A., Ripoll-Feliu, V. M., Fillol, A. G., Ensslin, S. R., & Ensslin, L. (2015). The construction of knowledge from the scientific literature about the theme seaport performance evaluation. *International Journal of Productivity and Performance Management*, 64(2), 243–269. <https://doi.org/10.1108/ijppm-01-2014-0015>
- Dyckhoff, H., & Souren, R. (2020). *Performance Evaluation Foundations and Challenges*. Springer Nature Switzerland AG.
- Dyson, R. G., Allen, R., Camanho, A. S., Podinovski, V. V., Sarrico, C. S., & Shale, E. A. (2001). Pitfalls and protocols in DEA. *European Journal of Operational Research*, 132(2001), 245–259. [https://doi.org/10.1016/S0377-2217\(00\)00149-1](https://doi.org/10.1016/S0377-2217(00)00149-1)
- Elferjani, A. (2015). *Examination of port performance in a developing economy : A case study of Libyan ports* (Issue April). RMIT University.

- Erkoc, T. (2012). Estimation Methodology of Economic Efficiency: Stochastic Frontier Analysis vs Data Envelopment Analysis. *International Journal of Academic Research in Economics and Management Sciences*, 1(1), 1–23.
- Erwin Diewert, W., & Schreyer, P. (2008). Capital Measurement. In *The New Palgrave Dictionary of Economics* (pp. 1–6). Palgrave Macmillan UK.
https://doi.org/10.1057/978-1-349-95121-5_2429-1
- Esmer, S. (2008). Performance Measurements of Container Terminal Operations. *DOKUZ EYLÜL Üniversitesi*, 10(1), 238–255.
<http://www.sbe.deu.edu.tr/dergi/cilt10.say1/10.1/esmer.pdf>
- Esser, A., Sys, C., Vanellander, T., & Verhetsel, A. (2020). The labour market for the port of the future. A case study for the port of Antwerp. *Case Studies on Transport Policy*, 8(2), 349–360. <https://doi.org/10.1016/j.cstp.2019.10.007>
- Estache, A., de la Fé, B. T., & Trujillo, L. (2004). Sources of efficiency gains in port reform: A DEA decomposition of a Malmquist TFP index for Mexico. *Utilities Policy*, 12(4), 221–230. <https://doi.org/10.1016/j.jup.2004.04.013>
- Estache, A., González, M., & Trujillo, L. (2002). Efficiency gains from port reform and the potential for yardstick competition: Lessons from Mexico. *World Development*, 30(4), 545–560. [https://doi.org/10.1016/S0305-750X\(01\)00129-2](https://doi.org/10.1016/S0305-750X(01)00129-2)
- Farrell, M. J. (1957). The Measurement of Productive Efficiency. *Journal of Roayal Statistical Society*, 120(3), 253–290.
- Faust, A.-K., & Baranzini, A. (2014). The economic performance of Swiss drinking water utilities. *Journal of Productivity Analysis*, 41(3), 383–397.
<https://doi.org/10.1007/s11123-013-0344-0>
- Ferrari, C, Puliafito, P., & Tei, A. (2018). Port Terminal Concessions: Towards a Dynamic Concession Fee. *Journal of Transport Economics and Policy*, 52(2), 137–156.
- Ferrari, Claudio, Puliafito, P. P., & Tei, A. (2019). Dynamics in terminal concessions: the role of performances. *Maritime Economics and Logistics*, 21(1), 99–110.
<https://doi.org/10.1057/s41278-017-0073-9>
- Figueiredo De Oliveira, G., & Cariou, P. (2015). The impact of competition on container port (in)efficiency. *Transportation Research Part A: Policy and Practice*, 78, 124–133. <https://doi.org/10.1016/j.tra.2015.04.034>
- Forsund, F. R., & Hjalmarsson, L. (1974). On the Measurement of Productive Efficiency. *The Swedish Journal of Economics*, 76(2), 141–154.
- Forsund, F. R., & Hjalmarsson, L. (1979). Generalised Farrell Measures of Efficiency: An Application to Milk Processing in Swedish Dairy Plants. *The Economic Journal*, 89(354), 294. <https://doi.org/10.2307/2231603>
- Førsund, F. R., & Sarafoglou, N. (2002). On the Origins of Data Envelopment Analysis. In *Source: Journal of Productivity Analysis* (Vol. 17, Issue 1).
<https://www.jstor.org/stable/pdf/41770070.pdf?refreqid=excelsior%3Aa6f578903f18e6aa942eccc3653f1dcb>
- Franc, P., & Van der Horst, M. (2010). Understanding hinterland service integration by shipping lines and terminal operators: A theoretical and empirical analysis. *Journal of Transport Geography*, 18(4), 557–566.
<https://doi.org/10.1016/j.jtrangeo.2010.03.004>

- Gekara, V. O., & Thanh Nguyen, V. X. (2018). New technologies and the transformation of work and skills: a study of computerisation and automation of Australian container terminals. *New Technology, Work and Employment*, 33(3), 219–233. <https://doi.org/10.1111/ntwe.12118>
- Gharehgozli, A., Zaerpour, N., & de Koster, R. (2020). Container terminal layout design: transition and future. *Maritime Economics and Logistics*, 22(4), 610–639. <https://doi.org/10.1057/s41278-019-00131-9>
- González, M. M., & Trujillo, L. (2009). Efficiency measurement in the port industry: A survey of the empirical evidence. *Journal of Transport Economics and Policy*, 43(2), 157–192.
- Grosskopf, S. (2002). Some remarks on productivity and its decompositions. *Journal of Productivity Analysis*, 20(3), 459–474. <https://doi.org/10.1023/A:1027364119672>
- Grzelakowski, A. S. (2020). The concept of internalisation of the external costs of transport in the EU' and its impact on the efficiency of transport systems and the performance of logistics supply chains. *Prace Naukowe Uniwersytetu Ekonomicznego We Wrocławiu*, 64(3), 161–174. <https://doi.org/10.15611/pn.2020.3.13>
- Guerrero C., A., & Rivera T., C. (2009). Mexico: Total productivity changes at the principal container ports. *CEPAL Review*, 2009(99), 173–185. <https://doi.org/10.18356/81702067-en>
- Günther, H. O., & Kim, K. H. (2006). Container terminals and terminal operations. *OR Spectrum*, 28(4), 437–445. <https://doi.org/10.1007/s00291-006-0059-y>
- Gutenberg, E. (1951). *Grundlagen der Betriebswirtschaftslehre. Band I: Die Produktion* (24th editi). Springer.
- Haezendonck, E., Pison, G., Rousseeuw, P., Struyf, A., & Verbeke, A. (2016). THE CORE COMPETENCES OF THE ANTWERP SEAPORT : AN ANALYSIS OF " PORT SPECIFIC " ADVANTAGES. *International Journal of Transport Economics*, 28(3), 325–349.
- Haezendonck, Elvira, & Langenus, M. (2019). Integrated ports clusters and competitive advantage in an extended resource pool for the Antwerp Seaport. *Maritime Policy and Management*, 46(1), 74–91. <https://doi.org/10.1080/03088839.2018.1471535>
- Haezendonck, Elvira, van den Broeck, J., & Jans, T. (2011). Analysing the lobby-effect of port competitiveness' determinants: A stochastic frontier approach. *Journal of Productivity Analysis*, 36(2), 113–123. <https://doi.org/10.1007/s11123-011-0208-4>
- Han, C. (2014). *Port performance benchmarking and efficiency analysis*.
- Heaver, T., Meersman, H., & Van De Voorde, E. (2001). Co-Operation and Competition in International Container Transport Strategies for Ports. *Maritime Policy & Management*, 28(3), 293–305. <https://doi.org/10.1080/03088830110055693>
- Heilig, L., Stahlbock, R., & Voß, S. (2019). *From Digitalization to Data-Driven Decision Making in Container Terminals*. <http://arxiv.org/abs/1904.13251>
- Heilig, L., Stahlbock, R., & Voß, S. (2020). From Digitalization to Data-Driven Decision Making in Container Terminals. In *Handbook of Terminal Planning* (p. 125). Springer Nature Switzerland AG.

- Hlali, A., & Hammami, S. (2017). Seaport Concept and Services Characteristics: Theoretical Test. *The Open Transportation Journal*, 11(1), 120–129. <https://doi.org/10.2174/1874447801711010120>
- Hofbauer, F., & Putz, L. M. (2020). External costs in inland waterway transport: An analysis of external cost categories and calculation methods. *Sustainability (Switzerland)*, 12(14). <https://doi.org/10.3390/su12145874>
- Holger Schütt. (2020). Advanced Simulation Technology in Planning, Implementation, and Operation of Container Terminals to Cope with the Varying Challenges Caused by the Shipping Industry. In *Handbook of Terminal Planning* (p. 31). Springer Nature Switzerland AG.
- Huguenin, J.-M. (2012). *Data Envelopment Analysis (DEA) A pedagogical guide for decision makers in the public sector*. https://serval.unil.ch/resource/serval:BIB_0FC432348A97.P001/REF
- Hung, S. W., Lu, W. M., & Wang, T. P. (2010). Benchmarking the operating efficiency of Asia container ports. *European Journal of Operational Research*, 203(3), 706–713. <https://doi.org/10.1016/j.ejor.2009.09.005>
- Imai, A., Nishimura, E., & Papadimitriou, S. (2001). The dynamic berth allocation problem for a container port. *Transportation Research Part B: Methodological*, 35(4), 401–417. [https://doi.org/10.1016/S0191-2615\(99\)00057-0](https://doi.org/10.1016/S0191-2615(99)00057-0)
- Issam, E. H., Lajjam, A., El Merouani, M., & Tabaa, Y. (2019). A modified sailfish optimizer to solve dynamic berth allocation problem in conventional container terminal. *International Journal of Industrial Engineering Computations*, 10(4), 491–504. <https://doi.org/10.5267/j.ijiec.2019.4.002>
- Itoh, H. (2002). Efficiency changes at major container ports in Japan: A window application of data envelopment analysis. *Review of Urban and Regional Development Studies*, 14(2), 133–152. <https://doi.org/10.1111/1467-940X.00052>
- Jara-Diaz, S., Tovar, B., & Trujillo, L. (2011). On the proper modelling of multioutput port cargo handling costs. *Applied Economics*, 40(13), 1699–1705.
- Jarboui, S., Forget, P., & Boujelbene, Y. (2012). Public road transport efficiency: A literature review via the classification scheme. *Public Transport*, 4(2), 101–128. <https://doi.org/10.1007/s12469-012-0055-3>
- Johnes, G., & Johnes, J. (2004). International handbook on the economics of education. In *International Handbook on the Economics of Education* (Issue January). <https://doi.org/10.4337/9781845421694>
- Kammoun, R., & Abdenadhe, C. (2018). The Technical Efficiency of Tunisian Ports: Comparing Data Envelopment Analysis and Stochastic Frontier Analysis Scores. *Logistics & Sustainable Transport*, 9(2), 73–84. <https://doi.org/10.2478/jlst-2018-0011>
- Karafolas, S., & Mantakas, G. (1996). A note on cost structure and economies of scale in Greek banking. *Journal of Banking & Finance*, 20, 377–387. http://ac.els-cdn.com/0378426695000046/1-s2.0-0378426695000046-main.pdf?_tid=fc7af192-35cc-11e7-baab-00000aabb0f02&acdnat=1494454198_2b59165c171af93c5c44ebc9f72a4f90
- Kemme, N. (2013). *Design and Operation of Automated Container Storage Systems*. <https://doi.org/10.1007/978-3-7908-2885-6>

- Koopmans, C. T. (1951). Activity Analysis of Production and Allocation. *Activity Analysis of Production and Allocation*. <https://ci.nii.ac.jp/naid/10021314433>
- Kumar, S., & Gulati, R. (2008). An examination of technical, pure technical and scale efficiencies in Indian public sector banks using data Envelopment Analysis. *Eurasian Journal of Business and Economics*, 1(2), 33–69. <https://doi.org/10.1504/ajfa.2008.019950>
- Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (1997). Applied Linear Statistical Models. In *Journal of Quality Technology* (Vol. 29, Issue 2). <https://doi.org/10.1080/00224065.1997.11979760>
- Kuzmicz, K. A. (2020). Problems in Eurasian Container Supply Chains. *Eurasian Economic Perspectives, Eurasian Studies in Business and Economics*, 165.
- Kuzmicz, K. A., & Pesch, E. (2019). Approaches to empty container repositioning problems in the context of Eurasian intermodal transportation. *Omega (United Kingdom)*, 85, 194–213. <https://doi.org/10.1016/j.omega.2018.06.004>
- Lange, A.-K., Schwientek, A., & Jahn, C. (2017). Reducing Truck Congestion at Ports – Classification and Trends. *Proceedings of the Hamburg International Conference of Logistics (HICL)*, 24, 37–58.
- Langenus, M., & Dooms, M. (2015). Port industry performance management: a meso-level gap in literature and practice? *International Journal of Logistics Research and Applications*, 18(3), 251–275. <https://doi.org/10.1080/13675567.2015.1026885>
- Lecue, M., & Darbra, R. M. (2019). Accidents in European ports involving chemical substances: Characteristics and trends. *Safety Science*, 115(July 2018), 278–284. <https://doi.org/10.1016/j.ssci.2019.02.015>
- Lee, P. T. W., Lam, J. S. L., Lin, C. W., Hu, K. C., & Cheong, I. (2018). Developing the fifth generation port concept model: an empirical test. *International Journal of Logistics Management*, 29(3), 1098–1120. <https://doi.org/10.1108/IJLM-10-2016-0239>
- Levinson, M. (2016). *The box: How the shipping container made the World smaller and the World economy bigger* (Second edi). Princeton University Press.
- Li, H., & Lu, Y. (2019). Container Terminal Handling Technology and Its Development. *Academic Journal of Engineering and Technology Science*, 2(1), 144–152. <https://doi.org/10.25236/AJETS.020026>
- Lim, S. (1998). Economies of scale in container shipping. *Maritime Policy & Management*, 25(4), 361–373. <https://doi.org/10.1080/030888398000000059>
- Linna, P., Pekkola, S., Ukko, J., & Melkas, H. (2010). Defining and measuring productivity in the public sector: Managerial perceptions. *International Journal of Public Sector Management*, 23(5), 479–499. <https://doi.org/10.1108/09513551011058493>
- Liu, Q. (2010). *Efficiency Analysis of Container Ports and Terminals* [University College London]. <http://discovery.ucl.ac.uk/19215/1/19215.pdf>
- Loertscher, S., & Reisinger, M. (2014). Market structure and the competitive effects of vertical integration. *RAND Journal Of Economics*, 45(3), 471–494. <https://doi.org/10.1016/j.econlet.2014.12.008>

- López, F. J., Ho, J. C., & Ruiz-Torres, A. J. (2016). A computational analysis of the impact of correlation and data translation on DEA efficiency scores. *Journal of Industrial and Production Engineering*, 33(3), 192–204.
<https://doi.org/10.1080/21681015.2015.1136704>
- Lozano, S., & Villa, G. (2004). Centralized Resource Allocation Using Data Envelopment Analysis. *Journal of Productivity Analysis*, 22, 143–161.
<https://doi.org/https://doi.org/10.1023/B:PROD.0000034748.22820.33>
- Lu, B., & Wang, S. (2016). DEA Conceptual Exposition and Literature Review. In *Container Port Production and Management* (pp. 1–123).
<https://doi.org/10.1007/978-981-10-2428-3>
- Luna, J. H., Mar-Ortiz, J., Gracia, M. D., & Morales-Ramírez, D. (2017). An efficiency analysis of cargo-handling operations at container terminals. *Maritime Economics and Logistics*, 20(2), 190–210. <https://doi.org/10.1057/s41278-017-0074-8>
- Manaadiar, H. (2021). *Importance of proper lashing of containers on board ships*. Shipping and Freight Resource.
<https://www.shippingandfreightresource.com/importance-of-proper-container-lashing-on-board-ships/>
- Mardani, A., Zavadskas, E. K., Streimikiene, D., Jusoh, A., & Khoshnoudi, M. (2017). A comprehensive review of data envelopment analysis (DEA) approach in energy efficiency. In *Renewable and Sustainable Energy Reviews*.
<https://doi.org/10.1016/j.rser.2016.12.030>
- Marek, R. (2019). THE ANALYSIS OF FIVE COMPETITIVE FORCES OF MARINE CONTAINER TERMINAL INDUSTRY BASED ON POLISH MARKET. *45th International Scientific Conference on Economic and Social Development – XIX International Social Congress*, 2019(1).
<https://www.researchgate.net/publication/341443992>
- Martín-Soberón, A. M., Monfort, A., Sapiña, R., Monterde, N., & Calduch, D. (2014). Automation in port container terminals. *Procedia - Social and Behavioral Sciences*, 160, 195–204. <https://doi.org/10.1016/j.sbspro.2014.12.131>
- Martinez-Budria, E., Coto-Millan, P., & Navarro- Ibanez, M. (2001). TRAFFIC REVENUES OF SPANISH PORT AUTHORITIES. *International Journal of Transport Economics*, 28(2), 249–262.
- MARTÍNEZ-BUDRÍA, E., DÍAZ-ARMAS, R., NAVARRO-IBAÑEZ, M., & MESA, T. R.-. (1999). A STUDY OF THE EFFICIENCY OF SPANISH PORT AUTHORITIES USING DATA ENVELOPMENT ANALYSIS. *International Journal of Transport Economics*, 26(2), 237–253.
- Medda, F., & Liu, Q. (2013). Determinants and strategies for the development of container terminals. *Journal of Productivity Analysis*, 40(1), 83–98.
<https://doi.org/10.1007/s11123-012-0291-1>
- Meersman, H., Pauwels, T., Van De Voorde, E., & Vanelslander, T. (2008). The relation between port competition and hinterland connections the case of the “Iron Rhine” and the “Betuweroute.” *Proceedings of International Forum on Shipping, Ports and Airports, IFSPA 2008 - Trade-Based Global Supply Chain and Transport Logistics Hubs: Trends and Future Development*, 358–376.

- Meersman, H., Strandenes, S. P., & van de Voorde, E. (2014). Port Pricing: Principles, Structure and Models. *SSRN Electronic Journal*, April. <https://doi.org/10.2139/ssrn.2432939>
- Meersman, H., Van de Voorde, E., & Vanellander, T. (2010). Port Competition Revisited. *Review of Business and Economic Literature Intersentia*, 55(2), 210–33.
- Merchan, A. L., Léonard, A., Limbourg, S., & Mostert, M. (2019). Life cycle externalities versus external costs: The case of inland freight transport in Belgium. *Transportation Research Part D: Transport and Environment*, 67(January), 576–595. <https://doi.org/10.1016/j.trd.2019.01.017>
- Merk, O. (2020). CHINA ' S PARTICIPATION IN EUROPEAN CONTAINER PORTS : DRIVERS AND POSSIBLE FUTURE SCENARIOS. *Revue Internationale et Stratégique*, 17, 41–53. <https://doi.org/10.3917/ris.117.0041>.
- Merkel, A. (2018). Competitive intensity and inefficiency in European container ports: An empirical investigation using SFA. *Maritime Business Review*, 3(2), 165–184. <https://doi.org/10.1108/mabr-01-2018-0003>
- Merkel, A. (2019). *Container port infrastructure in the European Union: Inter-port competition, investment appraisal and the measurement of efficienc* (Issue May). Molde University College.
- Min, H., & Park, B. I. (2005). Evaluating the inter-temporal efficiency trends of international container terminals using data envelopment analysis. *International Journal of Integrated Supply Management*, 1(3), 258–277. <https://doi.org/10.1504/IJISM.2005.005950>
- Mohseni, S. A., van Hassel, E., Sys, C., & Vanellander, T. (2019). Economic evaluation of alternative technologies to mitigate Sulphur emissions in maritime container transport from both the vessel owner and shipper perspective. *Journal of Shipping and Trade*, 4(1). <https://doi.org/10.1186/s41072-019-0051-8>
- Moschouli, E., Soecipto, R. M., & Vanellander, T. (2019). Cost performance of transport infrastructure projects before and after the global financial crisis (GFC): Are differences observed in the conditions of project performance? *Research in Transportation Economics*, 75(December 2016), 21–35. <https://doi.org/10.1016/j.retrec.2019.02.004>
- Munisamy, S., & Singh, G. (2011). Benchmarking the Efficiency of Asian Container Ports. *African Journal of Business Management*, 5(4), 1397–1407. <https://doi.org/10.15735/kl.2019.27.1.003>
- Musso, E., Ferrari, C., & Benacchio, M. (2000). Co-operation in maritime and port industry and its effects on markets structure. *Maritime Engineering and Ports II*. <https://doi.org/10.1017/CBO9781107415324.004>
- Nachar, N. (2008). The Mann-Whitney U: A Test for Assessing Whether Two Independent Samples Come from the Same Distribution. *Tutorials in Quantitative Methods for Psychology*, 4(1), 13–20. <https://doi.org/10.20982/tqmp.04.1.p013>
- Onghena, E. (2013). *From cost structure to strategy: The impact of the cost structure on the strategic behaviour of integrators*. University of Antwerp.
- Ostertagová, E., Ostertag, O., & Kováč, J. (2014). Methodology and application of the Kruskal-Wallis test. *Applied Mechanics and Materials*, 611(August 2014), 115–120.

<https://doi.org/10.4028/www.scientific.net/AMM.611.115>

- Oum, T. H., Tretheway, M. W., & Zhang, Y. (1991). A Note on Capacity Utilization and Measurement of Scale Economies. *Journal of Business & Economic Statistics*, 9(1), 119–123. <http://www.jstor.org/stable/1391947>
- Palmer, S., & Torgerson, D. J. (1999). Definitions of efficiency. *Bmj*, 318(7191), 1136. <https://doi.org/10.1136/bmj.318.7191.1136>
- Panayides, P. M., Maxoulis, C. N., Wang, T. F., & Ng, K. Y. A. (2009). A critical analysis of DEA applications to seaport economic efficiency measurement. *Transport Reviews*, 29(2), 183–206. <https://doi.org/10.1080/01441640802260354>
- Paradi, J. C., Sherman, H. D., & Tam, F. K. (2018). *DEA Models Overview*. In: *Data Envelopment Analysis in the Financial Services Industry. International Series in Operations Research & Management Science* (Issue 266). Springer, Cham.
- Park, N. K., & An, Y. (2020). Financial analysis of automated container terminal capacity from the perspective of terminal operating company. *Journal of Marine Science and Engineering*, 8(11), 1–18. <https://doi.org/10.3390/jmse8110954>
- Park, R., & De, P. (2004). An Alternative Approach to Efficiency Measurement of Seaports. In P. Macmillan (Ed.), *Palgrave Readers in Economics* (Haralambid, pp. 273–292). Palgrave Macmillan, London.
- Park, R. K., & De Prabir, P. (2004). An alternative approach to efficiency measurement of seaports. *Maritime Economics and Logistics*, 6(1), 53–69. <https://doi.org/10.1057/palgrave.mel.9100094>
- Plasschaert, K., Derudder, B., Dullaert, W., & Witlox, F. (2011). Redefining the Hamburg – Le Havre range in maritime networks. *BIVE-GIBET Transport Research Day*, 240–244.
- Poitras, G., Tongzon, J., & Li, H. (1996). *Measuring Port Efficiency: An Application of Data Envelopment Analysis*. <https://pdfs.semanticscholar.org/27fd/fff745c9f9f319ab28614386117fa4dafec1.pdf>
- Poole, M. A., & O'Farrell, P. N. (1971). The Assumptions of the Linear Regression Model. *Transactions of the Institute of British Geographers*, 52(52), 145. <https://doi.org/10.2307/621706>
- Port of Antwerp. (2020). *Online programme Port & Terminal Management, May 6th*.
- Pulley, L. B., & Braunstein, Y. M. (1992). A composite cost function for multiproduct firms with an application to economies of scope in banking. *MIT Press Journals*, 74(2), 221. <http://eds.a.ebscohost.com/eds/pdfviewer/pdfviewer?sid=6992f063-0d98-4ea8-81e1-923d6cdb3057%40sessionmgr4007&vid=18&hid=4105>
- Rahman, A. T., Sarno, R., & Effendi, Y. A. (2018). Goal programming to optimize time and cost for each activity in port container handling. *2018 International Conference on Information and Communications Technology, ICOIACT 2018, 2018-Janua*, 866–871. <https://doi.org/10.1109/ICOIACT.2018.8350808>
- Rajabi, A., Khodadad Saryazdi, A., Belfkih, A., & Duvallet, C. (2019). Towards Smart Port: An Application of AIS Data. *Proceedings - 20th International Conference on High Performance Computing and Communications, 16th International Conference on Smart City and 4th International Conference on Data Science and Systems, HPCC/SmartCity/DSS 2018*, 1414–1421.

<https://doi.org/10.1109/HPCC/SmartCity/DSS.2018.00234>

- Ramalho, E. A., Ramalho, J. J. S., & Henriques, P. D. (2010). Fractional regression models for second stage DEA efficiency analyses. *Journal of Productivity Analysis*, 34(3), 239–255. <https://doi.org/10.1007/s11123-010-0184-0>
- Rashed, Y., Meersman, H., Sys, C., Van de Voorde, E., & Vanelslender, T. (2018). A combined approach to forecast container throughput demand: Scenarios for the Hamburg-Le Havre range of ports. *Transportation Research Part A: Policy and Practice*, 117, 127–141. <https://doi.org/10.1016/j.tra.2018.08.010>
- Rezaei, J., van Wulfften Palthe, L., Tavasszy, L., Wiegmans, B., & van der Laan, F. (2019). Port performance measurement in the context of port choice: an MCDA approach. *Management Decision*, 57(2), 396–417. <https://doi.org/10.1108/MD-04-2018-0482>
- Rios, L. R., & Maçada, A. C. G. (2006). Analysing the relative efficiency of container terminals of mercosur using DEA. *Maritime Economics and Logistics*, 8(4), 331–346. <https://doi.org/10.1057/palgrave.mel.9100168>
- Rogerson, S., Santén, V., Svanberg, M., Williamsson, J., & Woxenius, J. (2020). Modal shift to inland waterways: dealing with barriers in two Swedish cases. *International Journal of Logistics Research and Applications*, 23(2), 195–210. <https://doi.org/10.1080/13675567.2019.1640665>
- Roghianian, P., Rasli, A., & Gheysari, H. (2012). Productivity Through Effectiveness and Efficiency in the Banking Industry. *Procedia - Social and Behavioral Sciences*, 40, 550–556. <https://doi.org/10.1016/j.sbspro.2012.03.229>
- Roll, Y., & Hayuth, Y. (1993). Port performance comparison applying data envelopment analysis (DEA). *Maritime Policy and Management*, 20(2), 153–161. <https://doi.org/10.1080/030888393000000025>
- Romano, G., & Guerrini, A. (2011). Measuring and comparing the efficiency of water utility companies: A data envelopment analysis approach. *Utilities Policy*, 19(3), 202–209. <https://doi.org/10.1016/j.jup.2011.05.005>
- Saavedra, J. P. (2002). Cargo Theft in Latin America. *International Union of Marine Insurance Conference*. [https://iumi.com/images/stories/IUMI/Pictures/Conferences/NewYork2002/Wednesday/03 saavedra cargo lp.pdf](https://iumi.com/images/stories/IUMI/Pictures/Conferences/NewYork2002/Wednesday/03%20saavedra%20cargo%20lp.pdf)
- Saeedi, H., Behdani, B., Wiegmans, B., & Zuidwijk, R. A. (2018). Performance Measurement in Freight Transport Systems. *SSRN Electronic Journal*, 1–22. <https://doi.org/10.2139/ssrn.3122275>
- Schroten, A., van Essen, H., van Wijngaarden, L., Sutter, D., & Andrew, E. (2019). *Sustainable transport infrastructure charging and internalisation of transport externalities: Executive summary* (Issue May 2019).
- Seedah, D., Harrison, R., Boske, L., & Kruse, J. (2013). *CONTAINER TERMINAL AND CARGO-HANDLING COST ANALYSIS TOOLKIT*. Texas A&M Transportation Institute.
- Senarak, C. (2021). The Asian Journal of Shipping and Logistics Port cybersecurity and threat : A structural model for prevention and policy development &. *The Asian Journal of Shipping and Logistics*, 37(1), 20–36. <https://doi.org/10.1016/j.ajsl.2020.05.001>

- Seo, Y.-J., & Park, J. S. (2016). The estimation of minimum efficient scale of the port industry. *Transport Policy*, 49, 168–175.
<https://doi.org/10.1016/j.tranpol.2016.04.012>
- Shephard, R. (1953). *Cost and production functions*. Princeton University Press. Princeton, NJ.
- Shephard, R. W. (1970). Theory of Cost and Production Functions. In *The Journal of Economic History* (Issue 3). Princeton University Press.
<https://doi.org/10.1017/s002205070007457x>
- Simar, L., & Wilson, P. W. (2007). Estimation and inference in two-stage, semi-parametric models of production processes. *Journal of Econometrics*, 136(1), 31–64. <https://doi.org/10.1016/j.jeconom.2005.07.009>
- Siminică, M., Berceanu, D., & Cîrciumaru, D. (2008). THE PERFORMANCES OF INDUSTRIAL FIRMS FROM ROMANIA. CORRELATION DIMENSION – INDICATORS OF RESULTS. *Financial Innovation and Competition in European Union*, 12(517), 71–78.
- Simões, P., & Marques, R. C. (2010). Influence of congestion efficiency on the european seaports performance: Does it matter? *Transport Reviews*, 30(4), 517–539.
<https://doi.org/10.1080/01441640903175592>
- Steenken, D., Voß, S., & Stahlbock, R. (2004). Container terminal operation and operations research – a classification and literature review. *Operations Research-Spektrum*, 26, 3–49. <https://doi.org/10.1007/s00291-003-0157-z>
- Sterlacchini, A. (2001). The determinants of export performance: A firm-level study of italian manufacturing. *Weltwirtschaftliches Archiv*, 137(3), 450–472.
<https://doi.org/10.1007/bf02707626>
- Stojaković, M., & Twardy, E. (2021). Determining the optimal number of yard trucks in smaller container terminals. *European Transport Research Review*, 13(1).
<https://doi.org/10.1186/s12544-021-00482-6>
- Struyf, E., Sys, C., Van de Voorde, E., & Vanellander, T. (2020). Calculating the cost of congestion to society: A case study application to Flanders. *Research in Transportation Business and Management*, March, 100573.
<https://doi.org/10.1016/j.rtbm.2020.100573>
- Svindland, M. (2018). The environmental effects of emission control area regulations on short sea shipping in Northern Europe: The case of container feeder vessels. *Transportation Research Part D: Transport and Environment*, 61(2018), 423–430.
<https://doi.org/10.1016/j.trd.2016.11.008>
- Sys, C., Blauwens, G., Omeij, E., Van de voorde, E., & Witlox, F. (2008). In Search of the Link between Ship Size and Operations. *Transportation Planning and Technology*, 31(4), 435–463. <https://doi.org/10.1080/03081060802335109>
- Taouab, O., & Issor, Z. (2019). Firm Performance: Definition and Measurement Models. *European Scientific Journal ESJ*, 15(1), 93–106.
<https://doi.org/10.19044/esj.2019.v15n1p93>
- The World bank. (2007). Port Reform Toolkit. In *Port Reform Toolkit*.
<https://doi.org/10.1596/0-8213-5046-3>
- The World Bank. (2007a). Port Redorm toolkit The evolution of ports in a competitive

- world. In *Port Reform Toolkit* (2nd Editio). <https://doi.org/10.1596/978-0-8213-6607-3>
- The World Bank. (2007b). Port Reform Toolkit Module 1: Framework for Port Reform. In *Port Reform Toolkit*.
- Tongzon, J. (2001). Efficiency measurement of selected Australian and other international ports using data envelopment analysis. *Transportation Research Part A*, 35, 107–122. file:///C:/Users/snduna/Downloads/2043.pdf
- Tongzon, J., & Heng, W. (2005). Port privatization, efficiency and competitiveness: Some empirical evidence from container ports (terminals). *Transportation Research Part A: Policy and Practice*, 39(5), 405–424. <https://doi.org/10.1016/j.tra.2005.02.001>
- Tovar, B. (2003). *A Multioutput Cost Function for Port Terminals Some Guidelines for Regulation*. <http://econ.worldbank.org>.
- Tran, N. K., & Haasis, H. D. (2015). An empirical study of fleet expansion and growth of ship size in container liner shipping. *International Journal of Production Economics*, 159, 241–253. <https://doi.org/10.1016/j.ijpe.2014.09.016>
- Turner, H., Windle, R., & Dresner, M. (2004). North American containerport productivity: 1984-1997. *Transportation Research Part E: Logistics and Transportation Review*, 40(4), 339–356. <https://doi.org/10.1016/j.tre.2003.06.001>
- UNCTAD, U. (1975). *Port Pricing & Concession*.
- Valentine, V. F., & Gray, R. (2001). The measurement of port efficiency using Data Envelopment Analysis. *Ninth World Conference on Transport Research, March*, 1–15.
- Valle, D. (2015). Agglomeration and technical efficiency : A survey of literature in frontier studies. *11th Annual International Beta Confrence for Ph.D Students and Young Researchers, February*.
- Van De Voorde, E., & Vanelslander, T. (2010). Market power and vertical and horizontal integration in the maritime shipping and port industry. In *Integration and Competition between Transport and Logistics Businesses*. OECD Publishing. <https://doi.org/https://doi.org/10.1787/9789282102619-3-en>
- Van Hassel, E., Meersman, H., Van de Voorde, E., & Vanelslander, T. (2016). Impact of scale increase of container ships on the generalised chain cost. *Maritime Policy & Management*, 43(2), 192–208. <https://doi.org/10.1080/03088839.2015.1132342>
- Vanelslander, T. (2005). *The Economics behind Co-operation and Competition in Sea-Port Container Handling*. University of Antwerp.
- Vanelslander, T., & Sys, C. (2014). *Port Business: Market Challenges and Management Actions* (eds). UPA University Press Antwerp.
- Varbanova, A. (2018). FEASIBILITY STUDY FOR THE IMPLEMENTATION OF EDI SYSTEMS FOR INFORMATION EXCHANGE BETWEEN BULGARIAN BLACK SEA PORTS AND. *International Scientific Journal "Industry 4.0,"* 147(3), 144–147.
- Walsh, C., & Mander, S. (2016). Contextualising the drivers for trade: Some lessons from historical case studies. *Marine Policy*, 75(2017), 290–299. <https://doi.org/10.1016/j.marpol.2016.04.004>

- Wang, N., Chang, D., Shi, X., Yuan, J., & Gao, Y. (2019). Analysis and design of typical automated container terminals layout considering carbon emissions. *Sustainability (Switzerland)*, 11(10), 1–40. <https://doi.org/10.3390/su11102957>
- Wang, T. F., & Cullinane, K. (2015). The efficiency of European container terminals and implications for supply chain management. In *Maritime Economics and Logistics* (Haralambid, pp. 253–272). Palgrave Readers in Economics. <https://doi.org/10.1057/palgrave.mel.9100151>
- Wanke, P., & Barros, C. P. (2016). New evidence on the determinants of efficiency at Brazilian ports: a bootstrapped DEA analysis. *International Journal of Shipping and Transport Logistics*, 8(3), 250. <https://doi.org/10.1504/IJSTL.2016.076240>
- Wanke, P. F. (2013). Physical infrastructure and shipment consolidation efficiency drivers in Brazilian ports: A two-stage network-DEA approach. *Transport Policy*, 29, 145–153. <https://doi.org/10.1016/j.tranpol.2013.05.004>
- Wanke, P. F., Barbastefano, G. R., & Hijjar, F. M. (2011). Determinants of Efficiency at Major Brazilian Port Terminals. *Transport Reviews*. <https://doi.org/10.1080/01441647.2011.547635>
- Watfa, M. K., & Karmadi, K. Al. (2019). Connected Automated Guided Vehicles in a Smart Container Terminal. *IOP Conference Series: Materials Science and Engineering*, 533(1). <https://doi.org/10.1088/1757-899X/533/1/012029>
- Wiegman, B., & Behdani, B. (2017). A review and analysis of the investment in, and cost structure of, intermodal rail terminals. *Transport Reviews*. <https://doi.org/10.1080/01441647.2017.1297867>
- Wiegman, B., & Dekker, S. (2014). Benchmarking deep-sea port performance in the Hamburg-Le Havre range. *Benchmarking*, 23(1), 96–112. <https://doi.org/10.1108/BIJ-04-2013-0050>
- Wiegman, B. W., Nijkamp, P., & Masurel, E. (1999). Intermodal Freight Terminals: Terminal Handling Costs. *Transportation Planning and Technology*, 23(2), 105–128. https://doi.org/10.2208/jscej.1999.632_1
- Wiegman, B., & Witte, P. (2017). Efficiency of inland waterway container terminals: Stochastic frontier and data envelopment analysis to analyze the capacity design- and throughput efficiency. *Transportation Research Part A: Policy and Practice*, 106(September), 12–21. <https://doi.org/10.1016/j.tra.2017.09.007>
- Wiese, J., Suhl, L., & Kliwer, N. (2011). Planning container terminal layouts considering equipment types and storage block design. In J. Böse (Ed.), *Handbook of Terminal Planning* (pp. 219–245). Springer.
- Wilmsmeier, G., & Monios, J. (2020). Container shipping: Beyond the era of maturity? *Geographies of Maritime Transport: Transport, Mobilities and Spatial Change*, January, 192–209. <https://doi.org/10.4337/9781788976640.00016>
- WIŚNICKI, B., CHYBOWSKI, L., & CZARNECKI, M. (2017). Management Systems in Production Engineering ANALYSIS OF THE EFFICIENCY OF PORT CONTAINER TERMINALS WITH THE USE OF THE DATA ENVELOPMENT ANALYSIS METHOD. *Management Systems in Production Engineering*, 1(251), 9–15.
- Woo, S. H., Pettit, S. J., Kwak, D. W., & Beresford, A. K. C. (2011). Seaport research: A structured literature review on methodological issues since the 1980s. *Transportation Research Part A: Policy and Practice*, 45(7), 667–685.

<https://doi.org/10.1016/j.tra.2011.04.014>

- Wu, W.-M., & Lin, J.-R. (2015). Productivity growth, scale economies, ship size economies and technical progress for the container shipping industry in Taiwan. *Transportation Research Part E: Logistics and Transportation Review*, 73, 1–16. <https://doi.org/10.1016/j.tre.2014.10.011>
- Wu, Y. C. J., & Goh, M. (2010). Container port efficiency in emerging and more advanced markets. *Transportation Research Part E: Logistics and Transportation Review*, 46(6), 1030–1042. <https://doi.org/10.1016/j.tre.2010.01.002>
- Yip, T. L., Liu, J. J., Fu, X., & Feng, J. (2014). Modeling the effects of competition on seaport terminal awarding. *Transport Policy*, 35, 341–349. <https://doi.org/10.1016/j.tranpol.2014.04.007>
- Yip, T. L., Sun, X., & Liu, J. J. (2010). Evolution of global container operator efficiency: A DEA approach. *2010 8th International Conference on Supply Chain Management and Information*, January 2010, 1–7.
- Young Kim, K., & Hwan Kim, K. (1999). Routing algorithm for a single straddle carrier to load export containers onto a containership. *International Journal of Production Economics*, 59(1), 425–433. [https://doi.org/10.1016/S0925-5273\(98\)00108-X](https://doi.org/10.1016/S0925-5273(98)00108-X)
- Zehendner, E. (2013). *Operations management at container terminals using advanced information technologies présentée par*. <https://doi.org/NNT : 2013EMSE0714>
- Zeng, Q., Feng, Y., & Chen, Z. (2017). Optimizing berth allocation and storage space in direct transshipment operations at container terminals. *Maritime Economics and Logistics*, 19(3), 474–503. <https://doi.org/10.1057/mel.2016.2>
- Zenzerović, Z., Vilke, S., & Antonini, N. (2013). COST MODEL IN FUNCTION OF OPTIMAL CAPACITY PLANNING OF PORT CONTAINER TERMINAL. In *Ministry of Science, Education and Sport of the Republic of Croatia*. https://bib.irb.hr/datoteka/655219.Zenzerovic_Vilke_Antonini_Cost_Model.pdf
- Zhang, J., & Liu, Y. (2020). Study on Estimation Method of Port Container Handling Cost Based on Interval Analysis. *Journal of Coastal Research*, 95(sp1), 979–984. <https://doi.org/10.2112/SI95-191.1>
- Zhang, X., Zeng, Q., & Yang, Z. (2019). Optimization of truck appointments in container terminals. *Maritime Economics and Logistics*, 21(1), 125–145. <https://doi.org/10.1057/s41278-018-0105-0>
- Zheng, X. Bin, & Park, N. K. (2016). A study on the efficiency of container terminals in Korea and China. *The Asian Journal of Shipping and Logistics*, 32(4), 213–220. http://ac.els-cdn.com/S2092521216300803/1-s2.0-S2092521216300803-main.pdf?_tid=e4fe5e16-88db-11e7-aa91-00000aacb360&acdnat=1503586549_db9e3bf3340144e074d7434cbf4fde18
- Zheng, S., & Yin, C. (2015). Technical, allocative and cost efficiencies of Chinese ports. *Maritime Policy & Management*, 42(8), 746–758. <https://doi.org/10.1080/03088839.2015.1040860>
- Zhou, Y., & Kim, K. H. (2020). A game theoretic model and a coevolutionary solution procedure to determine the terminal handling charges for container terminals. *Computers and Industrial Engineering*, 144(April), 106466. <https://doi.org/10.1016/j.cie.2020.106466>

Zhu, S., Zheng, S., Ge, Y. E., Fu, X., Sampaio, B., & Jiang, C. (2019). Vertical integration and its implications to port expansion. *Maritime Policy and Management*, 46(8), 920–938. <https://doi.org/10.1080/03088839.2019.1594426>

Appendix

Appendix 4.1: Container terminal characteristics

Sea-side subsystem (Berth & Quay)		
Automated	Semi-Automated	Non-Automated
EDI (TOS)		
Assigning vessels to berth		
Characteristics can differ according to: - Berth length - Berth depth -Berth design: *traditional (Parallel to shoreline) *Indented (Perpendicular to shoreline)- e.g Amsterdam container terminal therefore able to handle different vessels according to: -Vessel dimensions -Vessel draft		
Quay crane allocation and scheduling: -Dedicated allocation → single cycle - Pooled allocation → Dual cycle mode		
Lock Operations: - Access limitations		
Quay crane type: -RMGC/mobile Harbor	- RMGC / RTGC / Mobile harbor	- RTGCs/ Mobile harbor

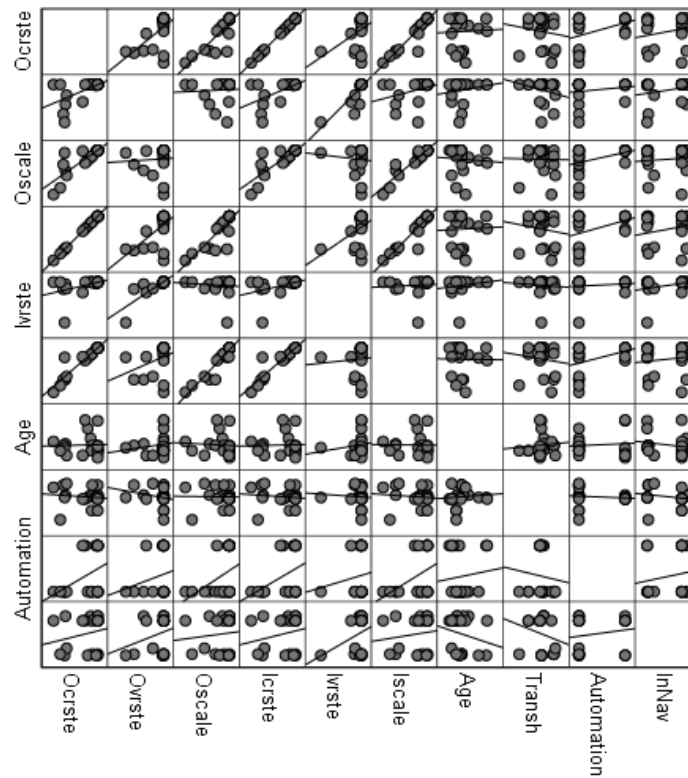
Quay crane features: - Multi-lift spreaders or single lift - Single / Double trolley - Floating/non-floating platforms		
Labour: - No labour (extreme cases) - Or Labour only on QC	-Labour on QC	-Labour on QC
Horizontal transportation		
Equipment type: Automated guided vehicle (AGV); Automated straddles (AutoStrad) also known as automated lifting vehicles (ALVs)		Straddle carriers (SCs); trucks with trailers (Chassis) also known as tractor-trailer unit (TTU); Multi Trailer System trains (MTS); reachstacker; empty/loaded container handler; shuttle carrier(ShC); fork lift;RTG; RMG
Transport cycles and QC allocation: --Dedicated allocation → single cycle - Pooled allocation → Dual cycle mode		
Direction of containers: - QC → storage = No sequence - QC ← storage = Stowage plan to follow		

Labour: -No human operation (remote controllers)		-Human operation: *checkers, tallyman, drivers, Vessel dispatcher, Dockers, etc
Storage yard sub-system		
Ways of storage: - Stacking (typical in Europe and Asia)		- Chassis (Typical in the United States) -Stacking
Stacking equipment: Automated Stacking crane (ASC); Automated rail mounted gantries (ARMG); automated rubber- Tyred gantries (ARTG); Automated straddles (AutoStrad) also known as automated lifting vehicles (ALVs)		Straddle carriers(SCs), Forklifts, reachstacker; top lifter; Overhead bridge crane (OHBC); Chassis*;
Stacking densities: -High		-Low
Yard blocks: -Perpendicular to quay	-Depending on equipment	-parallel to quay

Yard division:		
- Main storage, empty, CFS, Reefer, dangerous, maintenance, repair		
Terminal offered services:		
- Short sea services		
- Deep sea services		
- Empty depot		
Transfer possibilities:		
- Transfer lanes		- Transfer lanes
- Transfer points		
Storage capacity		
Storage space assignment		
Hinterland connection subsystem		
Mode of transport:		
- Truck (street)		
- Train (rail)		
- Barge (waterways)		
- Feeder vessel (waterways)		
Equipment type:		
Gantry cranes; Trucks; trucks with trailers; Multi Trailer System trains (MTS); reach		Fork lifts; trucks with trailers; SC; Multi Trailer System trains (MTS); reachstacker; RTG; RMG

stacker;		
In rail terminals: -When trains and trucks are used * Containers are buffered on trailers alongside rails - When vehicles with lifting capacity are used: *Containers are buffered on containers are buffered on container stacks alongside rails *Containers are directly loaded /unloaded from train		
Labour: -Artificial intelligence (Majority) *Computer controls -Human (minority)		-Artificial intelligence (Minority) *Computer controls -Human (Majority)
Gate procedures -More restrictions -Less restrictions		

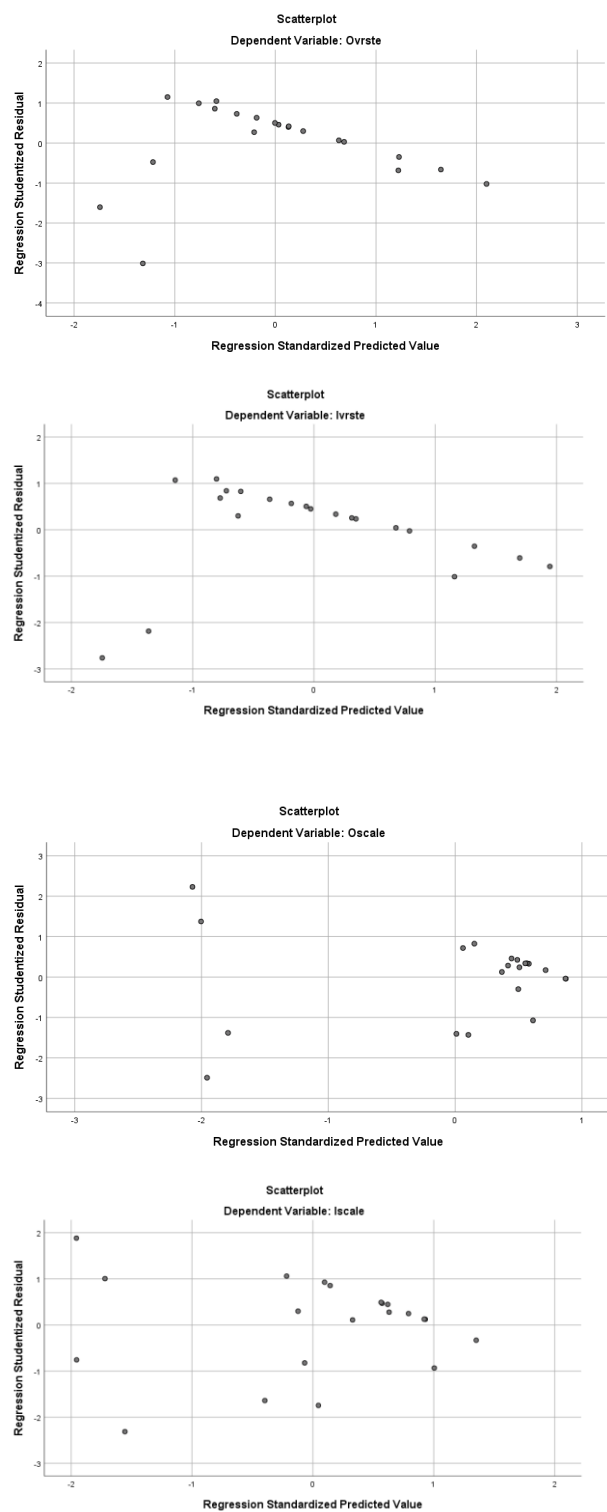
Appendix 6.1: Linearity



Appendix 6.2 :Pearson Correlation

Variable	Automation	Transshipment	Inland Nav	Age
Age				1.00
Inland Nav			1.00	0.17
Transshipment		1.00	0.44	0.15
Automation	1.00	0.68	0.20	0.10
Oscale	0.39	0.33	0.12	0.06
Ovrste	0.19	0.46	0.26	0.18
Ocrste	0.41	0.51	0.24	0.05
Iscale	0.40	0.42	0.16	0.01
Ivrste	0.12	0.38	0.28	0.16
Icrste	0.41	0.51	0.24	0.05

Appendix 6.3: Scatterplot of dependent variables



Appendix 6.4: The normality test

Tests of Normality

Kolmogorov-Smirnov ^a			Shapiro-Wilk		
Statistic	df	Sig.	Statistic	df	Sig.

Ovrste	.42	21	.00	.48	21	.00
Oscale	.26	21	.00	.71	21	.00
Ivrste	.41	21	.00	.44	21	.00
Iscale	.25	21	.00	.77	21	.00

a. Lilliefors Significance Correction

Appendix 6.5 :Summary of container terminal efficiency variables

Higher efficiency	Outside the CTO's control	Within the CTO's control	
	<u>Container trade</u>	<u>Quay side</u>	<u>Stacking</u>
	• High split of FEUs • High transshipment • Stowage concentrated • Larger vessels • Uniform arrival pattern	• Longer quay walls • High usage of quay cranes	• Higher stacking heights
		<u>Horizontal transportation</u>	<u>Hinterlands connections</u>
		• High use of RMGs,AMGs,ASCs	• 24/7 operation

lower efficiency	<u>Container trade</u>	<u>Quay side</u>	<u>Stacking</u>
	• High split of TEUs • High gateway • Stowage dispersed • Smaller vessels • Random arrival pattern	• Shorter quay walls vs large vessels • Lower quay crane use	• Lower stacking height
		<u>Horizontal transportation</u>	<u>Hinterlands connections</u>
		• High use of reach stackers • High use of SC	• Lower than 24/7 operation

